

SECTION 230503 - GENERAL PROVISIONS FOR HVAC WORK**PART 1 - GENERAL****1.1 SUMMARY**

- A. The work under Division 23 "Heating, Ventilating, and Air Conditioning (HVAC)" shall include all labor, services, materials and equipment and performance of all work required for the installation of all mechanical work as shown on the Drawings and herein specified in the following Sections.
- B. Should there be any discrepancies or a question of intent, refer the matter to the Architect/Engineer for decision before ordering any equipment or materials or before starting any related work.
- C. Where work connects to that of another trade, or to piping or equipment in place, take measurements in the field to make connecting work come true and line up with the item being connected.
- D. Where work specified under other Divisions of the Specifications connects to equipment which is a part of Division 23, provide proper connection(s) to such equipment.
- E. Minor items and accessories or devices reasonably inferable as necessary, to the complete and proper installation and operation of any system, shall be provided by the Trade Contractor for such system whether or not they are specifically called for by the Specifications or Drawings.
- F. The Drawings and Specifications are to be taken together. Work specified and not shown or work shown and not specified shall be performed or furnished as though mentioned in both Specifications and Drawings. If there is a discrepancy between the Drawings and Specifications as to the quantity or quality to be provided, the greater quantity or the better quality shall be provided.

1.2 DEFINITIONS

- A. "Piping" includes, in addition to pipe, all fittings, valves, hangers, and other supports and accessories related to such piping.
- B. "Ductwork" includes, in addition to ducts, all fittings, transitions, dampers, hangers and other supports and accessories related to such ductwork.
- C. "Concealed" means hidden from sight in chases, furred spaces, shafts, hung ceilings, embedded in construction, in crawl spaces or buried.
- D. "Exposed" means not installed underground or "concealed" as defined above.
- E. "Invert Elevations" means the elevation of the inside bottom of pipe or duct.
- F. "HVAC Work" is all of the work in Division 23.

1.3 QUALITY ASSURANCE

- A. Each major component of equipment to have the manufacturer's name, address, model number and rating on a plate securely affixed in a conspicuous place.

- B. Code Ratings, labels or other data which are die-stamped or otherwise affixed to the surface of the equipment shall be in visible location.
- C. All equipment provided under Division 23 to perform with the least possible noise and vibration consistent with its duty. Quietness of operation of all equipment is a requirement. Any equipment, as determined by the Owner's Representative or Architect/Engineer to be producing objectionable noise or transmitting noise or vibration to the building to be repaired or removed and replaced.
- D. All workmanship shall be first class in every respect and shall be performed only by skilled mechanics.
- E. Shutdown and Notifications:
 - 1. It is imperative that service interruptions on the various existing utilities be held to an absolute minimum. Wherever possible provide suitable temporary services or connections, where continuity of service for essential systems can be maintained by this means. It will be the Owner's final prerogative to decide which systems are to be considered as essential, and to establish the maximum allowable shutdown time, if any, for each system.
 - 2. Owner will require not less than 72 hours advance notice, in writing, that an interruption of service in any system is desired. Such notice shall identify the system or systems involved, and shall be submitted in duplicate, one copy of which will be signed and returned by the Owner's authorized representative stating whether the requested shutdown will be permitted or not.
- F. Existing Utilities:
 - 1. Location of utilities as shown on the drawings has been determined from the best available information and is given for convenience; however, Owner does not assume responsibility in the event that during construction, utilities other than those shown may be encountered, and that the actual location of those which are shown may be different from the location as shown on the plans.
 - 2. Assume responsibility for interference with or damage to any existing utilities, and repair or replace same with the least possible delay.
- G. Notify Architect of broken or open pipes discovered during construction.
- H. Layout and establish the lines and levels necessary for work.
- I. The following Standards shall be used where referenced by the following abbreviations:
 - 1. AABC: Associated Air Balance Council
 - 2. ADC: Air Diffusion Council
 - 3. AGA: American Gas Association
 - 4. AIA: American Institute of Architects
 - 5. AMCA: Air Moving and Conditioning Association
 - 6. ANSI: American National Standards Institute
 - 7. ARI: Air Conditioning and Refrigeration Institute
 - 8. ASE: Association of Safety Engineers
 - 9. ASHRAE: American Society of Heating, Refrigeration and Air Conditioning Engineers
 - 10. ASME: American Society of Mechanical Engineers
 - 11. ASPE: American Society of Plumbing Engineers
 - 12. ASTM: American Society of Testing and Materials
 - 13. AWPB: American Wood Preserves Bureau
 - 14. AWS: American Welding Society

15. AWWA: American Water Works Association
16. CSA: Canadian Standards Association
17. CISPI: Cast Iron Soil Pipe Institute
18. EIA: Electronic Industries Association
19. EPA: Environmental Protection Agency
20. FDA: Food and Drug Administration
21. FM: Factory Mutual Insurance Association
22. HIS: Hydraulic Institute Standards
23. IRI: Industrial Risk Insurers
24. IBR: Institute of Boiler and Radiator Manufacturers
25. IEEE: Institute of Electrical and Electronics Engineers
26. MCAA: Mechanical Contractors' Association of America
27. NIST: National Institute of Standards and Testing
28. NEBB: National Environmental Balancing Bureau
29. NEC: National Electric Code
30. NECA: National Electric Contractors Association
31. NEMA: National Electrical Manufacturers Association
32. NFPA: National Fire Protection Association
33. NSC: National Safety Council
34. NSF: National Sanitation Foundation
35. OSHA: Occupational Safety & Health Administration
36. SAE: Society of Automotive Engineers
37. SBI: Steel Boiler Institute Industry
38. SMACNA: Sheet Metal and Air Conditioning Contractors National Association
39. TIMA: Thermal Insulation Manufacturers Association
40. UL: Underwriters' Laboratories
41. USDA: United States Department of Agriculture

J. Project Certification:

1. Each trade shall submit a project certification, guaranteeing that this project was constructed and will operate in accordance with the performance requirements of the Drawings and Specifications. This certification shall be signed by a principal of the firm and shall be delivered to the Architect/Engineer prior to final payment.

K. Drawings:

1. The Drawings are essentially diagrammatic in nature and show general arrangement of the equipment, piping, ductwork, accessories, etc. Because of the small scale of the Drawings, it is not possible to show each offsets, fittings, and accessories, which may be required. Carefully investigate the structural conditions, Architectural Drawings, Equipment Drawings, and the finished conditions of the work and arrange such work accordingly, furnish any fittings, pipe accessories that may be required to meet such conditions.
2. Any changes from the plans necessary to make the work conform to building as constructed and to fit work of other trades, or to conform to rules of the governing authorities and regulations, shall be met without extra cost to the Owner.
3. The layout of the piping, ductwork, equipment, etc., as shown on the Drawings shall be checked and exact locations shall be determined by the dimensions of equipment approved and Contractor shall obtain the Architect's approval for revised layout before the apparatus is installed. Consult the Architectural, Structural, and Equipment Drawings for the dimensions, locations of partitions, locations and sizes of structural supports, foundations, etc.
4. Refer to the Architectural Plans for details and large scale Drawings and to approved Shop Drawing of equipment furnished under other Contracts or Sections of the Specifications for exact location of service connections. The equipment Shop Drawings

will be furnished to the Contractor before roughing in. Contractor shall not install any piping or ductwork for said equipment until they have received approved Coordination Drawings for same.

L. Minor Deviations:

1. The dimensions of equipment hereinafter specified or indicated on the Drawings are intended to establish the outlines and characteristics of such equipment in general. Minor deviations in dimensions will be permitted to allow the manufacturers specified to bid on their nearest stock equipment, provided the specified ratings are met or exceeded.
2. Where manufacturers' catalog numbers or types are mentioned in the Specifications or indicated on the Drawings, they are intended to be used as a guide only and shall not be interpreted as taking precedence over the basic rating and duty specified. In all cases, manufacturers shall verify the duty specified with particular characteristics of the equipment they intend to offer for approval and shall also pay the additional charges as may be required under other Divisions.

M. Interferences:

1. Before making any installation, the work of the trades must be coordinated and the necessary changes shall be made to avoid interferences or improper effect on work to be performed by any other Section. In the event that interferences develop, the Architect's decision will be final and no additional compensation will be allowed for moving of misplaced piping, ducts, conduit and/or equipment.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Unless otherwise specified, all material and equipment incorporated in the work under the contract shall be new.
- B. Material and equipment specified by one or more manufacturer's name, trade name and/or model number does not limit a bidder from bidding on other equipment providing the procedure set forth in the Conditions of the Contract and hereinafter specified is followed.
- C. The various mechanical systems have been engineered and designed on equipment name and catalog numbers specified or designated on the Drawings.
- D. A Contractor who intends to furnish equipment listed as approved equal shall proceed as follows:
 1. Obtain Architect/Engineer's approval of said equipment.
 2. Be fully responsible for said equipment.
 3. Include in the Base Bid, all cost for any changes that may be required in his work and/or work of other trades for the proper installation and functioning of said equipment.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Each Trade Contractor shall be responsible for all of his work fitting into place in a satisfactory and neat workmanlike manner acceptable to the Architect/Engineer.

- B. Confer with other Trade Contractors regarding the location and size of pipes, equipment, fixtures, conduit, ducts, openings, switches, outlets, etc., in order that there may be no interference between the installation of the progress of the work of any Trade Contractor on the project. The Architectural Drawings shall take precedence over the Mechanical and Electrical Drawings.
- C. The Mechanical Drawings are diagrammatic and shall be followed as closely as actual construction of the building and the work of other trade contractors will permit. All changes from Drawings necessary to make the work of each Contractor conform to the building construction and the work of other trade contractors shall be done at the appropriate Trade Contractor's expense.
- D. Unless explicitly stated to the contrary, each Trade Contractor shall furnish and install each item of equipment or material hereinafter specified, complete with all necessary fittings, supports, trim, piping, insulation, etc., as required for a complete and operating installation.
- E. All equipment and materials shall be installed according to the manufacturer's instructions unless otherwise specifically directed by the Trade Contract Documents. All piping, valves, connections, and other like items recommended by the manufacturer or required for proper operation shall be provided without additional cost to the Owner.
- F. All references to Contractors in Specifications and Drawings shall refer to the respective Trade Contractor performing that portion of the work.
- G. In general, all piping, ductwork and similar items shall be installed concealed from view above the ceiling, in partitions, shafts, chases, unless otherwise indicated.
- H. Locations of items not definitely fixed by dimensions are approximate only and exact locations necessary to secure the best conditions and results shall be determined at the site, subject to review.
- I. Where pipes are in partitions, furred out spaces and chases, obtain information as to their exact location and size and install work so as to be entirely concealed in the allotted space. If conflicts arise making this impossible, obtain instructions from the Architect/Engineer before proceeding with the work.
- J. Wherever two or more pipes are to be installed in parallel, or parallel to the piping of other trades, the piping shall be installed with sufficient space between pipes to allow for the proper application of pipe covering, painting and servicing.
- K. Furnish advance information on locations and sizes of frames, boxes, sleeves and openings needed for the work, and also furnish information and shop drawings necessary to permit installation of other work without delay.
- L. Where there is evidence that parts of the Mechanical Work will interfere with other work, assist in working out space conditions and/or the structure, make necessary adjustments to accommodate the work.
- M. Mechanical Work installed before coordinating with other work so as to cause interference with other work shall be changed to correct such condition without additional cost to the Owner.
- N. In no case shall any pipe, conduit, duct, or item of equipment be installed where it is supported on or suspended from another pipe, conduit, duct or item of equipment.

- O. Where an item or task is specified to be provided "under this Section," it shall be understood that, that item or task is the responsibility of the trade responsible for that Section, but the work must be performed by qualified workmen of the appropriate trade.
- P. Accessibility:
 - 1. Install Mechanical work to permit removal (without damage to other parts) of coils, heat exchangers, pumps, fan shafts and wheels, belt guards, sheaves and drives, and other parts requiring periodic replacement or maintenance.
 - 2. Arrange pipes, ducts, and equipment to permit ready access to valves, cocks, traps, starters, motors, dampers, control components, and to clear the openings of swinging and overhead doors and of access panels.
 - 3. Change dimensions of ductwork when required to meet job conditions but maintain the same equivalent cross-sectional area. Based on equal friction to airflow.
 - 4. Provide access panels in equipment, ducts, and like items for inspection of interiors and proper maintenance.

END OF SECTION 230503

SECTION 230505 - BASIC HVAC MATERIALS AND METHODS**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Piping materials and installation instructions common to most piping systems.
2. Concrete base construction requirements.
3. Escutcheons.
4. Mechanical sleeve seals.
5. Equipment nameplate data requirements.
6. Nonshrink grout for equipment installations.
7. Field-fabricated metal and wood equipment supports.
8. Installation requirements common to equipment specification sections.
9. Cutting and patching.
10. Touchup painting and finishing.
11. Access Panels
12. Flashings
13. Cleaning

B. Pipe and pipe fitting materials are specified in Division 22.**1.2 DEFINITIONS**

- A. **Finished Spaces:** Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct shafts, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawl spaces, and tunnels.
- B. **Exposed, Interior Installations:** Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. **Exposed, Exterior Installations:** Exposed to view outdoors, or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.
- D. **Concealed, Interior Installations:** Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and in duct shafts.
- E. **Concealed, Exterior Installations:** Concealed from view and protected from weather conditions and physical contact by building occupants, but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.
- F. **The following are industry abbreviations for rubber materials:**
 1. BS: Acrylonitrile-butadiene-styrene plastic.
 2. CPVC: Chlorinated polyvinyl chloride plastic.
 3. CR: Chlorosulfonated polyethylene synthetic rubber.
 4. EPDM: Ethylene-propylene-diene terpolymer rubber.
 5. NBR: Acrylonitrile-butadiene rubber.
 6. PE: Polyethylene plastic.
 7. PVC: Polyvinyl chloride plastic.

1.3 SUBMITTALS

- A. Refer to Division 01 Section “Submittal Procedures” for administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other miscellaneous submittals.
- B. Product Data: For dielectric fittings, flexible connectors, mechanical sleeve seals, access panels and identification materials and devices.
- C. Coordination Drawings: For access panel and door locations.
- D. Coordination Drawings: Detail major elements, components, and systems of mechanical equipment and materials in relationship with other systems, installations, and building components. Show space requirements for installation and access. Indicate if sequence and coordination of installations are important to efficient flow of the Work. Use BIM software for coordination. Include the following:
 - 1. Planned piping layout, including valve and specialty locations and valve-stem movement.
 - 2. Planned duct systems layout, including elbow radii and duct accessories.
 - 3. Clearances for installing and maintaining insulation.
 - 4. Clearances for servicing and maintaining equipment, accessories, and specialties, including space for disassembly required for periodic maintenance.
 - 5. Equipment and accessory service connections and support details.
 - 6. Exterior wall and foundation penetrations.
 - 7. Fire-rated wall and floor penetrations.
 - 8. Sizes and location of required concrete pads and bases.
 - 9. Scheduling, sequencing, movement, and positioning of large equipment into building during construction.
 - 10. Floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
 - 11. Reflected ceiling plans to coordinate and integrate installation of air outlets and inlets, light fixtures, communication system components, sprinklers, and other ceiling-mounted items.
 - 12. Access panel locations in ceilings/walls/floors.

1.4 QUALITY ASSURANCE

- A. Requirements of Regulatory Agencies:
 - 1. American Society for Testing and Materials
 - a. ASTM A 53-98: Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
 - b. ASTM B 32-96: Specification for Solder Metal
 - c. ASTM B 813-93: Specification for Liquid and Paste Fluxes for Soldering Applications of Copper and Copper Alloy Tube
 - d. ASTM B 828-98: Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings
 - e. ASTM C 1107-97: Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
 - f. ASTM C 1173-97: Specification for Flexible Transition Couplings for Underground Piping Systems
 - g. ASTM D 1785-96b: Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120

- h. ASTM D 2235-96a: Specification for Solvent Cement for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe and Fittings
 - i. ASTM D 2564-96a: Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
 - j. ASTM D 2672-96a: Specification for Joints for IPS PVC Pipe Using Solvent Cement
 - k. ASTM D 2855-96: Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings
 - l. ASTM D 3139-98: Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals
 - m. ASTM F 402-93: Practice for Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermostatic Pipe and Fittings
 - n. ASTM F 493-97: Specification for Solvent Cements for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe and Fittings
 - o. ASTM F 656-96a: Specification for Primers for Use in Solvent Cement Joints of Poly(Vinyl Chloride) (PVC) Plastic Pipe and Fittings
2. American Water Works Association
- a. AWWA C110-98: Ductile-Iron and Gray-Iron Fittings, 3 In. through 48 In. (76 mm through 1219 mm), for Water and Other Liquids
 - b. AWWA C219-97: Bolted, Sleeve-Type Couplings for Plain-End Pipe
3. American Welding Society
- a. AWS A5.8-92: Specification for Filler Metals for Brazing and Braze Welding
 - b. AWS D1.1-98: Structural Welding Code--Steel
 - c. AWS D10.12-89: Recommended Practices and Procedures for Welding Low Carbon Steel Pipe
 - d. Brazing Handbook. 1991.
4. ASME International
- a. ASME B1.20.1-83 (Reaffirmed 1992): Pipe Threads, General Purpose (Inch)
 - b. ASME B16.21-92: Nonmetallic Flat Gaskets for Pipe Flanges
 - c. ASME B18.2.1-96: Square and Hex Bolts and Screws--Inch Series
 - d. ASME B31 Series: Code for Pressure Piping
 - e. 1998 ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications"
5. Copper Development Association Inc.
- a. Copper Tube Handbook. 1995.
6. Manufacturers Standardization Society of the Valve and Fittings Industry, Inc.
- a. MSS SP-107-91: Transition Union Fittings for Joining Metal and Plastic Products
- B. Steel Support Welding: Qualify processes and operators according to AWS D1.1, "Structural Welding Code--Steel."
- C. Steel Pipe Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."

1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.

- D. Equipment of higher electrical characteristics, physical dimensions, capacities, and ratings may be furnished provided such proposed equipment is approved in writing and connecting mechanical and electrical services, circuit breakers, conduit, motors, bases, and equipment spaces are increased. Additional costs shall be approved in advance by appropriate Contract Modification for these increases. If minimum energy ratings or efficiencies of equipment are specified, equipment must meet design and commissioning requirements.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and prevent entrance of dirt, debris, and moisture.
- B. Protect stored pipes and tubes from moisture and dirt. Elevate above grade. Do not exceed structural capacity of floor, if stored inside.
- C. Protect flanges, fittings, and piping specialties from moisture and dirt.

1.6 SEQUENCING AND SCHEDULING

- A. Coordinate phasing and sequencing of all work with the Owner and Building Engineer.
- B. Coordinate mechanical equipment installation with other building components.
- C. Arrange for pipe spaces, chases, slots, and openings in building structure during progress of construction to allow for mechanical installations.
- D. Coordinate installation of required supporting devices and sleeves in poured-in-place concrete and other structural components, as they are constructed.
- E. Sequence, coordinate, and integrate installations of mechanical materials and equipment for efficient flow of the Work. Coordinate installation of large equipment requiring positioning before closing in building.
- F. Coordinate connection of mechanical systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies.
- G. Coordinate requirements for access panels and doors if mechanical items requiring access are concealed behind finished surfaces. Access panels and doors are specified in Division 08 Section "Access Doors and Frames."
- H. Coordinate installation of identifying devices after completing covering and painting, if devices are applied to surfaces. Install identifying devices before installing acoustical ceilings and similar concealment.
- I. Coordinate connection of electrical services.

1.7 WARRANTY

- A. Provide warranty on materials and labor for 30 months starting from date of delivery, or two years from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:

- 1. Mechanical Sleeve Seals:

- a. Calpico, Inc.
 - b. Metraflex Co.
 - c. Thunderline/Link-Seal.

- 2. Metal, Flexible Connectors:

- a. Grinnell Corp.; Grinnell Supply Sales Co..
 - b. Mercer Rubber Co.
 - c. Metraflex Co.

- 3. Rubber, Flexible Connectors:

- a. General Rubber Corp.
 - b. Metraflex Co.
 - c. Red Valve Co., Inc.

2.2 PIPE, TUBE, AND FITTINGS

- A. Refer to individual Division 22 piping Sections for pipe, tube, and fitting materials and joining methods.
- B. Pipe Threads: ASME B1.20.1 for factory-threaded pipe and pipe fittings.

2.3 JOINING MATERIALS

- A. Refer to individual Division 22 piping Sections for special joining materials not listed below.
- B. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
 - 1. ASME B16.21, nonmetallic, flat, 1/8-inch maximum thickness unless thickness or specific material is indicated.
 - a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
 - b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.
 - 2. AWWA C110, rubber, flat face, 1/8 inch thick, unless otherwise indicated; and full-face or ring type, unless otherwise indicated.

- C. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.
- D. Plastic, Pipe-Flange Gasket, Bolts, and Nuts: Type and material recommended by piping system manufacturer, unless otherwise indicated.
- E. Solder Filler Metals: ASTM B 32, lead-free alloys. Include water-flushable flux according to ASTM B 813.
- F. Brazing Filler Metals: AWS A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing, unless otherwise indicated; and AWS A5.8, BAg1, silver alloy for refrigerant piping, unless otherwise indicated.
- G. Welding Filler Metals: Comply with AWS D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.
- H. Solvent Cements for Joining Plastic Piping:
 - 1. PVC Piping: ASTM D 2564. Include primer according to ASTM F 656.

2.4 MECHANICAL SLEEVE SEALS

- A. Description: Modular sealing element unit, designed for field assembly, to fill annular space between pipe and sleeve.
 - 1. Sealing Elements: EPDM interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 - 2. Pressure Plates: Carbon steel. Include two for each sealing element.
 - 3. Connecting Bolts and Nuts: Carbon steel with corrosion-resistant coating of length required to secure pressure plates to sealing elements. Include one for each sealing element.

2.5 FLEXIBLE CONNECTORS

- A. General: Fabricated from materials suitable for system fluid and that will provide flexible pipe connections. Include 125-psig minimum working-pressure rating, unless higher working pressure is indicated, and ends according to the following:
 - 1. 2-Inch NPS and Smaller: Threaded.
 - 2. 2-1/2-Inch NPS and Larger: Flanged.
 - 3. Option for 2-1/2-Inch NPS and Larger: Grooved for use with keyed couplings.

2.6 SLEEVES

- A. Galvanized-Steel Sheet: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.
- B. Steel Pipe: ASTM A 53, Type E, Grade B, Schedule 40, galvanized, plain ends.
- C. Cast Iron: Cast or fabricated "wall pipe" equivalent to ductile-iron pressure pipe, with plain ends and integral water-stop, unless otherwise indicated.
- D. Stack Sleeve Fittings: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring and bolts and nuts for membrane flashing.

1. Under-deck Clamp: Clamping ring with set screws.

2.7 ESCUTCHEONS

- A. Description: Manufactured wall and ceiling escutcheons and floor plates, with an ID to closely fit around pipe, tube, and insulation of insulated piping and an OD that completely covers opening.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with polished chrome-plated finish.
- C. One-Piece, Cast-Brass Type: With set screw.
 1. Finish: Rough brass.
- D. Split-Casting, Cast-Brass Type: With concealed hinge and set screw.
 1. Finish: Rough brass.
- E. One-Piece, Stamped-Steel Type: With set screw and chrome-plated finish.
- F. Split-Plate, Stamped-Steel Type: With concealed hinge, set screw, and chrome-plated finish.
- G. One-Piece, Floor-Plate Type: Cast-iron floor plate.
- H. Split-Casting, Floor-Plate Type: Cast brass with concealed hinge and set screw.

2.8 ACCESS PANELS

- A. General: Refer to Division 08 Section, "Access Doors and Frames," for access panel manufacturers and other requirements.

2.9 GROUT

- A. Description: ASTM C 1107, Grade B, nonshrink and nonmetallic, dry hydraulic-cement grout.
 1. Characteristics: Post-hardening, volume-adjusting, nonstaining, noncorrosive, nongaseous, and recommended for interior and exterior applications.
 2. Design Mix: 5000-psi, 28-day compressive strength.
 3. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 PIPING SYSTEMS - COMMON REQUIREMENTS

- A. Install piping according to the following requirements and Division 22 Sections specifying piping systems.
- B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Coordination Drawings.

- C. Install piping in concealed locations, unless otherwise indicated and except in equipment rooms and service areas.
- D. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- E. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- F. Install piping to permit valve servicing.
- G. Install piping at indicated slopes.
- H. Install piping free of sags and bends.
- I. Install fittings for changes in direction and branch connections.
- J. Install piping to allow application of insulation.
- K. Select system components with pressure rating equal to or greater than system operating pressure.
- L. Install escutcheons for penetrations of walls, ceilings, and floors according to the following:
 - 1. New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.
 - b. Chrome-Plated Piping: One-piece, cast-brass type with polished chrome-plated finish.
 - c. Insulated Piping: One-piece, stamped-steel type with spring clips.
 - d. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - e. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type and set screw.
 - f. Bare Piping in Unfinished Service Spaces: One-piece, stamped-steel type with concealed hinge and set screw.
 - g. Bare Piping in Equipment Rooms: One-piece, stamped-steel type with set screw.
 - h. Bare Piping at Floor Penetrations in Equipment Rooms: One-piece, floor-plate type.
- M. Sleeves are not required for core-drilled holes.
- N. Permanent sleeves are not required for holes formed by removable PE sleeves.
- O. Install sleeves for pipes passing through concrete and masonry walls, gypsum-board partitions, and concrete floor and roof slabs.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
 - 2. Install sleeves in new walls and slabs as new walls and slabs are constructed.

3. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation. Use the following sleeve materials:
 - a. Steel Pipe Sleeves: For pipes smaller than NPS 6.
 - b. Steel Sheet Sleeves: For pipes NPS 6 and larger, penetrating gypsum-board partitions.
 - c. Stack Sleeve Fittings: For pipes penetrating floors with membrane waterproofing. Secure flashing between clamping flanges. Install section of cast-iron soil pipe to extend sleeve to 2 inches above finished floor level. Refer to Division 07 Section "Sheet Metal Flashing and Trim" for flashing.
 - 1) Seal space outside of sleeve fittings with grout.
 4. Except for underground wall penetrations, seal annular space between sleeve and pipe or pipe insulation, using joint sealants appropriate for size, depth, and location of joint. Refer to Division 07 Section "Joint Sealants" for materials and installation.
- P. Aboveground, Exterior-Wall Pipe Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.
1. Install steel pipe for sleeves smaller than 6 inches in diameter.
 2. Install cast-iron "wall pipes" for sleeves 6 inches and larger in diameter.
 3. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
- Q. Underground, Exterior-Wall Pipe Penetrations: Install cast-iron "wall pipes" for sleeves. Seal pipe penetrations using mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
1. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
- R. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Refer to Division 07 Section "Penetration Firestopping" for materials.
- S. Verify final equipment locations for roughing-in.
- 3.2 PIPING JOINT CONSTRUCTION
- A. Join pipe and fittings according to the following requirements and Division 22 Sections specifying piping systems.
 - B. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
 - C. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

- D. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
1. Note internal length of threads in fittings or valve ends, and proximity of internal seat or wall, to determine how far pipe should be threaded into joint.
 2. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
 3. Align threads at point of assembly.
 4. Tighten joint with wrench. Apply wrench to valve end into which pipe is being threaded.
 5. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Welded Joints: Construct joints according to AWS D10.12, using qualified processes and welding operators according to Part 1 "Quality Assurance" Article.
- F. Plastic Piping Solvent-Cement Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:
1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
 2. PVC Pressure Piping: Join schedule number ASTM D 1785, PVC pipe and PVC socket fittings according to ASTM D 2672. Join other-than-schedule-number PVC pipe and socket fittings according to ASTM D 2855.
 3. PVC Non-pressure Piping: Join according to ASTM D 2855.
- G. Plastic Pressure Piping Gasketed Joints: Join according to ASTM D 3139.
- H. Plastic Non-pressure Piping Gasketed Joints: Join according to ASTM D 3212.
- I. PE Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D 2657.
1. Plain-End Pipe and Fittings: Use butt fusion.
 2. Plain-End Pipe and Socket Fittings: Use socket fusion.

3.3 PIPING CONNECTIONS

- A. Make connections according to the following, unless otherwise indicated:
1. Install unions, in piping NPS 2 and smaller, adjacent to each valve and at final connection to each piece of equipment.
 2. Install flanges, in piping NPS 2-1/2 and larger, adjacent to flanged valves and at final connection to each piece of equipment.
 3. Dry Piping Systems: Install dielectric unions and flanges to connect piping materials of dissimilar metals.
 4. Wet Piping Systems: Install dielectric coupling and nipple fittings to connect piping materials of dissimilar metals

3.4 EQUIPMENT INSTALLATION - COMMON REQUIREMENTS

- A. Install equipment to provide maximum possible headroom, if mounting heights are not indicated.

- B. Install equipment according to approved submittal data. Portions of the Work are shown only in diagrammatic form. Refer conflicts to Architect.
- C. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, unless otherwise indicated.
- D. Install mechanical equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference to other installations. Extend grease fittings to accessible locations.
- E. Install equipment giving right of way to piping installed at required slope.
- F. Install flexible connectors on equipment side of shutoff valves, horizontally and parallel to equipment shafts if possible.

3.5 PAINTING AND FINISHING

- A. Refer to Division 09 Section "Painting" for paint materials, surface preparation, and application of paint.
- B. Do not paint piping specialties with factory-applied finish.
- C. Damage and Touchup: Repair marred and damaged factory-painted finishes with materials and procedures to match original factory finish.

3.6 CUTTING AND PATCHING

- A. Cut, channel, chase, and drill floors, walls, partitions, ceilings, and other surfaces necessary for mechanical installations. Perform cutting by skilled mechanics of trades involved.
- B. Repair cut surfaces to match adjacent surfaces.

3.7 GROUTING

- A. Install nonmetallic, nonshrink, grout for mechanical equipment base bearing surfaces, pump and other equipment base plates, and anchors. Mix grout according to manufacturer's written instructions.
- B. Clean surfaces that will come into contact with grout.
- C. Provide forms as required for placement of grout.
- D. Avoid air entrapment during placing of grout.
- E. Place grout, completely filling equipment bases.
- F. Place grout on concrete bases to provide smooth bearing surface for equipment.
- G. Place grout around anchors.
- H. Cure placed grout according to manufacturer's written instructions.

3.8 CLEANING

- A. Coordinate general cleanup with the work as specified in Division 1.

3.9 ACCESS PANELS

- A. Where control valves, shut-off valves, drip traps, heating coils, dampers, pull boxes or other specialties, which require service or adjustment, are installed above inaccessible type furred ceilings or within furred walls, the Trade Contractor whose equipment is involved shall furnish and install access panels as required.
- B. Each Trade shall confer with other trades with respect to access panel locations, and shall wherever practical group valves, traps, dampers, etc. in such a way as to be accessible from a single panel and eliminate as many access panels as possible.

3.10 ERECTION

- A. Provide all necessary rigging, scaffolding, tools, tackle, labor and other like items necessary for the complete installation of the equipment.
- B. Adapt his work to job conditions and install his work to clear beams, joists and light fixtures, adjusting risers, avoiding interferences with windows and openings, raising or lowering work to permit the passing of ductwork or the work of other trades, all as required or as job conditions dictate, without additional costs to the Owner.
- C. Trade Contractor shall not rig, tie to, or rest weight upon any part of the building or make use of any stairway until specific permission is obtained.
 - 1. Permission to rig to or make use of any part of the building premises shall not relieve the contractor of responsibility for any damage.
- D. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor mechanical materials and equipment.
- E. Field Welding: Comply with AWS D1.1, "Structural Welding Code--Steel."

3.11 ERECTION OF WOOD SUPPORTS AND ANCHORAGE

- A. Cut, fit, and place wood grounds, nailers, blocking, and anchorage to support and anchor mechanical materials and equipment.
- B. Select fastener sizes that will not penetrate members if opposite side will be exposed to view or will receive finish materials. Tighten connections between members. Install fasteners without splitting wood members.
- C. Attach to substrates as required to support applied loads.

3.12 FLASHINGS

- A. Openings in roof shall be flashed with 40 lb. sheet lead, in one piece, extending 24" under roofing in all directions from drains, pipes, or ducts, with upper edge at least 8" above the roof and turned over and down into increaser or sleeve or over curb for at least 1" and tightly bound with nonferrous wire.

1. All lead flashings shall be entirely painted with a good coat of black asphaltum before installation.
2. Roof flashing must be approved in writing by the Roofing Trade contractor to the effect that such flashing will not void guarantees or bonds pertaining to the roofing contract.
3. Coordinate with the work under Division 07.

3.13 OPENINGS

- A. Where temporary openings are necessary thru walls and partitions of the building for the entry or installation of tanks, fans, or other machinery or apparatus, or for driveways and other facilities, the permanent work of the mechanical trades at said openings shall be temporarily omitted and installed after equipment is brought into the building or after temporary facilities are removed.
- B. Refer to other Sections of the Specifications for framing of openings for ducts, grilles, registers, etc., in walls, partitions, floors, roofs, etc. The trade for each service shall be responsible for locating and providing the proper dimensions for all required openings.
 1. Space between ducts and wall or floor openings shall be sealed as specified in Division 07 Section "Penetration Firestopping."
- C. No cutting or drilling of any building structural members will be permitted, unless the specific extent and limits are approved, in writing, by the Architect.
- D. All openings in the existing structure shall be core drilled with a diamond drill. The use of jackhammers will not be permitted.
- E. The Contractor shall notify the Architect if any existing openings are uncovered adjacent to location of a new opening. The new opening shall not be drilled if the existing can be used unless directed by the Architect.

END OF SECTION 230505

SECTION 230513 - COMMON MOTOR REQUIREMENTS FOR HVAC EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes basic requirements for factory- and field-installed motors.

1.2 DEFINITIONS

- A. **Factory-Installed Motor:** A motor installed by motorized-equipment manufacturer as a component of equipment.

1.3 SUBMITTALS

- A. **Product Data for Field-Installed Motors:** For each type and size of motor, provide nameplate data and ratings; shipping, installed, and operating weights; enclosure type and mounting arrangements; size, type, and location of winding terminations; conduit entry and ground lug locations; and information on coatings or finishes.
- B. **Qualification Data:** For testing agency.
- C. **Source quality-control test reports.**
- D. **Training Reports:** Submit reports on training documenting dates and attendance.

1.4 QUALITY ASSURANCE

- A. **Testing Agency Qualifications:** An independent agency, with the experience and capability to conduct the testing indicated, that is a member company of the InterNational Electrical Testing Association or is a nationally recognized testing laboratory (NRTL) as defined by OSHA in 29 CFR 1910.7, and that is acceptable to authorities having jurisdiction.
 - 1. **Testing Agency's Field Supervisor:** Person currently certified by the InterNational Electrical Testing Association or the National Institute for Certification in Engineering Technologies to supervise on-site testing specified in PART 3.
- B. **Electrical Components, Devices, and Accessories:** Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. **Comply with NFPA 70.**

1.5 DELIVERY, STORAGE AND HANDLING

- A. Follow manufacturer's instructions for storage and handling of motors.

1.6 COORDINATION

- A. **Coordinate features of motors, installed units, and accessory devices and features that comply with the following:**
 - 1. **Compatible with the following:**
 - a. **Magnetic controllers.**

- b. Multispeed controllers.
 - c. Reduced-voltage controllers.
 - 2. Designed and labeled for use with variable frequency controllers, and suitable for use throughout speed range without overheating.
 - 3. Matched to torque and horsepower requirements of the load.
 - 4. Matched to ratings and characteristics of supply circuit and required control sequence.
 - B. Coordinate motor support with requirements for driven load; access for maintenance and motor replacement; installation of accessories, belts, belt guards; and adjustment of sliding rails for belt tensioning.
 - C. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.
- 1.7 WARRANTY
- A. Written manufacturer's warranty covering parts and labor for a period of one year from substantial completion, or eighteen months from shipment, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Motors:
 - a. Baldor.
 - b. General Electric.
 - c. Lincoln Motors.
 - d. Marathon.
 - e. Reliance.
 - f. U.S. Motors.

2.2 MOTOR REQUIREMENTS

- A. Motor requirements apply to factory- and field-installed motors except as follows:
 - 1. Different ratings, performance, or characteristics for motor are specified in another Section.
 - 2. Motorized-equipment manufacturer requires ratings, performance, or characteristics, other than those specified in this Section, to meet performance specified.

2.3 MOTOR CHARACTERISTICS

- A. Motors 1/2 HP and Larger: Three phase.
- B. Motors Smaller Than 1/2 HP: Single phase.
- C. Frequency Rating: 60 Hz.

- D. Voltage Rating: NEMA standard voltage selected to operate on nominal circuit voltage to which motor is connected. 208V motors shall be rated for continuous operation at 200V.
- E. Service Factor: 1.15 for open dripproof motors; 1.0 for totally enclosed motors.
- F. Duty: Continuous duty at ambient temperature of 105 deg F and at altitude of 3300 feet above sea level.
- G. Capacity and Torque Characteristics: Sufficient to start, accelerate, and operate connected loads at designated speeds, at installed altitude and environment, with indicated operating sequence, and without exceeding nameplate ratings or considering service factor.
- H. Enclosure: Open dripproof.

2.4 POLYPHASE MOTORS

- A. Description: NEMA MG 1, Design B, medium induction motor.
- B. Efficiency: Premium, as defined in NEMA MG 1.
- C. Stator: Copper windings, unless otherwise indicated.
 - 1. Multispeed motors shall have separate winding for each speed.
- D. Rotor: Squirrel cage, unless otherwise indicated.
- E. Bearings: Double-shielded, pre-lubricated ball bearings suitable for radial and thrust loading.
- F. Temperature Rise: Match insulation rating, unless otherwise indicated.
- G. Insulation: Class F, unless otherwise indicated.
- H. Code Letter Designation:
 - 1. Motors 15 HP and Larger: NEMA starting Code F or Code G.
 - 2. Motors Smaller Than 15 HP: Manufacturer's standard starting characteristic.
- I. Enclosure: Cast iron for motors 7.5 hp and larger; rolled steel for motors smaller than 7.5 hp.
 - 1. Finish: Gray enamel.

2.5 POLYPHASE MOTORS WITH ADDITIONAL REQUIREMENTS

- A. Motors Used with Reduced-Inrush Controllers: Match wiring connection requirements for controller with required motor leads. Provide terminals in motor terminal box, suited to control method.
- B. Motors Used with Variable Frequency Controllers: Motors shall be inverter-duty. Ratings, characteristics, and features coordinated with and approved by controller manufacturer.
 - 1. Designed with critical vibration frequencies outside operating range of controller output.
 - 2. Temperature Rise: Matched to rating for Class B insulation.
 - 3. Insulation: Class F.

4. Thermal Protection: Comply with NEMA MG 1 requirements for thermally protected motors.
 5. Provide Aegis shaft grounding rings.
- C. Rugged-Duty Motors: Totally enclosed, with 1.25 minimum service factor, greased bearings, integral condensate drains, and capped relief vents. Windings insulated with non-hygroscopic material.
1. Finish: Chemical-resistant paint over corrosion-resistant primer.
- D. Source Quality Control for Field-Installed Motors: Perform the following tests on each motor according to NEMA MG 1:
1. Measure winding resistance.
 2. Read no-load current and speed at rated voltage and frequency.
 3. Measure locked rotor current at rated frequency.
 4. Perform high-potential test.

2.6 SINGLE-PHASE MOTORS

- A. Type: One of the following, to suit starting torque and requirements of specific motor application:
1. Permanent-split capacitor.
 2. Split-phase start, capacitor run.
 3. Capacitor start, capacitor run.
- B. Shaded-Pole Motors: For motors 1/20 hp and smaller only.
- C. Thermal Protection: Internal protection to automatically open power supply circuit to motor when winding temperature exceeds a safe value calibrated to temperature rating of motor insulation. Thermal-protection device shall automatically reset when motor temperature returns to normal range.
- D. Bearings: Ball type for belt-connected motors and other motors with high radial forces on motor shaft; sealed, pre-lubricated-sleeve type for other single-phase motors.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for conduit systems to verify actual locations of conduit connections before motor installation.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.
- D. Perform the following field tests and inspections and prepare test reports:

1. Perform electrical tests and visual and mechanical inspections including optional tests and inspections stated in NETA ATS on factory- and field-installed motors. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3.2 CLEANING

- A. Comply with applicable requirements in Division 23 Section "HVAC Equipment Cleaning."

3.3 CONTRACTOR STARTUP AND REPORTING

- A. Prepare for acceptance tests.

1. Align motors, bases, shafts, pulleys, and belts. Tension belts according to manufacturer's written instructions.
2. Verify bearing lubrication.
3. Run each motor with its controller. Demonstrate correct rotation, alignment, and speed at motor design load.
4. Test interlocks and control and safety features for proper operation.
5. Verify that current and voltage for each phase comply with nameplate rating and NEMA MG 1 tolerances.

- B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.

- C. Perform the following field tests and inspections and prepare test reports:

1. Perform electrical tests and visual and mechanical inspections including optional tests and inspections stated in NETA ATS on factory- and field-installed motors. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3.4 DEMONSTRATION AND COMMISSIONING

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain field-installed motors. Refer to Division 01 Section "Demonstration and Training."

1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining chillers. The training will occur after the startup report has been provided to the owner and the trainer will provide two (2) Installation and Operations manuals for the use of the owners personnel during training.
2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
3. Schedule training with Owner, through Architect, with at least seven days' advance notice.

- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements

under both local and building automation control, as well as any commissioning requirements in Division 01 or 23.

END OF SECTION 230513

SECTION 230517 - SLEEVES AND SLEEVE SEALS FOR HVAC PIPING**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Sleeves.
2. Sleeve-seal systems.
3. Grout.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product indicated.**PART 2 - PRODUCTS****2.1 SLEEVES**

- A. Cast-Iron Wall Pipes: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.
- B. Galvanized-Steel Wall Pipes: ASTM A 53/A 53M, Schedule 40, with plain ends and welded steel collar; zinc coated.
- C. Galvanized-Steel-Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.
- D. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- E. Galvanized-Steel-Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.

2.2 SLEEVE-SEAL SYSTEMS

- A. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.
 1. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 2. Pressure Plates: Plastic.
 3. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2.3 GROUT

- A. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

- B. Characteristics: Nonshrink; recommended for interior and exterior applications.
- C. Design Mix: 5000-psi, 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION

- A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.
- B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.
 - 1. Sleeves are not required for core-drilled holes.
- C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.
 - 2. Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.
- D. Install sleeves for pipes passing through interior partitions.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - 2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
 - 3. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint. Comply with requirements for sealants specified in Section 079200 "Joint Sealants."
- E. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

- A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building.
- B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.3 SLEEVE AND SLEEVE-SEAL SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:

1. Exterior Concrete Walls above Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves.
2. Exterior Concrete Walls below Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
3. Concrete Slabs-on-Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
4. Concrete Slabs above Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel-pipe sleeves.
5. Interior Partitions:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel-sheet sleeves.

END OF SECTION 230517

SECTION 230518 - ESCUTCHEONS FOR HVAC PIPING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section Includes:
 - 1. Escutcheons.
 - 2. Floor plates.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS**2.1 ESCUTCHEONS**

- A. One-Piece, Cast-Brass Type: With polished, chrome-plated finish and setscrew fastener.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with chrome-plated finish and spring-clip fasteners.
- C. One-Piece, Stamped-Steel Type: With chrome-plated finish and spring-clip fasteners.

2.2 FLOOR PLATES

- A. One-Piece Floor Plates: Cast-iron flange.

PART 3 - EXECUTION**3.2 INSTALLATION**

- A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.
- B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
 - 1. Escutcheons for New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.

- b. Chrome-Plated Piping: One-piece, cast-brass type with polished, chrome-plated finish.
 - c. Insulated Piping: One-piece, stamped-steel type.
 - d. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, cast- brass type with polished, chrome-plated finish.
 - e. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - f. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, cast-brass type with polished, chrome-plated finish.
 - g. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - h. Bare Piping in Unfinished Service Spaces: One-piece, cast-brass type with polished, chrome-plated finish.
 - i. Bare Piping in Unfinished Service Spaces: One-piece, stamped-steel type.
 - j. Bare Piping in Equipment Rooms: One-piece, cast-brass type with polished, chrome-plated finish.
 - k. Bare Piping in Equipment Rooms: One-piece, stamped-steel type.
- C. Install floor plates for piping penetrations of equipment-room floors.
- D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
1. New Piping: One-piece, floor-plate type.

3.3 FIELD QUALITY CONTROL

- A. Replace broken and damaged escutcheons and floor plates using new materials.

END OF SECTION 230518

SECTION 230529 - HANGERS AND SUPPORTS FOR PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following hangers and supports for mechanical system piping and equipment:
 - 1. Steel pipe hangers and supports.
 - 2. Trapeze pipe hangers.
 - 3. Metal framing systems.
 - 4. Thermal-hanger shield inserts.
 - 5. Fastener systems.
 - 6. Rooftop Pipe Supports.
 - 7. Pipe positioning systems.

1.2 DEFINITIONS

- A. MSS: Manufacturers Standardization Society for the Valve and Fittings Industry Inc.
- B. Terminology: As defined in MSS SP-90, "Guidelines on Terminology for Pipe Hangers and Supports."

1.3 SUBMITTALS

- A. Product Data: For the following:
 - 1. Steel pipe hangers and supports.
 - 2. Thermal-hanger shield inserts.
 - 3. Powder-actuated fastener systems.
 - 4. Pipe positioning systems.
- B. Shop Drawings: Signed and sealed by a qualified professional engineer. Show fabrication and installation details and include calculations for the following:
 - 1. Trapeze pipe hangers. Include Product Data for components.
 - 2. Metal framing systems. Include Product Data for components.
 - 3. Rooftop Pipe Supports. Include Product Data for components.
 - 4. Equipment supports.

- C. Welding certificates.

1.4 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code-Steel." and ASME Boiler and Pressure Vessel Code: Section IX.

1.5 PERFORMANCE REQUIREMENTS

- A. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:

1. Steel Pipe Hangers and Supports:
 - a. Anvil
 - b. B-Line Systems, Inc.; a division of Cooper Industries.
 - c. Carpenter & Paterson, Inc.
2. Powder-Actuated Fasteners:
 - a. Hilti, Inc.
 - b. ITW Ramset/Red Head.
3. Mechanical-Expansion Anchors:
 - a. B-Line Systems, Inc.; a division of Cooper Industries.
 - b. Hilti, Inc.
 - c. ITW Ramset/Red Head.
4. Rooftop Pipe Supports:
 - a. Pate
 - b. RPS
 - c. Thybar
5. Thermal-Hanger Shield Inserts:
 - a. Carpenter & Paterson, Inc.
 - b. PHS Industries, Inc.
 - c. Pipe Shields, Inc.
6. Pipe Positioning Systems:
 - a. C & S Mfg. Corp.
 - b. HOLDRITE Corp.; Hubbard Enterprises.
 - c. Samco Stamping, Inc.

2.2 STEEL PIPE HANGERS AND SUPPORTS

- A. Description: MSS SP-58, Types 1 through 58, factory-fabricated components. Refer to PART 3 "Hanger and Support Applications" Article for where to use specific hanger and support types.

- B. Corrosion Protection: Hangers and components shall be galvanized or painted with carbo-zinc #11.
- C. Threads: All threads shall be UNC unless otherwise specified.
- D. Heat Transmission: Supports, guides and anchors shall limit the amount of heat transmitted to the structural steel. Temperature of supporting parts shall be based on a 100°F per inch temperature gradient from the outside pipe surface.

2.3 TRAPEZE PIPE HANGERS

- A. Description: MSS SP-69, Type 59, shop- or field-fabricated pipe-support assembly made from structural-steel shapes with MSS SP-58 hanger rods, nuts, saddles, and U-bolts.

2.4 FASTENER SYSTEMS

- A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.
- B. Mechanical-Expansion Anchors: Insert-wedge-type zinc-coated steel, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2.5 ROOFTOP PIPE SUPPORTS

- A. Description: Provide factory-fabricated steel sheet structural mounting supports. Construction shall include a welded 18-gauge galvanized steel shell, base plate and removable counterflashing. Support shall include a factory-installed wood nailer and internal bulkhead reinforcement. End sections shall be fully mitered. Support shall be compatible with insulated roof decks and include a 3 inch cant and variable step to match deck insulation thickness. The pipe roller assembly shall have galvanized 18 inches long continuous threaded rods to permit 12 inch vertical adjustments and a galvanized removable pipe retainer bracket.

2.6 THERMAL-HANGER SHIELD INSERTS

- A. Description: 100 psig minimum, compressive-strength insulation insert encased in sheet metal shield.
- B. Insulation-Insert Material for Cold Piping: Water-repellent treated, ASTM C 533, Type I calcium silicate with vapor barrier.
- C. Insulation-Insert Material for Hot Piping: Water-repellent treated, ASTM C 533, Type I calcium silicate.
- D. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.
- E. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.
- F. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.7 PIPE POSITIONING SYSTEMS

- A. Description: IAPMO PS 42, system of metal brackets, clips, and straps for positioning piping in pipe spaces for plumbing fixtures for commercial applications.

2.8 MISCELLANEOUS MATERIALS

- A. Structural Steel: ASTM A 36/A 36M, steel plates, shapes, and bars; black and galvanized.
- B. Grout: ASTM C 1107, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.
 - 1. Properties: Nonstaining, noncorrosive, and nongaseous.
 - 2. Design Mix: 5000-psi, 28-day compressive strength.

PART 3 - EXECUTION

3.1 HANGER AND SUPPORT APPLICATIONS

- A. Specific hanger and support requirements are specified in Sections specifying piping systems and equipment.
- B. Comply with MSS SP-69 for pipe hanger selections and applications that are not specified in piping system Sections.
- C. Use hangers and supports with galvanized, metallic coatings for piping and equipment that will not have field-applied finish. Hangers installed outdoors shall have two coats of rust inhibitor paint after installation and adjustment.
- D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.
- E. Use padded hangers for piping that is subject to scratching.
- F. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
 - 1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of non-insulated or insulated stationary pipes, NPS 1/2 to NPS 30.
 - 2. U-Bolts (MSS Type 24): For support of heavy pipes, NPS 1/2 to NPS 30.
 - 3. Single Pipe Rolls (MSS Type 41): For suspension of pipes, NPS 1 to NPS 30, from 2 rods if longitudinal movement caused by expansion and contraction might occur.
 - 4. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes, NPS 2-1/2 to NPS 20, from single rod if horizontal movement caused by expansion and contraction might occur.
- G. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
 - 1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers, NPS 3/4 to NPS 20.

- H. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.
 2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
 3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11, split pipe rings.
 4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
 5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.
- I. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
 2. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
 3. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
 4. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
 5. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
 6. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
 7. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
 8. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
 9. Malleable Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
 10. Welded-Steel Brackets: For support of pipes from below, or for suspending from above by using clip and rod. Use one of the following for indicated loads:
 - a. Light (MSS Type 31): 750 lb (340 kg).
 - b. Medium (MSS Type 32): 1500 lb (680 kg).
 - c. Heavy (MSS Type 33): 3000 lb (1360 kg).
 11. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
 12. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
 13. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.
- J. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
 2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
 3. Thermal-Hanger Shield Inserts: For supporting insulated pipe.
- K. Comply with MSS SP-69 for trapeze pipe hanger selections and applications that are not specified in piping system Sections.

- L. Comply with MFMA-102 for metal framing system selections and applications that are not specified in piping system Sections.
- M. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.
- N. Use pipe positioning systems in pipe spaces behind plumbing fixtures to support supply and waste piping for plumbing fixtures.

3.2 HANGER AND SUPPORT INSTALLATION

- A. Steel Pipe Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments as required to properly support piping from building structure.
- B. Trapeze Pipe Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping and support together on field-fabricated trapeze pipe hangers.
 - 1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified above for individual pipe hangers.
 - 2. Field fabricate from ASTM A 36/A 36M, steel shapes selected for loads being supported. Weld steel according to AWS D1.1.
 - 3. Trapeze bars shall be tightly secured to structural members at two points with bolts or other similar mechanical fasteners. Hangers from bar joist and fabricated truss members shall be located at the panel points of the structural members. C-clamp type hangers attached to one side of double-angle bottom members are not allowed. Point loads shall not exceed the lesser of:
 - a. Manufacturer's certified recommendation for the component parts.
 - b. The following maximum point loads, and maximum hanger spacings as herein specified, for structural elements in any direction; except as specifically approved by the Structural Engineer of Record:

Structural Element Type	Maximum Hanger Point Load (lb)
Metal deck without concrete topping	50
Composite metal deck slab with concrete topping	50
Steel Beams:	
All channels, W4 through W8	100
W10 through W14	200
W16 through W24	400
W27 through W36	750
Built-up structural steel trusses	250

Reinforced post-tensioned concrete elements

Slabs up to 6 inches thick	150
Slabs over 6 inches thick	250
Joists 8 inches wide	250
Beam/girders 8 inches wide	500

C. Fastener System Installation:

1. Drive screws, pins, studs, etc., of the type which are secured in place by means of explosive force may be used as a means of securing any of the hangers subject to the following:
 - a. The stud, pin or fastener shall be caused to have a velocity not in excess of 300 feet per second when measured 6-1/2 feet from the muzzle of the tool by accepted ballistic test methods.
 - b. Only workmen qualified by instructions of the manufacturers representative and/or licensed by the state and local authorities shall be assigned to use a powder actuated fastening tool.
 - c. Where practical, tools of only one manufacturer shall be used on a project.
 - d. Only cartridges and fasteners supplied by the manufacturer of the tool shall be used to operate that tool.
 - e. Powder actuated fastening tools shall be handled with the same care as firearms.
 - f. All safety devices incorporated in the tool by the manufacturer shall be used at all times.
 - g. Acceptable types of powder actuated fastening tools are:
 - 1) Piston Tool - Low Velocity Type - is a tool utilizing a piston, activated by the power of a blank cartridge furnished by the manufacturer for use with it, to drive a stud, pin or fastener into a work surface.
 - 2) Powder Assisted Hammer Drive Tool - Low Velocity Type - is a tool utilizing a captive piston, activated by a blow from a 4 lb. hammer supplemented by the power of a blank cartridge furnished by the manufacturer for use with the tool, to drive a stud, pin or fastener into a work surface.
2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

- D. Pipe Positioning System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture. Refer to Division 22 Section "Plumbing Fixtures" for plumbing fixtures.
- E. Install hangers and supports complete with necessary inserts, bolts, rods, nuts, washers, and other accessories.
- F. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

- G. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.
- H. Install lateral bracing with pipe hangers and supports to prevent swaying.
- I. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.
- J. Load Distribution: Install hangers and supports so piping live and dead loads and stresses from movement will not be transmitted to connected equipment.
- K. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and so maximum pipe deflections allowed by ASME B31.1 (for power piping) and ASME B31.9 (for building services piping) are not exceeded.
- L. Insulated Piping: Comply with the following:
 - 1. Attach clamps and spacers to piping.
 - a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.
 - b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.
 - c. Do not exceed pipe stress limits according to ASME B31.1 for power piping and ASME B31.9 for building services piping.
 - 2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
 - 3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
 - 4. Shield Dimensions for Pipe: Not less than the following:
 - a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
 - b. NPS 4: 12 inches long and 0.06 inch thick.
 - c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.
 - d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.
 - 5. Pipes NPS 8 and Larger: Include wood inserts.
 - 6. Insert Material: Length at least as long as protective shield.
 - 7. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.3 EQUIPMENT SUPPORTS

- A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.
- B. Grouting: Place grout under supports for equipment and make smooth bearing surface.
- C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.4 METAL FABRICATIONS

- A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.
- B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.
- C. Field Welding: Comply with AWS D1.1 procedures for shielded metal arc welding, appearance and quality of welds, and methods used in correcting welding work, and with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. Finish welds at exposed connections so no roughness shows after finishing and contours of welded surfaces match adjacent contours.

3.5 ADJUSTING

- A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.
- B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.6 PAINTING

- A. Touch Up: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils.
 - 2. Use hangers and supports with galvanized, metallic coatings for piping and equipment that will not have field-applied finish. Hangers installed outdoors shall have two coats of rust inhibitor paint after installation and adjustment.
 - 3. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.
- B. Touch Up: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal are specified in Division 09 Section "Painting."
- C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 230529

SECTION 230548 - VIBRATION CONTROLS FOR HVAC PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. This Section includes the following:
 - 1. Vibration isolation for piping, duct and equipment.
 - 2. Equipment isolation bases.
 - 3. Flexible piping connections.
 - 4. Resilient pipe anchors and guides.

1.2 SUBMITTALS

- A. Catalog cut sheets that include rated load, rated deflection, and overload capacity for each vibration isolation device.
- B. Coordination Drawings: Show coordination of seismic bracing for HVAC piping and equipment with other systems and equipment in the vicinity, including other supports.
- C. Submit details of equipment bases including dimensions, structural member sizes and support point locations.
- D. Submit details of isolation hangers for ceiling hung equipment, piping and ductwork.
- E. Submit details of mountings for floor-supported equipment, piping and ductwork.
- F. All hanger, mounting or pad drawings shall indicate deflections and model numbers as well as any other requirements in the specifications.
- G. Spring diameters, rated loads and deflections, heights at rated load and closed height shall be provided for all springs shown in the submittals in tabular form.
- H. Provide complete flexible connector details.

1.3 QUALITY ASSURANCE

- A. The isolation materials manufacturer shall be responsible for the proper selection of spring rates to accomplish the specified minimum static deflections for all spring and pay type isolators based on the weight distribution of equipment to be isolated.
- B. The isolation materials manufacturer shall be responsible for the structural design of steel beam bases and concrete inertia bases to support mechanical equipment scheduled or specified to receive a supplementary base.
- C. Furnish a complete set of final Shop Drawings of all mechanical equipment to receive vibration isolation devices to the vibration isolation materials manufacturer. These Drawings will be the basis upon which the selection of vibration isolators and design of supplementary bases will be completed. The Shop Drawings to be furnished shall include operating weight of the equipment to be isolated and the distribution of weight to the support points.

- D. Furnish a complete layout of piping and ductwork to be isolated, including vertical risers, showing size or weight and support points of the piping system to the vibration isolation materials manufacturer for selection and layout of isolation hangers.

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store equipment in accordance with manufacturer's instructions.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide diffusers by one of the following:

1. Mason Industries.
2. Vibration Eliminator Company.
3. Kinetics Noise Control, Inc.
4. Vibration Mountings and Controls, Inc.

2.2 VIBRATION ISOLATOR SELECTION

- A. Noise and vibration isolator types, minimum operating static deflections, and supplemental bases shall be provided for individual mechanical equipment units according to selection criteria delineated in Table 1 incorporated as part of this specification or as tabulated in the equipment schedules of the project Drawings.
- B. Isolator types are scheduled to establish minimum standards. At the Contractor's option labor saving accessories can be an integral part of isolators supplied to provide initial lift of equipment to operating height, hold piping at fixed elevations during installation and initial system filling operations, and similar installation advantages, provided isolators supplied incorporate and specified isolator type, and do not degrade the noise and vibration isolation of equipment mounted.
- C. Supplemental equipment base types tabulated can be deleted for unitary packaged air handling equipment having a rigid frame and casing providing a distortion free platform for attachment of vibration isolators.
- D. Isolators exposed to the outdoors shall have weather-proof finish on all parts.

2.3 NEOPRENE MOUNTINGS

- A. Neoprene mountings shall have a minimum static deflection of 0.35". All metal surfaces shall be neoprene covered and have friction pads both top and bottom. Bolt holes shall be provided on the bottom and a tapped hole and cap screw on top. Steel rails shall be used above the mountings under equipment such as small vent sets to compensate for the overhang. Mountings shall be type ND or rails type DNR as manufactured by Mason Industries, Inc.

2.4 SPRING ISOLATORS

- A. General: Spring isolators shall be free standing and laterally stable without any housing and complete with a molded neoprene cup or 1/4"(6mm) neoprene acoustical friction pad between the baseplate and the support. All mountings shall have leveling bolts that must be rigidly bolted to the equipment. Installed and operating heights shall be equal. The ratio of the spring

diameter divided by the compressed spring height shall be no less than 0.8. Springs shall have a minimum additional travel to solid equal to 50% of the rated deflection. Submittals shall include spring diameters, deflection, compressed spring height and solid spring height. Mountings shall be type SLF, as manufactured by Mason Industries, Inc.

- B. Equipment with Significant Weight Variations: Equipment with significant variations in the operating and installed weight, and equipment exposed to the wind shall be mounted on spring mountings as described above. Provide a neoprene acoustical pad within a rigid-sided housing with vertical limit stops to prevent spring extension when weight is removed. Also provide temporary steel spacers between the upper and lower housings. Housings shall serve as blocking during erection. When the equipment is at full operating weight, the springs shall be adjusted to assume the weight and the spacers removed, without changing the installed and operating heights. All restraining bolts shall have large rubber grommets to provide cushioning in the vertical as well as horizontal modes. The hole through the bushing shall be a minimum of 0.75" larger in diameter than the restraining bolt. Horizontal clearance on the sides between the spring assembly and the housing shall be a minimum of 0.5" to avoid bumping and interfering with the spring action. Vertical limit stops shall be out of contact during normal operation. When there is no provision for direct mounting, cooling tower mounts are to be located between the supporting steel and the roof, or the grillage and dunnage as shown on the drawings. Housings and springs shall be powder coated and hardware electro-galvanized. Mountings shall be SLR as manufactured by Mason Industries, Inc.

2.5 HANGERS

- A. General: Hangers shall consist of rigid steel frames containing minimum 1-1/4" thick neoprene elements at the top, a steel spring with general characteristics as described in Paragraph 2.4, and shall be seated in a steel washer-reinforced neoprene cup on the bottom. The neoprene element and the cup shall have neoprene bushings projecting through the steel box. In order to maintain stability the boxes shall not be articulated. The neoprene element shall not be stacked on top of the spring. Spring diameters and hanger box lower hole sizes shall be large enough to permit the hanger rod to swing through a 30° arc from side to side before contacting the cup bushing. Submittals shall include a hanger drawing showing the 30° capability. Hangers shall be type 30N as manufactured by Mason Industries, Inc.
- B. Spring Locks: Hangers shall be precompressed and locked at the rated deflection by means of a resilient upstop to keep the piping or equipment at a fixed elevation during installation. The hangers shall be designed with a release mechanism to free the spring after the installation is complete and the hanger is subjected to its full load. Deflection shall be clearly indicated by means of a scale. Submittals shall include a drawing of the hanger showing the 30° capability. Hangers shall be type PC30N as manufactured by Mason Industries, Inc.
- C. Hangers shall be manufactured with minimum characteristics as described in Paragraph 2.4, but without the neoprene element. Springs shall be seated in a steel washer reinforced neoprene cup that has a neoprene bushing projecting through the bottom hole to prevent rod hanger contact. Spring diameters and the lower hole sizes shall be large enough to allow the hanger rod to swing through a 30° arc from side to side before contacting the cup bushing. For ducts suspended by flat strap iron, the hanger assembly shall be modified by the manufacturer with an eye on top of the box and on the bottom of the spring hanger rod to allow for bolting to the hanger straps. Submittals shall include a scale drawing of the hanger showing the 30° capability. Hangers for rods shall be Type 30 or for straps W30 as manufactured by Mason Industries, Inc.

2.6 HORIZONTAL THRUST RESTRAINTS

- A. When total air thrust exceeds 10% of the isolated weight, floor mounted or suspended air handling equipment shall be protected against excessive displacement by the use of horizontal thrust restraints. The restraint shall consist of a modified spring mounting. Restraint springs shall have the same deflection as the isolator springs. The assembly shall be preset at the factory and adjusted in the field to allow for a maximum of 1/4" movement from stop to maximum thrust. The assemblies shall be furnished with rod and angle brackets for attachment to both the equipment and duct work or structure. Attach restraints at the centerline of thrust and symmetrically on both sides of the unit. Horizontal thrust restraints shall be WB as manufactured by Mason Industries, Inc.

2.7 STEEL BASES

- A. General: Vibration isolation manufacturer shall furnish integral structural steel bases. Rectangular bases are standard for all equipment. Centrifugal refrigeration machines and pump bases may require T or L shapes. Pump bases for split case pumps shall be of the dimensions to support suction and discharge elbows. All perimeter members shall be steel beams with a minimum depth equal to 1/10 of the longest dimension of the base. Base depth shall not exceed 14" if the deflection and misalignment is kept within acceptable limits as determined by the manufacturer. Provide height saving brackets in all mounting locations for a base clearance of 1". Bases shall be type WF as manufactured by Mason Industries, Inc.
- B. Partial Bases: Vibration isolation manufacturer shall provide steel members welded to height saving brackets to cradle equipment having legs or bases that do not require a complete supplementary base. Members shall have sufficient rigidity to prevent distortion of equipment. Inverted saddles shall be type ICS, as manufactured by Mason Industries, Inc.
- C. Floating Bases: Manufacturer shall furnish rectangular steel concrete pouring forms for floating concrete bases. Bases for split case pumps shall be of sufficient size to provide support for suction and discharge elbows. Bases shall be a minimum of 1/12 of the longest dimension of the base but not less than 6". The base depth need not exceed 12" unless specifically recommended by the base manufacturer for mass or rigidity. Forms shall include minimum concrete reinforcing consisting of 1/2" bars welded in place on 6" centers running both ways in a layer 1-1/2" above the bottom. Furnish forms with steel templates to hold the anchor bolt sleeves and anchor bolts while pouring concrete. Employ height-saving brackets in all mounting locations to maintain a 1" clearance below the base. Wooden formed bases are not acceptable. Base shall be type BMK or K as manufactured by Mason Industries, Inc.

2.8 EXPANSION JOINTS

- A. Rubber expansion joints shall be peroxide cured EPDM throughout with Kevlar tire cord reinforcement. Substitutions must have certifiable equal or superior characteristics. The raised face rubber flanges must encase solid steel rings to prevent pull out. Flexible cable wire is not acceptable. Sizes 1-1/2" through 14" shall have a ductile iron external ring between the two spheres. Sizes 16" through 24" may be single sphere. Sizes 3/4" through 2" may have one sphere, bolted threaded flange assemblies and cable retention.
- B. Minimum ratings through 14"(350mm) shall be 250psi at 170°F and 215psi at 250°F., 16" through 24" 180psi at 170°F and 150psi at 250°F. Higher published rated connectors may be used where required.
- C. Safety factors shall be a minimum of 3/1. All expansion joints must be factory tested to 150% of maximum pressure for 12 minutes before shipment.

- D. The piping gap shall be equal to the length of the expansion joint under pressure. Control rods passing through 1/2" thick Neoprene washer bushings large enough to take the thrust at 1000 psi of surface area may be used on unanchored piping where the manufacturer determines the condition exceeds the expansion joint rating without them. Submittals shall include two test reports by independent consultants showing minimum reductions of 20 DB in vibration accelerations and 10 DB in sound pressure levels at typical blade passage frequencies on this or a similar product by the same manufacturer. All expansion joints shall be installed on the equipment side of the shut off valves. Expansion joints shall be SAFEFLEX SFDEJ, SFEJ, SFDCR or SFU and Control Rods CR as manufactured by Mason Industries, Inc.

2.9 SPLIT SEALS

- A. Split Seals consist of pipe halves with minimum 3/4" thick neoprene sponge cemented to the inner faces. The seal shall be tightened around the pipe to eliminate clearance between the inner sponge face and the piping. Concrete may be packed around the seal to make it integral with the floor, wall or ceiling if the seal is not in place prior to the construction of the building member. Seals shall project a minimum of 1" past either face of the wall. Where temperatures exceed 240°F, 10 lb. density fiberglass may be used in lieu of the sponge. Seals shall be Type SWS as manufactured by Mason Industries, Inc.

2.10 ACOUSTICAL PIPE ANCHORS

- A. All directional acoustical pipe anchors, consist of two sizes of steel tubing separated by a minimum 1/2" thickness of 60 duro or softer neoprene. Vertical restraint shall be provided by similar material arranged to prevent up or down vertical travel. Allowable loads on the isolation material shall not exceed 500 psi and the design shall be balanced for equal resistance in any direction. All directional anchors shall be type ADA as manufactured by Mason Industries, Inc.

2.11 PIPE GUIDES

- A. General: Pipe guides shall consist of a telescopic arrangement of two sizes of steel tubing separated by a minimum 1/2" thickness of 60 durometer or softer neoprene. The height of the guides shall be preset with a shear pin to allow vertical motion due to pipe expansion or contraction. Guides shall be capable of $\pm 1\text{-}5/8$ " motion, or to meet location requirements. Pipe guides shall be type VSG as manufactured by Mason Industries, Inc.

2.12 HORIZONTAL PIPE ISOLATION

- A. The first four pipe hangers in the main lines near the mechanical equipment shall be as described Paragraph 2.5B. Hangers supporting piping 2" and larger in all other locations throughout the building shall be isolated by hangers as described in Paragraph 2.5A. Floor supported piping shall rest on isolators as described in Paragraph 2.5B. Heat exchangers and expansion tanks are considered part of the piping run. The first four isolators from the isolated equipment shall have the same static deflection as specified for the mountings under the connected equipment. If piping is connected to equipment located in basements and hangs from ceilings under occupied spaces the first four hangers shall have 0.75" deflection for pipe sizes up to and including 3", 1-1/2" deflection for pipe sizes over 3" and up to and including 6", and 2-1/2" deflection thereafter. Where piping connects to mechanical equipment install expansion joints as shown in Paragraph 2.10 or stainless hoses as shown in Paragraph 2.11. All piping passing through the equipment walls, floors or ceilings shall be protected against sound leakage by means of an acoustical seal, as described in Paragraph 2.12.

2.13 VERTICAL PIPE ISOLATION

- A. Support vertical risers by spring isolators, designed to support the riser filled with water, if it is a water line. Assigned loads must be within the building design limits at the support points. Neutral central resilient anchors close to the center of the run shall direct movement up and down. The anchors shall be capable of holding an upward force equal to the water weight when the system is drained. If one level cannot accommodate this force, anchors can be located on 2 or 3 adjacent floors. Resilient guides shall be spaced and sized properly depending on the pipe diameter. Submittals must include the initial load, initial deflection, change in deflection, final load and change in load at all spring and anchor support locations, as well as guide spacing. The initial spring deflection shall be a minimum of 0.75", or four times the thermal movement at the isolator location, whichever is greater. Calculations shall include pipe stress at end conditions and branch off locations and the manufacturer must include installation instructions. Submittal must be stamped and signed by a licensed professional engineer in the employ of the vibration vendor for at least 5 years. Proper provision shall be made for seismic protection in seismic zones. The isolator manufacturer shall be the same firm supplying the mechanical contract. Support spring mountings, anchors, and telescoping guides shall be in accordance with previous paragraphs of this section.

2.14 DUCT ISOLATION

- A. All air ducts with a cross section of 2 square feet or larger shall be isolated from the building structure by hangers as shown in Paragraph 2.5C or floor supports with a minimum deflection of 0.75". Isolators shall continue for 50' from the equipment. If air velocity exceeds 1000 fpm, hangers or supports shall continue for an additional 50' or as shown on the drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and equipment to receive vibration isolation and seismic[- and wind]-control devices for compliance with requirements for installation tolerances and other conditions affecting performance.
- B. Examine roughing-in of reinforcement and cast-in-place anchors to verify actual locations before installation.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 GENERAL

- A. All vibration isolators must be installed in strict accordance with the manufacturers written instructions and all certified submittal data.
- B. Installation of vibration isolators must not cause any change of position of equipment, piping or duct work resulting in stresses or misalignment.
- C. No rigid connections between equipment and the building structure shall be made that degrades the noise and vibration control system herein specified.
- D. The contractor shall not install any equipment, piping, duct or conduit which makes rigid connections with the building. This includes, but is not limited to, slabs, beams, columns, studs and walls.

- E. Coordinate work with other trades to avoid rigid contact with the building.
- F. Any conflicts with other trades which will result in rigid contact with equipment or piping due to inadequate space or other unforeseen conditions should be brought to the architects/engineers attention prior to installation. Corrective work necessitated by conflicts after installation shall be at the responsible contractors expense.
- G. Bring to the architects/engineers attention any discrepancies between the specifications and the field conditions or changes required due to specific equipment selection, prior to installation. Corrective work necessitated by discrepancies after installation shall be at the responsible contractors expense.
- H. Correct, at no additional cost, all installations which are deemed defective in workmanship and materials at the contractors expense.
- I. Hand built elastomeric expansion joints may be used when pipe sizes exceed 24" or specified movements exceed specified capabilities.
- J. Where piping passes through walls, floors or ceilings the vibration isolation manufacturer shall provide seals as described in this specification.
- K. Locate isolation hangers as near to the overhead support structure as possible.
- L. Air handling equipment and centrifugal fans shall be protected against excessive displacement which results from high air thrust when thrust forces exceed 10% of the equipment weight. Horizontal thrust restraints shall be in accordance with this specification.
- M. Rooftop equipment isolators must be bolted to the equipment and structure. Mountings must be designed to resist 100 mph wind loads.

3.3 VIBRATION ISOLATION OF PIPING

- A. Horizontal pipe shall be installed in accordance with 2.15 Horizontal Pipe Isolation.
- B. Risers shall be installed in accordance with 2.16 Riser Isolation.

3.4 VIBRATION ISOLATION OF DUCTWORK

- A. All duct runs shall be installed in accordance with 2.17 Duct Isolation.

3.5 ISOLATOR SCHEDULE

Equipment	Basement Slab Or Slab On Grade		Upper Floors	
	<i>Paragraph</i>	<i>Min Static Deflect</i>	<i>Paragraph</i>	<i>Min Static Deflect</i>
Cabinet Fan				
Section High				
Pressure				
(5" & Above)				
Low Pressure				
(Less than 5")				
Boilers				

	Basement Slab Or Slab On Grade		Upper Floors	
Pumps				
Close Coupled				
Thru 5HP				
7-1/2HP & Over				
Rooftop Units		-	2.4B	0.75
Suspended				
Centrifugal	2.4A, 2.7C,	2.5	2.4A, 2.7C,	2.5
Blowers	2.11		2.11	
15HP & Less	2.4A, 2.7C,	1.5	2.4A, 2.7C,	2.5
	2.11		2.11	
Condensing	2.4A, 2.7C	1.0	2.4A, 2.7C	3.0
Units				
Power Roof Exhausters Roof				
Mounted				
	2.5A, 2.6	1.0	2.5A, 2.6	1.0
	2.3, 2.6	0.35	2.4A, 2.6	1.0
	2.3, 2.7C, 2.6	0.35	2.4A, 2.7C, 2.6	1.0
			2.4B	2.0
			2.8	1.0

END OF SECTION 230548

SECTION 230553 - IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Equipment labels.
2. Warning signs and labels.
3. Pipe labels.
4. Valve tags.
5. Warning tags.
6. Sample Schedules.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples: For color, letter style, and graphic representation required for each identification material and device.
- C. Equipment Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label. An electronic copy and hard copy will be provided to THE OWNER before preliminary acceptance. Furnish extra copies (in addition to mounted copies) for Maintenance.
- D. Valve numbering scheme.
- E. Valve Schedules: For each piping system to include in maintenance manuals. An electronic copy and hard copy will be provided to the Owner before preliminary acceptance. Furnish extra copies (in addition to mounted copies) for Maintenance.

1.3 QUALITY ASSURANCE

- A. Follow manufacturer's recommended installation procedures.
- B. As applicable, comply with ASME A13.1 "Scheme for the Identification of Piping System"

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store materials in a dry and secure area on-site and protect against dirt and moisture damage
- B. Do not apply or install damaged materials.

1.5 COORDINATION

- A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- B. Coordinate installation of identifying devices with locations of access panels and doors.
- C. Install identifying devices before installing acoustical ceilings and similar concealment.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Seton
 - 2. Brady
 - 3. Best
 - 4. Milwaukee

2.2 EQUIPMENT LABELS

- A. Metal Labels for Equipment:
 - 1. Material and Thickness: Brass, 0.032-inch or Stainless steel, 0.025-inch minimum thickness, and having predrilled or stamped holes for attachment hardware.
 - 2. Minimum Letter Size: 4 by 2 inch. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 - 3. Fasteners: Stainless-steel self-tapping screws.
 - 4. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- B. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified.
- C. Equipment Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data. See Sample Equipment Schedule at end of this Specification.

2.3 WARNING SIGNS AND LABELS

- A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.
- B. Letter Color: Black.
- C. Background Color: Yellow.
- D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
- E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

- F. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- G. Fasteners: Stainless-steel self-tapping screws.
- H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- I. Label Content: Include caution and warning information, plus emergency notification instructions. Coordinate with Architect and Owner for specific warning label information to be included.

2.4 PIPE LABELS

- A. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction. Do not provide labels for bare pipes conveying fluids at temperatures of 125 °F or higher.
- B. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- C. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.
 - 1. Flow-Direction Arrows: Integral with piping system service lettering to accommodate both directions, or as separate unit on each pipe label to indicate flow direction.
 - 2. Lettering Size: At least 1-1/2 inches high.

2.5 VALVE TAGS

- A. Valve Tags: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2-inch numbers.
 - 1. Tag Material: Brass, 0.032-inch or Stainless steel, 0.025-inch minimum thickness, and having predrilled or stamped holes for attachment hardware.
 - 2. Fasteners: Brass beaded chain.
- B. Valve Schedules: For each piping system, on 8-1/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.
 - 1. Valve-tag schedule shall be included in operation and maintenance data. See Sample Valve Tag Schedule at the end of this Specification.

2.6 WARNING TAGS

- A. Warning Tags: Preprinted or partially preprinted, accident-prevention tags, of plasticized card stock with matte finish suitable for writing.
 - 1. Size: 4 by 7 inches.
 - 2. Fasteners: Reinforced grommet and wire.
 - 3. Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."

4. Color: Yellow background with black lettering.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.
- B. Identification for piping, duct or equipment in sales floor should be installed on surfaces away from direct view of customers.

3.2 EQUIPMENT LABEL INSTALLATION

- A. Install or permanently fasten labels on each major item of mechanical equipment.
- B. Locate equipment labels where accessible and visible.

3.3 PIPE LABEL INSTALLATION

- A. Piping Color-Coding: Painting of piping is specified in Division 09 Section " Painting"
- B. Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:
 1. Near each valve and control device.
 2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
 3. Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
 4. At access doors, manholes, and similar access points that permit view of concealed piping.
 5. Near major equipment items and other points of origination and termination.
 6. Spaced at maximum intervals of 25 feet along each run. Reduce intervals to 15 feet in areas of congested piping and equipment.
 7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.
- C. Pipe Label Color Schedule:
 1. Refrigerant Piping:
 - a. Background Color: White.
 - b. Letter Color: Black.

3.4 DUCT LABEL INSTALLATION

- A. Install self-adhesive duct labels with permanent adhesive on air ducts in the following color codes:
 1. Blue: For cold-air supply ducts.
 2. Yellow: For hot-air supply ducts.
 3. Green: For exhaust-, outside-, relief-, return-, and mixed-air ducts.

- B. Locate labels near points where ducts enter into concealed spaces and at maximum intervals of 50 feet in each space where ducts are exposed or concealed by removable ceiling system.

3.5 VALVE-TAG INSTALLATION

INSTALL TAGS ON VALVES AND CONTROL DEVICES IN PIPING SYSTEMS, EXCEPT CHECK VALVES; VALVES WITHIN FACTORY-FABRICATED EQUIPMENT UNITS; SHUTOFF VALVES; FAUCETS; CONVENIENCE AND LAWN-WATERING HOSE CONNECTIONS; AND HVAC TERMINAL DEVICES AND SIMILAR ROUGHING-IN CONNECTIONS OF END-USE FIXTURES AND UNITS. LIST TAGGED VALVES IN A VALVE SCHEDULE. SEE SAMPLE VALVE TAG SCHEDULE AT END OF THIS SPECIFICATION.

- A. Valve-Tag Application Schedule: Tag valves according to size, shape, and color scheme and with captions similar to those indicated in the following subparagraphs:
1. Valve-Tag Size and Shape:
 - a. Refrigerant: 2 inches, round..
 - b. Gas: 2 inches, round..
 2. Valve-Tag Color:
 - a. Refrigerant: Natural.
 - b. Gas: Natural.
 3. Letter Color:
 - a. Refrigerant: Black.
 - b. Gas: Black.

3.6 WARNING-TAG INSTALLATION

- A. Write required message on, and attach warning tags to, equipment and other items where required.

3.7 SAMPLE SCHEDULES

- A. The following tables are examples of schedules required to be submitted by the Contractor. Examples of the kinds of information to be included are in the first row in these tables.

EQUIPMENT LABEL SCHEDULE				
EQUIPMENT IDENTIFICATION	EQUIPMENT LOCATION	SPECIFICATION SECTION		REMARKS
		NUMBER	TITLE	
AHU-xx	MECHANICAL ROOM M-xxx	23 xx xx		MODULAR INDOOR AIR-HANDLING UNITS

3.8 CLEANING

- A. Follow manufacturer's instructions for surface preparation prior to application or installation.

- B. Follow Manufacturer's instructions for cleaning labels and tags.

END OF SECTION 230553

SECTION 230593 - TESTING, ADJUSTING, AND BALANCING FOR HVAC**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes procedural requirements for testing, adjusting, and balancing (TAB) of new and existing systems to achieve the required flows within the limits of the fan and the motor HP. The testing, adjusting, and balancing work includes producing design objectives for the following:
 - 1. Air Systems:
 - a. Constant-volume air systems.
 - b. VRF systems
 - c. Roof Top Units
 - 2. Hydronic Piping Systems:
 - a. Condensing Units.
 - 3. HVAC equipment quantitative-performance settings.

1.2 DEFINITIONS

- A. AABC: Associated Air Balance Council.
- B. NEBB: National Environmental Balancing Bureau.
- C. TAB: Testing, Adjusting, and Balancing.
- D. TABB: Testing, Adjusting, and Balancing Bureau.
- E. TAB Firm: Entity responsible for performing and reporting TAB procedures.
- F. TAB Specialist: Entity engaged by TAB Firm to perform TAB work.

1.3 SUBMITTALS

- A. Qualification Data: Within **30** days of Notice to Proceed, submit documentation for the TAB contractor and the Project's TAB team members.
- B. Contract Documents Examination Report: Within **30** days of Notice to Proceed, submit the Contract Documents Examination Report.
- C. TAB Plan: Within **30** days of Notice to Proceed, submit TAB strategies and step-by-step procedures as specified in "Preparation" Article.
- D. Certified TAB reports. Submit two copies of the TAB reports prepared as specified in this specification on approved forms certified by the TAB contractor.
- E. Sample report forms. Submit two copies of the sample TAB report forms
- F. Instrument calibration reports, including the following:

1. Instrument type and make.
2. Serial number.
3. Application.
4. Dates of use.
5. Dates of calibration.

1.4 QUALITY ASSURANCE

- A. TAB Contractor Qualifications: Engage a TAB entity certified by AABC, NEBB, or TABB.
 1. TAB Field Supervisor: Employee of the TAB contractor and certified by AABC, NEBB or TABB.
 2. TAB Technician: Employee of the TAB contractor and who is certified by AABC, NEBB or TABB as a TAB technician.
 3. Agenda Items:
 - a. The Contract Documents examination report.
 - b. The TAB plan.
 - c. Coordination and cooperation of trades and subcontractors.
 - d. Coordination of documentation and communication flow.
 - e. Submittal distribution requirements.
 - f. Work Schedule and Project-Site requirements.
- B. Certification of TAB Reports: Certify TAB field data reports and perform the following:
 1. Review field data reports to validate accuracy of data and to prepare certified TAB reports.
 2. Certify that the TAB team complied with the approved TAB plan and the procedures specified and referenced in this Specification.
- C. TAB Report Forms: Use standard TAB forms from NEBB, AABC or TABB as well as providing any additional information required by this specification.
- D. Instrumentation Type, Quantity, Accuracy, and Calibration: As described in ASHRAE 111, Section 5, "Instrumentation."
- E. Owner reserves the right to select at random 10% of the TAB report data for field verification witnessed by the commission agent. The TAB contractor will be given sufficient notice of the date of field verification. The same instruments that were used when the original test was recorded shall be used. A failure of more than 10% of the selected field verification items shall result in a repeat of the testing of the entire system at the TAB contractor's expense. The repeated work is also subject to field verification.

1.5 COORDINATION

- A. Coordinate the efforts of factory-authorized service representatives for systems and equipment, HVAC controls installers, and other mechanics to operate HVAC systems and equipment to support and assist TAB activities.
- B. Notice: Provide seven days' advance notice for each test. Include scheduled test dates and times.
- C. Perform TAB after leakage and pressure tests on air and water distribution systems have been satisfactorily completed.

1.6 WARRANTY

A. Provide warranty in accordance with AABC, NEBB or TABB standards:

1. National Project Performance Guarantee: Provide a guarantee on AABC's "National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems" forms stating that AABC will assist in completing requirements of the Contract Documents if TAB firm fails to comply with the Contract Documents.
2. Special Guarantee: Provide a guarantee on NEBB forms stating that NEBB will assist in completing requirements of the Contract Documents if TAB firm fails to comply with the Contract Documents.
3. Guarantee shall include the following provisions:
 - a. The certified TAB firm has tested and balanced systems according to the Contract Documents.
 - b. Systems are balanced to optimum performance capabilities within design and installation limits.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine the Contract Documents and field conditions to become familiar with Project requirements and to discover conditions that may preclude proper TAB of systems and equipment.
- B. Examine the approved submittals for HVAC systems and equipment.
- C. Examine design data including HVAC system descriptions, statements of design assumptions for environmental conditions and systems' output, and statements of philosophies and assumptions about HVAC system and equipment controls.
- D. Examine ceiling plenums used for return air to verify that they meet the leakage class of connected ducts as specified in Division 23 Section "Metal Ducts," and are properly separated from adjacent areas. Verify that penetrations in plenum walls are properly sealed or, as required, fire-stopped.
- E. Examine system and equipment installations and verify that field quality-control testing, cleaning, and adjusting specified in individual Sections have been performed.
- F. Examine test reports specified in individual system and equipment Sections.
- G. Examine HVAC equipment and filters and verify that bearings are greased, belts are aligned and tight, and equipment with functioning controls is ready for operation.
- H. Examine heat-transfer coils for correct piping connections and for clean and straight fins.
- I. Examine equipment for installation and properly operating safety interlocks and controls.

- J. Report deficiencies discovered before and during performance of TAB procedures. Observe and record system reactions to changes in conditions. Record default set points if different from indicated values.

3.2 PREPARATION

- A. Prepare a TAB Plan that includes strategies and step-by-step procedures.
- B. Complete system-readiness checks and prepare reports. Verify the following:
 - 1. Permanent electrical-power wiring is complete.
 - 2. Automatic temperature-control systems are operational.
 - 3. Equipment and duct access doors are securely closed.
 - 4. Balance, smoke, and fire dampers are open.
 - 5. Isolating and balancing valves are open and control valves are operational.
 - 6. Ceilings are installed in critical areas where air-pattern adjustments are required and access to balancing devices is provided.
 - 7. Windows and doors can be closed so indicated conditions for system operations can be met.

3.3 GENERAL PROCEDURES FOR TESTING AND BALANCING

- A. Perform testing and balancing procedures on each system according to the procedures contained in AABC's "National Standards for Total System Balance," ASHRAE 111, NEBB's "Procedural Standards for Testing, Adjusting, and Balancing of Environmental Systems," or SMACNA's "HVAC Systems - Testing, Adjusting, and Balancing," and this Section.
 - 1. Comply with requirements in ASHRAE 62.1-2007, Section 7.2.2, "Air Balancing."
- B. Cut insulation, ducts, pipes, and equipment cabinets for installation of test probes to the minimum extent necessary for TAB procedures.
 - 1. After testing and balancing, patch probe holes in ducts with same material and thickness as used to construct ducts.
- C. Mark equipment and balancing devices, including damper-control positions, valve position indicators, fan-speed-control levers, and similar controls and devices, with paint or other suitable, permanent identification material to show final settings.
- D. Take and report testing and balancing measurements in inch-pound (IP) units.
- E. HVAC system lineup. The contractor will test the system in the normal system lineups.
 - 1. A hydronic system with two chillers designed to run at the same time, the system will be balanced with pumps flowing to both chillers. All pump data will also be collected when the pump is flowing to only one chiller.
 - 2. Air systems will be balanced while aligned for minimum outside air flow and the position of the outside air dampers will be recorded for the minimum position.
 - 3. It is not acceptable to balance air systems when the normal boundaries, doors, walls, or ductwork are not in their design configuration.

3.4 GENERAL PROCEDURES FOR BALANCING AIR SYSTEMS

- A. Prepare test reports for both fans and outlets. Obtain manufacturer's outlet factors and recommended testing procedures. Crosscheck the summation of required outlet volumes with required fan volumes.
- B. Prepare schematic diagrams of systems' "as-built" duct layouts.
- C. For variable-air-volume systems, develop a plan to simulate diversity.
- D. Determine the best locations in main and branch ducts for accurate duct-airflow measurements.
- E. Check airflow patterns from the outdoor-air louvers and dampers and the return- and exhaust-air dampers through the supply-fan discharge and mixing dampers.
- F. Locate start-stop and disconnect switches, electrical interlocks, and motor starters.
- G. Verify that motor starters are equipped with properly sized thermal protection.
- H. Check dampers for proper position to achieve desired airflow path.
- I. Check for airflow blockages.
- J. Check condensate drains for proper connections and functioning.
- K. Check for proper sealing of air-handling-unit components.
- L. Check for proper sealing of air duct system.

3.5 PROCEDURES FOR CONSTANT-VOLUME AIR SYSTEMS

- A. Adjust fans to deliver total indicated airflows within the maximum allowable fan speed listed by fan manufacturer and within the limits of the motor HP. All fans shall deliver the design air flow capacity at actual static pressure up to design static. Fan sheaves shall be replaced as necessary to obtain desired results.
 - 1. Measure total airflow.
 - a. Where sufficient space in ducts is unavailable for Pitot-tube-traverse measurements, measure airflow at terminal outlets and inlets and calculate the total airflow.
 - 2. Measure fan static pressures as follows to determine actual static pressure:
 - a. Measure outlet static pressure as far downstream from the fan as practical and upstream from restrictions in ducts such as elbows and transitions.
 - b. Measure static pressure directly at the fan outlet or through the flexible connection.
 - c. Measure inlet static pressure of single-inlet fans in the inlet duct as near the fan as possible, upstream from the flexible connection, and downstream from duct restrictions.
 - d. Measure inlet static pressure of double-inlet fans through the wall of the plenum that houses the fan.

3. Measure static pressure across each component that makes up an air-handling unit, rooftop unit, and other air-handling and -treating equipment.
 - a. Report the cleanliness status of filters and the time static pressures are measured.
 4. Measure static pressures entering and leaving other devices, such as sound traps, heat-recovery equipment, and air washers, under final balanced conditions.
 5. Review Record Documents to determine variations in design static pressures versus actual static pressures. Calculate actual system-effect factors. Recommend adjustments to accommodate actual conditions.
 6. Do not make fan-speed adjustments that result in motor overload. Consult equipment manufacturers about fan-speed safety factors. Modulate dampers and measure fan-motor amperage to ensure that no overload will occur. Measure amperage in full-cooling, full-heating, economizer, and any other operating mode to determine the maximum required brake horsepower.
- B. Adjust volume dampers for main duct, sub-main ducts, and major branch ducts to indicated airflows within specified tolerances.
1. Measure airflow of sub-main and branch ducts.
 - a. Where sufficient space in sub-main and branch ducts is unavailable for Pitot-tube-traverse measurements, measure airflow at terminal outlets and inlets and calculate the total airflow for that zone.
 2. Measure static pressure at a point downstream from the balancing damper, and adjust volume dampers until the proper static pressure is achieved.
 3. Re-measure each sub-main and branch duct after all have been adjusted. Continue to adjust sub-main and branch ducts to indicated airflows within specified tolerances.
- C. Measure air outlets and inlets without making adjustments.
1. Measure terminal outlets using a direct-reading hood or outlet manufacturer's written instructions and calculating factors.
- D. Adjust air outlets and inlets for each space to indicated airflows within specified tolerances of indicated values. Make adjustments using branch volume dampers rather than extractors and the dampers at air terminals.
1. Adjust each outlet in same room or space to within specified tolerances of indicated quantities without generating noise levels above the limitations prescribed by the Contract Documents.
 2. Adjust patterns of adjustable outlets for proper distribution without drafts.
- 3.6 PROCEDURES FOR VARIABLE-AIR-VOLUME SYSTEMS
- A. Compensating for Diversity: When the total airflow of all terminal units is more than the indicated airflow of the fan, place a selected number of terminal units at a minimum set-point airflow with the remainder at maximum-airflow condition until the total airflow of the terminal units equals the indicated airflow of the fan. Select the reduced-airflow terminal units so they are distributed evenly among the branch ducts.
1. Minimum outside air flow rate must be verified under three conditions, with the OA damper position recorded in each configuration.

- a. System operating with all boxes at minimum primary air flow.
 - b. System at maximum supply flow (diversity may not allow all boxes to be open).
 - c. System flow halfway between box minimum and maximum.
- B. Pressure-Independent, Variable-Air-Volume Systems: After the fan systems have been adjusted, adjust the variable-air-volume systems as follows:
1. Set outdoor-air dampers at minimum, and set return- and exhaust-air dampers at a position that simulates full-cooling load.
 2. Select the terminal unit that is most critical to the supply-fan airflow and static pressure. Measure static pressure. Adjust system static pressure so the entering static pressure for the critical terminal unit is not less than the sum of the terminal-unit manufacturer's recommended minimum inlet static pressure plus the static pressure needed to overcome terminal-unit discharge system losses.
 3. Measure total system airflow. Adjust to within indicated airflow.
 4. Set terminal units at maximum airflow and adjust controller or regulator to deliver the designed maximum airflow. Use terminal-unit manufacturer's written instructions to make this adjustment. When total airflow is correct, balance the air outlets downstream from terminal units the same as described for constant-volume air systems.
 5. Set terminal units at minimum airflow and adjust controller or regulator to deliver the designed minimum airflow. Check air outlets for a proportional reduction in airflow the same as described for constant-volume air systems.
 - a. If air outlets are out of balance at minimum airflow, report the condition but leave outlets balanced for maximum airflow.
 6. Re-measure the return airflow to the fan while operating at maximum return airflow and minimum outdoor airflow.
 - a. Adjust the fan and balance the return-air ducts and inlets the same as described for constant-volume air systems.
 7. Measure static pressure at the most critical terminal unit and adjust the static-pressure controller at the main supply-air sensing station to ensure that adequate static pressure is maintained at the most critical unit.
 8. Record final fan-performance data.

3.7 PROCEDURES FOR MOTORS

- A. Motors, 1/2 HP and Larger: Test at final balanced conditions and record the following data:
1. Manufacturer's name, model number, and serial number.
 2. Motor horsepower rating.
 3. Motor rpm.
 4. Efficiency rating.
 5. Nameplate and measured voltage, each phase.
 6. Nameplate and measured amperage, each phase.
 7. Starter thermal-protection-element rating.
- B. Motors Driven by Variable-Frequency Controllers: Test for proper operation at speeds varying from minimum to maximum. Test the manual bypass of the controller to prove proper operation. Record observations including name of controller manufacturer, model number, serial number, and nameplate data.

3.8 TEMPERATURE-CONTROL VERIFICATION

- A. Verify that controllers are calibrated and commissioned.
- B. Check transmitter and controller locations and note conditions that would adversely affect control functions.
- C. Record controller settings and note variances between set points and actual measurements.
- D. Check the operation of limiting controllers (i.e., high- and low-temperature controllers).
- E. Check free travel and proper operation of control devices such as damper and valve operators.
- F. Check the sequence of operation of control devices. Note air pressures and device positions and correlate with airflow and water flow measurements. Note the speed of response to input changes.
- G. Check the interaction of electrically operated switch transducers.
- H. Check the interaction of interlock and lockout systems.
- I. Check main control supply-air pressure and observe compressor and dryer operations.
- J. Record voltages of power supply and controller output. Determine whether the system operates on a grounded or nongrounded power supply.
- K. Note operation of electric actuators using spring return for proper fail-safe operations.

3.9 TOLERANCES

- A. Set HVAC system air flow rates and water flow rates within the following tolerances:
 - 1. Supply, Return, and Exhaust Fans and Equipment with Fans: Plus or minus 10 percent.
 - 2. Air Outlets and Inlets: Plus or minus 10 percent.
 - 3. Heating-Water Flow Rate: Plus or minus 10 percent.
 - 4. Cooling-Water Flow Rate: Plus or minus 5 percent.

3.10 REPORTING

- A. Initial Construction-Phase Report: Based on examination of the Contract Documents as specified in "Examination" Article, prepare a report on the adequacy of design for systems' balancing devices. Recommend changes and additions to systems' balancing devices to facilitate proper performance measuring and balancing. Recommend changes and additions to HVAC systems and general construction to allow access for performance measuring and balancing devices.

3.11 FINAL REPORT

- A. General: Prepare a certified written report; tabulate and divide the report into separate sections for tested systems and balanced systems.
 - 1. Include a certification sheet at the front of the report's binder, signed and sealed by the certified testing and balancing engineer.
 - 2. Include a list of instruments used for procedures, along with proof of calibration.

- B. Final Report Contents: In addition to certified field-report data, include the following:
1. Fan curves.
 2. Manufacturers' test data.
 3. Field test reports prepared by system and equipment installers.
 4. Other information relative to equipment performance; do not include Shop Drawings and product data.
- C. General Report Data: In addition to form titles and entries, include the following data:
1. Title page.
 2. Name and address of the TAB contractor.
 3. Project name.
 4. Project location.
 5. Architect's name and address.
 6. Engineer's name and address.
 7. Contractor's name and address.
 8. Report date.
 9. Signature of TAB supervisor who certifies the report.
 10. Table of Contents with the total number of pages defined for each section of the report. Number each page in the report.
 11. Summary of contents including the following:
 - a. Indicated versus final performance.
 - b. Notable characteristics of systems.
 - c. Description of system operation sequence if it varies from the Contract Documents.
 12. Nomenclature sheets for each item of equipment.
 13. Data for terminal units, including manufacturer's name, type, size, and fittings.
 14. Notes to explain why certain final data in the body of reports vary from indicated values.
 15. Test conditions for fans and pump performance forms including the following:
 - a. Settings for outdoor-, return-, and exhaust-air dampers.
 - b. Conditions of filters.
 - c. Cooling coil, wet- and dry-bulb conditions.
 - d. Face and bypass damper settings at coils.
 - e. Fan drive settings including settings and percentage of maximum pitch diameter.
 - f. Inlet vane settings for variable-air-volume systems.
 - g. Settings for supply-air, static-pressure controller.
 - h. Other system operating conditions that affect performance.
- D. System Diagrams: Include schematic layouts of air and hydronic distribution systems. Present each system with single-line diagram and include the following:
1. Quantities of outdoor, supply, return, and exhaust airflows.
 2. Duct, outlet, and inlet sizes.
 3. Pipe and valve sizes and locations.
 4. Terminal units.
 5. Balancing stations.
 6. Position of balancing devices.
- E. Roof Top Unit Test Reports: For air-handling units with coils, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and unit size.
 - e. Manufacturer's serial number.
 - f. Unit arrangement and class.
 - g. Discharge arrangement.
 - h. Sheave make, size in inches, and bore.
 - i. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - j. Number, make, and size of belts.
 - k. Number, type, and size of filters.
2. Motor Data:
 - a. Motor make, and frame type and size.
 - b. Horsepower and rpm.
 - c. Volts, phase, and hertz.
 - d. Full-load amperage and service factor.
 - e. Sheave make, size in inches, and bore.
 - f. Center-to-center dimensions of sheave, and amount of adjustments in inches.
3. Test Data (Indicated and Actual Values):
 - a. Total air flow rate in cfm.
 - b. Total system static pressure in inches wg.
 - c. Fan rpm.
 - d. Discharge static pressure in inches wg.
 - e. Filter static-pressure differential in inches wg.
 - f. Cooling coil static-pressure differential in inches wg.
 - g. Heating coil static-pressure differential in inches wg.
 - h. Outdoor airflow in cfm.
 - i. Return airflow in cfm.
 - j. Outdoor-air damper position.
 - k. Return-air damper position.

F. Apparatus-Coil Test Reports:

1. Test Data (Indicated and Actual Values):
 - a. Air flow rate in cfm.
 - b. Average face velocity in fpm.
 - c. Air pressure drop in inches wg.
 - d. Outdoor-air, wet- and dry-bulb temperatures in deg F.
 - e. Return-air, wet- and dry-bulb temperatures in deg F.
 - f. Entering-air, wet- and dry-bulb temperatures in deg F.
 - g. Leaving-air, wet- and dry-bulb temperatures in deg F.
 - h. Refrigerant expansion valve and refrigerant types.
 - i. Refrigerant suction pressure in psig.
 - j. Refrigerant suction temperature in deg F.

G. Gas- and Oil-Fired Heat Apparatus Test Reports: In addition to manufacturer's factory startup equipment reports, include the following:

1. Unit Data:
 - a. System identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and unit size.
 - e. Manufacturer's serial number.
 - f. Fuel type in input data.
 - g. Output capacity in Btu/h.
 - h. Ignition type.
 - i. Burner-control types.
 - j. Motor horsepower and rpm.
 - k. Motor volts, phase, and hertz.
 - l. Motor full-load amperage and service factor.
 - m. Sheave make, size in inches, and bore.
 - n. Center-to-center dimensions of sheave, and amount of adjustments in inches.

2. Test Data (Indicated and Actual Values):
 - a. Total air flow rate in cfm.
 - b. Entering-air temperature in deg F.
 - c. Leaving-air temperature in deg F.
 - d. Air temperature differential in deg F.
 - e. Entering-air static pressure in inches wg.
 - f. Leaving-air static pressure in inches wg.
 - g. Air static-pressure differential in inches wg.
 - h. Low-fire fuel input in Btu/h.
 - i. High-fire fuel input in Btu/h.
 - j. Manifold pressure in psig.
 - k. High-temperature-limit setting in deg F.
 - l. Operating set point in Btu/h.
 - m. Motor voltage at each connection.
 - n. Motor amperage for each phase.
 - o. Heating value of fuel in Btu/h.

H. Electric-Coil Test Reports: For electric furnaces, duct coils, and electric coils installed in central-station air-handling units, include the following:

1. Unit Data:
 - a. System identification.
 - b. Location.
 - c. Coil identification.
 - d. Capacity in kW
 - e. Number of stages.
 - f. Connected volts, phase, and hertz.
 - g. Rated amperage.
 - h. Air flow rate in cfm.
 - i. Face area in sq. ft.
 - j. Minimum face velocity in fpm.
2. Test Data (Indicated and Actual Values):
 - a. Heat output in kW
 - b. Air flow rate in cfm.

- c. Air velocity in fpm.
 - d. Entering-air temperature in deg F.
 - e. Leaving-air temperature in deg F.
 - f. Voltage at each connection.
 - g. Amperage for each phase.
- I. Fan Test Reports: For supply, return, and exhaust fans, include the following:
 - 1. Fan Data:
 - a. System identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and size.
 - e. Manufacturer's serial number.
 - f. Arrangement and class.
 - g. Sheave make, size in inches, and bore.
 - h. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - 2. Motor Data:
 - a. Motor make, and frame type and size.
 - b. Horsepower and rpm.
 - c. Volts, phase, and hertz.
 - d. Full-load amperage and service factor.
 - e. Sheave make, size in inches, and bore.
 - f. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - g. Number, make, and size of belts.
 - 3. Test Data (Indicated and Actual Values):
 - a. Total airflow rate in cfm.
 - b. Total system static pressure in inches wg.
 - c. Fan rpm.
 - d. Discharge static pressure in inches wg.
 - e. Suction static pressure in inches wg.
- J. Round, Flat-Oval, and Rectangular Duct Traverse Reports: Include a diagram with a grid representing the duct cross-section and record the following:
 - 1. Report Data:
 - a. System and air-handling-unit number.
 - b. Location and zone.
 - c. Traverse air temperature in deg F.
 - d. Duct static pressure in inches wg.
 - e. Duct size in inches.
 - f. Duct area in sq. ft.
 - g. Indicated air flow rate in cfm.
 - h. Indicated velocity in fpm.
 - i. Actual air flow rate in cfm.
 - j. Actual average velocity in fpm.
 - k. Barometric pressure in psig.

- K. Compressor and Condenser Reports: For refrigerant side of unitary systems, stand-alone refrigerant compressors, air-cooled condensing units, or water-cooled condensing units, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Location.
 - c. Unit make and model number.
 - d. Compressor make.
 - e. Compressor model and serial numbers.
 - f. Refrigerant weight in lb.
 - g. Low ambient temperature cutoff in deg F.
2. Test Data (Indicated and Actual Values):
 - a. Inlet-duct static pressure in inches wg.
 - b. Outlet-duct static pressure in inches wg.
 - c. Entering-air, dry-bulb temperature in deg F.
 - d. Leaving-air, dry-bulb temperature in deg F.
 - e. Condenser entering-water temperature in deg F.
 - f. Condenser leaving-water temperature in deg F.
 - g. Condenser-water temperature differential in deg F.
 - h. Condenser entering-water pressure in feet of head or psig.
 - i. Condenser leaving-water pressure in feet of head or psig.
 - j. Condenser-water pressure differential in feet of head or psig.
 - k. Control settings.
 - l. Unloader set points.
 - m. Low-pressure-cutout set point in psig.
 - n. High-pressure-cutout set point in psig.
 - o. Suction pressure in psig.
 - p. Suction temperature in deg F.
 - q. Condenser refrigerant pressure in psig.
 - r. Condenser refrigerant temperature in deg F.
 - s. Oil pressure in psig.
 - t. Oil temperature in deg F.
 - u. Voltage at each connection.
 - v. Amperage for each phase.
 - w. Kilowatt input.
 - x. Crankcase heater kilowatt.
 - y. Number of fans.
 - z. Condenser fan rpm.
 - aa. Condenser fan airflow rate in cfm.
 - bb. Condenser fan motor make, frame size, rpm, and horsepower.
 - cc. Condenser fan motor voltage at each connection.
 - dd. Condenser fan motor amperage for each phase.

- L. Condenser Test Reports: For condensers, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Make and type.
 - c. Model and serial numbers.
 - d. Nominal cooling capacity in tons.

- e. Refrigerant type and weight in lb.
- f. Water-treatment chemical feeder and chemical.
- g. Number and type of fans.
- h. Fan motor make, frame size, rpm, and horsepower.
- i. Fan motor voltage at each connection.
- j. Sheave make, size in inches, and bore.
- k. Sheave dimensions, center-to-center, and amount of adjustments in inches.
- l. Number of belts, make, and size.
- m. Pump make and model number.
- n. Pump manufacturer's serial number.
- o. Pump motor make and frame size.
- p. Pump motor horsepower and rpm.

M. Instrument Calibration Reports:

1. Report Data:

- a. Instrument type and make.
- b. Serial number.
- c. Application.
- d. Dates of use.
- e. Dates of calibration.

3.12 INSPECTIONS

A. Initial Inspection:

- 1. After testing and balancing are complete, operate each system and randomly check measurements to verify that the system is operating according to the final test and balance readings documented in the final report.
- 2. Check the following for each system:
 - a. Measure airflow of at least 10 percent of air outlets.
 - b. Measure water flow of at least 5 percent of terminals.
 - c. Measure room temperature at each thermostat/temperature sensor. Compare the reading to the set point.
 - d. Verify that balancing devices are marked with final balance position.
 - e. Note deviations from the Contract Documents in the final report.

B. Final Inspection:

- 1. After initial inspection is complete and documentation by random checks verifies that testing and balancing are complete and accurately documented in the final report, request that a final inspection be made by Owner's Representative.
- 2. The TAB contractor's test and balance engineer shall conduct the inspection in the presence of the Owner Authorized Representative.
- 3. The Owner shall randomly select measurements, documented in the final report, to be rechecked. Rechecking shall be limited to either 10 percent of the total measurements recorded or the extent of measurements that can be accomplished in a normal 8-hour business day.
- 4. If rechecks yield measurements that differ from the measurements documented in the final report by more than the tolerances allowed, the measurements shall be noted as "FAILED."

5. If the number of "FAILED" measurements is greater than 10 percent of the total measurements checked during the final inspection, the testing and balancing shall be considered incomplete and shall be rejected.
- C. TAB Work will be considered defective if it does not pass final inspections. If TAB Work fails, proceed as follows:
 1. Recheck all measurements and make adjustments. Revise the final report and balancing device settings to include all changes; resubmit the final report and request a second final inspection.
 2. If the second final inspection also fails, Owner may contract the services of another TAB contractor to complete TAB Work according to the Contract Documents and deduct the cost of the services from the original TAB contractor's final payment.
- D. Prepare test and inspection reports.

3.13 ADDITIONAL TESTS

- A. Within 90 days of completing TAB, perform additional TAB to verify that balanced conditions are being maintained throughout and to correct unusual conditions.
- B. Seasonal Periods: If initial TAB procedures were not performed during near-peak summer and winter conditions, perform additional TAB during near-peak summer and winter conditions.
- C. Include 80 hours of additional TAB time for use at the direction of the engineer.

END OF SECTION 230593

SECTION 230700 - HVAC INSULATION**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Insulation Materials:
 - a. Flexible elastomeric.
 - b. Mineral fiber.
 - c. Polyisocyanurate.
2. Fire-rated insulation systems.
3. Insulating cements.
4. Adhesives.
5. Mastics.
6. Lagging adhesives.
7. Sealants.
8. Factory-applied jackets.
9. Field-applied jackets.
10. Identification
11. Tapes.
12. Securements.
13. Corner angles.

1.2 DEFINITIONS

- A. Hot Surfaces: Normal operating temperatures of 100 deg F or higher.
- B. Dual-Temperature Surfaces: Normal operating temperatures that vary from hot to cold.
- C. Cold Surfaces: Normal operating temperatures less than 75 deg F.
- D. Thermal Resistivity: "R-values" represent the reciprocal of thermal conductivity (K-value). Thermal conductivity is the rate of heat flow through a homogeneous material exactly 1 inch thick. Thermal resistivities are expressed by the temperature difference in degrees F between two exposed faces required to cause one Btu to flow through one square foot of material, in one hour, at a given mean temperature.
- E. Density: Is expressed in lb/cu. ft.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include thermal conductivity, thermal resistivity (R-value), thickness, and jackets (both factory and field applied, if any).
- B. Shop Drawings:
 1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
 2. Detail attachment and covering of heat tracing inside insulation.
 3. Detail insulation application at pipe expansion joints for each type of insulation.

4. Detail insulation application at elbows, fittings, flanges, valves, and specialties for each type of insulation.
 5. Detail removable insulation at piping specialties, equipment connections, and access panels.
 6. Detail application of field-applied jackets.
 7. Detail application of Identification
 8. Detail application at linkages of control devices.
 9. Detail field application for each equipment type.
 10. Detail outdoor duct insulation installation.
- C. Samples: For each type of insulation jacket, and identification indicated. Identify each Sample, describing product and intended use.
1. Sample Sizes:
 - a. Preformed Pipe Insulation Materials: 12 inches long by NPS 2.
 - b. Sheet Form Insulation Materials: 12 inches square.
 - c. Jacket Materials for Pipe: 12 inches long by NPS 2.
 - d. Sheet Jacket Materials: 12 inches square.
 - e. Manufacturer's Color Charts: For products where color is specified, show the full range of colors available for each type of finish material.
- D. Material Test Reports: From a qualified testing agency acceptable to the authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.
- B. Fire-Test-Response Characteristics: Except where indicated otherwise, insulation and related materials shall have fire-test-response characteristics indicated, as determined by testing identical products per ASTM E 84, by UL or another testing and inspecting agency acceptable to the authorities having jurisdiction. Identify materials with appropriate markings of applicable testing and inspecting agency.
1. Insulation Installed Indoors:
 - a. Flame-Spread Index: 25 or less.
 - b. Smoke-Developed Index: 50 or less.
 2. Insulation Installed Outdoors:
 - a. Flame-Spread Index: 75 or less.
 - b. Smoke-Developed Index: 150 or less.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials to Project site in original packages and containers with seals unbroken and bearing manufacturer's original labels, including manufacturer's name, product name, and directions for storing, handling, and use.

- B. Store materials in a clean, dry, fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

- 1. Remove damaged or wet insulation from Project site.

1.6 COORDINATION

- A. Coordinate size and location of supports, hangers, and insulation shields specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Coordinate clearance requirements with piping Installer for piping insulation application, duct Installer for duct insulation application, and equipment Installer for equipment insulation application. Before preparing piping and ductwork Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.
- C. Coordinate installation and testing of heat tracing.

1.7 SCHEDULING

- A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.
- B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

1.8 WARRANTY

- A. Provide warranty on materials and labor for 60 months starting from date of delivery, or one year from date of Preliminary Acceptance or Substantial Completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers and Products: Subject to compliance with requirements, provide one of the following products for each of the products identified:
 - 1. Flexible Elastomeric Insulation:
 - a. Aeroflex USA Inc.; Aerocel.
 - b. Armacell LLC; AP Armaflex.
 - c. RBX Corporation; Insul-Sheet 1800 and Insul-Tube 180.
 - 2. Mineral-Fiber Blanket Insulation:
 - a. CertainTeed Corp.; Duct Wrap.
 - b. Johns Manville; Microlite.
 - c. Owens Corning; All-Service Duct Wrap.
 - 3. Mineral-Fiber Board Insulation:

- a. CertainTeed Corp.; Commercial Board.
 - b. Fibrex Insulations Inc.; FBX.
 - c. Johns Manville; 800 Series Spin-Glas.
 - d. Knauf Insulation; Insulation Board.
 - e. Manson Insulation Inc.; AK Board.
 - f. Owens Corning; Fiberglas 700 Series.
4. Fire-Rated Blanket Insulation:
 - a. CertainTeed Corp.; FlameChek.
 - b. Johns Manville; Firetemp Wrap.
 - c. Nelson Firestop Products; Nelson FSB Flameshield Blanket.
 - d. Thermal Ceramics; FireMaster Duct Wrap.
 - e. 3M; Fire Barrier Wrap Products.
 - f. Unifrax Corporation; FyreWrap.
 - g. Vesuvius; PYROSCAT FP FASTR Duct Wrap.
5. Mineral-Fiber, Preformed Pipe Insulation:
 - a. Fibrex Insulations Inc.; Coreplus 1200.
 - b. Johns Manville; Micro-Lok.
 - c. Knauf Insulation; 1000 Pipe Insulation.
 - d. Owens Corning; Fiberglas Pipe Insulation.
6. Mineral-Fiber, Pipe and Tank Insulation:
 - a. CertainTeed Corp.; CrimpWrap.
 - b. Johns Manville; MicroFlex.
 - c. Knauf Insulation; Pipe and Tank Insulation.
 - d. Manson Insulation Inc.; AK Flex.
 - e. Owens Corning; Fiberglas Pipe and Tank Insulation.
7. Polyisocyanurate Insulation:
 - a. Apache Products Company; ISO-25.
 - b. Dow Chemical Company (The); Trymer.
 - c. Duna USA Inc.; Corafoam.
 - d. Elliott Company; Elfoam.
8. Mineral-Fiber Insulating Cement:
 - a. Insulco, Division of MFS, Inc.; Triple I.
 - b. P. K. Insulation Mfg. Co., Inc.; Super-Stik.
9. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement:
 - a. Insulco, Division of MFS, Inc.; SmoothKote.
 - b. P. K. Insulation Mfg. Co., Inc.; PK No. 127, and Quik-Cote.
 - c. Rock Wool Manufacturing Company; Delta One Shot.
10. Polyisocyanurate, Adhesive:
 - a. Childers Products, Division of ITW; CP-96.
 - b. Foster Products Corporation, H. B. Fuller Company; 81-33.

11. Flexible Elastomeric and Polyolefin Adhesive:
 - a. Aeroflex USA Inc.; Aero seal.
 - b. Armacell LCC; 520 Adhesive.
 - c. Foster Products Corporation, H. B. Fuller Company; 85-75.
 - d. RBX Corporation; Rubatex Contact Adhesive.
12. Mineral-Fiber Adhesive:
 - a. Childers Products, Division of ITW; CP-82.
 - b. Foster Products Corporation, H. B. Fuller Company; 85-20.
 - c. ITW TACC, Division of Illinois Tool Works; S-90/80.
13. ASJ Adhesive, and FSK Adhesive:
 - a. Childers Products, Division of ITW; CP-82.
 - b. Foster Products Corporation, H. B. Fuller Company; 85-20.
 - c. ITW TACC, Division of Illinois Tool Works; S-90/80.
14. PVC Jacket Adhesive:
 - a. Dow Chemical Company (The); 739, Dow Silicone.
 - b. Johns-Manville; Zeston Perma-Weld, CEEL-TITE Solvent Welding Adhesive.
 - c. Speedline Corporation; Speedline Vinyl Adhesive.
15. Vapor-Barrier Mastic:
 - a. Childers Products, Division of ITW; CP-35.
 - b. Foster Products Corporation, H. B. Fuller Company; 30-90.
 - c. ITW TACC, Division of Illinois Tool Works; CB-50.
 - d. Vimasco Corporation; 749.
16. Lagging Adhesives:
 - a. Childers Products, Division of ITW; CP-52.
 - b. Foster Products Corporation, H. B. Fuller Company; 81-42.
 - c. Vimasco Corporation; 136.
17. Joint Sealants for Cellular-Glass, Phenolic, and Polyisocyanurate Products:
 - a. Childers Products, Division of ITW; CP-76.
 - b. Foster Products Corporation, H. B. Fuller Company; 30-45.
 - c. Pittsburgh Corning Corporation; Pittseal 444.
 - d. Vimasco Corporation; 750.
18. Metal Jacket Flashing Sealants:
 - a. Childers Products, Division of ITW; CP-76-8.
 - b. Foster Products Corporation, H. B. Fuller Company; 95-44.
 - c. Vimasco Corporation; 750.
19. ASJ Flashing Sealants and PVC Jacket Flashing Sealants:
 - a. Childers Products, Division of ITW; CP-76.

20. PVC Jacket:
 - a. Johns Manville; Zeston.
 - b. P.I.C. Plastics, Inc.; FG Series.
 - c. Proto PVC Corporation; LoSmoke.
 - d. Speedline Corporation; SmokeSafe.
21. Metal Jacket:
 - a. Childers Products, Division of ITW; Metal Jacketing Systems.
 - b. IDC Corporation.
 - c. RPR Products, Inc.; Insul-Mate.
22. Identification:
 - a. Seton.
 - b. Brady.
 - c. Best.
 - d. Milwaukee.
23. ASJ Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0835.
 - b. Compac Corp.; 104 and 105.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 428 AWF ASJ.
 - d. Venture Tape; 1540 CW Plus, 1542 CW Plus, and 1542 CW Plus/SQ.
24. FSK Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0827.
 - b. Compac Corp.; 110 and 111.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 491 AWF FSK.
 - d. Venture Tape; 1525 CW, 1528 CW, and 1528 CW/SQ.
25. PVC Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0555.
 - b. Compac Corp.; 130.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 370 White PVC tape.
 - d. Venture Tape; 1506 CW NS.
26. Bands:
 - a. Childers Products; Bands.
 - b. IDC Corporation.
 - c. RPR Products, Inc.; Bands.
27. Insulation Pins and Hangers:
 - a. AGM Industries, Inc.; CWP-1.
 - b. GEMCO; Cupped Head Weld Pin.
 - c. Midwest Fasteners, Inc.; Cupped Head.
 - d. Nelson Stud Welding; CHP

28. Wire:

- a. C & F Wire.
- b. Childers Products.
- c. PABCO Metals Corporation.
- d. RPR Products, Inc.

2.2 INSULATION MATERIALS

- A. Comply with requirements in PART 3 schedule articles for where insulating materials shall be applied.
- B. Products shall not contain asbestos, lead, mercury, or mercury compounds.
- C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.
- D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.
- E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.
- F. Flexible Elastomeric: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C 534, Type I for tubular materials and Type II for sheet materials.
- G. Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 553, Type II and ASTM C 1290, Type III with factory-applied FSK jacket. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- H. Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 612, Type IB. For duct and plenum applications, provide insulation with factory-applied FSK jacket. For equipment applications, provide insulation with factory-applied ASJ. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- I. Fire-Rated Blanket: High-temperature, flexible, blanket insulation with FSK jacket that is tested by UL, or another testing and inspecting agency acceptable to the authorities having jurisdiction, and certified to provide a 2-hour fire rating.
- J. Mineral-Fiber, Preformed Pipe Insulation:
- K. Mineral-Fiber, Pipe and Tank Insulation: Mineral or glass fibers bonded with a thermosetting resin. Semirigid board material with factory-applied FSK jacket complying with ASTM C 1393, Type II or Type IIIA Category 2. Nominal density is 2.5 lb/cu. ft. or more. Thermal conductivity (k-value) at 100 deg F is 0.29 Btu x in./h x sq. ft. x deg F or less. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- L. Polyisocyanurate: Unfaced, preformed, rigid cellular polyisocyanurate material intended for use as thermal insulation.
 1. Comply with ASTM C 591, Type I or Type IV, except thermal conductivity (k-value) shall not exceed 0.19 Btu x in./h x sq. ft. x deg F at 75 deg F after 180 days of aging.
 2. Flame-spread index shall be 25 or less and smoke-developed index shall be 50 or less for thickness up to 1-1/2 inches as tested by ASTM E 84.

3. Fabricate shapes according to ASTM C 450 and ASTM C 585.
4. Factory-Applied Jacket: Requirements are specified in "Factory-Applied Jackets" Article.
 - a. Pipe Applications: ASJ.
 - b. Equipment Applications: ASJ.

2.3 FIRE-RATED INSULATION SYSTEMS

- A. Fire-Rated Blanket: High-temperature, flexible, blanket insulation with FSK jacket that is tested and certified to provide a 2-hour fire rating by UL or another testing and inspecting agency acceptable to the authorities having jurisdiction.

2.4 INSULATING CEMENTS

- A. Mineral-Fiber Insulating Cement: Comply with ASTM C 195.
- B. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C 449/C 449M.

2.5 ADHESIVES

- A. Adhesives, General: All adhesives and mastics installed within the building envelope shall be shown to comply either with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers," including 2004 Addenda, or with the VOC limits established in South Coast Air Quality Management District (SCAQMD) Rule #1168.
 1. Aerosol adhesives shall comply with Green Seal Standard for Commercial Adhesives GS-36 (2000).
- B. Compatibility: Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.
- C. Polyisocyanurate Adhesive: Solvent-based resin adhesive, with a service temperature range of minus 75 to plus 300 deg F.
- D. Flexible Elastomeric Adhesive: Comply with MIL-A-24179A, Type II, Class I.
- E. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
- F. ASJ Adhesive, and FSK Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.
- G. PVC Jacket Adhesive: Compatible with PVC jacket.
- H. Mastics: Comply with MIL-C-19565C, Type II. Materials shall be compatible with insulation materials, jackets, and substrates.
 1. Vapor-Barrier Mastic: Water based; suitable for indoor and outdoor use on below ambient services.
 - a. Water-Vapor Permeance: ASTM E 96, Procedure B, 0.013 perm at 43-mildry film thickness.

- b. Service Temperature Range: Minus 20 to plus 180 deg F
- c. Solids Content: ASTM D 1644, 59 percent by volume and 71 percent by weight.
- d. Color: White.

2.6 Lagging Adhesives: Comply with MIL-A-3316C Class I, Grade A. Materials shall be compatible with insulation materials, jackets, and substrates.

- a. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over duct, equipment, and pipe insulation.
- b. Service Temperature Range: Minus 50 to plus 180 deg F.
- c. Color: White.

2.7 SEALANTS

A. Sealants, General: All sealants installed within the building envelope shall be shown to comply either with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers," including 2004 Addenda, or with the VOC limits established in South Coast Air Quality Management District (SCAQMD) Rule #1168.

B. Joint Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Permanently flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 100 to plus 300 deg F.
- 4. Color: White or gray.

C. Metal Jacket Flashing Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Fire- and water-resistant, flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 40 to plus 250 deg F
- 4. Color: Aluminum.

D. ASJ Flashing Sealants and PVC Jacket Flashing Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Fire- and water-resistant, flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 40 to plus 250 deg F.
- 4. Color: White.

2.8 JACKETS

A. Factory-Applied Jackets: Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

- 1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I..
- 2. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

B. Field-Applied Jackets: Comply with ASTM C 921, Type I, unless otherwise indicated. Field-applied jackets shall be provided on all piping below 8 ft. height.

1. PVC Jackets: High-impact-resistant, UV-resistant PVC complying with ASTM D 1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.
 - a. Adhesive: As recommended by jacket material manufacturer.
 - b. Color: White.
 - c. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.
 - 1) Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.
 - d. Factory-fabricated tank heads and tank side panels.
2. Metal Jackets:
 - a. Stainless-Steel Jackets: ASTM A 167 or ASTM A 240.
 - 1) Sheet and roll stock ready for shop or field sizing.
 - 2) Material, finish, and thickness are indicated in field-applied jacket schedules.
 - 3) Moisture Barrier for Indoor Applications: 1-mil-thick, heat-bonded polyethylene and kraft paper.
 - 4) Moisture Barrier for Outdoor Applications: 2.5-mil-thick Polysurlyn.
 - 5) Factory-Fabricated Fitting Covers: Fabricated from the same material, finish, and thickness as jacket. Provide factory-fabricated covers for preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows; tee covers; flange and union covers; end caps; beveled collars; and valve covers. Field fabricate fitting covers only if factory-fabricated fitting covers are not available.
 - b. Aluminum Jackets: Comply with ASTM B 209, Alloy 3003, Temper H-14.
 - 1) Sheet and roll stock ready for shop or field sizing.
 - 2) Finish and thickness are indicated in field-applied jacket schedule.
 - 3) Moisture Barrier for Indoor Applications: 3-mil thick Polysurlyn.
 - 4) Moisture Barrier for Outdoor Applications: 3-mil thick Polysurlyn.
 - 5) Factory-Fabricated Fitting Covers: Fabricated from the same material, finish, and thickness as jacket. Provide factory-fabricated covers for preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows; tee covers; flange and union covers; end caps; beveled collars; and valve covers. Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

2.9 IDENTIFICATION

- A. General: Products specified are manufacturer's standard products of categories and types required for each application as referenced in other Division 23 Sections. Where more than single type is specified for listed application, selection is Installer's option but provide single selection for each product category.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Seton.
 2. Brady.
 3. Best.
 4. Milwaukee.
- C. Snap-On Plastic Pipe Markers: Manufacturer's standard pre-printed, semi-rigid snap-on, color-coded pipe markers.
1. Lettering: Use piping system terms as indicated and abbreviate only as necessary for each application length.
 2. Arrows: Either integrally with piping system service lettering (to accommodate both directions), or as separate unit, on each pipe marker to indicate direction of flow.
- D. Lettering and Graphics: Coordinate names, abbreviations, and other designations used in mechanical identification, with corresponding designations indicated. Use numbers, letters, and terms indicated for proper identification, operation, and maintenance of mechanical systems and equipment.
1. Multiple Systems: Where multiple systems of same name are indicated, identify individual system number as well as service.
- E. Color Coding:
- | | <u>Background</u> | <u>Lettering</u> |
|---|-------------------|------------------|
| 1. Domestic Cold Water | White | Black |
| 2. Domestic Hot Water/ Hot Water Return | Blue | White |
| 3. Non Potable Water/ Makeup Water | Purple | White |
| 4. Storm Water | Grey | White |
| 5. Air Conditioning Condensate | Grey | White |
| 6. Refrigerant Suction/ Hot Gas | Clear | Black |
| 7. Natural Gas | Yellow | Black |
| 8. Fire Protection | Red | White |

2.10 TAPES

- A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.
1. Width: 3 inches.
 2. Thickness: 11.5 mils.
 3. Adhesion: 90 ounces force/inch in width.
 4. Elongation: 2 percent.
 5. Tensile Strength: 40 lbf/inch in width.
 6. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.
- B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C 1136.
1. Width: 3 inches.
 2. Thickness: 6.5 mils.
 3. Adhesion: 90 ounces force/inch in width.
 4. Elongation: 2 percent.
 5. Tensile Strength: 40 lbf/inch in width.
 6. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

- C. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive. Suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

2.11 SECUREMENTS

A. Bands:

1. Stainless Steel: ASTM A 167 or ASTM A 240/A 240M, Type 304; 0.020-inch thick, 3/4-inch wide with wing or closed seal.
2. Springs: Twin spring set constructed of stainless steel with ends flat and slotted to accept metal bands. Spring size determined by manufacturer for application.

B. Insulation Pins and Hangers:

1. Capacitor-Discharge-Weld Pins: Copper-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
2. Cupped-Head, Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch-diameter shank, length to suit depth of insulation indicated with integral 1-1/2-inch galvanized carbon-steel washer.
3. Metal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following:
 - a. Baseplate: Perforated, galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
 - b. Spindle: Copper- or zinc-coated, low carbon steel, fully annealed, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
 - c. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
4. Nonmetal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate fastened to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following:
 - a. Baseplate: Perforated, nylon sheet, 0.030 inch thick by 1-1/2 inches in diameter.
 - b. Spindle: Nylon, 0.106-inch-diameter shank, length to suit depth of insulation indicated, up to 2-1/2 inches.
 - c. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
5. Self-Sticking-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following requirements:

- a. Baseplate: Galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
 - b. Spindle: Copper- or zinc-coated, low carbon steel, fully annealed, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
 - c. Adhesive-backed base with a peel-off protective cover.
 - 6. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- thick, galvanized-steel sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
 - a. Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.
 - 7. Nonmetal Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick nylon sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
 - C. Staples: Outward-clinching insulation staples, nominal 3/4-inch- wide, stainless steel or Monel.
 - D. Wire: 0.062-inch soft-annealed, stainless steel.
- 2.12 CORNER ANGLES
- A. PVC Corner Angles: 30 mils thick, minimum 1 by 1 inch, PVC according to ASTM D 1784, Class 16354-C. White or color-coded to match adjacent surface.
 - B. Stainless-Steel Corner Angles: 0.024 inch thick, minimum 1 by 1 inch, stainless steel according to ASTM A 167 or ASTM A 240, Type 304 or Type 316.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions for compliance with requirements for installation and other conditions affecting performance of insulation application.
 - 1. Verify that systems and equipment to be insulated have been tested and are free of defects.
 - 2. Verify that surfaces to be insulated are clean and dry.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.
- B. Coordinate insulation installation with the trade installing heat tracing, if any. Comply with requirements for heat tracing that apply to insulation.
- C. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.3 INSTALLATION – GENERAL

- A. Install insulation materials, accessories, and finishes according to the manufacturer's written instructions with smooth, straight, and even surfaces; free of voids throughout the length of equipment, ducts and fittings, and piping including fittings, valves, and specialties.
- B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of equipment, duct system, and pipe system as specified in insulation system schedules.
- C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.
- D. Keep insulation materials dry during application and finishing.
- E. Install insulation with least number of joints practical.
- F. Install insulation with longitudinal seams at top and bottom of horizontal runs.
- G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.
- H. Install multiple layers of insulation with longitudinal and end seams staggered.
- I. Install vapor barriers on insulated pipes, ductwork, and equipment having surface operating temperatures below 60 deg F.
- J. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.
- K. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.
 - 1. Install insulation continuously through hangers and around anchor attachments.
 - 2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
 - 3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
 - 4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.
- L. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.
- M. Install insulation with factory-applied jackets as follows:
 - 1. Draw jacket tight and smooth.
 - 2. Cover circumferential joints with 3-inch- wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.

3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 2 inches o.c.
 - a. For below ambient services, apply vapor-barrier mastic over staples.
 - b. Exception: Do not staple longitudinal laps on insulation applied to piping systems with a surface temperature at or below 35 deg F.
 4. Cover joints and seams with tape as recommended by insulation material manufacturer to maintain vapor seal.
 5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to duct and pipe flanges and fittings.
 6. At penetrations in jackets for thermometers, and pressure gages, fill and seal voids with vapor barrier coating.
- N. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.
- O. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.
- P. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.
- Q. For above ambient services, do not install insulation to the following:
1. Vibration-control devices.
 2. Testing agency labels and stamps.
 3. Nameplates and data plates.
 4. Manholes.
 5. Handholes.
 6. Cleanouts.

3.4 PENETRATIONS

- A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.
1. Seal penetrations with flashing sealant.
 2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
 3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
 4. Seal jacket to roof flashing with flashing sealant.
- B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.
- C. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.
 2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
 3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.
 4. Seal jacket to wall flashing with flashing sealant.
- D. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.
- E. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions in accordance with requirements of approved and tested assembly indicated. Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches.
1. Comply with requirements in Division 07 Section "Penetration Firestopping" for firestopping and fire-resistive joint sealers.
- F. Insulation Installation at Floor Penetrations:
1. Duct: Install insulation continuously through floor penetrations that are not fire rated.
 - a. For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches.
 - b. Comply with requirements in Division 07 Section "Penetration Firestopping" for firestopping and fire-resistive joint sealers.
 2. Pipe: Install insulation continuously through floor penetrations.

3.5 PIPE INSULATION INSTALLATION – GENERAL

- A. General: Requirements in this article apply to all insulation materials, unless more specific requirements are indicated for individual pipe insulation types.
- B. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity, unless otherwise indicated.
 2. Pipe Elbows: Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
 3. Tee Fittings: Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
 4. Valves: Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe

- diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Strainers: Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below ambient services, provide a design that maintains vapor barrier.
 6. Fittings and Unions: Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
 7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below ambient services and a breather mastic for above ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
 8. For services not specified to receive a field-applied jacket except for flexible elastomeric, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
 9. Stencil or label the outside insulation jacket of each union with the word "UNION." Match size and color of pipe labels.
- C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes, vessels, and equipment. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.
- D. Install removable insulation covers at locations indicated on Drawings. Installation shall conform to the following:
1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
 2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel. Select band material compatible with insulation and jacket.
 3. Construct removable valve insulation covers in same manner as for flanges except divide the two-part section on the vertical center line of valve body.
 4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
 5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.6 FLEXIBLE ELASTOMERIC INSULATION INSTALLATION

- A. Seal longitudinal seams and end joints with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

B. Insulation Installation on Pipe Flanges:

1. Install pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of sheet insulation of same thickness as pipe insulation.
4. Secure insulation to flanges and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install mitered sections of pipe insulation.
2. Secure insulation materials and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed valve covers manufactured of same material as pipe insulation when available.
2. When preformed valve covers are not available, install cut sections of pipe and sheet insulation to valve body. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.
4. Secure insulation to valves and specialties and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

3.7 MINERAL-FIBER INSULATION INSTALLATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below ambient surfaces, do not staple longitudinal tabs but secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with mineral-fiber blanket insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed sections are not available, install mitered sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application.

E. Blanket Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.
2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:
 - a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.
 - b. On duct sides with dimensions larger than 18 inches place pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
 - c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
 - d. Do not over compress insulation during installation.
 - e. Impale insulation over pins and attach speed washers.
 - f. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.
4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from 1 edge and 1 end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.
 - a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.
 - b. Install vapor stops for ductwork and plenums operating below 50 deg Fat 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to 2 times the insulation thickness but not less than 3 inches

5. Overlap unfaced blankets a minimum of 2 inches on longitudinal seams and end joints. At end joints, secure with steel bands spaced a maximum of 18 inches o.c.
 6. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
 7. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.
- F. Board Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.
1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.
 2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
 3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:
 - a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.
 - b. On duct sides with dimensions larger than 18 inches space pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
 - c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
 - d. Do not over compress insulation during installation.
 - e. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.
 4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from 1 edge and 1 end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.
 - a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.
 - b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to 2 times the insulation thickness but not less than 3 inches.
 5. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Groove and score insulation to fit as closely as possible to outside and inside radius of elbows. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
 6. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

3.8 POLYISOCYANURATE INSULATION INSTALLATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of insulation to pipe with tape or bands and tighten without deforming insulation materials. Orient longitudinal joints between half sections in 3 o'clock and 9 o'clock positions on the pipe.
2. For insulation with factory-applied jackets with vapor barriers, do not staple longitudinal tabs but secure tabs with additional adhesive or tape as recommended by insulation material manufacturer and seal with vapor-barrier mastic.
3. All insulation shall be tightly butted and free of voids and gaps at all joints. Vapor barrier must be continuous. Before installing jacket material, install vapor-barrier system.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, same thickness of adjacent pipe insulation, not to exceed 1-1/2-inch thickness.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of polyisocyanurate block insulation of same thickness as pipe insulation.

C. Insulation Installation on Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation. Secure according to manufacturer's written instructions.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of polyisocyanurate insulation to valve body.
2. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.

3.9 WEATHERPROOF DUCT INSULATION INSTALLATION

A. Provide thickness of polyisocyanurate insulation as indicated in the Schedule at the end of PART 3. Provide on the exterior of all supply, and return, ducts exposed to the weather.

B. Cover insulation with a field-applied jacket.

C. Top of duct insulation shall overlap the side of duct insulation. Side of duct insulation shall overlap the bottom of duct insulation. Provide corner angles at each corner.

D. Top of duct insulation shall be pitched for drainage. Pitch shall be a minimum of 1/2 inch per foot from the high point at the duct horizontal longitudinal centerline to the low points at the outsides of the duct. Insulation thickness at the low points shall be the minimum indicated in the Schedule at the end of PART 3. Duct pitch shall be achieved by either of the following:

1. Tapered insulation.
2. Insulation with tapered fesco board to achieve the pitch.
3. For ducts over 6 ft. wide, tapered continuous wood shims (1 in. wide) running laterally from the duct centerline high point to the low points at the outsides. Shims shall be located at 12 inches o.c. along the length of the duct.

- E. Vapor retarder type mastic, or joint sealer, shall be applied on longitudinal and butt joints to prevent moisture and moisture vapor infiltration. Vapor retarder butt joints shall be sealed with 3 in. wide vapor retarder tape.
- F. Apply insulation in a solid bed of waterproof adhesive.
- G. Tightly wrap the insulation circumferentially with saran film. Overlap the seams by a minimum of 2 inches. Seal the overlapped seams with vapor retarder tape.
- H. Secure the insulation with stainless steel bands on 12-inch centers.
- I. Jacketing shall be secured with 1/2-inch wide stainless steel bands on 12 inch centers. Rivets, screws, staples, or any other fastener capable of penetrating the underlying vapor retarder shall not be used.

3.10 FIELD-APPLIED JACKET INSTALLATION

- A. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints; for horizontal applications, install with longitudinal seams along top and bottom of tanks and vessels. Seal with manufacturer's recommended adhesive.
 - 1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.
- B. Where Stainless Steel, or Aluminum, jackets are indicated, install with 2-inch overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

3.11 LABELING AND IDENTIFYING INSTALLATION

- A. Piping Systems: Install pipe markers on each system. Include arrows showing normal direction of flow.
 - 1. Plastic markers with application systems. Install on pipe insulation segment where required for hot non-insulated pipes.
 - a. Fasten markers on pipes smaller than 6-inch diameter by one of following methods:
 - 1) Snap-on application of pre-tensioned semi-rigid plastic pipe marker.
 - 2) Adhesive lap joint in pipe marker overlap.
 - 3) Laminated or bonded application of pipe marker to pipe (or insulation).
 - 4) Taped to pipe (or insulation) with color-coded plastic adhesive tape, not less than 3/4-inch wide, lapped 1-1/2 inches minimum at both ends of pipe marker, and covering full circumference of pipe.
 - b. Fasten markers on pipes 6-inch and larger diameter by one of following methods:
 - 1) Laminated or bonded application of pipe marker to pipe (or insulation).
 - 2) Taped to pipe (or insulation) with color-coded plastic adhesive tape, not less than 1-1/2-inches wide, lapped 3 inches minimum at both ends of pipe marker, and covering full circumference of pipe.

- 3) Strapped to pipe (or insulation) with manufacturer's standard stainless steel bands.
2. Locate pipe markers and color bands as follows wherever piping is exposed in finished spaces, machine rooms, accessible maintenance spaces (shafts, tunnels, plenums) and exterior non-concealed locations.
 - a. Near each valve and control device.
 - b. Near each branch connection, excluding short take-offs for fixtures and terminal units. Mark each pipe at branch, where flow pattern is not obvious.
 - c. Near penetrations through walls, floors, ceilings, or enter non-accessible enclosures.
 - d. At access doors, manholes, and similar access points that permit view of concealed piping.
 - e. Near major equipment items and other points of origination and termination.
 - f. Spaced at a maximum of 50-feet o.c. along each run. Reduce intervals to 25 feet in congested areas of piping and equipment.
 - g. On piping above removable acoustical ceilings, except omit intermediately spaced markers.

3.12 FIRE-RATED INSULATION SYSTEM INSTALLATION

- A. Where fire-rated insulation system is indicated, secure system to ducts and duct hangers and supports to maintain a continuous fire rating.
- B. Insulate duct access panels and doors to achieve same fire rating as duct.
- C. Install firestopping at penetrations through fire-rated assemblies in accordance with Division 07 Section "Penetration Firestopping."

3.13 FIELD QUALITY CONTROL

- A. Testing Agency: Contractor shall engage a qualified testing agency to perform tests and inspections.
- B. Tests and Inspections:
 1. Inspect ductwork, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location for each duct system defined in the Article "Duct Insulation Schedule, General".
 2. Inspect field-insulated equipment, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location for each type of equipment requiring insulation for this project. For large equipment, remove only a portion adequate to determine compliance.
 3. Inspect pipe, fittings, strainers, and valves, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, three locations of threaded fittings, three locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, three locations of threaded valves, and three locations of flanged valves for each pipe service defined in the Article "Piping Insulation Schedule, General".

- C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements. Architect may reject all work if sample work is found to be defective.

3.14 DUCT INSULATION PERFORMANCE , GENERAL

A. Plenums and Ducts Requiring Insulation:

1. Indoor, supply and outdoor air.
2. Indoor, concealed return located in non-conditioned space.
3. Indoor, return located in non-conditioned space.
4. Indoor, kitchen hood exhaust.
5. Indoor, exhaust between isolation damper and penetration of building exterior.
6. Outdoor, supply, return and outdoor air.

B. Items Not Insulated:

1. Factory-insulated flexible ducts.
2. Factory-insulated plenums and casings.
3. Flexible connectors.
4. Vibration-control devices.
5. Factory-insulated access panels and doors.

3.15 DUCT AND PLENUM INSULATION SCHEDULE

A. INSULATION APPLICATION SCHEDULE

1. General: Abbreviations used in the following schedules include:
 - a. Field-Applied Jackets: P - PVC, K - Foil and Paper, A - Aluminum, SS - Stainless Steel.

- B. Indoor, Concealed duct insulation shall be mineral fiber blanket unless indicated otherwise.

C. DUCT SYSTEMS INSULATION SCHEDULE

DUCT LOCATION	OUTSIDE AIR, SUPPLY AIR, EXHAUST AIR (DOWNSTREAM OF DAMPER) INSULATION INSTALLED R-VALUE (H·Ft ² ·deg F/BTU)	RETURN AIR INSULATION INSTALLED R-VALUE (H·Ft ² ·deg F/BTU)
Exterior of Building (Notes 5, 6, 8)	8.0	8.0
Ventilated Attic (Notes 5, 7)	8.0	8.0
Unvented Attic Above Insulated Ceiling (Notes 5, 7)	8.0	8.0
Unvented Attic with Roof Insulation (Notes 1, 5, 7)	6.0	N/A
Unconditioned Space (Notes 2, 5, 6, 7)	8.0	8.0
Indirectly Conditioned	N/A	N/A

Space (Notes 3, 7)		
Ceiling Cavity / Shafts / Soffits / Me- chanical Spaces and Rooms (Notes 4, 5, 6, 7)	6.0	N/A
Exposed Locations within Conditioned Space (Note 9)	N/A	N/A

NOTE 1: INSULATION R-VALUES, MEASURED IN (H X FT² X F)/BTU, ARE FOR THE INSULATION AS INSTALLED AND DO NOT INCLUDE FILM RESISTANCE. WHERE EXTERIOR WALLS ARE USED AS PLENUM WALLS, WALL INSULATION SHALL BE AS REQUIRED BY THE MOST RESTRICTIVE CONDITION OF ASHRAE 90.1-2004 SECTION 5 OR 6.4.4.2. INSULATION RESISTANCE MEASURED ON A HORIZONTAL PLANE IN ACCORDANCE WITH ASTM C518 AT A MEAN TEMPERATURE OF 75F AT THE INSTALLED THICKNESS.

NOTE 2: INCLUDING CRAWL SPACES (BOTH VENTILATED/NON-VENTILATED), FRAMED CAVITIES IN WALLS, FLOOR AND CEILING ASSEMBLIES WHICH (A) SEPARATE CONDITIONED SPACE FROM UNCONDITIONED SPACE OR OUTSIDE AIR, AND (B) ARE UNINSULATED ON THE SIDE FACING AWAY FROM CONDITIONED SPACE.

NOTE 3: RETURN AIR PLENUMS WITH OR WITHOUT EXPOSED ROOFS ABOVE.

NOTE 4: CAVITY CONTAINED WITHIN THE INSULATED BUILDING ENVELOPE.

NOTE 5: VAPOR BARRIER REQUIRED.

NOTE 6: FIELD APPLIED JACKET (STAINLESS STEEL, OR ALUMINUM, FOR EXTERIOR APPLICATIONS, PVC FOR INTERIOR EXPOSED LOCATIONS).

NOTE 7: PROVIDE MINERAL FIBER BOARD WITH FIELD APPLIED JACKET (SS, OR A, EXTERIOR, ALL SERVICE INTERIOR) IN EXPOSED LOCATIONS IN LIEU OF MINERAL FIBER BLANKET.

NOTE 8: PROVIDE POLYISOCYANURATE BOARD INSULATION FOR OUTDOOR DUCTWORK

NOTE 9: UNLESS NOTED OTHERWISE IN THE CONTRACT DOCUMENTS.

D. KITCHEN EXHAUST DUCTS

MATERIAL	FORM	THICKNESS (INCHES)	VAPOR BARRIER REQUIRED	FIELD-APPLIED JACKET
Fire-Rated Blanket	Blanket	Note 1	No	Aluminum

NOTE 1: AS REQUIRED TO PROVIDE A 2-HOUR FIRE RATING.

3.16 EQUIPMENT INSULATION SCHEDULE

- A. General: Insulation materials and thicknesses are identified below. If more than one material is listed for a type of equipment, selection from materials listed is Contractor's option.
- B. Insulate indoor and outdoor equipment in paragraphs below that are not factory insulated.

3.17 PIPING INSULATION SCHEDULE, GENERAL

- A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.
- B. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:
1. Drainage piping located in crawl spaces.
 2. Underground piping.
 3. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.18 PIPING INSULATION SCHEDULE

- A. Abbreviations used in the following schedule include:
1. Field-Applied Jackets: P - PVC, K - Foil and Paper, A - Aluminum, SS - Stainless Steel.
 2. Pipe Sizes: NPS - Nominal Pipe Size.
- B. Minimum HVAC Pipe Insulation Thickness Table :

HEATING SYSTEMS (Steam and Hot Water) (Note 1)	FLUID TEMPERATURE RANGE (deg F)	INSULATION TYPE / FIELD-APPLIED JACKET / IDENTIFICATION			PIPE SIZE AND INSULATION THICKNESS (INCHES) (Note 5, 6)				
		Glass Fiber	Polyiso-cyanurate	Flexible Elastomeric	<1" (Note 4)	1" to <1-1/2"	1-1/2" to <4"	4" to 6"	8" and Larger
COOLING SYSTEMS									
Refrigerant	40 – 60	X (Note 2, 7)	X (Note 2, 3, 7)	X (Note 7)	1/2	3/4	1	1	1
	Below 40	X (Note 2, 7)	X (Note 2, 3, 7)	X (Note 7)	1	1-1/2	1-1/2	1-1/2	1-1/2
Indoor Air-Conditioning Condensate Drains	40 – 60	X		X	1	1	1	1	1

NOTE 1: GLASS-FIBER INSULATION ONLY FOR HYDRONIC PIPING.

NOTE 2: (P), (A), OR (SS) FIELD-APPLIED JACKET ON OUTDOOR INSTALLATIONS, EXPOSED AND CONCEALED.

NOTE 3: FOR OUTDOOR USE ONLY.

NOTE 4: PIPING INSULATION IS NOT REQUIRED BETWEEN THE CONTROL VALVE AND COIL ON RUNOUTS, WHEN THE CONTROL VALVE IS WITHIN 4-FEET OF THE COIL AND THE PIPE SIZE IS 1-INCH OR LESS.

NOTE 5: FOR PIPING EXPOSED TO OUTDOOR AIR, INCREASE INSULATION THICKNESS BY 1-INCH.

NOTE 6: INSULATION THICKNESS IS BASED ON INSULATION HAVING A THERMAL CONDUCTIVITY OF 0.22 – 0.25 BTU·INCH/(H·FT²·deg F)

NOTE 7: VAPOR BARRIER REQUIRED.

3.19 INDOOR, FIELD-APPLIED JACKET SCHEDULE

- A. Install jacket over insulation material for all piping below 8 ft. height. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.
- B. Provide identification for all piping.
- C. Piping, Exposed:
 - 1. PVC, White – below 8 ft. height: 20 mils thick.

3.20 OUTDOOR, FIELD-APPLIED JACKET SCHEDULE

- A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.
- B. Ducts and Plenums, Exposed, up to 48 Inches in Diameter or with Flat Surfaces up to 72 Inches
 - 1. Stainless Steel, Type 304, Smooth 2B Finish: 0.020 inch thick.
 - 2. Aluminum, Alloy 3003, Temper H-14, Smooth Finish: 0.020 inch thick.
- C. Ducts and Plenums, Exposed, Larger Than 48 Inches in Diameter or with Flat Surfaces Larger Than 72 Inches:
 - 1. Stainless Steel, Type 304, Smooth Finish: 0.020 inch thick.
 - 2. Aluminum, Alloy 3003, Temper H-14, Smooth Finish: 0.020 inch thick.
- D. Piping, Exposed
 - 1. Stainless Steel, Type 304 Smooth 2B Finish 0.016 inch thick.

END OF SECTION 230700

SECTION 23 08 00 - COMMISSIONING OF HVAC**PART 1 – GENERAL****1.01 SUMMARY****A. Work Included:**

1. Definitions, warranties, test equipment requirements, and mechanical commissioning requirements.

1.02 RELATED SECTIONS

- A. Contents of Division 23, HVAC and Division 01, General Requirements apply to this Section.
- B. In addition, reference the following:
 1. Section 01 91 13, General Commissioning Requirements.

1.03 REFERENCES AND STANDARDS

- A. References and Standards as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, meet the following:
 1. Current edition of ASHRAE Guideline 0, The Commissioning Process.

1.04 SUBMITTALS

- A. Submittals as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, provide:
 1. Certificates of readiness.
 2. Certificates of completion of installation, prestart, and startup activities.
 3. Operation and Maintenance Manuals.
 4. Test reports.
 5. Control Drawings Submittal
 - a. Provide a key to abbreviations.
 - b. Provide graphic schematic depictions of the systems and each component.
 - c. Include the system and component layout of any equipment that the control system monitors, enables or controls, even if the equipment is primarily controlled by packaged or integral controls.
 - d. Provide a full points list with at least the following included for each point:
 - 1) Controlled system
 - 2) Point abbreviation

- 3) Point description
 - 4) Display unit
 - 5) Control point or set point (Yes / No)
 - 6) Monitoring point (Yes / No)
 - 7) Intermediate point (Yes / No)
 - 8) Calculated point (Yes / No)
6. Architect forwards one set of submittals for systems to be commissioned to Commissioning Agent at same time as design team.
 7. Commissioning Agent forwards comments to design team for consideration in their submittal response.
 8. Design team sends consolidated response to submittals and copies to Commissioning Agent.

1.05 QUALITY ASSURANCE

- A. Quality assurance as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. Test Equipment Calibration Requirements: Contractors will comply with test manufacturer's calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Commissioning Authority upon request.

1.06 WARRANTY

- A. Warranty of materials and workmanship as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, provide:
 1. Commissioning, inspecting, and testing will not modify terms or time periods of mechanical equipment, systems, and controls warranties including related equipment and systems, and adjacent work.
 2. Control system warranty period starts from date of Commissioning Agent acceptance.

1.07 COORDINATION

- A. Reference Section 01 91 13, General Commissioning Requirements, for requirements pertaining to coordination during the commissioning process.

1.08 PURPOSE

- A. Purpose of commissioning process is to provide Owner assurance that systems have been installed in prescribed manner and will operate within performance guidelines. Commissioning is intended to enhance quality of system startup and aid in orderly transfer of systems to beneficial use by Owner.
- B. Commissioning procedures and results will be observed by Commissioning Authority or Owner's staff. Contractor is expected to verify functional readiness of systems to be tested prior to performing the tests in presence of Owner's witness. A high rate of test failure will indicate that Contractor has not adequately verified readiness of systems.

PART 2 – PRODUCTS

2.01 TEST EQUIPMENT

- A. Provide standard testing equipment required to perform startup, initial checkout and functional performance testing for the equipment being tested. For example, the mechanical contractor of Division 23, HVAC will ultimately be responsible for standard testing equipment for the HVAC&R system and controls system in Division 23, HVAC, except for the equipment specific to and used by TAB in their commissioning responsibilities. Provide a sufficient quantity of two-way radios by each subcontractor.
- B. Include special equipment, tools and instruments (specific to a piece of equipment and only available from vendor) required for testing in the base bid price to the Owner and leave on site, except for stand-alone data logging equipment that may be used by the Commissioning Authority.
- C. Manufacturer of equipment to provide proprietary test equipment and software required for programming and/or start-up, whether specified or not. Manufacturer provides the test equipment, demonstrates its use, and assists in the commissioning process as needed. Proprietary test equipment (and software) become the property of the Owner upon completion of the commissioning process.
- D. Data logging equipment and software required to test equipment will be provided by the Commissioning Authority, and will not become the property of the Owner.
- E. Use only testing equipment of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the specifications. If not otherwise noted, the following minimum requirements apply: Temperature sensors and digital thermometers have a certified calibration within the past year to an accuracy of 0.5 degree F and a resolution of plus or minus 0.1 degree F. Pressure sensors have an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and have been calibrated within the last year.

PART 3 – EXECUTION

3.01 GENERAL DOCUMENTATION REQUIREMENTS

- A. With assistance from the installing contractors, the Commissioning Authority will prepare prefunctional checklists for commissioned components, equipment, and systems

B. Red-Lined Drawings:

1. Verify equipment, systems, instrumentation, wiring and components are shown correctly on red-lined drawings.
2. Preliminary red-lined drawings must be made available to the Commissioning Team for use prior to the start of Functional Performance Testing.
3. Changes, as a result of Functional Testing, must be incorporated into the final as-built drawings, which will be created from the red-lined drawings.
4. The contracted party, as defined in the Contract Documents will create the as-built drawings.

C. Operation and Maintenance (O&M) Data:

1. Contractor will provide a copy of O&M literature within 45 days of each submittal acceptance for use during the commissioning process for commissioned equipment and systems.
2. The Commissioning Authority will review the O&M literature once for conformance to project requirements.
3. The Commissioning Authority will receive a copy of the final approved O&M literature once corrections have been made by the Contractor.

D. Demonstration and Training:

1. Contractor will provide demonstration and training as required by the specifications.
2. A complete training plan and schedule must be submitted by the contractor to the Commissioning Authority four weeks prior to any training.
3. A training agenda for each training session must be submitted to the Commissioning Authority one week prior the training session.
4. Notify the Commissioning Authority at least 72 hours in advance of scheduled tests so that testing may be observed by the Commissioning Authority and Owner's Authorized Representative. Provide a copy of the test record to the Commissioning Authority, Owner, and Architect.
5. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain specific equipment.
6. Train Owner's maintenance personnel on procedures and schedules for starting and stopping, trouble shooting, servicing, and maintaining equipment.
7. Review data in O&M Manuals.

E. Systems Manual Requirements:

1. The Systems Manual is intended to be a usable information resource containing the information related to the systems, assemblies, and Commissioning Process in one place with indexes and cross references.
2. Include final approved versions of the following information for the Systems Manual:
 - a. Facility Description.

- b. Basis of Design (A/E).
 - c. A list of contractors, subcontractors, suppliers, architects, and engineers involved in the project along with their contact information.
 - d. Overview of each commissioned system including operational schedules and sequence of operations.
 - e. General maintenance recommendations and procedures.
 - f. Recommendations for recalibration frequency of sensors and actuators
 - g. Recommended best practices for keeping the system running efficiently.
 - h. Blank Functional Performance Tests so the Owner can recommission the facility at a later date.
3. Organize and arrange information by building system, such as fire alarm, chilled water, heating hot water, etc.
 4. Provide Information in an electronic version to the extent possible. Legible, scanned images are acceptable for non-electronic documentation to facilitate this deliverable.

3.02 CONTRACTOR'S RESPONSIBILITIES

- A. Mechanical, Controls and TAB Contractors. The commissioning responsibilities applicable to each of the mechanical, controls and TAB contractors of Division 23, HVAC are as follows (references apply to commissioned equipment only):
 1. Perform commissioning tests at the direction of the Commissioning ~~Authority~~Agent.
 2. Attend construction phase controls coordination meetings.
 3. Attend testing, adjusting, and balancing review and coordination meetings.
 4. Participate in HVAC&R systems, assemblies, equipment, and component maintenance orientation and inspection as directed by the Commissioning Authority.
 5. Provide information requested by the Commissioning Authority for final commissioning documentation.
 6. Include requirements for submittal data, operation and maintenance data, and training in each purchase order or subcontract written.
 7. Prepare preliminary schedule for mechanical system orientations and inspections, operation and maintenance manual submissions, training sessions, pipe and duct system testing, flushing and cleaning, equipment start-up, testing and balancing and task completion for owner. Distribute preliminary schedule to commissioning team members.
 8. Update schedule as required throughout the construction period.
 9. During the startup and initial checkout process, execute the related portions of the prefunctional checklists for commissioned equipment.

10. Contractor to participate and complete checklists using the Commissioning Authority's web based commissioning software ~~Facility Grid~~Checklisten App. A desktop, laptop, tablet, or iPad will be required.
 11. Assist the Commissioning Authority in verification and functional performance tests.
 12. Gather operation and maintenance literature on equipment, and assemble in binders as required by the specifications. Submit to Commissioning Authority 45 days after submittal acceptance.
- B. Coordinate with the Commissioning Authority to provide 48 hour advance notice so that the witnessing of equipment and system start-up and testing can begin.
- C. Notify the Commissioning Authority a minimum of two weeks in advance of the time for start of the testing and balancing work. Attend the initial testing and balancing meeting for review of the official testing and balancing procedures.
- D. Participate in, and schedule vendors and contractors to participate in the training sessions.
- E. Provide written notification to the Construction Manager/General Contractor (CM/GC) and Commissioning Authority that the following work has been completed in accordance with the Contract Documents, and that the equipment, systems, and sub-system are operating as required.
1. HVAC&R equipment including fans, air handling units, ductwork, dampers, terminals, and other equipment furnished under this Division.
 2. Fire stopping in the fire rated construction, including fire and smoke damper installation, caulking, gasketing and sealing of smoke barriers.
 3. Fire detection and smoke detection devices furnished under other divisions of the specification.
- F. Equipment supplier to document the performance of his equipment.
- G. Test, Adjust and Balance Contractor:
- ~~1. Attend initial commissioning coordination meeting scheduled by the Commissioning Authority.~~
 2. Participate in verification of the testing and balancing report, which will consist of repeating measurements contained in the testing and balancing reports. Assist in diagnostic purposes when directed.
- H. Provide training of the Owner's operating staff using expert qualified personnel, as specified.
- I. Equipment Suppliers:
1. Provide requested submittal data, including detailed start-up procedures and specific responsibilities of the Owner, to keep warranties in force.
 2. Assist in equipment testing per agreements with contractors.
 3. Provide information requested by Commissioning Authority regarding equipment sequence of operation and testing procedures.
- J. Reference Section 01 91 13, General Commissioning Requirements for additional contractor responsibilities.

3.03 OWNER'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Owner's Responsibilities.

3.04 DESIGN PROFESSIONAL'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Design Professional's Responsibilities.

3.05 COMMISSIONING AUTHORITY'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Commissioning Authority's Responsibilities.

3.06 TESTING PREPARATION

- A. Certify in writing to the Commissioning Authority that HVAC&R systems, subsystems, and equipment have been installed, calibrated, and started and are operating according to the Contract Documents.
- B. Certify in writing to the Commissioning Authority that HVAC&R instrumentation and control systems have been completed and calibrated, that they are operating according to the Contract Documents, and that pretest set points have been recorded.
- C. Certify in writing that testing, adjusting, and balancing procedures have been completed and that testing, adjusting, and balancing reports have been submitted, discrepancies corrected, and corrective work approved.
- D. Place systems, subsystems, and equipment into operating mode to be tested (e.g., normal shutdown, normal auto position, normal manual position, unoccupied cycle, emergency power, and alarm conditions).
- E. Inspect and verify the position of each device and interlock identified on checklists.
- F. Check safety cutouts, alarms, and interlocks with smoke control and life-safety systems during each mode of operation.
- G. Testing Instrumentation: Install measuring instruments and logging devices to record test data as directed by the Commissioning Authority.

3.07 TESTING, ADJUSTING AND BALANCING VERIFICATION

- A. Prior to performance of Testing, Adjusting and Balancing work, provide copies of reports, sample forms, checklists, and certificates to the Commissioning Authority.
- B. Notify the Commissioning Authority at least 10 days in advance of testing and balancing Work, and provide access for the Commissioning Authority to witness testing and balancing Work.

- C. Provide technicians, instrumentation, and tools to verify testing and balancing of HVAC&R systems at the direction of the Commissioning Authority.
 - 1. The Commissioning Authority will notify testing and balancing subcontractor 10 days in advance of the date of field verification. Notice will not include data points to be verified.
 - 2. Testing and balancing subcontractor to use the same instruments (by model and serial number) that were used when original data were collected.
 - 3. Failure of an item includes, other than sound, a deviation of more than 10 percent. Failure of more than 10 percent of selected items to result in rejection of final testing, adjusting, and balancing report. For sound pressure readings, a deviation of 3 dB to result in rejection of final testing. Variations in background noise must be considered.
 - 4. Remedy the deficiency and notify the Commissioning Authority so verification of failed portions can be performed.

3.08 GENERAL TESTING REQUIREMENTS

- A. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of the Commissioning Authority.
- B. Scope of HVAC&R testing to include entire HVAC&R installation, from central equipment for heat generation and refrigeration through distribution systems to each conditioned space. Testing to include measuring capacities and effectiveness of operational and control functions.
- C. Test operating modes, interlocks, control responses, and responses to abnormal or emergency conditions, and verify proper response of building automation system controllers and sensors.
- D. The Commissioning Authority along with the HVAC&R contractor, testing and balancing Subcontractor, and HVAC&R Instrumentation and Control Subcontractor to prepare detailed testing plans, procedures, and checklists for HVAC&R systems, subsystems, and equipment.
- E. Tests will be performed using design conditions whenever possible.
- F. Simulated conditions may need to be imposed using an artificial load when it is not practical to test under design conditions. Before simulating conditions, calibrate testing instruments. Provide equipment to simulate loads. Set simulated conditions as directed by the Commissioning Authority and document simulated conditions and methods of simulation. After tests, return settings to normal operating conditions.
- G. The Commissioning Authority may direct that set points be altered when simulating conditions is not practical.
- H. The Commissioning Authority may direct that sensor values be altered with a signal generator when design or simulating conditions and altering set points are not practical.
- I. If tests cannot be completed because of a deficiency outside the scope of the HVAC&R system, document the deficiency and report it to the Owner. After deficiencies are resolved, reschedule tests.
- J. If the testing plan indicates specific seasonal testing, complete appropriate initial performance tests and documentation and schedule seasonal tests.

3.09 HVAC&R SYSTEMS, SUBSYSTEMS, AND EQUIPMENT TESTING PROCEDURES

- A. Equipment Testing and Acceptance Procedures: Testing requirements are specified in individual Division 23, HVAC Sections. Provide submittals, test data, inspector record, and certifications to the Commissioning Authority.
- B. HVAC&R Instrumentation and Control System Testing: Field testing plans and testing requirements are specified in Division 23, HVAC Sections "Instrumentation and Control for HVAC" and "Sequence of Operations for HVAC Controls." Assist the Commissioning Authority with preparation of testing plans.
- C. Pipe System Cleaning, Flushing, Hydrostatic Tests, and Chemical Treatment: Test requirements are specified in Division 23, HVAC Piping Sections. HVAC&R Contractor to prepare a pipe system cleaning, flushing, and hydrostatic testing plan. Provide cleaning, flushing, testing, and treating plan and final reports to the Commissioning Authority. Plan to include the following:
 - 1. Sequence of testing and testing procedures for each section of pipe to be tested, identified by pipe zone or sector identification marker. Markers keyed to Drawings for each pipe sector, showing the physical location of each designated pipe test section. Provide drawings keyed to pipe zones or sectors formatted to allow each section of piping to be physically located and identified when referred to in pipe system cleaning, flushing, hydrostatic testing, and chemical treatment plan.
 - 2. Description of equipment for flushing operations.
 - 3. Minimum flushing water velocity.
 - 4. Tracking checklist for managing and ensuring that pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.
- D. The work included in the commissioning process involves a complete and thorough evaluation of the operation and performance of components, systems and sub-systems. Evaluate the following equipment and systems:
 - 1. HVAC Equipment and Systems (all)
 - 3. Building Automation System
 - 5. Computer Room AC
 - 6. Domestic Hot Water Systems
 - 8. Pumps

3.10 DEFICIENCIES/NONCONFORMANCE, COST OF RETESTING, FAILURE DUE TO MANUFACTURER DEFECT

- A. Reference Division 01, General Requirements for requirements pertaining to deficiencies/nonconformance, cost of retesting, or failure due to manufacturer defect.

3.11 OPERATION AND MAINTENANCE MANUALS

- A. The Operation and Maintenance Manuals to conform to Contract Documents requirements as stated in Division 23, HVAC.
- B. Provide an updated as-built version of the control drawings and sequences of operation in the final controls O&M manual submittal.

END OF SECTION 23 08 00

SECTION 23 09 20 – BUILDING AUTOMATION SYSTEM (BAS) - GENERAL**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the installation of a Building Automation/Management System (BAS/BMS) for control and monitoring of systems (HVAC, energy, and lighting) as set forth with Lidl National (NAT) & International (INT).

1.2 APPLICATION OF BUILDING AUTOMATION SYSTEM (BAS)

- A. The distributed digital control (DDC) and Building Automation System (BAS/BMS) defined herein shall provide a complete BMS for all mechanical systems and other facility systems as included in the project documents. The contractor shall provide a complete and operational system to perform all sequences of operations stated in Section 23 or as shown on the control drawings.
- B. The BMS Head Controller/Building Level Controller shall communicate with Lidl server in Germany via an approved "L-Box" or equal configuration.
- C. The BMS shall utilize electronic sensing, microprocessor-based digital control, and electronic actuation of dampers and valves to perform the control sequences and functions specified.
- D. The BMS for this project shall consist of monitoring and control of the systems indicated.
- E. All interlock wiring for mechanical system equipment shall be by this contractor unless specifically stated otherwise. This shall include, but not be limited to, items such as thermostats for unit heaters, interlock wiring to rooftop factory controllers, flow switches, and duct smoke detectors.
- F. System to be a fully integrated BMS incorporating direct digital control (DDC) for energy management, equipment monitoring and control, and subsystems with open communications capabilities as herein specified.
- G. All wiring, conduit, panels, and accessories for a complete operational system.
- H. BMS Contractor shall be responsible for all electrical work associated with the BMS:
 - 1. Perform all wiring in accordance with all local and national codes.
 - 2. All line voltage and low voltage control wiring, concealed or exposed, shall be in conduit in accordance with the division 26 specifications, NEC and local building code.
 - 3. Provide a maximum of 50 feet extension of 120 volt, 20 amp circuits and circuit breakers from Emergency power panels for all BMS equipment power. All control panels shall be wired to the UPS systems.
 - 4. Surge transient protection shall be incorporated in design of system to protect electrical components in all DDC Controllers and the operator interface.
 - 5. Provide installation kits for fluid flow meters, which allow service and removal under pressure.

1.3 APPLICATION OF ENERGY CONSERVATION INITIATIVES

- A. Night set back schedules for space unoccupied times shall be incorporated into design.

- B. Demand control ventilation shall be incorporated as a function of rooftop factory controls.
- C. Design systems for spaces that need 24 hours per day operations separate from spaces that can utilize night set back. Example, data centers shall be design separate from the building HVAC system.
- D. Incoming electrical and gas utilities shall be metered.

1.4 APPLICATION OF OPEN PROTOCOLS

- A. Provide open communications system. The system shall be an open architecture with the capabilities to support a multi-vendor environment. The BMS shall be capable of utilizing industry-standard protocols and as well as with integrate third-party systems as follows:
 - 1. Capable of Modbus RTU communication
 - 2. Capable of BACnet-MSTP communication according to ANSI/ASHRAE 135-2004.
 - 3. Capable of BACnet-IP communication according to ANSI/ASHRAE 135-2008.
 - 4. All supervisory (BC) and field-level controllers (AAC, ASC) shall be certified and listed under the appropriate device profile.
- B. Software Licensing:
 - 1. Include open/non-proprietary licensing for all software packages at the required Operator Interface located on the BMS panel.
 - 2. All operator interface, programming environment, networking, database management and any other software used by the Contractor to install the system or needed to operate the system to its full capabilities shall be open-licensed and provided to the Owner.
 - 3. Open/non-proprietary licensing shall allow access to all aspects of the system including system access, thin-client workstations, points, programming, database management, graphics etc. No restrictions shall be placed on the licensing.

1.5 APPLICATION OF HARDWARE

- A. Provide hardware, software, and wiring to provide communication and/or interfaces with each of the applicable major systems listed below:
 - 1. Rooftop Units
 - 2. AC Units
 - 3. Exhaust Fan Units
 - 4. Lighting Control System
 - 5. UPS System
 - 6. Generator System
 - 7. Gas Meter
 - 8. Electrical Meter
 - 9. Miscellaneous Monitoring
- B. Operator interface hardware and standard peripherals located on the BMS panel shall be furnished by the BMS Contractor and it shall match Lidl's IT Department standards.

1.6 APPLICATION OF GRAPHICS AND WEB-BASED SYSTEM

- A. Provide system graphics for each controlled device and/or integrated systems as required by the owner.

- B. Origin of information shall be transparent to the operator and shall be controlled, displayed, trended, etc. as if the points were hardwired to the BMS.
- C. System should be web based with the user interface software – including graphics - imbedded in each Building Controller.
- D. Any owner provided BMS computer(s) connected physically or virtually to the store's local BMS network shall have access to the web based user interface.

1.7 APPROVED CONTROL SYSTEM AND INTEGRATION CONTRACTORS

- A. The BMS system shall be furnished, engineered, installed and subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Emerson Climate Technologies, Inc.
 - b. No Substitutions/Exceptions without owner approval
- B. The BMS system shall be designed and installed, commissioned, and serviced by factory trained personnel. BMS contractor shall have an in-place support facility within 100 miles of the site with technical staff, spare parts inventory and necessary test and diagnostic equipment. The BMS contractor shall provide full time, on site, experienced project manager for this work, responsible for direct supervision of the design, installation, start up and commissioning of the BMS. The Bidder shall be regularly engaged in the installation and maintenance of BMS systems and shall have a minimum of ten (10) years of demonstrated technical expertise and experience in the installation and maintenance of BMS systems similar in size and complexity to this project.
- C. The BMS contractor shall maintain a service organization consisting of factory trained service personnel and provide a list of 10 projects, similar in size and scope to this project, completed within the last five years.

1.8 CODES, STANDARDS, AND REFERENCES

- A. The BMS design shall meet the most stringent of the Local Code, ASHRAE, or Lidl's Building Design Criteria Matrix conditions and shall meet latest the most stringent ASHRAE's standard 55 Thermal Environmental Conditions for occupant comfort zone or Lidl's operating temperatures.
- B. Work, materials, and equipment shall comply with the most restrictive of local, state, and federal authorities' codes and ordinances or these plans and specifications. As a minimum, the design shall comply with current editions of the following codes:
- C. National Electric Code (NEC), latest edition
- D. ANSI Standards, latest edition
- E. Local Building Code, latest edition (including mechanical, plumbing, fire protection, and energy codes)
- F. FM Global Property Loss Prevention Data Sheets and approved products as applicable.
- G. American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE) Standards, Latest edition.

1. 55: Thermal Environmental Conditions for Human Occupancy.
2. 62.1: Ventilation for Acceptable Indoor Air Quality.
3. 90.1: Energy Standard for Buildings.
4. 135: 2010: BACnet-A Data Communication Protocol for Building Automation and Control Networks.

H. Electronics Industries Alliance:

1. EIA-709.1-A-99: Control Network Protocol Specification.
2. EIA-709.3-99: Free-Topology Twisted-Pair Channel Specification.
3. EIA-232: Interface between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange.
4. EIA-458: Standard Optical Fiber Material Classes and Preferred Sizes.
5. EIA-485: Standard for Electrical Characteristics of Generator and Receivers for use in Balanced Digital Multipoint Systems.
6. EIA-472: General and Sectional Specifications for Fiber Optic Cable.
7. EIA-475: Generic and Sectional Specifications for Fiber Optic Connectors and all Sectional Specifications.
8. EIA-573: Generic and Sectional Specifications for Field Portable Polishing Device for Preparation Optical Fiber and all Sectional Specifications.
9. EIA-590: Standard for Physical Location and Protection of Below-Ground Fiber Optic Cable Plant and all Sectional Specifications.
10. EIA-852: Tunneling of Component Network Data over IP Channels.

I. Underwriters Laboratories:

1. UL 916: Energy Management Systems.
2. ULC C100: Smoke Control System Equipment.
3. UL 864: Control Unit and Accessories for Fire Alarm Systems.

J. NEMA Compliance:

1. NEMA 250: Enclosure for Electrical Equipment.
2. NEMA ICS 1: General Standards for Industrial Controls.

K. NFPA Compliance:

1. NFPA 90A: "Standard for the Installation of Air Conditioning and Ventilating Systems" where applicable to controls and control sequences.
2. NFPA 70: National Electrical Code (NEC).

L. Institute of Electrical and Electronics Engineers (IEEE):

1. IEEE 142: Recommended Practice for Grounding of Industrial and Commercial Power Systems.
2. IEEE 802.3: CSMA/CD (Ethernet – Based) LAN.
3. IEEE 802.4: Token Bus Working Group (ARCNET – Based) LAN.

M. European Electromagnetic Compatibility (EMC):

1. EN-50081-2: Emission Standard for Residential, Commercial, and Light-Industrial Environments.

2. EN-50082-2: Emission Standard for Industrial Environments.
- N. Federal Communications Commission (FCC):
1. 47-CFR-15: Regulations regarding Unlicensed Transmissions.
- O. International Organization for Standardizations (ISO):
1. ISO-9001:2008: Model for Quality Assurance in Production, Installation, and Servicing. Quality Management Systems.
 2. ISO-140001: Application of Well-Accepted Business Management Principles to the Environment.
- P. Materials and equipment shall be the catalogued products of manufacturers regularly engaged in production and installation of automatic temperature control systems and shall be manufacturer's latest standard design that complies with the specification requirements.
- Q. All BMS peer-to-peer building controllers and local user displays shall be UL Listed under Standard UL 916, category PAZX; Standard ULC C100, category UUKL7; and under Standard UL 864, categories UUKL, UDTZ, and QVAX and be so listed at the time of bid.
- R. The BMS peer-to-peer network controllers and local user display shall also comply with the European Electromagnetic Compatibility (EMC) Framework, and bear the C-Tic Mark to show compliance. The purpose of the regulation is to minimize electromagnetic interference between electronic products, which may diminish the performance of electrical products or disrupt essential communications.
- S. DDC peer-to-peer controllers shall be compliant with the European EMC Directive, Standards EN 50081-2 and EN 50082-2, at the Industrial Levels. Additionally the equipment shall be compliant with the European LVD Directive and bear the CE mark in order to show compliance to both directives.
- T. All field level controllers shall comply, at a minimum, with UL Standard UL 916 category PAZX; and be so listed at the time of Bid.
- U. All electronic equipment shall conform to the requirements of FCC Regulation, Part 15, and Governing Radio Frequency Electromagnetic Interference and be so labeled.
- V. The manufacturer of the building automation system shall provide documentation supporting compliance with ISO-9001:2008 and ISO-140001. The intent of this specification requirement is to ensure that the products from the manufacturer are delivered through a Quality System and Framework that will assure consistency in the products delivered for this project.
- 1.9 DEFINITIONS
- A. Acknowledged: Data is broadcast repeatedly until an acknowledgement is received. Used for critical data using one to one bindings only. This type of service shall not be used for one to many bindings.
- B. Adjustable (ADJ): A characteristic of a control logic parameter such that it can be varied by the operator without downloading the program.
- C. Advanced Application Controller (AAC): A device with limited resources relative to the Building Controller (BC). It may support a level of programming and may also be intended for

application specific applications. A fully programmable control module. This control module shall be capable of certain advanced features found in Building Controllers (e.g. storing trends, and initiating read and write requests) but it shall not serve as a master controller. Advanced Application Controllers may reside on either the Ethernet/IP backbone or on a subnet. A BACnet device to be used as an AAC will meet the requirements of ASHRAE 135, Annex L and will be listed as an AAC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.

- D. Application Programming Tool: A vendor unique software tool used to create applications for programmable controllers.
- E. Application Specific Controller (ASC): A device with limited resources relative to the Advanced Application Controller (AAC). A pre-programmed control module, intended for use in a specific application. ASCs shall have limited configurability, allowing the user to select various pre-programmed options, but it shall not be fully customizable. A BACnet device used as an ASC will meet the requirements of ASHRAE 135, Annex L and will be listed as an ASC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.
- F. BACnet/BACnet Standard: BACnet communication requirements, as defined by ASHRAE/ANSI 135, current version including all annexes and addenda.
- G. Bandwidth Utilization: The average utilization of the network capacity. Network loading is controlled by the use of event driven broadcast based data propagation and the use of appropriate binding services.
- H. Broadcasting: The propagation of data from a device to the control network. Software objects that broadcast data to the network shall include the following parameters:
- I. Building Automation System (BAS): Same as Building Management System (BMS). The entire integrated energy management and control system.
- J. Building Controller (BC) (also "Head Controller"): A fully programmable control module capable of storing trends and schedules, serving as a router to devices on a subnet, and initiating read and write requests to other controllers. Typically this controller is located on the Ethernet/IP backbone of the BMS. A BACnet device to be used as a BC shall meet the requirements of ASHRAE 135, Annex L and will be listed as a BC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.
- K. Change of Value (COV): An event that occurs when a measured or calculated analog value changes by a predefined amount (ASHRAE/ANSI 135-1995).
- L. Client: A device that is the requestor of services from a server. A client device makes requests of, and receives responses from, a server device.
- M. Configuration Parameter: An input network variable to a controller that is written to the EEPROM.
- N. Connection: Made when a device is physically connected to a communication cable. Devices that count against the number of connections limit include adapters, any sensor, actuator, or controller, and a router or repeater. Terminators are not considered to be a connection.
- O. Continuous Monitoring: Sampling and recording of a variable based on time or change of state (e.g. trending an analog value, monitoring a binary change of state).

- P. Web Server (WS): Provides access to the control system. This device will allow access to the control system with a web browser over the Owner's network. The webserver shall have NIC cards so that it functions as the bridge between the local supervisory LAN and the Owner.
- Q. Direct Digital Control (DDC): Microprocessor-based control including Analog/Digital conversion and program logic
- R. Media Access Control address (MAC): Hardware address that uniquely identifies each node of a network. Each different type of network medium requires a different MAC layer.
- S. Managed Communication: Transmission of data from a controller to a data manager, which in turn re-broadcasts the data to a second controller.
- T. Manual Control: Where the operator takes control of an end device and forces a specific position or state. The manual mode and the desired manual position or states are parameters that are set by the operator.
- U. Network Controller: Term used to describe a BMS hardware component that serves several key functions:
 - 1. Serve as the LANID.
 - 2. Transmission of data to an owner interface on the TCP/IP network.
 - 3. Capable of collecting/integrating with different protocol data such as BACnet IP, BACnet MS/TP, Modbus, LON, OPC, etc.
 - 4. Location for time schedules to support all of the devices.
 - 5. Location for trend logs for all data to be trended from the devices.
 - 6. Location for alarm handling software.
 - 7. Shall process event broadcasted data from the devices (alarm indication) and enter the appropriate alarm information in the alarm reporting system at the TCP/IP level.
- V. Stand-Alone Controller: A stand-alone controller has provisions for all of the physical inputs and physical outputs associated with a single mechanical component such as a terminal unit, air handling unit, chiller or boiler. The controller shall also have embedded in it all of the control logic that associated the physical inputs to the physical outputs. A stand-alone controller may rely on other networked devices for time schedule inputs and trend data storage.
- W. Test Mode: A concept where the operator from the operator interface can interrupt the flow of data from a sensor to the control logic and insert a mandatory test value or test state to be used by the control logic. The test mode and the desired test value or states are parameters that are set by the operator.
- X. Unacknowledged Repeated: The data being broadcast is sent three times and an acknowledgement of receipt is not required. This type of service shall be used for most process control related data requiring timely receipt of the data.
- Y. Unacknowledged: The data being broadcast is sent one time and an acknowledgement of receipt is not required. This type of service shall be used for non-critical data where there is no significant impact should the receiving device have to wait for the next broadcast.

1.10 FUNCTIONAL INTENT

- A. Where detailed functional or performance requirements are specified, products intended for the Project, conforming to the specified requirements, must be submitted to, and approved by, Owner prior to shipment to the Project site.

1.11 SUBMITTALS

- A. Electronic Submittals: While all requirements for hard copy submittal apply, control submittals and O&M information shall also be provided in electronic format as follows.
1. Drawings and Diagrams: Shop drawings shall be provided on electronic media as an AutoCAD drawing file and/or Adobe Portable Document Format (PDF) file. All 'X-Reference' and font files must be provided with AutoCAD files.
 2. Other Submittals: All other submittals shall be provided in Adobe PDF. Provide documents, such as Cheat Sheets and Trouble Shooting Guide, in rich text format (RTF) or Microsoft Word format as required.
 3. BIM modeling for temperature controls shall be furnished by the BMS Contractor. All control panels and raceways (at $\frac{3}{4}$ " or greater) shall be modeled for clash-detection efforts with the prime contractor.
 4. The naming convention for all physical control points and software objects shall be reviewed in a meeting with the Owner's Representative, Commissioning Agent, and Engineer before generation of control drawings and sequences of operation.
 5. Where applicable, any deviations against the design intent shall be summarized with associated description of said deviation and itemized with an associated credit for final approval. Deviation(s) not listed and submitted may warrant an automatic rejection.
 6. All sequences shall be reviewed with thought and care with respect to programming needs for the specific project's design. Insufficient or generic sequence details may warrant an automatic rejection.
 7. Generic, non-compliant submittals shall not be reviewed, and shall be returned for resubmittal.
 8. BMS Contractor shall not order material, begin fabrication or field installation until receiving authorization to proceed in the form of an approved submittal. BMS Contractor shall be solely responsible for the removal and replacement of any item not approved by submittal at no cost to the Owner.
 9. It is the BMS contractors' responsibility to coordinate and allow adequate time for an approval process. Where an expedited submittal review is required, the contractor(s) shall still be responsible for any associated project delays and pay all additional services incurred by the design engineer, at the design engineer's current standard rate.
 10. The design engineer shall perform no more than two reviews of each submittal item. Should the submittals not be acceptable after the second review, the Contractor shall pay all additional expenses incurred by the design engineer in reviewing additional submittals at the design engineer's current standard rate.
 11. At the time of the 23 09 00 series submittal, the BMS Contractor shall submit a copy of the following with a response of "comply" or "non-comply" for every bullet point for:
 - a. 23 09 20, Section 1.9 "SUBMITTALS"
 - b. Entirety of the 23 09 00 series and sub-sections, upon further request
- B. Shop Drawings: Submit shop drawings for each control system, including a complete drawing for each air handling unit, system, pump, device, etc. with all point descriptors, addresses and point names indicated. Each shop drawing shall contain the following information:
1. System Architecture and System Layout:

- a. One-line diagram indicating schematic locations of all control units, owner interface, LAN interface devices, gateways, etc. Indicate network number, device ID, address, device instance, MAC address, drawing reference number, and controller type for each control unit. Indicate media, protocol, baud rate, and type of each LAN. All optical isolators, repeaters, end-of-line resistors, junctions, terminators, ground locations etc. shall be located on the diagram.
 - b. Provide floor plans locating all control units, owner interfaces, head controller, LAN interface devices, gateways, etc. Include all WAN and LAN communication wiring routing, power wiring, power originating sources, and low voltage power wiring. Indicate network number, device ID, address, device instance, MAC address, drawing reference number, and controller type for each control unit. All optical isolators, repeaters, end-of-line resistors, junctions, ground locations etc. shall be located on the floor plans. Wiring routing as-built conditions shall be maintained accurately throughout the construction period and the drawing shall be updated to accurately reflect accurate, actual installed conditions.
2. Schematic flow diagram of each air and water system showing fans, coils, dampers, valves, pumps, heat exchange equipment and control devices. Include contractor written description of sequence of operation.
3. All physical points on the schematic flow diagram shall be indicated with names, descriptors, and point addresses identified as listed in the point summary table.
4. Provide a dedicated shop drawings of all space sensors, thermostats, OA sensors, and space pressure sensors for final EOR and AOR review.
5. With each schematic, provide a point summary table listing building number and abbreviation, system type, equipment type, full point name, point description, Ethernet backbone network number, network number, device ID, object ID (object type, instance number). Refer to other Building Automation System (BAS) 23 09 00 specification series for additional requirements.
6. Label each control device with setting.
7. Label each input and output with the appropriate range.
8. Provide a Bill of Materials with each schematic. Indicate device identification to match schematic and actual field labeling, quantity, actual product ordering number, manufacturer, description, size, voltage range, pressure range, temperature range, etc. as applicable. Also identify the specification section and specification reference.
9. Indicate all required electrical wiring. Electrical wiring diagrams shall include both ladder logic type diagram for motor starter, RTU, control, and safety circuits and detailed digital interface panel point termination diagrams with all wire numbers and terminal block numbers identified. Provide panel termination drawings on separate drawings. Ladder diagrams shall appear on system schematic. Clearly differentiate between portions of wiring, factory-installed and portions to be field-installed. For all devices with safety circuits, field wiring will be labeled and all added devices will be properly mounted. Any internal wiring changes shall be approved by the manufacturer in writing. If for example a gas booster needs to be tied into the burner circuit the manufacturer shall identify the terminal points and provide an updated control diagram.
10. Sample Operator Interface Graphic Screens for each unique type of system, with final screens to be received 60 days prior to system startup.
11. Details of control panels, including controls, instruments, and labeling shown in plan or elevation indicating the installed locations.
12. Sheets shall be consecutively numbered.
13. Each sheet shall have a title indicating the type of information included and the HVAC system controlled.
14. Table of Contents listing sheet titles and sheet numbers.
15. Legend and list of abbreviations.

16. Provide an operating schedule for review. The schedule shall have a schedule for each AHU/RTU and the associated equipment. Terminal units serving offices will have a separate schedule from the labs.
- C. Product Data:
1. For each control device, panel, and accessory indicated or furnished. Include dimensions, capacities, performance and electrical characteristics, and material finishes.
 2. Include installation and start-up instructions.
 3. When manufacturer's cutsheets apply to a product series rather than a specific product, clearly indicate applicable data by highlighting or by other means general catalogs shall not be accepted as cutsheets to fulfill submittal requirements.
- D. Product Line Demonstrated History: The product line being proposed for the project must have an installed history of demonstrated satisfactory operation for a length of one (1) year since date of final completion in at least ten (10) installations of comparative size and complexity. Submittals shall document this requirement with references.
- E. Qualifications: For manufacturer, companies, Contractor and key personnel.
- F. Checkout and Testing Forms: Submit a blank copy of the forms that will be used during Point-to-Point Checkout, Prefunctional Checkout, and Functional Performance Testing. Those forms should be structured to capture the following information at a minimum during each particular testing phase.
1. Point-to-Point Checkout Form containing the following information:
 - a. Each point is addressed, labeled and that proper communication exists between the controller and the field device.
 - b. Documents that installed condition match the control drawings and that any changes or differences are noted on the drawings.
 2. Prefunctional Checkout Forms containing the following information:
 - a. Documents correct voltage and or current present as well as verifying circuits are free from grounds or faults for each control device.
 - b. Obtain and Record Test and Balance settings and incorporate into the BMS. Information from the TAB contractor shall include:
 - 1) Air system differential pressure and flow settings.
 - 2) RTU minimum outside air control point or damper setting.
 - c. Calibration data for all sensing and actuating devices recording final measured and displayed value. Record the type and model of the meter(s) that determined the measured value for analog inputs.
 - d. For analog outputs record both the displayed output as well state of the receiving device.
 - e. For digital input/outputs record the signal at the controller and the state of the sensing/control device.
 - f. For actuators:
 - 1) Check to ensure that actuated device moves smoothly and results are repeatable thru full range and seals tightly when the appropriate signal is applied to the operator.

- 2) Check for appropriate fail position, and that the stroke and range is as required.
 - 3) For sequenced electronic actuators, calibrate in accordance with manufacturer's instructions to required ranges. Record final settings.
 - g. For all actuators, verify the actual position against the Operator Interface readout.
 - h. For air flow measuring stations the data recorded shall include the independent flow measurement, area, and the independently measured output of the flow station. The BMS input from the flow station and any factors used to calculate the flow including area and any constants used in the calculation of flow. Two sets of data shall be collected. The first at design flow and the second at 50% of design flow. It is not acceptable to simply add a correction factor to address differences between the flow station and the independent reading.
 - i. For Operator Interfaces and Web accessible display:
 - 1) Verify that all elements on the graphics are functional and are properly bound to physical devices and/or virtual points, and that hot links or page jumps are functional and logical.
 - 2) Output all specified BMS reports for review and approval.
 - 3) Verify that the alarm pop ups, printing, and logging are functional and in accordance with requirements.
 - 4) Verify that all points are trended and are archiving to disk. Provide a sample to the Commissioning Authority and owner for review.
 - 5) Verify that paging/dial-out alarm annunciation is functional.
 - 6) Verify the functionality of remote Operator Interfaces and that a robust connection can be established consistently.
 - 7) Verify that required third party software applications required with the bid are installed and are functional.
 - j. For all actuating devices record final settings for device.
 - k. Document verification of point to graphics binding for all points displayed on the owner interface and that webserver display have been mapped correctly, and display the correct information.
 - l. Document that the webserver is on LAN and can be viewed from off site, that the modem is connected, and that the BMS is accessible by the contractor.
3. Functional Performance Forms shall contain:
- a. List of all sequences, modes of operation and setpoint that initiates each sequence and/or mode. For each confirm that proper sequence of operation. Document any variance between designed sequence and actual condition.
 - b. Record tuning parameters and response time for each control loop.
 - c. Document all alarm and safeties test and final results.
 - d. Results of trends including controlled points, setpoints, actual readings, and other point defined by the Boards Authorized Representative.

G. Testing Plan:

1. Submit a plan for executing all phases of testing and completion of checkout forms. This includes the following: manufacturers' normal testing, point-to-point testing, pre-functional testing, and functional performance testing. The testing plan shall show the overall milestones of the controls work and testing of the controls system.
2. Provide the schedule for completing each phase of testing for each system or set of equipment including, but not limited to, rooftop units, exhaust fans, miscellaneous

monitoring network wiring, and operator interface. Schedules shall show the time frame needed to complete the tasks.

3. The testing plan shall identify other trade milestones that impact the successful completion of during each phase of testing.
4. This plan is not meant to take precedence over any other plan but is intended to provide coordination assistance to all trades as the project is scheduled.

H. Open Protocol Information:

1. General: Provide all information necessary for review of the proposed system, including information required by the authority maintaining the protocol standard to determine if the product selected for implementation complies with the protocol standards specified.
2. BACnet Systems:
 - a. BACnet object description, object ID, and device ID, for each I/O point.
 - b. Documentation for any non-standard BACnet objects, properties, or enumerations used detailing their structure, data types, and any associated lists of enumerated values.
 - c. Documentation of all explicit messaging.
 - d. Submit PICS indicating the standardized BACnet device profile, functionality and configuration of each controller along with proof of BTL listing.

I. Framed Control Drawings: After completion of installation and check out, but prior to training, laminated control drawings including system control schematics, sequences of operation, and panel termination drawings, shall be provided in panels for major pieces of equipment. Terminal unit drawings shall be located in the central plant equipment panel or mechanical room panel.

J. Control Logic Documentation:

1. Documents are to be submitted, recieved, and approved prior to training.
2. Submit control logic program listings (for graphical programming) and logic flow charts illustrating (for line type programs) to document the control software of all control units.
3. Control logic shall be annotated to describe how it accomplishes the sequence of operation. Annotations shall be sufficient to allow an operator to relate each program component (block or line) to corresponding portions of the specified Sequence of Operation.
4. Include written description of each control sequence.
5. Include control response, settings, setpoints, throttling ranges, gains, reset schedules, adjustable parameters, and limits.
6. Sheets shall be consecutively numbered.
7. Each sheet shall have a title indicating the controller designations and the HVAC system controlled.
8. Include Table of Contents listing sheet titles and sheet numbers
9. Submit one complete set of programming and operating manuals for all digital controllers concurrently with control logic documentation.

K. Operation and Maintenance Manual:

1. Completion of project shall include Operation and Maintenance (O&M) Manuals. Consultant shall outline requirements for Contractor's submission and review with Lidl.
2. Operations and Maintenance Manuals shall be submitted during construction. Manuals shall be formatted in current CSI Master Specification format. Reviewed O&M Manuals shall be submitted in full prior to Lidl's facilities operation participation in start-up activities

- near the end of construction and to be coordinated with Information Resources standards.
3. In addition to other copies required, submit one copy of the approved materials directly for building operation staff.
 4. Submit maintenance instructions and spare parts lists for each type of control device, control unit, and accessory.
 5. Submit BMS User's Guides (Operating Manuals) for each controller type and for all operator interface hardware and software and peripherals.
 6. Submit BMS advanced Programming Manuals for each controller type and for all operator interface software.
 7. Include all as-built submittals (product data, shop drawings, control logic documentation, hardware manuals, software manuals, installation guides or manuals, maintenance instructions, and spare parts lists) in maintenance manual, in accordance with requirements of Division 01 Section "Operations and Maintenance Manual."
- L. Video Training: The BMS contractor shall adhere to the Demonstration and Training Videos requirements detailed in section 01 79 00.
- M. Demonstration of operator interface device display capabilities.
- N. Manufacturers Certificates: For all listed and/or labeled products, provide certificate of conformance.
- O. Product Warranty Certificates: Submit manufacturers product warranty certificates covering the hardware provided.

1.12 PROJECT RECORD DOCUMENTS

- A. The Project Record documents that have not already been submitted as part of the Operating and Maintenance Manual or Training Manual are to be submitted with the Record Documents. Any documents in the Operating and Maintenance Manual or Training Manual that have changed since they were submitted will need to be re-submitted as part of the Project record documents. All of these documents may be submitted electronically.
- B. Record copies of product data and control shop drawings updated to reflect the final installed condition.
- C. Record copies of approved control logic programming and database on USB drives. The USB drives shall contain all information required to reinstall the control system program. It will include actual setpoints and settings of controls, final sequence of operation, including changes to programs made after submission and approval of shop drawings and including changes to programs made during specified testing. One set of USB drives will be stored at the building in the main control panel and the second set will be provided to Owner's facilities department. A third set shall be provided to the EOR for record keeping only.
- D. Graphic Software: Record copies of approved project specific graphic software on USB drives.
- E. Provide as-built network architecture drawings showing all BACnet nodes including a description field with specific controller identification, description and location information.
- F. Include individual floor plans with controller locations with all interconnecting wiring routing including space sensors, LAN wiring, power wiring, low voltage power wiring. Indicate device instance, MAC address and drawing reference number.

- G. Provide record riser diagram showing the location of all controllers.
- H. Maintain project record documents throughout the warranty period and submit final documents at the end of the warranty period

1.13 OPERATOR INTERFACE

- A. The Operator Interface (OI) shall provide for overall system supervision, graphical user interface, management report generation, alarm annunciation, remote monitoring, and trend reporting. Refer to other Building Automation System (BAS) 23 09 00 specification series for additional requirements.

1.14 SYSTEM ARCHITECTURE

- A. Application of Open Protocols:
 - 1. Subject to the detailed requirements provided throughout the specifications, the BMS and digital control and communications components installed, as work of this contract shall be an integrated distributed processing system utilizing the following standards:
 - a. BACnet or Hybrid System: The system architecture shall consist of a BACnet IP Router, a single Local Area Network (LAN) or two-level LANs that support BCs, AACs, ASCs, Operator Interface, Smart Devices (SD), and Remote Communication Devices (RCDs) as applicable. In no event shall there be more than two levels of LAN topology within the system, excluding wiring to sensors with no control intelligence.
- B. The system provided shall incorporate hardware resources sufficient to meet the functional requirements specified. The Contractor shall include all items not specifically itemized in these Specifications that are necessary to implement, maintain, and operate the system in compliance with the functional intent of these Specifications.
- C. The system shall be configured as a distributed processing network(s) capable of expansion as specified below. Refer to the network architecture on the BMS drawings for other requirements and details.
- D. The system architecture shall consist of an Ethernet-based, wide area network (WAN), a single Local Area Network (LAN) or multi-leveled LANs that support BCs, Operator Interface, and Remote Communication Devices (RCDs) as applicable. The following indicates a functional description of the BMS structure.
 - 1. Owner WAN: Intranet-based network connecting multiple facilities with a central data warehouse and BMS interface/server, accessible via standard web-browser.
 - 2. Local BMS Supervisory LAN: The Local BMS Supervisory LAN shall be an Ethernet-based, 100 Mbps LAN connecting Primary Control LANs and OWSs. The LAN serves as the inter-BC gateway and OWS-to-BC gateway and communications path and as the connection point for the Owner WAN. LAN shall be IEEE 802.3 Ethernet over Fiber or Category 5 cable with switches and routers that support 100 Mbps throughput. Power-line carrier communication shall not be acceptable for communications. The higher level layers of this network shall be BACnet/IP of the BACnet standard, and shall share a common network number for the Ethernet backbone, as defined in BACnet.
Primary Controller LAN ('Primary LAN'): High-speed, peer-to-peer communicating LAN used to connect and BC and communicate exclusively control information. A BACnet network shall be used to connect AACs, ASCs or SDs. These can be Master Slave/

Token Passing or polling, or Modbus RTU, in addition to those allowed for Primary Controller LANs. Network speed vs. the number of controllers on the LAN shall be dictated by the response time and trending requirements. The primary network shall communicate at a minimum of 38 kbps. Each secondary network may support up to 32 communicating devices without segmentation or repeaters subject to the requirements for response time, trending and bandwidth utilization.

- E. Dynamic Data Access: Any data throughout any level of the network shall be available to and accessible by all other devices, Controllers and OWS, whether directly connected or connected remotely.
- F. Remote Data Access: The system shall support the following methods of remote access to the building data.
 - 1. Browser-based access: A remote user, connecting via the Owner WAN and using a standard browser shall be able access all control system facilities and graphics with proper password. The remote access user will require a password to enable applications to view the web pages.
- G. Network Performance: The communication speed between the controllers, control LAN interface devices, and operator interface devices shall be sufficient to ensure fast system response time under any loading condition. Contractor shall submit guaranteed response times with shop drawings including calculations to support the guarantee. Contractor shall reconfigure LAN as necessary to accomplish these performance requirements. The performance shall also include the trending of all AI, AO and DI points at 15-minute intervals.
- H. Data Storage and Operator Interface:
 - 1. The BMS contractor shall provide a Temperature Control Panel (TCP) with a DDC user LCD interface and a Building Level Controller (BLC) with Web Server (WS) functionalities. These two items shall maintain the systems configuration and programming database and is the operating platform BMS System.
 - 2. It shall hold the backup files of the information downloaded into the individual controllers and as such support uploading and downloading that information directly to/from the controllers.
 - 3. It shall be located within each facility.
 - 4. It shall also act as a control information server to non-HVAC system based controls.
 - 5. It shall allow secure multiple-access to the control information.
 - 6. Refer to Division 23 Section "Building Automation System (BAS)" sections and drawings for further requirements.
- I. The BC shall monitor, control, and provide the field interface for all points specified. Each BC shall be capable of performing all specified energy management functions, and all DDC functions, independent of other BCs and operator interface devices as more fully specified in Division 23 "Building Automation System (BAS)" sections.
- J. Systems Configuration Database: The system architecture shall support maintaining the systems configuration database on a BMS Operator Interface or BC on the Local Supervisory LAN. User tools provided to Owner shall allow configuring, updating, and maintaining current configurations and settings at the BMS Operator Interface or BC.
 - 1. Database Schema shall be published and provided to Owner to facilitate easy access to the data.

2. Database shall be ODBC compliant or a data access driver shall be provided to act as an ODBC or OLE DB data provider.
 - K. Interruptions or fault at any point on any Primary Controller LAN shall not interrupt communications between other nodes on the network. If a LAN is severed, two separate networks shall be formed and communications within each network shall continue uninterrupted.
 - L. All line drivers, repeaters, terminators, signal boosters, and signal conditioners shall be provided as necessary for proper data communication.
 - M. Anytime any controller's database or program is changed in the field, the controller shall be capable of automatically uploading the new data to the BMS Operator Interface and BC controller.
- 1.15 WARRANTY MAINTENANCE
- A. Warranty shall include materials and labor during first two (2) years of operation from substantial completion date or completion of all punch list items, whichever is later.
 - B. The Owner reserves the right to make changes to the BMS during the warranty period. Such changes do not constitute a waiver of warranty. The Contractor shall warrant parts and installation work regardless of any such changes made by owner, unless the Contractor provides clear and convincing evidence that a specific problem is the result of such changes to the BMS.
 - C. At no cost to the Owner during the warranty period, the Contractor shall provide maintenance services for software and hardware components as specified below:
 1. Maintenance services shall be provided for all devices and hardware specified in Division 23 Section "Building Automation System (BMS)" Sections. Service all equipment per the manufacturer's recommendations. All devices shall be calibrated within the last month of the warranty period.
 2. Emergency Service: Any malfunction, failure, or defect in any hardware component or failure of any control programming that would result in property damage or loss of comfort control shall be corrected and repaired following telephonic notification by Owner to the Contractor.
 - a. Response by telephone to any request for service shall be provided within two (2) hours of the initial telephone request for service.
 - b. In the event that the malfunction, failure, or defect is not corrected through the telephonic communication, at least one (1) hardware and software technician, trained in the system to be serviced, shall be dispatched to the site within eight (8) hours of the initial telephone request for such services, as specified.
 - c. Emergency service shall be available on a 24-hour, 7-day-a-week basis.
 3. Normal Service: Any malfunction, failure, or defect in any hardware component or failure of any control programming that would not result in property damage or loss of comfort control shall be corrected and repaired following telephonic notification by Owner to the Contractor.
 - a. Response by telephone to any request for service shall be provided within eight (8) working hours (contractor specified 40 hr/week normal working period) of the initial telephone request for service.

- b. In the event that the malfunction, failure, or defect is not corrected through the telephonic communication, at least one (1) hardware and software technician, trained in the system to be serviced, shall be dispatched to the site within three (3) working days of the initial telephone request for such services, as specified.
- 4. Telephonic Request for Service: Contractor shall provide up to three telephone numbers for Owner to call in the event of a need for service. At least one of the lines shall be attended 24 hours a day, 7 days a week. Alternatively, pagers can be used for technicians trained in system to be serviced. A technician shall respond to every call within 15 minutes.
- 5. Technical Support: Contractor shall provide technical support by telephone throughout the warranty period.
- 6. Preventive maintenance shall be provided throughout the warranty period in accordance with the hardware component manufacturer's requirements.

1.16 DELIVERY, STORAGE, AND HANDLING

- A. Provide factory-shipping cartons for each piece of equipment and control device. Maintain cartons during shipping, storage and handling as required to prevent equipment damage, and to eliminate dirt and moisture from equipment. Store equipment and materials inside and protect from weather.

1.17 LISTING AND LABELING

- A. The BMS and components shall be listed by Underwriters Laboratories (UL 916) as an Energy Management System.
- B. BACnet controllers, B-BC, B-AAC, B-ASC, etc. shall carry the BTL Mark for their device profile.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Materials shall be new, the best of their respective kinds without imperfections or blemishes, and not be damaged in any way. Used equipment shall not be used in any way for the permanent installation, no exception.

2.2 UNIFORMITY

- A. To the extent practical, all equipment of the same type serving the same function shall be identical and from the same manufacturer.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected.
- B. Where a discrepancy exist among details, schematic, specification, or sequence, the most stringent shall apply to complete a fully functional system.

3.2 INSTALLATION OF CONTROL SYSTEMS

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings.

- B. Factory/Owner Acceptance Test: a test shall be conducted by this contractor with the owner and designer with a minimum of three months prior to onsite installation. The intent of this verification/approval is to determine that all custom built panels, programs, graphical user interface screens, PID loops for Rooftop Units equipment comply with specifications and construction codes. Approval by Owner and EOR are required prior to installation. All delays incurred due to non-compliance shall be at the cost of this contractor.
- C. The Contractor shall furnish and install any power supply surge protection, filters, and other equipment as necessary for proper operation and protection of all BCs, operator interfaces, routers, gateways and other hardware and interface devices. All equipment shall be capable of handling voltage variations 10% above or below measured nominal value, with no effect on hardware, software, communications, and data storage.

3.3 CONTROL POWER SOURCE AND SUPPLY

- A. Extend all power source wiring required for operation of all equipment and devices provided under Division 23 "Building Automation System (BAS)" sections.

3.4 STARTUP, COMMISSIONING, AND TRAINING

- A. Refer to Division 01 "Demonstration and Training" sections.
- B. Refer to Division 23 "Building Automation System (BAS)" sections.
- C. At the direction of the engineer during construction, testing implementation, and/or after acceptance, allow for a minimum of (24) hours of application programming modifications to the sequence of operation to accommodate tuning of controlled system devices to match intended functional performance.

3.5 SEQUENCE OF OPERATION

- A. Refer to Division 23 "Building Automation System (BAS)" sections or control drawings.

3.6 IDENTIFICATION STANDARDS

- A. Controller Identification. All controllers shall be identified by a plastic engraved nameplate securely fastened to the outside of the controller enclosure.
- B. Panel Identification. All local control panels shall be identified by a plastic engraved nameplate securely fastened to the outside of the controller enclosure.
- C. Field Devices. All field devices shall be identified by a typed (not handwritten) securely attached tag label.
- D. Panel Devices. All panel devices shall be identified by a typed label securely fastened to the backplane of the local control panel.
- E. Raceway Identification. All the covers to junction and pull boxes of the control system raceways shall be painted blue or have identification labels stating "Control System Wiring" affixed to the covers. Labels shall be typed, not hand written.
- F. Wire Identification. All low and line voltage control wiring shall be identified by a number, as referenced to the associated control diagram, at each end of the conductor or cable. Identification number shall be permanently secured to the conductor or cable and shall be typed

END OF SECTION 230920

SECTION 230921 - BUILDING AUTOMATION SYSTEM (BAS) – INTERFACE AND COMMUNICATION DEVICES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes network integration devices.

1.2 DESCRIPTION OF WORK

- A. The BMS head controller and interface controllers (e.g. BX-300) shall be approved by LIDL.
- B. The BMS Contractor shall install a LCD Operator Interfaces and BC controller with web serving capabilities for the BMS functions specified.
- C. Contractor shall provide all interface devices and software to provide an integrated system connecting Building Controllers, Advanced Application Controllers, Application Specific Controllers, Gateways, and Web Server. The BMS System shall be connected to the Owner's WAN.
- D. The BMS Contractor shall provide all software and licenses for the BAS System to integrate with project specific equipment with factory provided BACnet MSTP, BACnet IP, Modbus RTU, or LON.
- E. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.

PART 2 - PRODUCTS**2.1 GENERAL**

- A. Materials:
 - 1. All products used in this project installation shall be new and currently manufactured and shall have been applied in similar installations. Do not use this installation as a product test site unless explicitly approved in writing by Owner or Owner's representative. Spare parts shall be available for at least five years after completion of this contract.
- B. The Owner's WAN:
 - 1. Internet-based network connecting multiple facilities with a central data warehouse and server, accessible via standard web-browser.
 - 2. This is a new infrastructure and Contractor is required to configure any components of this WAN. However, coordination with LIDL's BMS team is required for final connectivity.
 - 3. Contractor shall provide Modbus data to the Owner's WAN.

2.2 NETWORK CONNECTION AND COMMUNICATION

- A. The design of the BMS shall support networking of an owner's thin-client connectivity, a local Operator Interface, a Building Controller (BC), and field level controllers (AAC or ASC). The network architecture shall consist of two levels, an Ethernet-based primary network for the owner's connectivity and Building DDC controllers along with secondary Field Level Networks (FLN) for terminal equipment application specific controllers (ASC).

- B. Access to system data shall not be restricted by the hardware configuration of the BMS system. The hardware configuration of the BMS network shall be totally transparent to the user when accessing data or developing control programs.
- C. Operator GUI Communication:
 - 1. All color graphic shall reside on the interface or BC via the ethernet network and the consoles shall be set up in a client/server configuration.
 - 2. The web servers shall act as the central database for system graphics and databases to provide consistency throughout all system.
 - 3. The network shall allow concurrent use of multiple BAS software site licenses.
 - 4. Provide a web server and a shared hard disk drive array. The web server shall be an enterprise level fault-tolerant server.
- D. Management Level Network Communication (MLN)
 - 1. Any owner remote BMS workstations shall be simultaneously direct connect to the Ethernet Management Level Network without the use of an interposing device.
 - 2. Operator Interface and/or BC controller shall be capable of simultaneous direct connection and communication with BACnet/IP, Modbus, OPC, and TCP/IP corporate level networks without the use of interposing devices.
 - 3. The Management Level Network shall not impose a maximum constraint on the number of owner remote BMS workstations.
 - 4. Any controller residing on the Building level networks shall connect to Ethernet network without the use of a PC or a gateway with a hard drive.
 - 5. Any owner remote BMS workstations on the Management Level Network shall have transparent communication with controllers on the building level networks connected via Ethernet.
 - 6. Any break in Ethernet communication from an owner's IT network to the web server on the building level networks shall result in a notification at the web server.
 - 7. The standard client and BMS web server on the Management Level Network shall reside on industry standard Ethernet utilizing standard TCP/IP, IEEE 802.3.
 - 8. System software applications shall run as a service to allow communication with Building Controllers without the need for user log in. Closing the application or logging off shall not prevent the processing of alarms, network status, panel failures, and trend information.
 - 9. Access to the system database shall be available from any standard owner provided workstation on the Management Level Network.
 - 10. Client access to web server BC configurations over the Internet network shall be available via Web browser interface.
 - 11. Thin-client remote access to a web server BC or operator interface shall be available and provided via web browser interface. Remote access setup shall provide multiple, independent sessions without the need to install a dedicated workstation software on a local or remote hard drive.
- E. Primary Network - Panel to Panel Communication:
 - 1. Only one (1) BMS BC controller shall directly reside on the primary IP Ethernet network.
 - 2. Systems that operate via polled response or other types of protocols that rely on a central processor, file server, or similar device to manage panel- to-panel or device-to-device communications shall not be acceptable.
 - 3. All operator interfaces shall have the ability to access all point status and application report data or execute control functions for any and all other devices. Access to data shall be based upon logical identification of building equipment. No hardware or software limits shall be imposed on the number of devices with global access to the network data.

4. The primary network shall use Modbus over Ethernet. All devices must:
 - a. Auto-sense 10/100 Mbps networks.
 - b. Receive an IP Address from a Dynamic Host Configuration Protocol (DHCP) Server or be configured with a Fixed IP Address.
 - c. Resolve Name to IP Addresses for devices using a Domain Name Service (DNS) Server on the Ethernet network.
 - d. Allow MMI access to an individual Primary Network Controller using industry standard Telnet software to view and edit entire Primary Network.
 5. The primary network shall provide the following minimum performance:
 - a. Provide high-speed data transfer rates for alarm reporting, report generation from a BC controller and upload/download efficiency between network devices.
 - b. System performance shall insure that an alarm occurring at the BC is displayed at any owner's remote BMS workstations, all BCs, and other interfaces within 15 seconds.
 - c. Message and alarm buffering to prevent information from being lost.
 - d. Error detection, correction, and re-transmission to guarantee data integrity.
 - e. Synchronization of real-time clocks, including automatic daylight savings time corrections
 - f. The primary network shall allow the BC to access any data from, or send control commands and alarm reports directly to, any other controllers on the network without dependence upon a central or intermediate processing device. The network shall also allow any Building controller to access, edit, modify, add, delete, back up, restore all system point database and all programs.
 - g. The primary network controllers shall back-up and restore their own current database including programs, and points without the requirement for connection to a mass storage device.
 - h. The primary network controllers shall provide system-wide wild card point search, command, and access direct from any building controller on the network.
 - i. The primary network shall allow the BC to access on-demand display and reports regarding system-wide information including point names, point status, present value, command priority array, trend information, and field panel configuration information.
 - j. The primary network shall allow the BC to be configured system-wide by software based tools, and by direct access from any Building Controller on the network. Proprietary vendor specific software shall not be required for system configuration.
 - k. The primary network shall allow the BC to assign password access and control priorities to each point individually. The logon password (at any PC workstation or portable operator terminal) shall enable the operator to monitor, adjust and control only the points that the operator is authorized for. All other points shall not be displayed at the PC workstation or portable terminal. (e.g. all base building and all tenant points shall be accessible to any base building operators, but only certain base building and tenant points shall be accessible to tenant building operators). Passwords and priorities for every point shall be fully programmable and adjustable.
 - l. Devices containing custom programming may reside on the Primary Network.
- F. Secondary Network – Advanced Application Controller (AAC) and Application Specific Controller (ASC) Communication:
1. The BC controller shall directly connect to all pertinent sub-networks such that communications may be executed directly between BC controller and Field Level Network controllers on a peer-to-peer basis.

2. Communication over the secondary network shall be either BACnet MS/TP or Modbus RTU protocols.
3. This level communication shall support a family of AAC for pieces of equipment requiring full programmability and ASC for terminal equipment.
4. The AACs and ASCs shall communicate bi-directionally with the BC controllers for transmission of global data.
5. A maximum of 96 terminal equipment controllers may be configured on individual secondary network trunks to insure adequate global data and alarm response times.

G. Internet Based Communication:

1. Web Based Operator Interface

- a. The BMS shall provide a HTML-5 web based graphical interface that allows users to access the BMS data via the Internet, extranet, or Intranet. The interface shall use HTML based ASP pages to send and receive data from the BMS to a web browser.
- b. All information exchanged over Internet shall be optionally encrypted and secure via SSL (provided by Owner).
- c. Access to the web interface must be password protected. Users' rights and privileges to points and graphics shall be the same as those assigned at the BMS system. An option shall exist to only allow users "read" access via the web browser, while maintaining "command" privileges via any owner's remote BMS workstation.
- d. The web interface shall not require modification or creation of HTML or ASP pages using an HTML editor.
- e. The web based interface shall provide the following functionality to users, based on their access and privilege rights:
 - 1) Logon Screen - allows the user to enter their name, password and domain name for logging into the web server.
 - 2) Alarm Display - a display of current BMS System alarms to which the user has access will be displayed. Users will be able to acknowledge and erase active alarms, and link to additional alarm information including alarm messages. Any alarm acknowledgments initiated through the web interface will be recorded to the BAS System activity log.
 - 3) Graphic Display - Display of system graphics, including animated motion, shall be available for viewing over the web browser. A graphic selector list will allow users to select any graphics to which they have access. Graphics displays will automatically refresh with the latest change of values. Users will have the ability to command and override points from the graphic display as determined by their user account rights.
 - 4) Point Details - users will have access to point detail information including operational status, operational priority, physical address, and alarm limits, for point objects to which they have access.
 - 5) Point Commanding - users will be able to override and command points they have access to via the web browser interface. Any commands or overrides initiated via the web browser interface will be written to the BMS system's central workstation activity log.
- f. The web server licensing options shall allow concurrent access by a minimum of five (5) browser connections.
- g. Internet connections, ISP services, as well as necessary firewalls or proxy servers shall be provided by the owner as required to support the web access feature.

H. Remote Notification Paging System

1. The BMS web serving BC shall be configured to send out messages to numeric pagers, alphanumeric pagers, phones (via text to speech technology), SMS (Simple Messaging Service, text messaging) Devices, and email accounts based on a point's alarm condition.
2. There shall be no limit to the number of points that can be configured for remote notification of alarm conditions and no limit on the number of remote devices which can receive messages from the system.
3. On a per point basis, system shall be configurable to send messages to an individual or group and shall be configurable to send different messages to different remote devices based on alarm message priority level.
4. Remote devices may be scheduled as to when they receive messages from the system to account for operators' work schedules.
5. System must be configurable to send messages to an escalation list so that if the first device does not respond, the message is sent on to the next device after a configurable time has elapsed.
6. Message detail shall be configurable on a per user basis.
7. During a "flood" of alarms, remote notification messages shall have the ability to optimize several alarms into an individual remote notification message.
8. The BMS web serving BC shall have the ability to send manual messages allowing an operator to type in a message to be sent immediately.
9. The BMS web serving BC shall have a feature to send a heartbeat message to periodically notify users that they have communication with the system.

2.3 COMMUNICATION AND CONTROL WIRING

A. General:

1. Provide copper wiring, plenum cable, and raceways as specified in the applicable sections of Division 16 unless otherwise noted herein.
2. All insulated wire to be copper conductors, UL labeled for 90°C minimum service.

B. Wire Sizing and Insulation

1. Wiring shall comply with minimum wire size and insulation based on services listed below:

a.	Service	Minimum Gage/Type Insulation Class
b.	AC 24V Power	12 Ga Solid, Shielded 600 Volt
c.	DC 24V Power	10 Ga Solid, Shielded 600 Volt
d.	Class 1	14 Ga Stranded, Shielded 600 Volt
e.	Class 2	18 Ga Stranded, Shielded 300 Volt
f.	Class 3	18 Ga Stranded, Shielded 300 Volt
2. Provide plenum-rated cable when open cable is permitted in supply or return air plenum where allowed per execution specifications defined in Paragraph 3.07

C. Power Wiring:

1. 115V power circuit wiring above 100 feet distance shall use minimum 10 gage.
2. 24V control power wiring above 200 feet distance shall use minimum 12 gage.

D. Control Wiring:

1. Digital Input/output wiring shall use Class 2 twisted pair, insulated and jacketed and require a grounded shield.
2. Analog inputs shall use Class 2 twisted shielded pair, insulated, and jacketed and require a grounded shield.
3. Actuators with tri-state control shall use 3 conductor, twisted shielded, insulated, and jacketed and require a grounded shield.

E. Communication Wiring

1. Ethernet Cable shall be minimum CAT5
2. Secondary level network shall be 18-24 gage, TSP, low capacitance cable

F. Approved Cable Manufacturers:

1. Wiring from the following manufacturers which meet the above criteria shall be acceptable:
 - a. Anixter
 - b. Belden

G. Data Information/Connection Outlet

1. Technology contractor is responsible for furnishing and installing the Category 6A Ethernet cable, network switches and routers, and network data “drops” required for the BMS system. Follow all LIDL Facilities Maintenance Management IT services and standards for network hardware requirements. The BMS Contractor shall follow the installation requirements as stated below:
 - a. Ethernet cable shall be Commscope Uniprise Category 6A, non-plenum, 4-pair cable supporting 10GB/s Ethernet requirements.
 - b. At the IDF end, the cable will be terminated at a Category 6A RJ45 Uniprise patch panel.
 - c. Each data outlet shall have two Category 6A cables installed. One white cable and one gray cable.
 - d. At the data outlet end, the cable will be terminated at Category 6A 8- position 568B/RJ45 jacks, gray in color, and terminated in a 2-port face plate
 - e. Horizontal face plate configuration:
 - 1) In a 2-port face plate, the “C1” data jack (first, white cable) shall be in the bottom position and the “C2” data jack (second, gray cable) shall be in the top position.
 - 2) In a 4-port face plate, the “C1” data jack shall be in the bottom left position, the “C2” data jack shall be in the top left position, the “C3” data jack shall be in the bottom right position, and the “C4” data jack shall be in the top right position.
 - f. Vertical face plate configuration:
 - 1) In a 2-port face plate, the “C1” data jack (first, white cable) shall be in the left position and the “C2” data jack (second, gray cable) shall be in the right position.
 - 2) In a 4-port face plate, the “C1” data jack shall be in the top left position, the “C2” data jack shall be in the top right position, the “C3” data jack shall be in

the bottom left position, and the “C4” data jack shall be in the bottom right position.

- g. All cable will be secured and dressed with Velcro tie-wraps.
- h. Do not install compressed, kinked, scored, deformed, or abraded cable. If any cable is, or becomes damaged, replace the entire length of damaged cable.
- i. All data outlets shall be serviced with 1” conduit to the basket tray, or pull box via horizontal conduit system. Conduits shall service only one data outlet each. No daisy-chaining or looping between data outlets is allowed.
- j. Labeling:
 - 1) For vertical face plates, the location labels shall be installed on the top label holder on the face plate.
 - 2) For horizontal face plates, the location labels shall be installed on the left label holder on the face plate.
 - 3) For vertical face plates, the IDF labels shall be installed on the bottom label holder on the face plate. This label will contain the IDF room number only.
 - 4) For horizontal face plates, the location labels shall be installed on the right label holder on the face plate. This label will contain the IDF room number only.

2.4 INTEGRATION:

A. BACnet IP Node:

- 1. Equipment with factory-furnished BACnet IP communication shall be reside on a sub-network under the BMS BC/head controller. No exceptions.
- 2. The web server with BMS enterprise software shall perform information translation between the BACnet IP node and other devices on other BMS sub-networks.
- 3. The server shall communicate points as required for the project to a BC to support interrogation, full control, and/or utilize information to command all associated AACs and ASCs under the Primary Controller LAN.

B. Modbus Node:

- 1. An equipment with factory-furnished Modbus-RTU communication shall be reside on a sub-network under the BMS BC/head controller. No exceptions.
- 2. The web server with BMS enterprise software shall perform information translation between the Modbus nodes and other devices on other BMS sub-networks.
- 3. BC shall support interrogation, full control, and/or utilize information to command all associated AACs and ASCs under the Primary Controller LAN.

PART 3 - EXECUTION

3.1 INSPECTION:

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Installer.

3.2 INSTALLATION OF CONTROL SYSTEMS:

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings.

- B. BMS web server shall be approved by Lidl IT group.
- C. BMS Contractor shall install and fully configure and provide all interface devices and software to provide an integrated system.
- D. Contractor shall closely coordinate with the Owner, or designated representative, to establish IP addresses and communications to assure proper operation of the building control system with the owner's WAN and owner's operator workstation.
- E. No license, software component, key, etc or any piece of information required to install, configure, operate, diagnose and maintain the system shall be withheld from the Owner.
- F. Install electronic control system Operation and Maintenance Manuals, programming guides, network configuration tools, and control shop drawings etc on the web server or as directed by the owner. Provide interface or shortcuts to guide user to the appropriate information.
- G. Install systems and materials in accordance with manufacturer's instructions.
- H. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.

END OF SECTION 230921

SECTION 230922 - BUILDING AUTOMATION SYSTEM (BAS) - BASIC MATERIALS, INTERFACE DEVICES, AND SENSORS**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Specified in this section are the following hard wired input/output devices connected to a BC, AAC, or ASC.
 - a. Binary Temperature Device
 - b. Temperature Sensor
 - c. Dew Point/Humidity Sensor
 - d. Indoor Air Quality (CO2) Sensor
 - e. Differential Pressure Transmitter
 - f. Light Level Sensor
 - g. Current Switches
 - h. Relays
 - i. Voltage Transformers
 - j. Water Leak Detector
 - k. Duct Smoke Detector
 - l. Electric Control Components
 - m. Audio/Visual Alarm Units
 - n. Nameplates.
 - o. Testing Equipment.

1.2 DESCRIPTION OF WORK

- A. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.
- B. Refer to other Division 23 Sections for installation of instruments in mechanical systems; not work of this Section.
- C. Provide the following electrical work as work of this Section, complying with requirements of Division 26 Sections:
 1. Control wiring between field-installed controls, indicating devices, and unit control panels.
 2. Interlock wiring between electrically interlocked devices, sensors, and between a hand or auto position of motor starters as indicated for all mechanical and controls.
 3. Wiring associated with indicating and alarm panels (remote alarm panels) and connections to their associated field devices.
 4. All other necessary wiring for fully complete and functional control system as specified.
 5. Power wiring from spare circuits in electrical panels to BMS Field Panels.

1.3 WORK BY OTHERS

- A. Flow Switches, Flow Meters, etc. that will have wet surfaces, shall be installed under the applicable piping Section under the direction of the BMS Contractor who will be fully responsible for the proper installation and application.
- B. Controlled Equipment Power Wiring shall be furnished and installed under Division 26. Where control involves 120V control devices controlling 120V equipment, Division 26 Contractor shall extend power wiring to the equipment. Building Automation System Contractor shall extend it from the equipment to the control device.
- C. BMS Contractor shall provide adequate time to coordinate and work with Equipment Manufacturer where factory-controls are provided to achieve the design sequence of operations.
- D. BMS Contractor shall provide adequate time to coordinate and work with Test and Balancing Contractor to achieve the design sequence of operations.

PART 2 - PRODUCTS

2.1 DDC FUNCTIONALITY

- A. General: These requirements clarify the requirement for stand-alone DDC functionality relative to packaging I/O devices with a controller. Stand-alone functionality is specified with the controller and for each Application Category specified in PART 3. This item refers to acceptable paradigms for associating the points with the processor.
- B. Functional Boundary: Provide controllers so that all points associated with and common to one unit or other complete system/equipment shall reside within a single control unit. The boundaries of a standalone system shall be as dictated in the contract documents. Generally, systems specified for the Application Category will dictate the boundary of the standalone control functionality. See related restrictions below. When referring to the controller as pertains to the standalone functionality, reference is specifically made to the processor. One processor shall execute all the related I/O control logic via one operating system that uses a common programming and configuration tool.
- C. The following configurations are considered acceptable with reference to a controller's standalone DDC functionality:
 - 1. Points packaged as integral to the controller such that the point configuration is listed as an essential piece of information for ordering the controller (having a unique ordering number).
 - 2. Controllers with processors and modular back planes that allow plug in point modules as an integral part of the controller.
 - 3. I/O point expander boards, plugged directly into the main controller board to expand the point capacity of the controller.
 - 4. I/O point expansion devices connected to the main controller board via wiring and as such shall be remote from the controller and that communicate via a sub LAN protocol. These arrangements to be considered standalone shall have a sub LAN that is dedicated to that controller and include no other controller devices. All wiring to interconnect the I/O expander board shall be:
 - a. Contained in the control panel enclosure;
 - b. Or run in conduit. Wiring shall only be accessible at the terminations.

5. General purpose I/O or Smart Devices racked with a processor module shall be in the same contiguous physical enclosure. The controller shall also include its own dedicated processor module and bridge or router making the controllers LAN communication a subnet or LAN segment dedicated to that controller as specified under Application Categories below. The following are additional requirements of this configuration:
 - a. Configuration must meet the requirements for battery back up.
 - b. If processor fails, the I/O devices shall go to their fail condition.
 - c. Logic must provide for orderly sequencing of I/O during a power interruption and restart of program logic upon restoration of power.
 - d. Programming must facilitate a robust uploading scheme and limit available bandwidth during upload.
 - e. Trending shall be buffered in the processor or dedicated data logging module and uploaded to a Building or Network Controller.

D. Building Controller (BC)

E. General Requirements:

1. The BC(s) shall provide fully distributed control independent of the operational status of any workstation/server. All necessary calculations required to achieve control shall be executed within the BC independent of any other device.
2. BCs shall be connected to a controller network that qualifies as a Primary Controlling LAN. BCs shall share information with the entire network of BCs and AACs/ASCs for full global control.
3. BCs shall be programmable from an operator workstation or an operator's terminal device. Each controller shall permit multi-user operation from multiple owner interfaces connected either locally or over the Primary Controller LAN.
4. All control strategies performed by the BCs shall be both operator definable and modifiable through the Operator Interfaces. All point data, algorithms and application software within each BC shall be modifiable from an owner workstation.
5. Each BC shall be capable of standalone direct digital operation utilizing its own processor, non-volatile memory, input/output, wiring terminal strips, A/D converters, and voltage transient and lightning protection devices.
6. BCs shall perform overall system coordination, accept control programs, perform automated HVAC functions, control peripheral devices and perform all necessary mathematical and logical functions.
7. Each unit shall have its own internal RAM, non-volatile memory, microprocessor, battery backup, regulated power supply, power conditioning equipment, ports for connection of operating interface devices, and control enclosure.
8. BC shall contain sufficient memory for all specified global control strategies, user defined reports and trending, communication programs, and central alarming.
9. All BCs shall be protected from any memory loss due to a loss of power by one or a combination of the following:
 - a. Volatile RAM shall have a battery backup using a lithium battery with a rated service life of fifty (50) hours, and a rated shelf life of at least five years. Self-diagnostic routine shall report an alarm for a low battery condition.
 - b. EEPROM, EPROM, or NOVRAM non-volatile memory
10. Each BC shall provide for point mix flexibility and expandability. This requirement shall be met via either a family of expander boards, modular input/output configuration, or a combination thereof.
11. Each BC shall execute application programs, calculations, and commands via a microprocessor resident in the BC.

12. The database and all application programs for each BC shall be stored in non-volatile or battery backed volatile memory within the BC and shall be able to upload/download to/from an owner workstation.
13. BCs shall provide buffer for holding alarms, messages, trends etc.
14. Each BC shall contain software to perform full DDC/PID control loops.
15. For systems requiring end-of-line resistors those resistors shall be located in the BC, if it has I/O capability.
16. A communication port for operator interface through a terminal shall be provided in each BC. It shall be possible to perform all program and database back-up, system monitoring, control functions, and BC diagnostics through this port. Standalone BC panels shall allow temporary use of portable devices without interrupting the normal operation of permanently connected modems, printers, or workstations.
17. Each BC shall be equipped with loop tuning algorithm for precise proportional, integral, derivative (PID) control. Loop tuning tools provided with a workstation software is acceptable. Tools to support loop tuning must be provided such that P, I, and D gains are automatically calculated.
18. Slope intercepts and gain adjustments shall be available on a per-point basis.
19. Each BC shall be at least 32 bit, multi-tasking, multi-user, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
20. Each BC shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, operator I/O, and dial-up communications.
21. Each BC shall have an integral real-time clock and calendar.
22. Each BC shall support firmware upgrades without the need to change hardware.

F. Each Building Controller shall support:

1. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the Building Controller cabinet.
2. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the Building Controller cabinet.

G. Spare Point Capacity. Each Building Controller shall have a minimum of 20% spare point capacity.

1. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
2. Provide all processors, power supplies, and communication controllers so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - a. Expansion modules
 - b. Sensor/actuator
 - c. Field wiring/tubing.

H. Serial Communication. Building Controllers shall provide at least one data communication port for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. Building Controllers shall allow temporary use of portable devices without interrupting the normal operation of permanently connected terminals. A USB port shall alternatively be available to support local HMI tools connection.

- I. Self-Diagnostics. Each Building Controller shall continuously perform self-diagnostics, communication diagnosis, and diagnosis of all panel components. The Building Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
- J. Power loss:
 - 1. In the event of the loss of power, there shall be an orderly shutdown of all Building Controllers to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 100 hours.
 - 2. Upon a loss of power to any BC, the other units on the primary controlling network shall not in any way be affected.
 - 3. Upon a loss of power to any BC, the UPS backup shall ensure that the energy management control software, the Direct Digital Control software, the database parameters, and all other programs and data stored in the RAM are retained.
 - 4. Upon restoration of power within the specified UPS backup period, the BC shall resume full operation without operator intervention. The BC shall automatically reset its clock such that proper operation of any time dependent function is possible without manual reset of the clock. All monitored functions shall be updated.
 - 5. Should the duration of a loss of power exceed the specified UPS backup period or BC panel memory be lost for any reason, the panel shall automatically report the condition (upon resumption of power) and be capable of receiving a download via the network, and connected workstation.
 - 6. The owner shall be able to upload the most current versions of all energy management control programs, DDC programs, database parameters, and all other data and programs in the memory of each BC to an owner workstation via the local area network, or via the telephone line dial-up modem where applicable, or to the laptop PC via the local RS-232C port.
- K. BC Failure:
 - 1. Building Controller LAN Data Transmission Failure: BC shall continue to operate in stand-alone mode. BC shall store loss of communication alarm along with the time of the event. All control functions shall continue with the global values programmable to either last value or a specified value. Peer BCs shall recognize the loss, report alarm and reconfigure the LAN.
 - 2. BC Hardware Failure: BC shall cease operation and terminate communication with other devices.
- L. Environment.
 - 1. Controller hardware shall be suitable for the anticipated ambient conditions for most mechanical spaces.
 - 2. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
- M. Immunity to power and noise.
 - 1. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 - 2. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).

3. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - a. RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3 V
 - b. Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact
 - c. Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500 V signal, 1 kV power
 - d. Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max)
4. Isolation shall be provided at all Building Controller's AC input terminals to suppress induced voltage transients consistent with:
 - a. IEEE Standard 587 1980
 - b. UL 864 Supply Line Transients
 - c. Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000-4-11)
5. All products shall be BACnet certified, and shall be designed according to the BACnet Interoperability Guidelines. Product documentation and devices shall display the BACnet symbol, indicating conformance to the BACnet Testing Laboratory (BTL) Standards.

N. Advanced Application Controller (AAC)

1. Each Building Controller shall be able to communicate with Advanced Application Controllers (AACs) over the Secondary Network.
2. AACs shall be at least 32 bit, multi-tasking, multi-user, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
3. Each AAC shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, operator I/O, and dial-up communications.
4. AACs shall have an integral real-time clock.
5. Each AAC shall support firmware upgrades without the need to change hardware.
6. Each AAC shall support:
 - a. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the AAC cabinet.
 - b. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the AAC cabinet.
7. Spare Point Capacity. Each AAC shall have a minimum of 20% spare point capacity.
 - a. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
 - b. Provide all processors, power supplies, and communication controllers so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - 1) Expansion modules
 - 2) Sensor/actuator

- 3) Field wiring/tubing.
8. Serial Communication. AACs shall provide at least one data communication port for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. AACs shall allow temporary use of portable devices without interrupting the normal operation of permanently connected terminals. A USB port shall alternatively be available to support local HMI tools connection.
9. Self-Diagnostics. Each AAC shall continuously perform self-diagnostics, communication diagnosis, and diagnosis of all panel components. The Building Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
10. Power loss. In the event of the loss of power, there shall be an orderly shutdown of all AACs to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 100 hours.
11. Environment.
 - a. Controller hardware shall be suitable for the anticipated ambient conditions for most mechanical spaces.
 - b. Controllers used outdoors and/or in wet ambient conditions shall be mounted within waterproof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
 - c. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
12. Immunity to power and noise.
 - a. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 - b. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).
 - c. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - 1) RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3 V
 - 2) Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact
 - 3) Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500 V signal, 1 kV power
 - 4) Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max)
 - d. Isolation shall be provided at all AAC's AC input terminals to suppress induced voltage transients consistent with:
 - 1) IEEE Standard 587 1980
 - 2) UL 864 Supply Line Transients
 - 3) Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000-4-1)

O. APPLICATION SPECIFIC CONTROLLER (ASC)

1. Each Building Controller shall be able to communicate with application specific controllers (ASCs) over the Secondary Network to control terminal equipment only.

2. The use of Secondary Network controllers with custom program applications to control RTU's, AC units, etc. is not acceptable.
3. Each ASC shall operate as a stand-alone controller capable of performing its specified control responsibilities independently of other controllers in the network. Each ASC shall be a microprocessor-based, multi-tasking, real-time digital control processor.
4. Each ASC shall include all point inputs and outputs necessary to perform the specified control sequences. The ASC shall accept input and provide output signals that comply with industry standards. Controllers utilizing proprietary control signals shall not be acceptable. Outputs utilized either for two-state, modulating floating, or proportional control, allowing for additional system flexibility.
5. Space Temperature Sensors. Each controller performing space temperature control shall be provided with a matching room temperature sensor.
 - a. Wired Sensor specifications. The sensor may be either RTD or thermistor type providing the following.
 - 1) Accuracy: + .5 F
 - 2) Operating Range: 35 to 115° F
 - 3) Set Point Adjustment Range: 55 to 95° F
 - 4) Calibration Adjustments: None required
 - 5) Installation: Up to 100 ft. from controller
 - 6) Auxiliary Communications Port: as required
 - b. Space temperature sensors are to be provided with neither display nor setpoint adjustment. The owner and authorized facility operator have the option of modifying individual sensor types and/or functionality. Coordinate with the owner's staff and authorized facility operator prior to the installation of the space temperature sensors.
6. Communication. Each controller shall perform its primary control function independent of other Secondary Network communication, or if Secondary Network communication is interrupted. Reversion to a fail- safe mode of operation during Secondary Network interruption is not acceptable.
7. Control Algorithms. The controller shall receive its real-time data from the Building Controller time clock to insure Secondary Network continuity. Each controller shall include algorithms incorporating proportional, integral and derivative (PID) gains for all applications. All PID gains and biases shall be field-adjustable by the user via the operator's terminal. Controllers that incorporate proportional and integral (PI) control algorithms only shall not be acceptable.
8. Control Applications. Operating programs shall be field-selectable for specific applications. In addition, specific applications may be modified to meet the user's exact control strategy requirements, allowing for additional system flexibility. Controllers that require factory changes of all applications are not acceptable.
9. Memory:
 - a. Provide each ASC with sufficient memory to accommodate point databases, operating programs, local alarming and local trending. All databases and programs shall be stored in non-volatile EEPROM, EPROM and PROM, or minimum of 72-hour battery backup shall be provided. The controllers shall be able to return to full normal operation without user intervention after a power failure of unlimited duration.
 - b. Upon replacement, new ASCs shall recover control function and site specific defaults automatically and resume normal operation.

10. Power Supply. The ASCs shall be powered from a 24 VAC source and shall function normally under an operating range of 18 to 28 VAC, allowing for power source fluctuations and voltage drops. Power supply for the ASC must be rated at a minimum of 125% of ASC power consumption and shall be of the fused or current limiting type. The BMS contractor shall provide 24 VAC power to the terminal units by utilizing a dedicated line voltage power source and isolation transformers at a central location and installing 24VAC power trunk to supply multiple ASCs in the area.
11. Environment. The controllers shall function normally under ambient conditions of 32 to 122 F (0 to 50 C) and 10% to 95%RH (non- condensing). Provide each controller with a suitable cover or enclosure to protect the circuit board assembly.
12. Immunity to noise. Operation shall be protected against electrical noise of 5-120 Hz and from keyed radios up to 5 W at 1 m (3 ft).

P. INPUT-OUTPUT INTERFACES

1. Hardwired inputs and outputs may tie into the system through building or application specific controllers.
2. All input points and output points shall be protected such that shorting of the point to itself, to another point, or to ground will cause no damage to the controller. All input and output points shall be protected from voltage up to 24 V of any duration, such that contact with this voltage will cause no damage to the controller.
3. Binary inputs shall allow the monitoring of On/Off signals from remote devices. The binary inputs shall provide a wetting current of at least 12 mA to be compatible with commonly available control devices and shall be protected against the effects of contact bounce and noise. Binary inputs shall sense “dry contact” closure without external power (other than that provided by the controller) being applied.
4. Pulse accumulation input objects. This type of object shall conform to all the requirements of binary input objects and also accept up to 10 pulses per second for pulse accumulation.
5. Analog inputs shall allow the monitoring of low-voltage (0 to 10 VDC), current (4 to 20 mA), or resistance signals (thermistor, RTD). Analog inputs shall be compatible with— and field configurable to— commonly available sensing devices.
6. Binary outputs shall provide for On/Off operation or a pulsed low-voltage signal for pulse width modulation control. Binary outputs on building and custom application controllers shall have three-position (On/Off/Auto) override switches and status lights. Outputs shall be selectable for either normally open or normally closed operation.
7. Analog outputs shall provide a modulating signal for the control of end devices. Outputs shall provide either a 0 to 10 VDC, 4 to 20 mA or 0-20 PSI signal as required to provide proper control of the output device. Analog outputs on building or custom application controllers shall have status lights and a two-position (AUTO/MANUAL) switch and manually adjustable potentiometer for manual override. Analog outputs shall not exhibit a drift of greater than 0.4% of range per year.
8. System Object Capacity. The system size shall be expandable to at least twice the number of input/ output objects required for this project. Additional controllers (along with associated devices and wiring) shall be all that is necessary to achieve this capacity requirement. The operator interfaces installed for this project shall not require any hardware additions or software revisions in order to expand the system.

Q. Power Supplies and Line Filtering

1. Control transformers shall be UL listed. Furnish Class 2 current-limiting type or furnish over-current protection in both primary and secondary circuits for Class 2 service in accordance with NEC requirements. Limit connected loads to 80% of rated capacity.
2. DC power supply output shall match output current and voltage requirements. Unit shall be full-wave rectifier type with output ripple of 5.0 mV maximum peak-to-peak. Regulation

shall be 1.0% line and load combined, with 100-microsecond response time for 50% load changes. Unit shall have built-in over-voltage and over-current protection and shall be able to withstand a 150% current overload for at least three seconds without trip-out or failure.

3. Unit shall operate between 0°C and 50°C (32°F and 120°F). EM/RF shall meet FCC Class B and VDE 0871 for Class B and MILSTD 810C for shock and vibration.
4. Line voltage units shall be UL recognized and CSA approved.
5. Power line filtering and surge protection.
 - a. Provide transient voltage and surge suppression for network controllers and other critical panels either internally or as an external component. Surge protection shall have the following at a minimum:
 - b. Dielectric strength of 1000 volts minimum
 - c. Response time of 10 nanoseconds or less
 - d. Transverse mode noise attenuation of 65 dB or greater
 - e. Common mode noise attenuation of 150 dB or better at 40 Hz to 100 Hz.
 - f. Hot swappable batteries
 - g. Automatic voltage regulation
 - h. Automatic bypass
 - i. Energy efficient design
 - j. Remote on/off
 - k. Web/SNMP enabled

2.2 MATERIALS AND EQUIPMENT

- A. General: Provide electronic control products in sizes and capacities indicated, consisting of valves, dampers, thermostats, clocks, controllers, sensors, and other components as required for complete installation. Except as otherwise indicated, provide manufacturer's standard materials and components as published in their product information; designed and constructed as recommended by manufacturer, and as required for application indicated.
- B. Communication Wiring: All wiring shall be in accordance with National Electrical Codes and Division 26 of this specification.
 1. Contractor shall supply all communication wiring between Building Controllers (BC), Routers, Gateways, Advanced Application Controllers (AAC), Application Specific Controllers (ASC) and local and remote peripherals (e.g., workstations, printers, and modems).
 2. Local Supervisory LAN: For any portions of this network required under this Section of the specification, contractor shall use Category 5 of standard TIA/EIA 68 (10BaseT). Network shall be run with no splices and separate from any wiring over thirty (30) volts.
 3. Primary and Secondary Controller LANs: Communication wiring shall be individually 100% shielded pairs per manufacturers recommendations for distances installed, with overall PVC cover, Class 2, plenum-rated run with no splices and separate from any wiring over thirty (30) volts. Shield shall be terminated and wiring shall be grounded as recommended by BC manufacturer.
- C. Control Panels: Provide control panels with suitable brackets for wall mounting for each control system. Locate panel adjacent to systems served.

1. Fabricate panels of 16-gage furniture-grade steel, or 6063-T5 extruded aluminum alloy, totally enclosed on four sides, with hinged door and keyed lock, with manufacturer's standard shop-painted finish and color.
 2. Provide UL-listed cabinets for use with line voltage devices.
 3. All gauges and control components shall be identified by means of nameplates.
 4. All control tubing and wiring shall be run neatly and orderly in open slot wiring duct with cover.
 5. Complete wiring and tubing termination drawings shall be mounted in or adjacent to panel.
- D. Signal Wiring: BMS Contractor shall run all signal wiring in accordance with National Electric Codes and Division 26 of this Specification.
1. Signal wiring to all field devices, including, but not limited to, all sensors, transducers, transmitters, switches, etc. shall be twisted, 100% shielded pair, minimum 18-gauge wire, with PVC cover. Signal wiring shall be run with no splices and separate from any wiring above thirty (30) volts.
 2. Signal wiring shield shall be grounded at controller end only unless otherwise recommended by the controller manufacturer.
- E. Low Voltage Analog Output Wiring: Contractor shall run all low voltage control wiring in accordance with National Electric Codes and Division 26 of this Specification.
1. Low voltage control wiring shall be minimum 16-gauge, twisted pair, 100% shielded, with PVC cover, Class 2 plenum-rated. Low voltage control wiring shall be run with no splices separate from any wiring above thirty (30) volts.
- F. Instrument Pipe and Tube (Low Pressure Air Instrument Sensing Lines)
1. Connections: Use suitable bulkhead type fitting and static sensing tip for static pressure connections. Adapt tubing to instrument connection.
 2. Tubing: Virgin polyethylene non-metallic tubing type FR, ASTM D 2737, and with flame-retardant harness for multiple tubing. Use compression or push-on brass fittings.

2.3 GENERAL FIELD DEVICES

- A. Provide field devices for input and output of digital (binary) and analog signals into controllers (BCs, AACs, ASCs). Provide signal conditioning for all field devices as recommended by field device manufacturers, and as required for proper operation in the system.
- B. It shall be the Contractor's responsibility to assure that all field devices are compatible with controller hardware and software.
- C. Field devices specified herein are generally 'two-wire' type transmitters, with power for the device to be supplied from the respective controller. If the controller provided is not equipped to provide this power, or is not designed to work with 'two-wire' type transmitters, or if field device is to serve as input to more than one controller, or where the length of wire to the controller will unacceptably affect the accuracy, the Contractor shall provide 'four-wire' type equal transmitter and necessary regulated DC power supply or 120 VAC power supply, as required.

- D. For field devices specified hereinafter that require signal conditioners, signal boosters, signal repeaters, or other devices for proper interface to controllers, Contractor shall furnish and install proper device, including 120V power as required. Such devices shall have accuracy equal to, or better than, the accuracy listed for respective field devices.
- E. Accuracy: As stated in this Section, accuracy shall include combined effects of nonlinearity, non-repeatability and hysteresis.

2.4 TEMPERATURE SENSORS

- A. Provide the following instrumentation as required by the monitoring, control and optimization functions. All temperature sensor shall use platinum or nickel RTD elements only, silicon elements are not acceptable. All control signals shall be via a 4-20 mA loop. For critical system applications, only temperature transmitters will be accepted. RTDs and transmitters are acceptable for non-critical applications.

- B. Room Temperature:

- | | |
|--------------------------------------|---|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 °F |
| 6. Accuracy at calibration point | +0.5 °F |
| 7. Wall mounted Sensor Type: | Flat Plate |
| 8. Sensor Body/Cover Color: | White |
| 9. Approved Manufacturer: Subject to | compliance with requirements, provided on LiDL |
| National Account List: | BAPI, Emerson, or equal (|

- C. Combined Room Temperature and Humidity Sensor:

- | | |
|----------------------------------|---|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10 VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 (degree symbol) F |
| 6. Accuracy at calibration point | +0.5 (degree symbol) F |
| 7. Data-Column Sensor Type: | Flush Button Type |
| 8. Sensor Body/Cover Color: | White |
| 9. Approved manufacturer: | Subject to compliance with requirements, provided on LiDL |
| National Account List: | Elan, Emerson, or equal |

- D. Slab Temperature:

- | | |
|--------------------------------------|---|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 °F |
| 6. Accuracy at calibration point | +0.5 °F |
| 7. Wall mounted Sensor Type: | Flat Plate |
| 8. Data-Column Sensor Type: | Flush Button Type |
| 9. Approved Manufacturer: Subject to | compliance with requirements, provided on LiDL |
| National Account List: | BAPI, Emerson, or equal |

E. Duct (Single Point) Temperature

- | | | |
|----|-----------------------------------|---|
| 1. | Temperature monitoring range | +20/+120 F |
| 2. | Output signal | 4-20 mA, 0-10VDC, 1k ohm |
| 3. | Installation adjustments | none required |
| 4. | Calibration adjustments | zero & span |
| 5. | Factory calibration point | 70°F |
| 6. | Accuracy at calibration point | +0.5 F |
| 7. | Sensor Body/Cover Color: | White |
| 8. | Approved Manufacturer: Subject to | compliance with requirements, provided on LiDL |
| | National Account List: | BAPI, RTU manufacturer, or equal |

2.5 DEW POINT/HUMIDITY SENSORS

A. Duct Relative Humidity:

- | | | |
|-----|--------------------------|---|
| 1. | Sensor Humidity range | 0 to 100% |
| 2. | Operating temperature | 15°F to +170°F |
| 3. | Accuracy | +2% RH |
| 4. | Sensing element | Capacitive sensor |
| 5. | Output signal | 4-20 mA DC |
| 6. | Installation adjustments | zero & span |
| 7. | Operating temperature | 15°F to +170°F |
| 8. | Voltage requirement | 12-36 VDC |
| 9. | Sensor Body/Cover Color: | White |
| 10. | Approved Manufacturer: | BAPI, RTU manufacturer, Emerson, or equal |

2.6 INDOOR AIR QUALITY (CO2) SENSORS

- A. Provide indoor air quality duct-mounted sensors to monitor Carbon Dioxide (CO2) levels.
- B. The sensors shall utilize single-beam, dual-wavelength, non-dispersive infrared technology.
- C. The sensor shall meet the following requirements:

- | | | |
|-----|--------------------------|----------------------------------|
| 1. | Operating voltage: | 24 VAC/DC |
| 2. | Accuracy: | ± 2% of range + 2% reading |
| 3. | Power consumption: | <2.5W |
| 4. | Measuring range: | 0 – 2000 ppm |
| 5. | Tolerance: | +/- 100 ppm |
| 6. | Output: | 4-20mA, 0 – 10 VDC |
| 7. | Calibration: | every 5 years recommended |
| 8. | Long-term Stability: | <± 5% of range/5 years |
| 9. | Response time: | 1 minute |
| 10. | Sensor Body/Cover Color: | White |
| 11. | Approved Manufacturer: | BAPI, RTU manufacturer, or equal |

2.7 DIFFERENTIAL PRESSURE TRANSMITTERS (DP)

A. General Air Pressure Sensor Requirements:

1. Pressure sensors shall be constructed to withstand 100% pressure over-range without damage and to hold calibrated accuracy when subject to a momentary 40% over-range input.

2. Pressure transmitters shall transmit a 0 to 5 VDC, 0 to 10 VDC, or 4 to 20 mA output signal.
3. Differential pressure transmitters used for flow measurement shall be sized to the flow sensing device and shall be supplied with Tee fittings and shut-off valves in the high and low sensing pick-up lines to allow the balancing contractor and Owner permanent easy-to-use connection.
4. Provide a minimum of a NEMA 1 housing for the transmitter. Locate transmitters in accessible local control panels wherever possible.
5. Provide a two year warranty for each transmitter. Replace all transmitters found to be defective at no cost to the Owner during the warranty period.
6. Operating voltage: 24 VAC/DC
7. Accuracy: ± 0.1 " w.c.
8. Measuring range: ± 0.25 " to 40" w.c.
9. Output: 4-20 mA, 0 – 10 VDC
10. Long-term Stability: $<\pm 5\%$ of range/5 years
11. Response time: 8 seconds
12. Output Update Rate: 1 second
13. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
14. Sensor Body/Cover Color: White
15. Approved Manufacturer: BAPI, RTU manufacturer, or equal

2.8 LIGHT LEVEL SENSOR

A. Photodiode Sensor:

1. Measuring Range: 0 – 10,000 FC
2. Output: 0-9 Volts
3. Response Time: Instant
4. Listings: UL 916, Title 24, RoHS compliant
5. Sensor Body/Cover Color: White
6. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
7. Approved Manufacturer: PLC Sensor or equal

2.9 CURRENT SWITCHES (CS)

A. Current-operated switches shall be self-powered, solid-state with adjustable trip current. The switches shall be selected to match the current of the application and output requirements of the DDC system.

1. Sensor Power: 24 VAC/DC
2. Minimum Trip Point: 0.5 A
3. Adjustable Trip Point: Required
4. Status LEDs: Required
5. Insulation Class: ± 0.1 " w.c.
6. Frequency Range: 50/60 Hz
7. Temperature Range: 5 to 140 °F
8. Humidity Range: 10 to 90% RH, Non-condensing
9. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
10. Approved Manufacturer: Veris, or equal

2.10 RELAYS

- A. Control relays shall be UL listed plug-in type with dust cover and LED “energized” indicator. Contact rating, configuration, and coil voltage shall be suitable for application.
- B. Time delay relays shall be UL listed solid-state plug-in type with adjustable time delay. Delay shall be adjustable $\pm 200\%$ (minimum) from set point shown on plans. Contact rating, configuration, and coil voltage shall be suitable for application. Provide NEMA 1 enclosure when not installed in local control panel.
- C. Where required, provide “delay on break” relays where the output remains energized for the pre-determined time delay after the input has been de-energized. Reclosing the input switch during timing resets the time delay. Loss of input voltage resets the time delay as well as the output.
- D. Approved Manufacturer: Idec, Functional Devices, or equal

2.11 VOLTAGE TRANSFORMER

- A. AC voltage transformers shall be UL/CSA Recognized, 600 VAC rated, complete with built-in fuse protection.
- B. Transformers shall be suitable for ambient temperatures of 4°C to 55°C (40°F to 130°F) and shall provide $\pm 0.5\%$ accuracy at 24 VAC and a 5 VA load.
- C. Windings (except for terminals) shall be completely enclosed with metal or plastic material.

2.12 WATER LEAK DETECTOR

- A. Single point leak detector (external powered models)
 - 1. Service: Water or conductive fluids
 - 2. Minimum Sensing Gap: 1/32"
 - 3. Switch Type: DPDT
 - 4. Audible Alarm: at least 85 dB @ 1' distance
 - 5. Visual Alarm: Red LED for water level, Green LED for power condition
 - 6. Temperature Limits: 32 to 122 °F
 - 7. Power Requirements: 24 VAC or 11 to 27 VDC
 - 8. Listings: CE, RoHS
 - 9. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
 - 10. Approved Manufacturer: Dwyer (WD3 series) or equal

2.13 DUCT SMOKE DETECTOR

- A. Photoelectric detector with sampling tube that spans the entire width of duct. .
- B. Velocity Rating: 100 to 4000 fpm or 500 to 4000fpm depending on the minimum velocity in the duct. Provide the 100 to 4000 fpm detector if the min duct velocity is below 550 fpm.
- C. Output Contact: Alarm, two sets form “C” rated at 10amps 115V resistive. One set of alarm contacts for BMS monitoring and fan shutdown. Trouble, one set of contacts.

D. Temperature & RH limits: 32 to 120°F and 10 to 85% relative humidity.

E. Acceptable Manufacturer:

1. Invensys FIREX model 2650
2. Sensor Systems DH100ACDCLP
3. Air Products and Controls SL-2000

2.14 ELECTRIC CONTROL COMPONENTS

A. Horn & Strobe:

1. Description: Xenon strobe light source with horn
2. Flash Rate: 60 FPM
3. Sound: 100 dB at 1 meter/90 dB at 10 ft. (measured in anechoic chamber)
4. Wiring: Terminals
5. Housing: Gray, engineered thermoplastic housing
6. Operating Temp Range: Indoor: 85% relative humidity at 86 °F, 32 °F to 120 °F variable ambient temperature
Outdoor: 95% relative humidity at 86 °F, -31 °F to 150 °F variable ambient temperature
7. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
8. Approved Manufacturer: Edwards Signal (869STR series) or equal

B. Timer with Relay:

1. Description: Multifunction timer
2. Input Voltage: 24, 120, 230 VAC, 50/60 Hz
3. Output Current: 10 A resistive, max.
4. Time Delay: Adjustable from 0.1 second to 1023 minutes
5. Repeatability: 0.1% or 20 mS, whichever is greater
6. Time Delay Accuracy: 2.5% or 50 mS, whichever is greater
7. Recycle / Start-up Time Delay: 250 mS, typ.
8. Mechanical: 2.5" x 2.5" case, Surface mount
9. Protection: Encapsulated circuitry, MOV transient protection
10. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
11. Approved Manufacturer: Airotronics (MC363 series) or equal

2.15 NAMEPLATES

- A. Provide engraved phenolic or micarta nameplates for all equipment, components, and field devices furnished. Nameplates shall be 1/8 thick, black, with white center core, and shall be minimum 1" x 3", with minimum 1/4" high block lettering. Nameplates for devices smaller than 1" x 3" shall be attached to adjacent surface.
- B. Each nameplate shall identify the function for each device.

2.16 TESTING EQUIPMENT

- A. BMS Contractor shall test and calibrate all signaling circuits of all field devices to ascertain that required digital and accurate analog signals are transmitted, received, and displayed at system operator terminals, and make all repairs and recalibrations required to complete test. Contractor shall be responsible for test equipment required to perform these tests and calibrations. Test equipment used for testing and calibration of field devices shall be at least twice as accurate as respective field device (e.g., if field device is $\pm 0.5\%$ accurate, test equipment shall be $\pm 0.25\%$ accurate over same range).

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Contractor.

3.2 INSTALLATION OF CONTROL SYSTEMS

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings. Install electrical components and use electrical products complying with requirements of National Electric Code and all local codes.
- B. Control Wiring: The term "control wiring" is defined to include providing of wire, conduit and miscellaneous materials as required for mounting and connection of electric control devices.
 - 1. Wiring System: Install complete wiring system for electric control systems. Install all control wiring external to panels in electric metallic tubing or raceway. On Renovation projects, wiring in finished areas shall be routed in wire mold. The routing of wiring in finished areas must be specifically approved by the AOR/EOR. Installation of wiring shall generally follow building lines. Install in accordance with National Electrical Code and Division 26 of this Specification. Fasten flexible conductors bridging cabinets and doors, neatly along hinge side, and protect against abrasion. Tie and support conductors neatly.
 - 2. Control Wiring Conductors: Install control wiring conductors, without splices between terminal points, color-coded. Install in neat workmanlike manner, securely fastened. Install in accordance with National Electrical Code and Division 26 of this Specification.
 - 3. Communication wiring, signal wiring and low voltage control wiring shall be installed separate from any wiring over thirty (30) volts. Signal wiring shield shall be grounded at controller end only, unless otherwise recommended by the controller manufacturer.
 - 4. All WAN and LAN Communication wiring shield shall be terminated as recommended by controller manufacturer. All WAN and LAN Communication wiring shall be labeled with a network number, device ID at each termination and shall correspond with the WAN and LAN system architecture and floor plan submittals. All WAN and LAN cabling shall comply with applicable Division 26 requirements.
 - 5. Number-code or color-code conductors appropriately for future identification and servicing of control system. Code shall be as indicated on approved installation drawings.

- C. Room Temperature Sensors: Install sensors as shown on the drawings. Seal conduit penetrations at the wall box airtight. Install batt insulation in the wall box to completely fill the box. Electrical connections shall be made using a twist-on sealant filled connectors suitable for the installation.
- D. Relative Humidity Sensors: Provide element guard as recommended by manufacturer for high velocity installations. For high limit sensors, position remote enough to allow full moisture absorption into the air stream before reaching the sensor.
- E. Current Switches for Motor Status Monitoring: Adjust so that setpoint is below minimum operating current and above motor no load current.

3.3 DDC CONTROLLERS:

A. INSPECTION:

- 1. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Contractor.

B. System Access:

- 1. Provide an Ethernet connection and 4-port hub at the panel for the BC controller or controllers that provides access to the Local Supervisory LAN and to the BMS web server for all controllers, other than an Application Category-1 controllers. The user shall be able to access each controller on the system using this connection via the Control System Server database for graphics, schedules, programming, controller configuration etc.

C. INSTALLATION OF CONTROL SYSTEMS:

- 1. General: Install systems and materials in accordance with manufacturer's instructions, specifications roughing-in drawings and details shown on drawings. BMS Contractor shall install all controllers in accordance with manufacturer's installation procedures and practices.

D. HARDWARE APPLICATION REQUIREMENTS

- 1. General: The functional intent of this specification is to allow cost effective application of manufacturers standard products while maintain the integrity and reliability of the control functions. Specific requirements indicated below are required for the respective application. Manufacturer shall apply the most cost-effective unit that meets the requirement of that application.
- 2. Standalone Capability: Each Control Unit shall be capable of performing the required sequence of operation for the associated equipment. All physical point data and calculated values required to accomplish the sequence of operation shall originate within the associated control unit with only the exceptions enumerated below.
- 3. Where associated control functions involve functions from different categories identified below, the requirements for the most restrictive category shall be met.
- 4. Primary DDC panels shall be as follows:
 - a. Application Category-0: Distributed Monitoring
 - 1) Applications in this category include the following:

- a) Monitoring of variables that are not used in a control loop, sequence logic, or safety.
 - 2) Points on BCs, AACs, and ASCs may be used in these applications as well as Ds and/or general-purpose I/O modules.
 - 3) Where these points are trended, contractor shall verify and document that the network bandwidth is acceptable for such trends and is still capable of acceptable and timely control function.
 - 4) LAN Restrictions: These points may reside on any controller
- b. Application Category-1: Application Specific Controller (ASC)
 - 1) Minimum one (1) ASC for each mechanical equipment:
 - a) Rooftop Unit
 - b) Air Conditioning Unit
 - c) Unitary equipment <15 tons
 - 2) LAN Segment Restrictions: BACnet Systems: Limit the number of AAC's/ASC's servicing any one of these applications on the LAN Segment to 40.
- c. Application Category-2: Advance Application Controller (AAC)
 - 1) Minimum one (1) AAC for building control of the following:
 - a) Lighting Control
 - b) Exhaust Fan Control & Monitoring
 - c) Slab Monitoring
 - d) UPS Monitoring
 - e) Leak Detection Monitoring
 - f) Fire Alarm System Monitoring
 - g) Intrusion Controls Monitoring
 - h) Generator Monitoring
 - i) Misc. Monitoring and Meters (not directly associated with a control sequence and where trending is not critical)
 - 2) LAN Segment Restrictions: BACnet Systems: Limit the number of AAC's servicing any one of these applications on the LAN Segment to 40.
- d. Application Category-3: Building Controller (BC) with Advance Application Controller (AAC) or Expansion I/O Modules (IOM)
 - 1) Minimum one (1) BC for global operations of the following:
 - a) Occupancy Scheduling
 - b) Modbus-RTU Communications
 - c) BACnet-MS/TP Communications
 - d) BACnet-IP Communications
- 5. Motors in motor control centers shall be controlled from the DDC controller associated with HVAC system. It shall not be acceptable to control all motors in a MCC from one DDC controller dedicated to the MCC. The intent of this specification is that the loss of

any one DDC controller shall not affect the operation of other HVAC systems, only for the points connected to the DDC controller.

END OF SECTION 230922

SECTION 230923 – BUILDING AUTOMATION SYSTEM (BAS) – GENERAL SEQUENCE**PART 1 - GENERAL****1.1 SYSTEM DESCRIPTION**

- A. Refer to Division 23 Section and Control Drawings for a description of the systems to be controlled.
- B. Refer to Control Drawings for detailed sequence of operations for major HVAC equipment.
- C. This Section contains general sequence and alarming method for Illinois Institute of Technologies.

1.2 SUBMITTALS

- A. Refer to Division 23 Section "Building Automation System (BAS)" and Division 01 Sections for requirements for control shop drawings, product data, Users Manual, etc.
- B. Programming Manual: Provide DDC system programming manual as well as documentation of site specific programming prior to the start of Acceptance Phase.

1.3 PROJECT RECORD DOCUMENTS

- A. Within two weeks of the completion of commissioning, provide record documents to represent the final control configuration with actual setpoints and tuning parameters as existed at acceptance.
- B. Record documents shall be modified control drawings with the actual installed information. Drawings shall be delivered in both reproducible hard copy and electronic format in AutoCAD v13 or later. Provide all supporting files, blocks, fonts, etc. required by the drawings.
- C. Provide final points list
- D. Provide final detailed wiring diagrams with all wire numbers and termination points indicated
- E. Accurately record final sequences and control logic made after submission of shop drawings.

1.4 DEFINITIONS/ABBREVIATIONS

- A. Absolute Minimum OA: Minimum flow rate setpoint to which the OA or primary air may throttle down. This value is acceptable as long as CO₂ levels are within acceptable limits.
- B. DDC: Direct Digital Control
- C. Design Minimum OA: Minimum flow rate setpoint based on code requirements or designed system and coil capacities.
- D. BAS: Building Automation System (Same as "Building Management System (BMS)")
- E. MVR: Minimum required ventilation rate
- F. OA: Outdoor Air

- G. Physical Point: A point on the BMS that is physically connected to an I/O device such that a hardware point exists
- H. Virtual Point: A point to store values (i.e.: a setpoint) that do not represent a physical device

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 GENERAL SEQUENCE NOTES

- A. Sequences specified herein indicate the functional intent of the systems operation and may not fully detail every aspect of the programming that may be required to obtain the indicated operation. Contractor shall provide all programming necessary to obtain the sequences/system operation indicated.
- B. When a rooftop unit is not in operation, control devices shall remain in their "off" positions. "Off" positions may differ from the "normal" (meaning failed) position.
- C. Except as specified otherwise, throttling ranges, proportional bands, and cycle differentials shall be centered on the associated setpoint. All modulating feedback control loops shall include the capability of having proportional, integral, and derivative action. Unless the loop is specified "proportional only" or "P+I", Contractor shall apply appropriate elements of integral and derivative gain to each control loop which shall result in stable operation, minimum settling time, and shall maintain the primary variable within the specified maximum allowable variance.
- D. Scheduling Terminology: When rooftop units are scheduled, the following defines the terminology used:
 - 1. Occupied Period: period of time when the building is in use and occupied. Generally systems will be fully operational throughout this period and ventilation air shall be continuously introduced. Space temperature setpoints will generally be in the "normal" range of 69°-76°F.
 - 2. Unoccupied period: period of time when the building or zone is not in use and unoccupied. Ventilation air shall not be introduced.
 - 3. Preoccupancy Period: Time prior to the Occupied period when the systems are returning the space temperatures from setback to "normal" or occupied setpoints (warm-up and cool-down). Ventilation air shall not be introduced unless outside air conditions permit free-cooling. Time period shall be determined by an optimum start strategy unless otherwise specified.
 - 4. Setback Period: Setback start will typically coincide with the end of the occupied period and end with the start of the preoccupancy period, however it shall be provided with its own schedule. Generally systems will be off except to maintain a "setback" temperature.
- E. Automatic Daylight Savings Time Switchover
 - 1. The system shall provide automatic time adjustment for switching to/from Daylight Savings Time.
- F. Night Setback Control
 - 1. The system shall provide the ability to automatically adjust setpoints for night control.

G. Enthalpy Switchover (Economizer)

1. The Building Controller Software shall control the position of the rooftop unit relief, return, and outside air dampers. If the outside air enthalpy falls below changeover set point the BCS shall modulate the dampers to provide 100% outside air. The user shall be able to quickly changeover to an economizer system based on dry bulb temperature and shall be able to override the economizer cycle and return to minimum outside air operation at any time.

H. Loop Control

1. A Model-Free Adaptive Control algorithm or alternatively a PID (proportional-integral-derivative) closed-loop control algorithm with direct or reverse action and anti-windup shall be supplied. The algorithm shall calculate a time-varying analog value that is used to position an output or stage a series of outputs. The controlled variable, set point, and weighting parameters shall be user-selectable.

I. Sequencing

1. Provide application software based upon the sequences of operation specified to properly sequence equipment.

J. All setpoints shall be adjustable (adj.), they shall be modifiable, with the proper password level, from the operator interface or via a function block menu. For these points, it is unacceptable to have to modify programming statements to change the setpoint.

K. Where “prove operation” of a device (generally controlled by a digital output) is indicated in the sequence, it shall require that the BMS, after an adjustable time delay & after the device is commanded to operate (feedback delay,) confirm that the device is operational via the status input. If the status point does not confirm operation after the time delay or anytime thereafter for an adjustable time delay (debounce delay) while the device is commanded to run, an alarm shall be enunciated audibly and via an alarm message at the operator interface and print at the alarm printers. A descriptive message shall be attached to the alarm message indicating the nature of the alarm and actions to be taken. Contractor shall provide messages to meet this intent.

L. Wherever a value is indicated to be dependent on another value (i.e.: setpoint plus 5°F) the BMS shall use that equation to determine the value. Simply providing a virtual point that the operator must set is unacceptable. In this case three virtual points shall be provided. One to store the parameter (5°F), one to store the setpoint, and one to store the value which is the result of the equation.

M. Some sequences rely on integration with third-party manufacturer control equipment. See the control equipment specifications, equipment schedules and equipment specifications for more information on this integration.

3.2 GENERAL ALARMING NOTES

A. Alarming and faults shall be a function of rooftop unit factory controller BACnet object data to a BMS System (Subject to compliance with requirements, provided on **LiDL National Account List**. basis-of-design: Emerson). The BMS system shall read all space temperature (where applicable), duct CO2 (where applicable), and duct relative humidity (where applicable), and place high and low limits for consumer comfort and product tempering.

- B. Analog Input Alarms: For each analog input, program an alarm message for reporting whenever the analog value is outside of the programmed alarm limits. Report a 'Return-to-Normal' message after the analog value returns to the normal range, using a programmed alarm differential. The alarm limits shall be individually selected by the Contractor based on the following criteria:
1. Space temperature, except as otherwise stated in sequence of operation:
 - a. Low alarm: 5°F (adj) below setpoint
 - b. High alarm: 5°F (adj) above setpoint
 2. Space humidity:
 - a. Low alarm: 2°F (adj) below setpoint
 - b. High alarm: 2°F (adj) above setpoint
- C. Maintenance Alarms: Enunciate alarms when runtime accumulation exceeds a value specified by the operator.
1. *DEVICE XXXX REQUIRES MAINTENANCE.* Runtime has exceeded specified value since last reset.

END OF SECTION 230923

SECTION 231123 - NATURAL GAS PIPING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Pipes, tubes, and fittings.
 - 2. Piping specialties.
 - 3. Joining materials for piping and tubing.
 - 4. Valves.
 - 5. Pressure regulators.
 - 6. Service meters.
 - 7. Mechanical sleeve seals.

1.2 DEFINITIONS

- A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct shafts, unheated spaces immediately below roofs, spaces above ceilings, unexcavated spaces, crawlspaces, and tunnels.
- B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.
- D. Anodless Risers and Sweeps: Fittings that can be installed without additional cathodic protection.

1.3 PERFORMANCE REQUIREMENTS

- A. Natural-Gas System Pressure within Buildings: 2-psig or less.
- B. Operating Pressure Ratings:
 - 1. Piping and Valves: 100-psig minimum unless otherwise indicated.
 - 2. Service Regulators: 65-psig minimum unless otherwise indicated.
 - 3. Operating Pressure of Service Meter: 5-psig minimum unless otherwise indicated.

1.4 SUBMITTALS

- A. Product Data: For each type of the following:
 - 1. Piping specialties.
 - 2. Corrugated, stainless-steel tubing with associated components.
 - 3. Valves. Include pressure rating, capacity, settings, and electrical connection data of selected models.
 - 4. Pressure regulators. Indicate pressure ratings and capacities.
 - 5. Dielectric fittings.
 - 6. Mechanical sleeve seals.
 - 7. Escutcheons.

- B. Shop Drawings: Include plans, piping layout and elevations, sections, and details for fabrication of pipe anchors, hangers, supports for multiple pipes, alignment guides, expansion joints and loops, and attachments of the same to building structure. Detail locations of anchors, alignment guides, and expansion joints and loops. Include a site plan, on which natural-gas piping is indicated and coordinated with other services and utilities.
 - 1. Shop Drawing Scale: Not less than $1/4" = 1'-0"$.
- C. Coordination Drawings: Plans and details, drawn to scale, on which natural-gas piping is shown and coordinated with other installations, using input from installers of the items involved.
- D. Installer Qualifications.
- E. Welding Certificates.
- F. Operation and Maintenance Data: For motorized gas valves, pressure regulators, and service meters, include data in operation, and maintenance manuals.
- G. Record Drawings: Indicate, at $1/4" = 1'-0"$ scale, the actual natural-gas piping installation layout and elevations, sections, fabrication of pipe anchors, hangers, supports for multiple pipes, alignment guides, expansion joints and loops, and attachments of the same to the building structure. Detail location of anchors, alignment guides, and expansion joints and loops.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an installer that is experienced in the installation of natural-gas piping of the types required for the Project, and whose work has resulted in a record of successful in-service performance. Installer licensed to perform natural-gas piping installation and maintenance in the project jurisdiction.
- B. Steel Support Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- C. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- E. Regulatory Requirements: Natural-gas piping installations shall comply with the International Fuel Gas Code, 2000 edition.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Handling Flammable Liquids: Remove and dispose flammable liquids from existing natural-gas piping according to requirements of authorities having jurisdiction.
- B. Deliver pipes and tubes with factory-applied end caps. Maintain end caps throughout shipping, storage, and handling to prevent pipe end damage and to prevent entrance of dirt, debris, and moisture.
- C. Store and handle pipes and tubes having factory-applied protective coatings to avoid damaging coating, and protect from direct sunlight.

- D. Protect stored polyethylene pipes and valves from direct sunlight.

1.7 PROJECT CONDITIONS

- A. Perform site survey, research public utility records, and verify existing utility locations. Contact utility-locating service for area where Project is located.

1.8 COORDINATION

- A. Coordinate sizes and locations of concrete bases with actual equipment provided.
- B. Coordinate installation of anchorages for natural-gas systems. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete. Deliver such items to Project site in time for installation and to not cause a delay in the Work.
- C. Coordinate requirements for access panels and doors for valves and controls installed concealed behind finished surfaces, and for areas requiring routine maintenance. Comply with requirements in Division 08 Section "Access Doors and Frames."

1.9 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Special Warranty: Submit written warranty, executed by the manufacturer and Installer, agreeing to repair or replace components that develop defects in material or workmanship within the specified warranty period.
 - 1. Warranty Period for Natural-Gas Systems: One year from date of Substantial Completion, or eighteen months from date of shipment, whichever is greater.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the manufacturers indicated for each component:
 - 1. Appliance Flexible Connector: No preference.
 - 2. Two-Piece, Full-Port, Bronze Ball Valves with Bronze Trim:
 - a. Conbraco Industries, Inc.; Apollo Div.
 - b. Lyall, R. W. & Company, Inc.
 - c. McDonald, A. Y. Mfg. Co.
 - d. Perfection Corporation; a subsidiary of American Meter Company.
 - 3. Bronze Plug Valves:
 - a. Lee Brass Company.
 - b. McDonald, A. Y. Mfg. Co.
 - 4. Cast-Iron, Non-lubricated Plug Valves:

- a. McDonald, A. Y. Mfg. Co.
 - b. Mueller Co.; Gas Products Div.
 - c. Xomox Corporation; a Crane company.
5. Cast-Iron, Lubricated Plug Valves:
- a. Flowserve.
 - b. McDonald, A. Y. Mfg. Co.
 - c. Mueller Co.; Gas Products Div.
 - d. R&M Energy Systems, a Unit of Robbins & Myers, Inc.
6. Automatic Gas Valves:
- a. ASCO Power Technologies, LP; Division of Emerson.
 - b. Eaton Corporation; Controls Div.
 - c. Eclipse Combustion, Inc.
 - d. Honeywell International Inc.
 - e. Johnson Controls.
7. Electrically Operated Valves:
- a. ASCO Power Technologies, LP; Division of Emerson.
 - b. Eclipse Combustion, Inc.
 - c. Goyen Valve Corp.; Tyco Environmental Systems.
 - d. Magnatrol Valve Corporation.
 - e. Parker Hannifin Corporation; Climate & Industrial Controls Group; Skinner Valve Div.
 - f. Watts Regulator Co.; Division of Watts Water Technologies, Inc.
8. Line Pressure Regulators:
- a. Eclipse Combustion, Inc.
 - b. Fisher Control Valves and Regulators; Division of Emerson Process Management.
 - c. Invensys.
 - d. Maxitrol Company.
 - e. Richards Industries; Jordan Valve Div.
9. Appliance Pressure Regulators:
- a. Eaton Corporation; Controls Div.
 - b. Harper Wyman Co.
 - c. Maxitrol Company.
 - d. SCP, Inc.
10. Dielectric Unions: No preference.
11. Dielectric Flanges: No preference.
12. Mechanical Sleeves: No preference.

2.2 PIPES, TUBES, AND FITTINGS

- A. Steel Pipe: ASTM A53/A53M, black steel, Schedule 40, Type E or S, Grade B.
- 1. Malleable-Iron Threaded Fittings: ASME B16.3, Class 150, standard pattern.
 - 2. Wrought-Steel Welding Fittings: ASTM A234/A234M for butt welding and socket welding.

3. Unions: ASME B16.39, Class 150, malleable iron with brass-to-iron seat, ground joint, and threaded ends.
4. Forged-Steel Flanges and Flanged Fittings: ASME B16.5, minimum Class 150, including bolts, nuts, and gaskets of the following material group, end connections, and facings:
 - a. Material Group: 1.1.
 - b. End Connections: Threaded or butt welding to match pipe.
 - c. Lapped Face: Not permitted underground.
 - d. Gasket Materials: ASME B16.20, metallic, flat, asbestos free, aluminum o-rings, and spiral-wound metal gaskets.
 - e. Bolts and Nuts: ASME B18.2.1, carbon steel aboveground and stainless steel underground.

B. Polyethylene (PE) Pipe: ASTM D2513, SDR 11 (Underground Piping Only).

1. PE Fittings: ASTM D2683, socket-fusion type or ASTM D3261, butt-fusion type with dimensions matching PE pipe.
2. PE Transition Fittings: Factory-fabricated fittings with PE pipe complying with ASTM D2513, SDR 11; and steel pipe complying with ASTM A53/A53M, black steel, Schedule 40, Type E or S, Grade B.
3. Anodeless Service-Line Risers: Factory fabricated and leak tested.
 - a. Underground Portion: PE pipe complying with ASTM D2513, SDR 11 inlet. Underground piping to receive factory-applied, three layer coating of epoxy, adhesive, and polyethylene.
 - b. Casing: Steel pipe complying with ASTM A53/A53M, Schedule 40, black steel, Type E or S, Grade B, with corrosion-protective coating covering. Vent casing aboveground.
 - c. Aboveground Portion: PE transition fitting.
 - d. Outlet shall be threaded or flanged or suitable for welded connection.
 - e. Tracer wire connection.
 - f. Ultraviolet shield.
 - g. Stake supports with factory finish to match steel pipe casing or carrier pipe.
 - h. Joint Cover Kits: Epoxy paint, adhesive, and heat-shrink polyethylene sleeves.
4. Protective Coating for Underground Piping: Factory-applied, three-layer coating of epoxy, adhesive and polyethylene.

2.3 PIPING SPECIALTIES

A. Appliance Flexible Connectors:

1. Indoor, Fixed-Appliance Flexible Connectors: Comply with ANSI Z21.24.
2. Indoor, Movable-Appliance Flexible Connectors: Comply with ANSI Z21.69.
3. Outdoor, Appliance Flexible Connectors: Comply with ANSI Z21.75.
4. Corrugated stainless-steel tubing with polymer coating.
5. Operating-Pressure Rating: 0.0.5-psig End Fittings: Zinc-coated steel.
6. Threaded Ends: Comply with ASME B1.20.1.
7. Maximum Length: 72-inches.

B. Y-Pattern Strainers:

1. Body: ASTM A126, Class B, cast iron with bolted cover and bottom drain connection.

2. End Connections: Threaded ends for NPS 2 and smaller; flanged ends for NPS 2-1/2 and larger.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 50-percent free area.
4. CWP Rating: 125-psig.

C. Basket Strainers:

1. Body: ASTM A126, Class B, high-tensile cast iron with bolted cover and bottom drain connection.
2. End Connections: Threaded ends for NPS 2 and smaller; flanged ends for NPS 2-1/2 and larger.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 50 percent free area.
4. CWP Rating: 125-psig.

D. T-Pattern Strainers:

1. Body: Ductile or malleable iron with removable access coupling and end cap for strainer maintenance.
2. End Connections: Grooved ends.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 57-percent free area.
4. CWP Rating: 750-psig.

- E. Weatherproof Vent Cap: Cast- or malleable-iron increaser fitting with corrosion-resistant wire screen, with free area at least equal to cross-sectional area of connecting pipe and threaded-end connection.

2.4 JOINING MATERIALS

- A. Joint Compound and Tape: Suitable for natural-gas pipe.
- B. Welding Filler Metals: Comply with AWS D10.12/D10.12M for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.5 MANUAL GAS SHUTOFF VALVES

- A. Refer to Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule" for where each valve type is applied in various services.
- B. General Requirements for Metallic Valves, NPS 2 and Smaller: Comply with ASME B16.33.
1. CWP Rating: 125-psig.
 2. Threaded Ends: Comply with ASME B1.20.1.
 3. Dryseal Threads on Flare Ends: Comply with ASME B1.20.3.
 4. Tamperproof Feature: Locking feature for valves indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 5. Listing: Listed and labeled by an NRTL acceptable to authorities having jurisdiction for valves 1 inch diameter and smaller.
 6. Service Mark: Valves NPS 1-1/4 to NPS 2 shall have initials "WOG" permanently marked on valve body.

- C. General Requirements for Metallic Valves, NPS 2-1/2 and Larger: Comply with ASME B16.38.
1. CWP Rating: 125-psig.
 2. Flanged Ends: Comply with ASME B16.5 for steel flanges.
 3. Tamperproof Feature: Locking feature for valves indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 4. Service Mark: Initials "WOG" shall be permanently marked on valve body.
- D. Two-Piece, Full-Port, Bronze Body: Complying with ASTM B584.
1. Ball: Chrome-plated bronze.
 2. Stem: Bronze; blowout proof.
 3. Seats: Reinforced TFE; blowout proof.
 4. Packing: Threaded-body packnut design with adjustable-stem packing.
 5. Ends: Threaded, flared, or socket as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 6. CWP Rating: 600-psig.
 7. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 8. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- E. Bronze Plug Valves: MSS SP-78.
1. Body: Bronze, complying with ASTM B584.
 2. Plug: Bronze.
 3. Ends: Threaded, socket, or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 4. Operator: Square head or lug type with tamperproof feature where indicated.
 5. Pressure Class: 125-psig.
 6. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 7. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- F. Cast-Iron, Non-lubricated Plug Valves: MSS SP-78.
1. Body: Cast iron, complying with ASTM A126, Class B.
 2. Plug: Bronze or nickel-plated cast iron.
 3. Seat: Coated with thermoplastic.
 4. Stem Seal: Compatible with natural-gas.
 5. Ends: Threaded or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 6. Operator: Square head or lug type with tamperproof feature where indicated.
 7. Pressure Class: 125-psig.
 8. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 9. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- G. Cast-Iron, Lubricated Plug Valves: MSS SP-78.
1. Body: Cast iron, complying with ASTM A126, Class B.
 2. Plug: Bronze or nickel-plated cast iron.
 3. Seat: Coated with thermoplastic.
 4. Stem Seal: Compatible with natural-gas.

5. Ends: Threaded or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
6. Operator: Square head or lug type with tamperproof feature where indicated.
7. Pressure Class: 125- psig.
8. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
9. Service: Suitable for natural-gas service with "WOG" indicated on valve body.

H. Valve Boxes:

1. Cast-iron, two-section box.
2. Top section with embossed cover with letters "GAS".
3. Bottom section with base to fit over valve and a minimum 5-inches diameter barrel.
4. Adjustable cast-iron extensions of length required for depth of embedment.
5. Include tee-handle, steel operating wrench with socket end fitting, either valve nut or flat head, with stem of length required to operate valve.

2.6 MOTORIZED GAS VALVES

A. Electrically Operated Valves: Comply with UL 429.

1. Pilot operated.
2. Body: Brass or aluminum.
3. Seats and Disc: Nitrile rubber.
4. Springs and Valve Trim: Stainless steel.
5. 120-V ac, 60-Hz, Class B, continuous-duty molded coil, and replaceable.
6. NEMA ICS 6, Type 4, coil enclosure.
7. Normally closed.
8. Visual position indicator.

2.7 PRESSURE REGULATORS

A. General Requirements:

1. Single stage and suitable for natural-gas.
2. Steel jacket and corrosion-resistant components.
3. Elevation compensator.
4. End Connections: Threaded for regulators NPS 2 and smaller; flanged for regulators NPS 2-1/2 and larger.

B. Line Pressure Regulators: Comply with ANSI Z21.80.

1. Body and Diaphragm Case: Cast iron or die-cast aluminum.
2. Springs: Zinc-plated steel; interchangeable.
3. Diaphragm Plate: Zinc-plated steel.
4. Seat Disc: Nitrile rubber resistant to gas impurities, abrasion, and deformation at the valve port.
5. Orifice: Aluminum; interchangeable.
6. Seal Plug: Ultraviolet-stabilized, mineral-filled nylon.
7. Single-port, self-contained regulator with orifice no larger than required at maximum pressure inlet, and no pressure sensing piping external to the regulator.
8. Pressure regulator shall maintain discharge pressure setting downstream, and not exceed 1.5 times the design discharge pressure at shutoff.
9. Overpressure Protection Device: Factory mounted on pressure regulator.

10. Atmospheric Vent: Factory- or field-installed, stainless-steel screen in opening if not connected to vent piping.
11. Maximum Inlet Pressure: 2-psig.

C. Appliance Pressure Regulators: Comply with ANSI Z21.18.

1. Body and Diaphragm Case: Die-cast aluminum.
2. Springs: Zinc-plated steel; interchangeable.
3. Diaphragm Plate: Zinc-plated steel.
4. Seat Disc: Nitrile rubber.
5. Seal Plug: Ultraviolet-stabilized, mineral-filled nylon.
6. Factory-Applied Finish: Minimum three-layer polyester and polyurethane paint finish.
7. Regulator may include vent limiting device, instead of vent connection, if approved by authorities having jurisdiction.
8. Maximum Inlet Pressure: 2-psig

2.8 DIELECTRIC FITTINGS

A. Dielectric Unions:

1. Minimum Operating-Pressure Rating: 150-psig.
2. Combination fitting of copper alloy and ferrous materials.
3. Insulating materials suitable for natural-gas.
4. Combination fitting of copper alloy and ferrous materials with threaded, brazed-joint, plain, or welded end connections that match piping system materials.

B. Dielectric Flanges:

1. Minimum Operating-Pressure Rating: 150-psig
2. Combination fitting of copper alloy and ferrous materials.
3. Insulating materials suitable for natural-gas.
4. Combination fitting of copper alloy and ferrous materials with threaded, brazed-joint, plain, or welded end connections that match piping system materials.

2.9 SLEEVES

A. Steel Pipe Sleeves: ASTM A53, Type E, Grade B, Schedule 40, galvanized steel, plain ends.

2.10 MECHANICAL SLEEVE SEALS

A. Description: Modular sealing element unit, designed for field assembly, to fill annular space between pipe and sleeve.

1. Sealing Elements: EPDM interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe and sleeve.
2. Pressure Plates: Stainless steel.
3. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements. Include one nut and bolt for each sealing element.

2.11 ESCUTCHEONS

A. General Requirements for Escutcheons: Manufactured wall and ceiling escutcheons and floor plates, with ID to fit around pipe or tube, and OD that completely covers opening.

- B. One-Piece, Deep-Pattern Escutcheons: Deep-drawn, box-shaped brass with polished chrome-plated finish.
- C. One-Piece, Stamped-Steel Escutcheons: With set screw and chrome-plated finish.
- D. Split-Plate, Stamped-Steel Escutcheons: With concealed hinge, set screw, and chrome-plated finish.
- E. One-Piece, Floor-Plate Escutcheons: Cast-iron floor plate.
- F. Split-Casting, Floor-Plate Escutcheons: Cast brass with concealed hinge and set screw.

2.12 LABELING AND IDENTIFYING

- A. Detectable Warning Tape: Acid- and alkali-resistant, PE film warning tape manufactured for marking and identifying underground utilities, a minimum of 6-inches wide and 4-mils thick, continuously inscribed with a description of utility and the word "CAUTION", and colored yellow.
- B. Safety Tracer Wire: Metallic core (copper minimum AWG 14) encased in a protective insulated jacket for corrosion protection, detectable by metal detector when tape is buried up to 36-inches deep; colored yellow.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for natural-gas piping system to verify actual locations of piping connections before equipment installation.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Close equipment shutoff valves before turning off natural-gas to premises or piping section.
- B. Inspect natural gas piping according to the International Fuel Gas Code to determine that natural gas utilization devices are turned off in piping section(s) affected.
- C. Comply with the International Fuel Gas Code requirements for prevention of accidental ignition.

3.3 OUTDOOR PIPING INSTALLATION

- A. Comply with the International Fuel Gas Code for installation and purging of natural-gas piping.
- B. Underground Installations: Install underground natural-gas piping according to ASTM D2774.
 - 1. Locate natural-gas piping between 18-and 30-inches below finished grade, except in areas subjected to vehicular traffic, where natural-gas piping shall be located between 24- and 30-inches below finished grade. In all cases, when the backfilled trench is to be compacted, the natural-gas piping shall be placed at a depth sufficient to ensure the piping will be covered by not less than 18-inches of backfill material. Comply with requirements in Division 31 Section "Earthwork" for excavating, trenching, and backfilling.

2. At building, PE piping shall extend to a point 12-inches above finished grade with shut-off valve on either end.
 3. Install a safety tracer wire along the entire length of underground piping. Extend safety tracer wire to a point 12-inches above finished grade on the outside gas riser.
 4. Provide a continuous detectable warning tape 12-inches below finished grade, directly above - the entire length of the underground piping. Install detectable warning tape 6-inches below subgrade under paved areas and slabs.
 5. Line the trench bottom with a 3-inches thick bedding layer of clean sand, taking care to ensure the pipe will be uniformly and continuously supported over its entire length. Place PE pipe. Cover pipe with clean sand, not less than 6-inches above top of pipe, prior to backfilling the trench. Cover the sand with not less than 12-inches of excavated soil, free of rocks, frozen clods, or debris greater than 3-inches diameter, prior to compacting the backfilled trench.
- C. Steel Piping with Protective Coating (Above Ground Only):
1. Apply joint cover kits to pipe after joining to cover, seal, and protect joints.
 2. Replace pipe having damaged protective coating with new pipe.
- D. Aboveground Natural-Gas Piping:
1. Pipe Diameter – 2-inches Diameter and Smaller: Steel pipe with malleable-iron fittings and threaded joints.
 2. Pipe Diameter – Larger than 2-inches Diameter: Steel pipe with wrought-steel fittings and welded joints.
 3. Concealed Piping: All concealed natural-gas piping shall have welded joints.
- E. Install fittings for changes in direction and branch connections.
1. Anodeless risers, pieces, or sweeps shall be used when rising up from soil and when penetrating basement walls. Anodeless components shall be secured to the foundation with proper brackets when used as risers.
- F. Above-ground, Exterior Wall Pipe Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
1. Install steel pipe for sleeves smaller than 6-inches diameter.
 2. Install cast-iron "wall pipes" for sleeves 6-inches diameter and larger.
- G. Exterior Wall Pipe Penetrations: Install cast-iron "wall pipes" for sleeves. Seal pipe penetrations using mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
- H. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
- I. Exterior Rooftop Pipe Installation:
1. Rooftop piping should meet the requirements of above ground natural gas piping in the paragraphs above.

2. Support steel piping above roof with pillow block support system equal to Miro Industries model 3- R or 3-RAH. Support system shall consist of polycarbonate base mounted on elastomeric walk pad or support pad. Support system should include pipe rollers to allow movement.
 3. Exterior Roof Pipe Penetrations: Penetration of all piping through the roof shall be through pipe curb cap assembly with sealed neoprene boots equal to Pate Company model pca-2 or equal.
- J. Install pressure gage upstream and downstream from each service regulator. Pressure gages are specified in Division 23 Section "Meters and Gages for HVAC Piping."

3.4 INDOOR PIPING INSTALLATION

- A. Comply with the International Fuel Gas Code for installation and purging of natural-gas piping.
 - B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements are used to size pipe and calculate friction loss, expansion, and other design considerations. Install piping as indicated unless deviations to layout are approved by Architect on Coordination Drawings.
 - C. Arrange for pipe spaces, chases, slots, sleeves, and openings in building structure during progress of construction, to allow for mechanical installations.
 - D. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.
 - E. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building surfaces. Diagonal runs are prohibited unless specifically indicated otherwise.
 - F. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
 - G. Locate valves for easy access.
 - H. Install piping free of sags and bends.
 - I. Install fittings for changes in direction and branch connections.
 - J. Install escutcheons at penetrations of interior walls, ceilings, and floors.
1. New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.
 - b. Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - c. Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type or split-plate, stamped-steel type with concealed hinge and set screw.
 - d. Piping in Unfinished Service Spaces: One-piece, stamped-steel type with concealed hinge and set screw.
 - e. Piping in Equipment Rooms: One-piece, stamped-steel type with set screw.
 - f. Piping at Floor Penetrations in Equipment Rooms: One-piece, floor-plate type.

- K. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire stop materials. Comply with requirements in Division 07 Section "Penetration Firestopping."
- L. Verify final equipment locations for roughing-in.
- M. Comply with requirements in Sections specifying gas-fired appliances and equipment for roughing-in requirements.
- N. Drips and Sediment Traps: Provide drips at points where condensate may collect, including service-meter outlets. Locate where accessible to permit cleaning and emptying. Do not install where condensate is subject to freezing.
 - 1. Construct drips and sediment traps using tee fitting with bottom outlet plugged or capped. Use nipple a minimum length of 3 pipe diameters, but not less than 3 inches long and same size as connected pipe. Install with space below bottom of drip to remove plug or cap.
- O. Extend relief vent connections for service regulators, line regulators, and overpressure protection devices to outdoors and terminate with weatherproof vent cap.
- P. Concealed natural-gas piping shall be installed according to the International Fuel Code.
- Q. Natural-gas piping valves and regulators shall not be installed in concealed spaces.
 - 1. In Floor Channels: Install natural-gas piping in floor channels. Channels must have cover and be open to space above cover for ventilation.
 - 2. Prohibited Locations:
 - a. Do not install natural-gas piping in or through circulating air ducts, chimneys or gas vents (flues), ventilating ducts, or dumbwaiter or elevator hoistways.
 - b. Do not install natural-gas piping in solid walls or partitions.
 - c. Do not install natural-gas piping underground below building slabs.
- R. Use eccentric reducer fittings to make reductions in pipe sizes. Install fittings with level side down.
- S. Connect branch piping from top or side of horizontal piping.
- T. Provide unions in pipes NPS 2 and smaller, adjacent to each valve, at final connection to each piece of equipment. Unions are not required at flanged connections.
- U. Do not use natural-gas piping as grounding electrode.
- V. Provide strainer on inlet of each line-pressure regulator and automatic or electrically operated valve.
- W. Provide pressure gage upstream and downstream from each pipe regulator. Pressure gages are specified in Division 23 Section "Meters and Gages for HVAC Piping."

3.5 VALVE INSTALLATION

- A. Provide manual gas shutoff valve for each gas appliance ahead of appliance flexible connector. Valve shall be located in same room as equipment.

- B. Provide outdoor underground valves with valve boxes.
- C. Provide a shutoff valve, acceptable to the authority having jurisdiction, on the outside riser.
- D. Install regulators and overpressure protection devices with maintenance access space adequate for servicing and testing.
- E. Provide anode for metallic valves in underground PE piping.
- F. Provide pressure reducing valve at each equipment and appliance with relief piped outdoors.
- G. As a minimum install automatic gas shut off valve with alarm or remote operation for kitchen equipment motor hood with fire suppression system.

3.6 THERMAL MASS NATURAL GAS FLOW METER

- A. Type: Insertion Flow Meter, Outdoor Conditions.
- B. Flow Meter shall be complete with all installation hardware necessary to enable insertion and removal of the meter without system interruption.
- C. Fitment: Meter shall be installed in accordance with the manufacturer's installation guide including meter orientation and straight pipe recommendations. Applicable pipe size shall be from 0.5 to 8 inches.
- D. Maximum Operating Pressure: The flow meter shall be able to operate at up to 60 psig (standard. High pressure to 150 psig).
- E. Materials: Wetted metal components shall be 316 Stainless Steel.
- F. Electrical Connections: Enclosed terminal blocks, cable access through two ¾" NPT conduit fittings
- G. Flow Accuracy (natural gas / propane gas):
 - 1. ± 1.0% of reading from 500 – 7,000 SFPM.
 - 2. ± 2.0% of reading from 100 – 500 SFPM.
- H. Temperature Accuracy: + 1.0 °F over the range of -40 °F to 250 °F.
- I. Ambient Temperature Range: -40 °F to 158 °F.
- J. Output Signals: One (1) 4-20 mA analog output. Include Modbus RTU connectivity.
- K. Installation: Coordinate manufacturer's recommended installation requirements with piping contractor to ensure proper installation.
- L. Power: 12-28VDC, 6W minimum power.
- M. Contractor shall provide written coordination on capacity, performance, and location with trade contractor during the submittal phase.
- N. Acceptable Manufacturer: Onicon F-5500 Series or Sierra Instruments QuadraTherm-640i Series. No exception.

3.7 PIPING JOINT CONSTRUCTION

- A. Ream ends of pipes and remove burrs.
- B. Remove scale, slag, dirt, and debris from inside and outside of pipes and fittings before assembly.
- C. Threaded Joints:
 - 1. Thread pipe with tapered pipe threads according to ASME B1.20.1.
 - 2. Cut threads full and clean using sharp dies.
 - 3. Ream threaded pipe ends to remove burrs and restore full inside diameter of pipe.
 - 4. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
- D. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds. Welded Joints:
 - 1. Construct joints according to AWS D10.12/D10.12M, using qualified processes and welding operators.
 - 2. Bevel plain ends of steel pipe.
 - 3. Patch factory-applied protective coating as recommended by manufacturer at field welds and where damage to coating occurs during construction.
- E. Flanged Joints: Provide gasket material, size, type, and thickness appropriate for natural-gas service. Install gasket concentrically positioned.
- F. PE Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D2657.
 - 1. Plain-End Pipe and Fittings: Use butt fusion.
 - 2. Plain-End Pipe and Socket Fittings: Use socket fusion.

3.8 HANGER AND SUPPORT INSTALLATION

- A. Comply with requirements for pipe hangers and supports specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Provide hangers for horizontal steel piping with the following maximum spacing and minimum rod sizes:
 - 1. NPS 1 and Smaller: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 2. NPS 1-1/2: Maximum span, 108 inches; minimum rod size, 3/8 inch.
 - 3. NPS 1-1/2 and NPS 2: Maximum span, 108 inches; minimum rod size, 3/8 inch.
 - 4. NPS 2-1/2 to NPS 3-1/2: Maximum span, 10 feet; minimum rod size, 1/2 inch.
 - 5. NPS 4 and Larger: Maximum span, 10 feet; minimum rod size, 5/8 inch.

3.9 CONNECTIONS

- A. Install natural-gas piping electrically continuous, and bonded to gas appliance equipment grounding conductor of the circuit powering the appliance according to NFPA 70.
- B. Install piping adjacent to appliances to allow service and maintenance of appliances.

- C. Connect piping to appliances using manual gas shutoff valves and unions. Install valve within 72 inches of each gas-fired appliance and equipment. Install union between valve and appliances or equipment.
- D. Sediment Traps: Provide tee fitting with capped nipple in bottom to form drip, as close as practical to inlet of each appliance.
- E. Connect wiring according to Division 26 Section "Conductors and Cables".

3.10 LABELING AND IDENTIFYING

- A. Comply with requirements in Division 23 Section "Identification for HVAC" for piping and valve identification.
- B. Below Grade: Provide detectable warning tape continuously and directly above gas piping, 12-inches below finish grade; 6-inches below sub-grade under pavements and concrete slabs.
- C. Labeling and identification materials are specified in Division 22 "Identification for Plumbing Piping and Equipment".

3.11 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Tests and Inspections:
 - 1. Test, inspect, and purge natural-gas according to the International Fuel Gas Code and authorities having jurisdiction.
- C. Natural-gas piping will be considered defective if it does not pass tests and inspections.
- D. Prepare test and inspection reports.

3.12 OUTDOOR PIPING SCHEDULE

- A. Underground Natural-Gas Piping: Polyethylene (PE) pipe and fittings joined by heat fusion, or mechanical coupling. Service-line riser with tracer wire shall be terminated in an accessible location.
- B. Aboveground Natural-Gas Piping: Steel pipe, with joints as indicated in Article, "Outdoor Piping Installation."

3.13 INDOOR PIPING SCHEDULE FOR SYSTEM PRESSURES LESS THAN 0.5 PSIG

- A. Aboveground, branch piping NPS 1 and smaller shall be the following:
 - 1. Steel pipe with malleable-iron fittings and threaded joints.
- B. Aboveground, distribution piping shall be the following:
 - 1. Steel pipe with malleable-iron fittings and threaded joints NPS 2 and smaller.
 - 2. Steel pipe with wrought-steel fittings and welded joints NPS 2-1/2 and larger.
 - 3. All concealed natural-gas piping shall be welded.
 - 4. Underground, below building, piping shall not be allowed.

3.14 INDOOR PIPING SCHEDULE FOR SYSTEM PRESSURES MORE THAN 0.5 PSIG ND LESS THAN 2 PSIG

A. Aboveground piping shall be the following:

1. Steel pipe with malleable-iron fittings and threaded joints smaller or equal to NPS 2.
2. Steel pipe with steel welding fittings and welded joints NPS 2-1/2 and larger.
3. All concealed natural-gas piping shall be in a double wall configuration and vented to the outdoors.

3.15 ABOVEGROUND MANUAL GAS SHUTOFF VALVE SCHEDULE

A. Valves for pipe sizes NPS 2 and smaller at service meter shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

B. Valves for pipe sizes NPS 2-1/2 and larger at service meter shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.
3. Cast-iron, non-lubricated plug valve.

C. Distribution piping valves for pipe sizes NPS 2 and smaller shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

D. Distribution piping valves for pipe sizes NPS 2-1/2 and larger shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.
3. Cast-iron, lubricated plug valve.

E. Valves in branch piping for single appliance shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

END OF SECTION 231123

SECTION 232300 - REFRIGERANT PIPING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes refrigerant piping used for air-conditioning applications.

1.2 SUBMITTALS

- A. Product Data: For each type of valve and refrigerant piping specialty indicated. Include pressure drop, based on manufacturer's test data, for the following:
 - 1. Thermostatic expansion valves.
 - 2. Solenoid valves.
 - 3. Hot-gas bypass valves.
 - 4. Filter dryers.
 - 5. Strainers.
 - 6. Pressure-regulating valves.
- B. Shop Drawings: Show layout of refrigerant piping and specialties, including pipe, tube, and fitting sizes, flow capacities, valve arrangements and locations, slopes of horizontal runs, oil traps, double risers, wall and floor penetrations, and equipment connection details. Show interface and spatial relationships between piping and equipment.
 - 1. Shop Drawing Scale: 1/4 inch equals 1 foot.
- C. Welding certificates.
- D. Field quality-control test reports.
 - 1. Submit written reports documenting the activities required to be performed in PART 3. These reports are to be submitted two weeks after the startup is completed.
- E. Operation and Maintenance Data: For refrigerant valves and piping specialties to include in maintenance manuals.

1.3 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
- B. Comply with ASHRAE 15, "Safety Code for Refrigeration Systems."
- C. Comply with ASME B31.5, "Refrigeration Piping and Heat Transfer Components."

1.4 DELIVERY STORAGE AND HANDLING

- A. Store piping in a clean and protected area with end caps in place to ensure that piping interior and exterior are clean when installed.

1.5 COORDINATION

- A. Coordinate size and location of roof curbs, equipment supports, and roof penetrations. These items are specified in Division 07 Section "Roof Accessories."

1.6 PERFORMANCE REQUIREMENTS

- A. Line Test Pressure for Refrigerant R-410A:
 - 1. Suction Lines for Air-Conditioning Applications: 300 psig.
 - 2. Suction Lines for Heat-Pump Applications: 535 psig.
 - 3. Hot-Gas and Liquid Lines: 535 psig.

PART 2 - PRODUCTS

2.1 MANUFACTURERS:

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Refrigerants:
 - a. Atofina Chemicals, Inc.
 - b. DuPont Company; Fluorochemicals Div.
 - c. Honeywell, Inc.; Genetron Refrigerants.
 - d. INEOS Fluor Americas LLC.

2.2 COPPER TUBE AND FITTINGS

- A. Copper Tube: ASTM B 280, Type ACR or type K complying with ASTM B88 or ASTM B819.
- B. Wrought-Copper Fittings: ASME B16.22.
- C. Wrought-Copper Unions: ASME B16.22.
- D. Brazing Filler Metals: AWS A5.8.
- E. Flexible Connectors:
 - 1. Body: Tin-bronze bellows with woven, flexible, tinned-bronze-wire-reinforced protective jacket.
 - 2. End Connections: Socket ends.
 - 3. Offset Performance: Capable of minimum 3/4-inch (20-mm) misalignment in minimum 7-inch- (180-mm-) long assembly.
 - 4. Pressure Rating: Factory test at minimum 500 psig (3450 kPa).
 - 5. Maximum Operating Temperature: 250 deg F (121 deg C).

2.3 VALVES AND SPECIALTIES

- A. Diaphragm Packless Valves:
 - 1. Body and Bonnet: Forged brass or cast bronze; globe design with straight-through or angle pattern.

2. Diaphragm: Phosphor bronze and stainless steel with stainless-steel spring.
3. Operator: Rising stem and hand wheel.
4. Seat: Nylon.
5. End Connections: Socket, union, or flanged.
6. Working Pressure Rating: 500 psig.
7. Maximum Operating Temperature: 275 deg F.

B. Packed-Angle Valves:

1. Body and Bonnet: Forged brass or cast bronze.
2. Packing: Molded stem, back seating, and replaceable under pressure.
3. Operator: Rising stem.
4. Seat: Nonrotating, self-aligning polytetrafluoroethylene.
5. Seal Cap: Forged-brass or valox hex cap.
6. End Connections: Socket, union, threaded, or flanged.
7. Working Pressure Rating: 500 psig.
8. Maximum Operating Temperature: 275 deg F.

C. Check Valves:

1. Body: Ductile iron, forged brass, or cast bronze; globe pattern.
2. Bonnet: Bolted ductile iron, forged brass, or cast bronze; or brass hex plug.
3. Piston: Removable polytetrafluoroethylene seat.
4. Closing Spring: Stainless steel.
5. End Connections: Socket, union, threaded, or flanged.
6. Maximum Opening Pressure: 0.50 psig.
7. Working Pressure Rating: 500 psig.
8. Maximum Operating Temperature: 275 deg F.

D. Service Valves:

1. Body: Forged brass with brass cap including key end to remove core.
2. Core: Removable ball-type check valve with stainless-steel spring.
3. Seat: Polytetrafluoroethylene.
4. End Connections: Copper spring.
5. Working Pressure Rating: 500 psig.

E. Solenoid Valves: Comply with ARI 760 and UL 429; listed and labeled by an NRTL.

1. Body and Bonnet: Plated steel.
2. Solenoid Tube, Plunger, Closing Spring, and Seat Orifice: Stainless steel.
3. Seat: Polytetrafluoroethylene.
4. End Connections: Threaded.
5. Electrical: Molded, watertight coil in NEMA 250 enclosure of type required by location with 1/2-inch (16-GRC) conduit adapter, and 115-V ac coil.
6. Working Pressure Rating: 400 psig.
7. Maximum Operating Temperature: 240 deg F.

F. Safety Relief Valves: Comply with ASME Boiler and Pressure Vessel Code; listed and labeled by an NRTL.

1. Body and Bonnet: Ductile iron and steel, with neoprene O-ring seal.
2. Piston, Closing Spring, and Seat Insert: Stainless steel.
3. Seat Disc: Polytetrafluoroethylene.

4. End Connections: Threaded.
 5. Working Pressure Rating: 400 psig.
 6. Maximum Operating Temperature: 240 deg F.
- G. Thermostatic Expansion Valves: Comply with ARI 750.
1. Body, Bonnet, and Seal Cap: Forged brass or steel.
 2. Diaphragm, Piston, Closing Spring, and Seat Insert: Stainless steel.
 3. Packing and Gaskets: Non-asbestos
 4. Capillary and Bulb: Copper tubing filled with refrigerant charge.
 5. Suction Temperature: See Equipment Schedules
 6. Superheat: Adjustable.
 7. Reverse-flow option (for heat pump applications).
 8. End Connections; Socket, flare, or threaded union.
 9. Working Pressure Rating: 700 psig.
- H. Hot-Gas Bypass Valves: Comply with UL 429, listed and labeled by NRTL.
1. Body, Bonnet, and Seal Cap: Ductile iron or steel.
 2. Diaphragm, Piston, Closing Spring, and Seat Insert: Stainless steel.
 3. Packing and Gaskets: Non-asbestos.
 4. Solenoid Tube, Plunger, Closing Spring, and Seat Orifice: Stainless steel.
 5. Seat: Polytetrafluoroethylene.
 6. Equalizer: Internal or External.
 7. Electrical: Molded, watertight coil in NEMA 250 enclosure of type required by location with 1/2-inch conduit adapter, and [24] [115] [208]-V ac coil.
 8. End connections: Socket.
 9. Set Pressure: As required.
 10. Throttling Range: Maximum 5 psig.
 11. Working Pressure Range: 500 psig.
 12. Maximum Operating temperature: 240 deg F.
- I. Straight-Type Strainers:
1. Body: Welded steel with corrosion-resistant coating.
 2. Screen: 100-mesh stainless steel.
 3. End Connections: Socket or flare.
 4. Working Pressure Rating: 500 psig.
 5. Maximum Operating Temperature: 275 deg F.
- J. Angle-Type Strainers:
1. Body: Forged brass or cast bronze.
 2. Drain Plug: Brass hex plug.
 3. Screen: 100-mesh monel.
 4. End Connections: Socket or flare.
 5. Working Pressure Rating: 500 psig.
 6. Maximum Operating Temperature: 275 deg F.
- K. Moisture/Liquid Indicators:
1. Body: Forged brass.
 2. Window: Replaceable, clear, fused glass window with indicating element protected by filter screen.

3. Indicator: Color coded to show moisture content in ppm.
 4. Minimum Moisture Indicator Sensitivity: Indicate moisture above 60 ppm.
 5. End Connections: Socket or flare.
 6. Working Pressure Rating: 500 psig.
 7. Maximum Operating Temperature: 240 deg F.
- L. Replaceable-Core Filter Dryers: Comply with ARI 730.
1. Body and Cover: Painted-steel shell with ductile-iron cover, stainless-steel screws, and neoprene gaskets.
 2. Filter Media: 10 micron, pleated with integral end rings; stainless-steel support.
 3. Desiccant Media: Activated charcoal.
 4. End Connections: Socket.
 5. Access Ports: NPS 1/4 connections at entering and leaving sides for pressure differential measurement.
 6. Maximum Pressure Loss: 2 psig.
 7. Rated Flow: See equipment sizing
 8. Working Pressure Rating: 500 psig.
 9. Maximum Operating Temperature: 240 deg F.
- M. Permanent Filter Dryers: Comply with ARI 730.
1. Body and Cover: Painted-steel shell.
 2. Filter Media: 10 micron, pleated with integral end rings; stainless-steel support.
 3. Desiccant Media: Activated charcoal.
 4. End Connections: Socket.
 5. Access Ports: NPS 1/4 connections at entering and leaving sides for pressure differential measurement.
 6. Maximum Pressure Loss: 2 psig.
 7. Rated Flow: See equipment sizing
 8. Working Pressure Rating: 500 psig.
 9. Maximum Operating Temperature: 240 deg F.
- N. Mufflers:
1. Body: Welded steel with corrosion-resistant coating.
 2. End Connections: Socket or flare.
 3. Working Pressure Rating: 500 psig.
 4. Maximum Operating Temperature: 275 deg F.
- O. Receivers: Comply with ARI 495.
1. Comply with ASME Boiler and Pressure Vessel Code; listed and labeled by an NRTL.
 2. Comply with UL 207; listed and labeled by an NRTL.
 3. Body: Welded steel with corrosion-resistant coating.
 4. Tappings: Inlet, outlet, liquid level indicator, and safety relief valve.
 5. End Connections: Socket or threaded.
 6. Working Pressure Rating: 500 psig.
 7. Maximum Operating Temperature: 275 deg F.
- P. Liquid Accumulators: Comply with ARI 495.
1. Body: Welded steel with corrosion-resistant coating.
 2. End Connections: Socket or threaded.

3. Working Pressure Rating: 500 psig.
4. Maximum Operating Temperature: 275 deg F.

2.4 REFRIGERANTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Atofina Chemicals, Inc.
 2. DuPont Company; Fluorochemicals Div.
 3. Honeywell, Inc.; Genetron Refrigerants.
 4. INEOS Fluor Americas LLC.
- B. ASHRAE 34, R-410A: Pentafluoroethane/Difluoromethane.

PART 3 - EXECUTION

3.1 REFRIGERANT PIPE SIZING:

- A. All refrigerant pipe sizing shall be the responsibility of the Contractor in accordance with the equipment manufacturer's recommendations.
- B. Pipe sizing shall be in accordance with the recommendations in the 2010 ASHRAE Handbook – Refrigeration, Chapter 2 – System Practices for Halocarbon Refrigerants.
- C. Refrigerant piping indicated on Drawings is schematic only. Size piping and design actual piping layout, including oil traps, double risers, specialties, and pipe and tube sizes to accommodate, as a minimum, equipment provided, elevation difference between compressor and evaporator, and length of piping to ensure proper operation and compliance with warranties of connected equipment.

3.2 PIPING APPLICATIONS FOR REFRIGERANT R-410A

- A. Suction Lines NPS 3-1/2 and Smaller for Conventional Air-Conditioning Applications: Copper, Type ACR, drawn-temper tubing and wrought-copper fittings with brazed or soldered joints.
- B. Safety-Relief-Valve Discharge Piping: Copper, Type ACR, annealed- or drawn-temper tubing and wrought-copper fittings with brazed joints.

3.3 VALVE AND SPECIALTY APPLICATIONS

- A. Install packed-angle valves in suction and discharge lines of compressor.
- B. Install service valves for gage taps at inlet and outlet of hot-gas bypass valves and strainers if they are not an integral part of valves and strainers.
- C. Install a check valve at the compressor discharge and a liquid accumulator at the compressor suction connection.
- D. Except as otherwise indicated, install packed-angle valves on inlet and outlet side of filter dryers.
- E. Install a full-sized, three-valve bypass around filter dryers.

- F. Install solenoid valves upstream from each expansion valve and hot-gas bypass valve. Install solenoid valves in horizontal lines with coil at top.
- G. Install thermostatic expansion valves as close as possible to distributors on evaporators.
 - 1. Install valve so diaphragm case is warmer than bulb.
 - 2. Secure bulb to clean, straight, horizontal section of suction line using two bulb straps. Do not mount bulb in a trap or at bottom of the line.
 - 3. If external equalizer lines are required, make connection where it will reflect suction-line pressure at bulb location.
- H. Install safety relief valves where required by ASME Boiler and Pressure Vessel Code. Pipe safety-relief-valve discharge line to outside according to ASHRAE 15.
- I. Install moisture/liquid indicators in liquid line at the inlet of the thermostatic expansion valve or at the inlet of the evaporator coil capillary tube.
- J. Install strainers upstream from and adjacent to the following unless they are furnished as an integral assembly for device being protected:
 - 1. Solenoid valves.
 - 2. Thermostatic expansion valves.
 - 3. Compressor.
- K. Install filter dryers in liquid line between compressor and thermostatic expansion valve, and in the suction line at the compressor.
- L. Install receivers sized to accommodate pump-down charge.
- M. Install flexible connectors at compressors.

3.4 PIPING INSTALLATION

- A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems; indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Shop Drawings.
- B. Install refrigerant piping according to ASHRAE 15.
- C. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.
- D. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- E. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- F. Install piping adjacent to machines to allow service and maintenance.
- G. Install piping free of sags and bends.
- H. Install fittings for changes in direction and branch connections.

- I. Select system components with pressure rating equal to or greater than system operating pressure.
 - J. Refer to Division 23 Sections "Building Automation System (BAS)" and "Building Automation System (BAS) - Sequence of Operation" for solenoid valve controllers, control wiring, and sequence of operation.
 - K. Install piping as short and direct as possible, with a minimum number of joints, elbows, and fittings.
 - L. Arrange piping to allow inspection and service of refrigeration equipment. Install valves and specialties in accessible locations to allow for service and inspection. Install access doors or panels as specified in Division 08 Section "Access Doors and Frames" if valves or equipment requiring maintenance is concealed behind finished surfaces.
 - M. Install refrigerant piping in protective conduit where installed belowground.
 - N. Install refrigerant piping in rigid or flexible conduit in locations where exposed to mechanical injury.
 - O. Slope refrigerant piping as follows:
 - 1. Install horizontal suction lines with a uniform slope downward to compressor.
 - 2. Install traps and double risers to entrain oil in vertical runs.
 - 3. Liquid lines may be installed level.
 - P. When brazing , remove solenoid-valve coils and sight glasses; also remove valve stems, seats, and packing, and accessible internal parts of refrigerant specialties. Do not apply heat near expansion-valve bulb.
 - Q. Install pipe sleeves at penetrations in exterior walls and floor assemblies.
 - R. Seal penetrations through fire and smoke barriers according to Division 07 Section "Penetration Firestopping."
 - S. Install piping with adequate clearance between pipe and adjacent walls and hangers or between pipes for insulation installation.
 - T. Install sleeves through floors, walls, or ceilings, sized to permit installation of full-thickness insulation.
 - U. Seal pipe penetrations through exterior walls according to Division 07 Section "Joint Sealants" for materials and methods.
 - V. Identify refrigerant piping and valves according to Division 23 Section "Identification for HVAC."
- 3.5 PIPE JOINT CONSTRUCTION
- A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
 - B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

- C. Brazed Joints: Construct joints according to AWS's "Brazing Handbook," Chapter "Pipe and Tube."
 - 1. Use Type BcuP, copper-phosphorus alloy for joining copper socket fittings with copper pipe.
 - 2. Use Type BAg, cadmium-free silver alloy for joining copper with bronze or steel.
- D. Threaded Joints: Thread steel pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads unless dry-seal threading is specified.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Steel pipe can be threaded, but threaded joints must be seal brazed or seal welded.
- F. Welded Joints: Construct joints according to AWS D10.12/D10.12M.
- G. Flanged Joints: Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Use suitable lubricants on bolt threads.

3.6 HANGERS AND SUPPORTS

- A. Hanger, support, and anchor products are specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Install the following pipe attachments:
 - 1. Adjustable steel clevis hangers for individual horizontal runs less than 20 feet (6 m) long.
 - 2. Roller hangers and spring hangers for individual horizontal runs 20 feet (6 m) or longer.
 - 3. Pipe Roller: MSS SP-58, Type 44 for multiple horizontal piping 20 feet (6 m) or longer, supported on a trapeze.
 - 4. Spring hangers to support vertical runs.
 - 5. Copper-clad hangers and supports for hangers and supports in direct contact with copper pipe.
- C. Install hangers for copper tubing with the following maximum spacing and minimum rod sizes:
 - 1. NPS 1/2: Maximum span, 60 inches; minimum rod size, 1/4 inch.
 - 2. NPS 5/8: Maximum span, 60 inches; minimum rod size, 1/4 inch.
 - 3. NPS 1: Maximum span, 72 inches; minimum rod size, 1/4 inch.
 - 4. NPS 1-1/4: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 5. NPS 1-1/2: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 6. NPS 2: Maximum span, 96 inches; minimum rod size, 3/8 inch.
- D. Support multifloor vertical runs at least at each floor.

3.7 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.
- B. Tests and Inspections:

1. Comply with ASME B31.5, Chapter VI.
2. Test refrigerant piping, specialties, and receivers. Isolate compressor, condenser, evaporator, and safety devices from test pressure if they are not rated above the test pressure.
3. Test high- and low-pressure side piping of each system separately at not less than the pressures indicated in Part 1 "Performance Requirements" Article.
 - a. Fill system with nitrogen to the required test pressure.
 - b. System shall maintain test pressure at the manifold gage throughout duration of test.
 - c. Test joints and fittings with electronic leak detector or by brushing a small amount of soap and glycerin solution over joints.
 - d. Remake leaking joints using new materials, and retest until satisfactory results are achieved.

3.8 SYSTEM CHARGING

A. Charge system using the following procedures:

1. Install core in filter dryers after leak test but before evacuation.
2. Evacuate entire refrigerant system with a vacuum pump to 500 micrometers (67 Pa). If vacuum holds for 12 hours, system is ready for charging.
3. Break vacuum with refrigerant gas, allowing pressure to build up to 2 psig (14 kPa).
4. Charge system with a new filter-dryer core in charging line.

3.9 ADJUSTING

- A. Adjust thermostatic expansion valve to obtain proper evaporator superheat.
- B. Adjust high- and low-pressure switch settings to avoid short cycling in response to fluctuating suction pressure.
- C. Adjust set-point temperature of air-conditioning or chilled-water controllers to the system design temperature.
- D. Perform the following adjustments before operating the refrigeration system, according to manufacturer's written instructions:
 1. Open shutoff valves in condenser water circuit.
 2. Verify that compressor oil level is correct.
 3. Open compressor suction and discharge valves.
 4. Open refrigerant valves except bypass valves that are used for other purposes.
 5. Check open compressor-motor alignment and verify lubrication for motors and bearings.
- E. Replace core of replaceable filter dryer after system has been adjusted and after design flow rates and pressures are established.

END OF SECTION 232300

SECTION 233113 - METAL DUCTS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes metal ducts for supply, return, outside, transfer and exhaust air-distribution systems in pressure classes from minus four (4)- inch wg to plus ten (10)-inch wg. Metal ducts include the following:
 - 1. Rectangular ducts and fittings.
 - 2. Single-wall, round, and flat-oval spiral-seam ducts and formed fittings.
 - 3. Duct Liner
- B. Sealing Requirements Definitions: For the purposes of duct systems sealing requirements specified in this Section, the following definitions apply:
 - 1. Seams: A seam is defined as joining of two longitudinally (in the direction of airflow) oriented edges of duct surface material occurring between two joints. All other duct surface connections made on the perimeter are deemed to be joints.
 - 2. Joints: Joints include girth joints; branch and sub-branch intersections; so-called duct collar tap-ins; fitting subsections; louver and air terminal connections to ducts; access door and access panel frames and jambs; duct, plenum, and casing abutments to building structures.

1.2 PERFORMANCE REQUIREMENTS

- A. Delegated Hanger and Support Design: Hangers and supports shall comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

1.3 SUBMITTALS

- A. Shop Drawings: Drawn at a scale of not less than 1/4" = 1'-0". Show fabrication and installation details for metal ducts.
 - 1. Fabrication, assembly, and installation, including plans, elevations, sections, components, and attachments to other work.
 - 2. Duct layout indicating sizes and pressure classes.
 - 3. Elevations of top and bottom of ducts.
 - 4. Dimensions of main duct runs from building grid lines.
 - 5. Fittings.
 - 6. Reinforcement and spacing.
 - 7. Seam and joint construction.
 - 8. Penetrations through fire-rated and other partitions.
 - 9. Equipment installation based on equipment being used on Project.
 - 10. Duct accessories, including access doors and panels.
 - 11. Hangers and supports, including methods for duct and building attachment and vibration isolation.
- B. Delegated-Design Submittal:
 - 1. Spacing of hangers and supports.

2. Design calculations: Calculations, including analysis data, signed and sealed by the qualified professional engineer responsible for their preparation for selecting hangers and supports.
 - C. Coordination Drawings: Reflected ceiling plans and BIM model, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 1. Ceiling suspension assembly members.
 2. Other systems installed in same space as ducts.
 3. Ceiling- and wall-mounting access doors and panels required to provide access to dampers and other operating devices.
 4. Ceiling-mounting items, including lighting fixtures, diffusers, grilles, speakers, sprinklers, access panels, and special moldings.
 - D. Field quality-control test reports.
- 1.4 QUALITY ASSURANCE
- A. Welding: Qualify procedures and personnel according to AWS D9.1, "Sheet Metal Welding Code," for duct joint and seam welding.
 - B. NFPA Compliance:
 1. NFPA 90A, "Installation of Air Conditioning and Ventilating Systems."
 2. NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."
 - C. Comply with NFPA 96, "Ventilation Control and Fire Protection of Commercial Cooking Operations," Ch. 3, "Duct System," for range hood ducts, unless otherwise indicated.
- 1.5 DELIVERY, STORAGE AND HANDLING
- A. All materials shall be stored in a designated area and protected from inclement weather.
 - B. All materials shall be secured so as not to be a hazard during the construction process.
 - C. Store ductwork with tight-fitting seals on open ends to ensure ductwork is free of all dirt, debris and moisture during the installation process.

PART 2 - PRODUCTS

2.1 SHEET METAL MATERIALS

- A. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods, unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.
- B. Galvanized Sheet Steel: Lock-forming quality; complying with ASTM A 653/A 653M and having G90 coating designation; ducts shall have mill-phosphatized finish for surfaces exposed to view.
- C. Carbon-Steel Sheets: ASTM A 366/A 366M, cold-rolled sheets; commercial quality; with oiled, matte finish for exposed ducts.

- D. Stainless Steel: ASTM A 480/A 480M, Type 304 & 316.
- E. Aluminum Sheets: ASTM B 209, alloy 3003, temper H14; with mill finish for concealed ducts and standard, 1-side bright finish for exposed ducts.
- F. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts.
- G. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.
- H. Insulated Flexible Ducts: Flexible ducts wrapped with flexible glass fiber insulation, enclosed by seamless aluminum pigmented plastic vapor barrier jacket; maximum 0.23 K value at 75 degrees F.

2.2 SEALANT MATERIALS

- A. Joint and Seam Sealants, General: The term "sealant" is not limited to materials of adhesive or mastic nature but includes combinations of open-weave fabric strips and mastics.
- B. Water-Based Joint and Seam Sealant: Flexible, adhesive sealant, resistant to UV light when cured, UL 723 listed, and complying with NFPA requirements for Class 1 ducts.
- C. Flanged Joint Mastic: One-part, acid-curing, silicone, elastomeric joint sealant complying with ASTM C 920, Type S, Grade NS, Class 25, Use O.
- D. Flange Gaskets: Butyl rubber or EPDM polymer with polyisobutylene plasticizer.

2.3 HANGERS AND SUPPORTS

- A. Building Attachments: Concrete inserts, powder-actuated fasteners, or structural-steel fasteners appropriate for construction materials to which hangers are being attached.
 - 1. Use powder-actuated concrete fasteners for standard-weight aggregate concretes or for slabs more than 4 inches thick.
 - 2. Exception: Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.
- B. Hanger Materials: Galvanized sheet steel or threaded steel rod.
 - 1. Hangers Installed in Corrosive Atmospheres: All-thread rods used in pool areas, pool equipment rooms, and pool supporting spaces shall be aluminum if the ducts are aluminum and stainless steel if the ducts are stainless steel.
 - 2. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for steel sheet width and thickness and for steel rod diameters.
 - 3. Galvanized-steel straps attached to aluminum ducts shall have contact surfaces painted with zinc-chromate primer.
- C. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials.
- D. Trapeze and Riser Supports: Steel shapes complying with ASTM A 36/A 36M.
 - 1. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.

2. Supports for Stainless-Steel Ducts: Stainless-steel support materials.
3. Supports for Aluminum Ducts: Aluminum support materials..

2.4 RECTANGULAR DUCT FABRICATION

- A. Fabricate ducts, elbows, transitions, offsets, branch connections, and other construction according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" and complying with requirements for metal thickness, reinforcing types and intervals, tie-rod applications, and joint types and intervals.
 1. Lengths: Fabricate rectangular ducts in lengths appropriate to reinforcement and rigidity class required for pressure class.
 2. Deflection: Duct systems shall not exceed deflection limits according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible."
- B. Transverse Joints: Prefabricated slide-on joints and components constructed using manufacturer's guidelines for material thickness, reinforcement size and spacing, and joint reinforcement.
- C. Formed-On Flanges: Construct according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," Figure 1-4, using corner, bolt, cleat, and gasket details.
 1. Duct Size: Maximum 30 inches wide and up to 2-inch wg pressure class.
 2. Longitudinal Seams: Pittsburgh lock sealed with noncuring polymer sealant.
- D. Cross Breaking or Cross Beading: Cross break or cross bead duct sides 19 inches and larger and 0.0359 inch thick or less, with more than 10 sq. ft. of nonbraced panel area unless ducts are lined.

2.5 ROUND AND FLAT-OVAL DUCT AND FITTING FABRICATION

- A. Manufacturers: applicable to factory-fabricated duct and fittings:
- B. Subject to compliance with requirements, provide products by one of the following manufacturers:
 1. Ductmate Industries, Inc.
 2. Lindab.
 3. Lockformer.
 4. McGill Airflow.
 5. Nexus Inc.
 6. Semco, Inc.
 7. Ward Industries.
- C. Diameter as applied to flat-oval ducts in this Article is the diameter of a round duct with a circumference equal to the perimeter of a given size of flat-oval duct.
- D. Round, Longitudinal- and Spiral Lock-Seam Ducts: Fabricate supply ducts of galvanized steel according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible."
- E. Flat-Oval, Longitudinal- and Spiral Lock-Seam Ducts: Fabricate supply ducts according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible. Fabricate ducts larger than 72 inches in diameter with butt-welded longitudinal seams.

F. Duct Joints:

1. Ducts up to 20 Inches in Diameter: Interior, center-beaded slip coupling, sealed before and after fastening, attached with sheet metal screws.
2. Ducts 21 to 72 Inches in Diameter: Three-piece, gasketed, flanged joint consisting of two internal flanges with sealant and one external closure band with gasket.
3. Ducts Larger Than 72 Inches in Diameter: Companion angle flanged joints per SMACNA "HVAC Duct Construction Standards--Metal and Flexible," Figure 3-2.
4. Round Ducts: Prefabricated connection system consisting of double-lipped, EPDM rubber gasket. Manufacture ducts according to connection system manufacturer's tolerances.
5. Flat-Oval Ducts: Prefabricated connection system consisting of two flanges and one synthetic rubber gasket.

G. 90-Degree Tees and Laterals and Conical Tees: Fabricate to comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," with metal thicknesses specified for longitudinal-seam straight ducts.

H. Diverging-Flow Fittings: Fabricate with reduced entrance to branch taps and with no excess material projecting from fitting onto branch tap entrance.

I. Fabricate elbows using die-formed, gored, pleated, or mitered construction. Bend radius of die-formed, gored, and pleated elbows shall be 1-1/2 times duct diameter. Unless elbow construction type is indicated, fabricate elbows as follows:

1. Mitered-Elbow Radius and Number of Pieces: Welded construction complying with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated.
2. Round Mitered Elbows: Welded construction with the following metal thickness for pressure classes from minus 2- to plus 2-inch wg:
 - a. Ducts 3 to 36 Inches in Diameter: 0.034 inch.
 - b. Ducts 37 to 50 Inches in Diameter: 0.040 inch.
 - c. Ducts 52 to 60 Inches in Diameter: 0.052 inch.
 - d. Ducts 62 to 84 Inches in Diameter: 0.064 inch.
3. Round Mitered Elbows: Welded construction with the following metal thickness for pressure classes from 2- to 10-inch wg:
 - a. Ducts 3 to 26 Inches in Diameter: 0.034 inch.
 - b. Ducts 27 to 50 Inches in Diameter: 0.040 inch.
 - c. Ducts 52 to 60 Inches in Diameter: 0.052 inch.
 - d. Ducts 62 to 84 Inches in Diameter: 0.064 inch.
4. 90-Degree, 2-Piece, Mitered Elbows: Use only for supply systems or for material-handling Class A or B exhaust systems and only where space restrictions do not permit using radius elbows. Fabricate with single-thickness turning vanes.
5. Round Elbows 8 Inches and Less in Diameter: Fabricate die-formed elbows for 45- and 90-degree elbows and pleated elbows for 30, 45, 60, and 90 degrees only. Fabricate nonstandard bend-angle configurations or nonstandard diameter elbows with gored construction.
6. Round Elbows 9 through 14 Inches in Diameter: Fabricate gored or pleated elbows for 30, 45, 60, and 90 degrees unless space restrictions require mitered elbows. Fabricate

nonstandard bend-angle configurations or nonstandard diameter elbows with gored construction.

7. Round Elbows Larger Than 14 Inches in Diameter and All Flat-Oval Elbows: Fabricate gored elbows unless space restrictions require mitered elbows.
8. Die-Formed Elbows for Sizes through 8 Inches in Diameter and All Pressures 0.040 inch thick with 2-piece welded construction.
9. Pleated Elbows for Sizes through 14 Inches in Diameter and Pressures through 10-Inch wg: 0.022 inch.

2.6 DUCTWORK LINING:

- A. Ductwork liner shall be Johns Manville Linacoustic RC. Ductwork liner as manufactured by Knauf, Owens-Corning or Certain-Teed will be acceptable if equal. The product name is to be imprinted on the air stream surface.
- B. Ductwork Liner shall be installed in accordance with the Sheet Metal and Air Conditioning Contractor's National Association Duct Liner Application Standard, Second Edition, shall meet ASTM C1071 standards and liner facing must meet EPA registration requirements for antimicrobial protection.
- C. Transfer air duct shall be provided with ½" duct lining with reinforced air side coating and anti-microbial protection.
 1. 1/2" thick liner to be 1-1/2 pcf density, shall have a minimum sound NRC of 0.55 based on a Type "A" mounting, and a "K" factor not exceeding 0.46 at 75° F mean temperature.
- D. Supply, return and exhaust air duct shall be provided with 1 ½" duct lining with reinforced air side coating and anti-microbial protection.
 1. 1 1/2" thick liner to be 1-1/2 pcf density, shall have a minimum sound NRC of 0.85 based on a Type "A" mounting, and a "K" factor not exceeding 0.16 at 75° F mean temperature.
- E. Unless otherwise indicated, the duct dimensions given on the drawings are to be clear inside dimensions after insulation is applied.
- F. Round ductwork indicated to be lined shall be lined with 1" thick "Permacote Spiracoustic Plus" insulation manufactured by Johns Manville. Liner shall be installed per manufacturer's recommendations and requirements.

PART 3 - EXECUTION

3.1 DUCT APPLICATIONS

- A. Static-Pressure Classes: Unless otherwise indicated, construct ducts according to the following:
 1. Supply Ducts (constant volume units): +2".
 2. Supply Ducts (between fan and first system fire damper): +8".
 3. Return Ducts (Negative Pressure): -2".
 4. Return Ducts (between nearest fire damper and return fan inlet): -4".
 5. Return Ducts (return fan discharge and AHU intake / exhaust damper): -4".
 6. Exhaust Ducts (Negative Pressure): -2".

- B. All ducts shall be galvanized steel except as follows:
 - 1. Range Hood Exhaust Ducts: Comply with NFPA 96.
 - a. Type 430 Stainless Steel Double wall with 20 gauge interior and 24 gauge exterior
 - b. Grease duct joints shall be held together by means of formed V clamps and sealed with 3M Fire Barrier 2000+.
 - 2. Exposed Supply Ducts In Occupied Spaces (Sales areas / Lobbies / back-of-house etc.):
 - a. Spiral round/oval galvanized sheet steel unless otherwise indicated as rectangular with paint grip finish.

3.2 DUCT INSTALLATION

- A. Construct and install ducts according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated.
- B. Install round and flat-oval ducts in lengths not less than 12 feet unless interrupted by fittings.
- C. Install ducts with fewest possible joints.
- D. Install fabricated fittings for changes in directions, size, and shape and for connections.
- E. Install couplings tight to duct wall surface with a minimum of projections into duct. Secure couplings with sheet metal screws. Install screws at intervals of 12 inches, with a minimum of 3 screws in each coupling.
- F. Install ducts, unless otherwise indicated, vertically and horizontally and parallel and perpendicular to building lines; avoid diagonal runs.
- G. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.
- H. Install ducts with a clearance of 1 inch, plus allowance for insulation thickness.
- I. Conceal ducts from view in finished spaces. Do not encase horizontal runs in solid partitions unless specifically indicated.
- J. Coordinate layout with suspended ceiling, fire- and control dampers, lighting layouts, and similar finished work.
- K. Seal all joints and seams. Apply sealant to male end connectors before insertion, and afterward to cover entire joint and sheet metal screws.
- L. Electrical Equipment Spaces: Route ducts to avoid passing through transformer vaults and electrical equipment spaces and enclosures.
- M. Non-Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls and are exposed to view, conceal spaces between construction openings and ducts or duct insulation with sheet metal flanges of same metal thickness as ducts. Overlap openings on 4 sides by at least 1-1/2 inches.

- N. Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls, install appropriately rated fire dampers, sleeves, and firestopping sealant. Fire and smoke dampers are specified in Division 23 Section "Air Duct Accessories." Firestopping materials and installation methods are specified in Division 07.
- O. Protect duct interiors from moisture, construction debris and dust, and other foreign materials. Comply with SMACNA's "IAQ Guidelines for Occupied Buildings Under Construction," Appendix G, "Duct Cleanliness for New Construction Guidelines."

3.3 RANGE HOOD EXHAUST DUCTS, SPECIAL INSTALLATION REQUIREMENTS

- A. Install ducts to allow for thermal expansion through 2000 deg F temperature range.
- B. Install ducts without dips or traps that may collect residues unless traps have continuous or automatic residue removal.
- C. Install access panels at each change in direction and at intervals defined by NFPA 96; locate on sides of duct a minimum of 1-1/2 inches from bottom; and fit with grease-tight covers of same material as duct.
- D. Do not penetrate fire-rated assemblies except as permitted by applicable building codes.

3.4 SEAM AND JOINT SEALING

- A. Seal all duct seams and joints to the most severe requirement between the latest Chicago Building Code and SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for duct pressure class indicated.
- B. Utilize sealant designed for outdoor use with ductwork exposed to the outdoors.
- C. Seal ducts before external insulation is applied.

3.5 HANGING AND SUPPORTING

- A. Support horizontal ducts within 24 inches of each elbow and within 48 inches of each branch intersection.
- B. Support vertical ducts at maximum intervals of 16 feet and at each floor.
- C. Install upper attachments to structures with an allowable load not exceeding one-fourth of failure (proof-test) load.
- D. For concrete structure installations: Install concrete inserts before placing concrete.
- E. For concrete structure installations: Install powder-actuated concrete fasteners after concrete is placed and completely cured.
 - 1. Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.

3.6 CONNECTIONS

- A. Make connections to equipment with flexible connectors according to Division 23 Section "Air Duct Accessories."

- B. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for branch, outlet and inlet, and terminal unit connections.

3.7 DUCTWORK LINER:

- A. All rectangular and square ductwork shall be lined single wall. Liner shall be applied to ductwork indicated on plans for lining application.
- B. Duct liner shall be cut to assure overlapped and compressed longitudinal corner joints. Cut edges shall be properly sealed with mastic to assure no erosion of insulation.
- C. Fasteners shall start within 3" of the upstream transverse edges of the duct liner and 3" from the longitudinal joints and shall be spaced at a maximum of 12" o.c. except that they shall be placed not more than 6" from a longitudinal joint of the liner nor 12" from a corner break.
- D. Duct liner shall be adhered to the sheet metal (with 100% coverage) and edges coated with one (1) of the adhesives conforming to the Standard for Adhesives for Duct Liner, ASC-A-7001C - 1972 of the Adhesive and Sealant Council, Inc.
- E. Duct liner shall be further secured with fasteners conforming to the Mechanical Fastener Standard, MF-1-1975 on Page 22 of the Duct Liner Application Standard, Second Edition, of the Sheet Metal and Air Conditioning Contractor's National Association.
- F. Duct liner in medium pressure ductwork shall be installed per manufacturer's recommendations for velocities above 2,000 fpm.
- G. The leading edge of duct liner joints (in direction of air flow) shall be provided with metal nosing or a hardening mastic to prevent lining from coming loose.

3.8 FIELD QUALITY CONTROL

- A. Perform the following field tests and inspections according to SMACNA's "HVAC Air Duct Leakage Test Manual" and prepare test reports:
 - 1. Disassemble, reassemble, and seal segments of systems to accommodate leakage testing and for compliance with test requirements.
 - 2. Conduct tests at static pressures equal to maximum design pressure of system or section being tested. If pressure classes are not indicated, test entire system at maximum system design pressure. Do not pressurize systems above maximum design operating pressure. Give seven days' advance notice for testing.
 - 3. Maximum Allowable Leakage: Comply with requirements for Leakage Class 3 for round and flat-oval ducts, Leakage Class 12 for rectangular ducts in pressure classes lower than and equal to 2-inch wg (both positive and negative pressures), and Leakage Class 6 for pressure classes from 2- to 10-inch wg.
 - 4. Remake leaking joints and retest until leakage is equal to or less than maximum allowable.

END OF SECTION 233113

SECTION 233300 - AIR DUCT ACCESSORIES**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Backdraft dampers.
2. Volume dampers.
3. Fire dampers.
4. Turning vanes.
5. Duct-mounting access doors.
6. Flexible connectors.
7. Flexible ducts.
8. Duct accessory hardware.

1.2 SUBMITTALS**A. Product Data: For the following:**

1. Backdraft dampers.
2. Volume dampers.
3. Fire dampers.
4. Turning vanes.
5. Duct-mounting access doors.
6. Flexible connectors.
7. Flexible ducts.

B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.

1. Special fittings.
2. Backdraft dampers.
3. Manual-volume damper installations.
4. Motorized-control damper installations.
5. Fire-damper and combination fire- and smoke-damper installations, including sleeves and duct-mounting access doors.
6. Duct mounted access doors.
7. Flexible connectors.
8. Flexible ducts.
9. Wiring Diagrams: Power, signal, and control wiring.

C. Coordination Drawings: Reflected ceiling plans, drawn to scale and coordinating penetrations and ceiling-mounting items. Show ceiling-mounting access panels and access doors required for access to duct accessories.**1.3 QUALITY ASSURANCE****A. Comply with NFPA 90A, "Installation of Air Conditioning and Ventilating Systems," and NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."**

1.4 DELIVERY, STORAGE AND HANDLING

- A. Materials delivered to the site must be coordinated with the site supervisor prior to delivery.
- B. All materials shall be stored in a designated area and protected from the environment.
- C. All materials shall be secured so as not to be a hazard during the construction process.
- D. All materials must be free of all dirt, debris and moisture during the installation process.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Backdraft Dampers:
 - a. Greenheck.
 - b. Ruskin.
 - c. Vent Products Company.
 - 2. Volume Dampers
 - a. Nailor.
 - b. Ruskin.
 - c. Vent Products Company.
 - 3. Fire Dampers:
 - a. Greenheck.
 - b. Ruskin.
 - c. Vent Products Company.
 - 4. Duct-Mounting Access Doors:
 - a. CESCO Products.
 - b. Ductmate Industries.
 - c. Greenheck.
 - 5. Flexible Connectors:
 - a. Ductmate Industries.
 - b. Ventfabrics, Inc.
 - c. Ward Industries.
 - 6. Flexible Ducts:
 - a. Flexmaster USA.(MCA-TLD)
 - b. Clevaflex (DBA)
 - c. Novaflex (T/L-AM)

2.2 SHEET METAL MATERIALS

- A. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods, unless otherwise indicated.
- B. Galvanized Sheet Steel: Lock-forming quality; complying with ASTM A 653/A 653M and having G90coating designation; ducts shall have mill-phosphatized finish for surfaces exposed to view.
- C. Stainless Steel: ASTM A 480/A 480M Type 304 (specify Type 314 as required by the application).
- D. Aluminum Sheets: ASTM B 209 alloy 3003, temper H14; with mill finish for concealed ducts and standard, 1-side bright finish for exposed ducts.
- E. Extruded Aluminum: ASTM B 221, alloy 6063, temper T6.
- F. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts; compatible materials for aluminum and stainless-steel ducts.
- G. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

2.3 BACKDRAFT DAMPERS

- A. Description: Multiple-blade, parallel action gravity balanced, with center-pivoted blades of maximum 6-inch width, with sealed edges, assembled in rattle-free manner with 90-degree stop, steel ball bearings, and axles; adjustment device to permit setting for varying differential static pressure.
- B. Frame: 0.063-inch thick extruded aluminum, with welded corners and mounting flange.
- C. Blades: 0.050-inch thick aluminum sheet.
- D. Blade Seals: Neoprene.
- E. Blade Axles: Galvanized steel.
- F. Tie Bars and Brackets: Galvanized steel.
- G. Return Spring: Adjustable tension.

2.4 VOLUME DAMPERS

- A. General Description: Factory fabricated, with required hardware and accessories. Stiffen damper blades for stability. Include locking device to hold single-blade dampers in a fixed position without vibration. Close duct penetrations for damper components to seal duct consistent with pressure class.
 - 1. Pressure Classes of 3-Inch wg or Higher: End bearings or other seals for ducts with axles full length of damper blades and bearings at both ends of operating shaft.

- B. Standard Volume Dampers: Multiple- or single-blade, parallel- or opposed-blade design as indicated, standard leakage rating, with linkage outside airstream and suitable for horizontal or vertical applications.
1. Steel Frames (For use in steel ductwork): Hat-shaped, galvanized sheet steel channels, minimum of 0.064 inch thick, with mitered and welded corners; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 2. Roll-Formed Steel Blades (For use with steel frames): 0.064-inch- thick, galvanized sheet steel.
 3. Aluminum Frames (For use in stainless steel or aluminum ductwork): Hat-shaped, 0.10-inch- thick, aluminum sheet channels; frames with flanges where indicated for attaching to walls; and flangeless frames where indicated for installing in ducts.
 4. Roll-Formed Aluminum Blades (For use with aluminum frames): 0.10-inch- thick aluminum sheet.
 5. Extruded-Aluminum Blades (For use with aluminum frames): 0.050-inch- thick extruded aluminum.
 6. Blade Axles: Galvanized steel. Drive shaft will be the full length of the blade.
 7. Bearings: Stainless-steel sleeve.
 8. Tie Bars and Brackets: Aluminum (aluminum or stainless steel ductwork applications), Galvanized steel (galvanized steel ductwork applications).
- C. Low-Leakage Volume Dampers: Multiple- or single-blade, parallel- or opposed-blade design as indicated, low-leakage rating, with linkage outside airstream, and suitable for horizontal or vertical applications.
1. Steel Frames (For use in steel ductwork) : galvanized sheet steel channels, minimum of 0.064 inch thick, with mitered and welded corners; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 2. Roll-Formed Steel Blades (For use with steel frames): 0.064-inch- thick, galvanized sheet steel.
 3. Aluminum Frames (For use in stainless steel or aluminum ductwork) : 0.10-inch- thick, aluminum sheet channels; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 4. Roll-Formed Aluminum Blades (For use with aluminum frames): 0.10-inch- thick aluminum sheet.
 5. Extruded-Aluminum Blades (For use with aluminum frames): 0.050-inch- thick extruded aluminum.
 6. Blade Axles: Galvanized steel. Drive shaft will be the full length of the blade.
 7. Bearings: Stainless-steel sleeve thrust or ball.
 8. Blade Seals: Neoprene.
 9. Jamb Seals: Cambered stainless steel.
 10. Tie Bars and Brackets: Aluminum (aluminum or stainless steel ductwork applications), Galvanized steel (galvanized steel ductwork applications).
- D. Jackshaft: 1-inch- diameter, galvanized-steel pipe rotating within pipe-bearing assembly mounted on supports at each mullion and at each end of multiple-damper assemblies.
1. Length and Number of Mountings: Appropriate to connect linkage of each damper in multiple-damper assembly.
- E. Damper Hardware: Zinc-plated, die-cast core with dial and handle made of 3/32-inch- thick zinc-plated steel, and a 3/4-inch hexagon locking nut. Include center hole to suit damper operating-rod size. Include elevated platform for insulated duct mounting.

2.5 FIRE DAMPERS

- A. Fire dampers shall be labeled according to UL 555.
- B. Fire Rating: Insert rating as required by application hours.
- C. Frame: Curtain type with blades outside airstream fabricated with roll-formed, 0.034-inch- thick galvanized steel; with mitered and interlocking corners.
- D. Mounting Sleeve: Factory- or field-installed, galvanized sheet steel.
 - 1. Minimum Thickness: 0.052 or 0.138 inch thick as indicated and of length to suit application.
 - 2. Exceptions: Omit sleeve where damper frame width permits direct attachment of perimeter mounting angles on each side of wall or floor, and thickness of damper frame complies with sleeve requirements.
- E. Mounting Orientation: Vertical or horizontal as indicated.
- F. Blades: Roll-formed, interlocking, 0.034-inch- thick, galvanized sheet steel. In place of interlocking blades, use full-length, 0.034-inch- thick, galvanized-steel blade connectors.
- G. Horizontal Dampers: Include blade lock and stainless-steel closure spring.
- H. Fusible Links: Replaceable, 165 deg F rated.

2.6 TURNING VANES

- A. Fabricate to comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for vanes and vane runners. Vane runners shall automatically align vanes.
- B. Manufactured Turning Vanes: Fabricate 1-1/2-inch- wide, single-vane, curved blades of galvanized sheet steel set 3/4 inch o.c.; support with bars perpendicular to blades set 2 inches o.c.; and set into vane runners suitable for duct mounting.
- C. Acoustic Turning Vanes: Fabricate airfoil-shaped aluminum extrusions with perforated faces and fibrous-glass fill.

2.7 DUCT-MOUNTING ACCESS DOORS

- A. General Description: Fabricate doors airtight and suitable for duct pressure class.
- B. Door: Double wall, duct mounting, and rectangular; fabricated of galvanized sheet metal with insulation fill and thickness as indicated for duct pressure class. Include vision panel where indicated. Include piano hinge and cam latches.
 - 1. Frame: Galvanized sheet steel, with bend-over tabs and foam gaskets.
 - 2. Provide number of hinges and locks as follows:
 - a. Less than 12 Inches Square: Secure with two sash locks.
 - b. Up to 18 Inches Square: Continuous hinge and two sash locks.
 - c. Larger than 18 inches square: Continuous hinge and two compression latches with outside and inside handles.

- C. Door: Double wall, duct mounting, and round; fabricated of galvanized sheet metal with insulation fill and 1-inch thickness. Include cam latches.
 - 1. Frame: Galvanized sheet steel, with spin-in notched frame.
- D. Pressure Relief Access Door: Double wall and duct mounting; fabricated of galvanized sheet metal with insulation fill and thickness as indicated for duct pressure class. Include vision panel where indicated, latches, and retaining chain.
 - 1. Frame: Galvanized sheet steel, with bend-over tabs and foam gaskets.
- E. Seal around frame attachment to duct and door to frame with neoprene or foam rubber.
- F. Insulation: 1-inch- thick, fibrous-glass or polystyrene-foam board.

2.8 FLEXIBLE CONNECTORS

- A. General Description: Flame-retardant or noncombustible fabrics, coatings, and adhesives complying with UL 181, Class 1.
- B. Metal-Edged Connectors: Factory fabricated with a fabric strip 5-3/4 inches wide attached to two strips of 2-3/4-inch- wide, 0.028-inch- thick, galvanized sheet steel or 0.032-inch- thick aluminum sheets. Select metal compatible with ducts.
- C. Indoor System, Flexible Connector Fabric: Glass fabric double coated with neoprene.
 - 1. Minimum Weight: 26 oz./sq. yd..
 - 2. Tensile Strength: 480 lbf/inch in the warp and 360 lbf/inch in the filling.
 - 3. Service Temperature: Minus 40 to plus 200 deg F.
- D. Outdoor System, Flexible Connector Fabric: Glass fabric double coated with weatherproof, synthetic rubber resistant to UV rays and ozone.
 - 1. Minimum Weight: 24 oz./sq. yd..
 - 2. Tensile Strength: 530 lbf/inch in the warp and 440 lbf/inch in the filling.
 - 3. Service Temperature: Minus 50 to plus 250 deg F.

2.9 FLEXIBLE DUCTS

- A. Insulated-Duct Connectors: UL 181, Class 1, 2-ply vinyl film supported by helically wound, spring-steel wire; fibrous-glass insulation; aluminized vapor barrier film.
 - 1. Pressure Rating: 10-inch wg positive and 1.0-inch wg negative.
 - 2. Maximum Air Velocity: 4000 fpm.
 - 3. Temperature Range: Minus 10 to plus 160 deg F
- B. Flexible Duct Clamps: Stainless-steel band with cadmium-plated hex screw to tighten band with a worm-gear action, in sizes 3 through 18 inches to suit duct size.

2.10 DUCT ACCESSORY HARDWARE

- A. Instrument Test Holes: Cast iron or cast aluminum to suit duct material, including screw cap and gasket. Size to allow insertion of pitot tube and other testing instruments and of length to suit duct insulation thickness.

- B. Adhesives: High strength, quick setting, neoprene based, waterproof, and resistant to gasoline and grease.

PART 3 - EXECUTION

3.1 APPLICATION AND INSTALLATION

- A. Install duct accessories according to applicable details in SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for metal ducts and in NAIMA AH116, "Fibrous Glass Duct Construction Standards," for fibrous-glass ducts.
- B. Provide duct accessories of materials suited to duct materials; use galvanized-steel accessories in galvanized-steel and fibrous-glass ducts, stainless-steel accessories in stainless-steel ducts, and aluminum accessories in aluminum ducts.
- C. Install backdraft dampers on exhaust fans or exhaust ducts nearest to outside and where indicated.
- D. Provide balancing dampers at points on supply, return, and exhaust systems where branches lead from larger ducts as required for air balancing. Install at a minimum of two duct widths from branch takeoff.
- E. Provide test holes at fan inlets and outlets and elsewhere as indicated.
- F. Install fire dampers, with fusible links, according to manufacturer's UL-approved written instructions.
- G. Install duct silencers rigidly to ducts.
- H. Install duct access doors to allow for inspecting, adjusting, and maintaining duct accessories, control devices - sensors and terminal units as follows:
 - 1. On both sides of duct coils. On terminal units coordinate upstream coil access door with equipment supplier.
 - 2. Downstream from volume dampers, turning vanes, and duct mounted equipment.
 - 3. Adjacent to fire dampers, providing access to reset or reinstall fusible links.
 - 4. To interior of ducts for cleaning; before and after each change in direction, at maximum 50-foot spacing.
 - 5. On sides of ducts where adequate clearance is available.
 - 6. Where indicated on plans.
 - 7. Upstream and downstream of ducted fans.
- I. Label access doors according to Division 23 Section "Identification for HVAC."
- J. Install flexible connectors immediately adjacent to equipment in ducts associated with fans and motorized equipment supported by vibration isolators.
- K. For fans developing static pressures of 5-inch wg and higher, cover flexible connectors with loaded vinyl sheet held in place with metal straps.
- L. Connect terminal units to supply ducts directly or with maximum 12-inch lengths of flexible duct. Do not use flexible ducts to change directions.

- M. Connect diffusers or light troffer boots to low pressure ducts directly or with maximum 60-inch lengths of flexible duct clamped or strapped in place.
- N. Connect flexible ducts to metal ducts with adhesive plus sheet metal screws.
- O. Install duct test holes where indicated and required for testing and balancing purposes.
- P. Provide turning vanes in all short radius / square elbows (>45 degrees) and tees.

3.2 ADJUSTING

- A. Adjust duct accessories for proper settings.
- B. Adjust fire dampers for proper action.
- C. Final positioning of manual-volume dampers is specified in Division 23 Section "Testing, Adjusting, and Balancing for HVAC."

3.3 DEMONSTRATION AND COMMISSIONING

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain the duct accessories.
 - 1. Train Owner's maintenance personnel on troubleshooting, servicing, and maintaining duct accessories. The training will occur after all devices are installed including all access doors. The trainer will provide two (2) Installation and Operations manuals for the use of the owners personnel during training.
 - 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
 - 3. Schedule training with Owner, through Architect, with at least seven days' advance notice.
 - 4. Training will occur in one (1) two (2) hour session and will include the dropping and resetting of 3 fire dampers selected by the owner. This portion of the training may not take longer than 30 minutes of the training session.
- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements under both local and building automation control as well as any commissioning requirements in Division 01 or 23.
 - 1. For all fire dampers or access doors for fire dampers installed on this project the Contractor will demonstrate that any fire dampers selected by the owner can be dropped and reset using the provided access doors.

END OF SECTION 233300

SECTION 233423 - HVAC POWER VENTILATORS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Upblast propeller roof exhaust fans.
 - 2. In-line centrifugal fans.
 - 3. Destratification fans.

1.2 SUBMITTALS

- A. Product Data: Include rated capacities, furnished specialties, and accessories for each type of product indicated and include the following:
 - 1. Certified fan performance curves with system operating conditions indicated.
 - 2. Certified fan sound-power ratings.
 - 3. Motor ratings and electrical characteristics, plus motor and electrical accessories.
 - 4. Material thickness and finishes, including color charts.
 - 5. Dampers, including housings, linkages, and operators.
- B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection. Provide wiring Diagrams: Power, signal, and control wiring.
- C. Operation and Maintenance Data: For power ventilators to include in emergency, operation, and maintenance manuals.
- D. Field Quality-Control Test Reports: Submit reports documenting the activities required to be performed in PART 3. These reports are to be submitted two weeks after the startup is completed.
- E. Training Reports: Submit reports on training documenting dates and attendance.

1.3 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. AMCA Compliance: Products shall comply with performance requirements and shall be licensed to use the AMCA-Certified Ratings Seal.
- C. NEMA Compliance: Motors and electrical accessories shall comply with NEMA standards.
- D. UL Standard: Power ventilators shall comply with UL 705.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Deliver fans as factory-assembled unit, to the extent allowable by shipping limitations, with protective crating and covering.

- B. Disassemble and reassemble units, as required for moving to final location, according to manufacturer's written instructions.
- C. Lift and support units with manufacturer's designated lifting or supporting points.

1.5 PERFORMANCE REQUIREMENTS

- A. Project Altitude: Base fan-performance ratings on sea level.
- B. Operating Limits: Classify according to AMCA 99.

1.6 COORDINATION

- A. Coordinate size and location of structural-steel support members.
- B. Coordinate installation of roof curbs, equipment supports, and roof penetrations.

1.7 EXTRA MATERIALS

- A. Furnish one set of belts that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1.8 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide diffusers by one of the following:
 - 1. Upblast Centrifugal Roof Ventilators:
 - a. Greenheck.
 - b. Loren Cook Company.
 - c. Penn Ventilation.
 - d. Twin City
 - 2. In-line Centrifugal Fans:
 - a. Greenheck.
 - b. Loren Cook Company.
 - c. Penn Ventilation.
 - d. Twin City
 - 3. Destratification Fans:
 - a. Air-Row

2.2 CENTRIFUGAL ROOF VENTILATORS

- A. Description: Direct- or belt-driven centrifugal fans consisting of housing, wheel, fan shaft, bearings, motor and disconnect switch, drive assembly, curb base, and accessories.
- B. Housing: Removable, spun-aluminum, dome top and outlet baffle; square, one-piece, aluminum base with venturi inlet cone.
 - 1. Upblast Units: Provide spun-aluminum discharge baffle to direct discharge air upward, with rain and snow drains.
- C. Fan Wheels: Aluminum hub and wheel with backward-inclined blades.
- D. Belt-Driven Drive Assembly (Provide direct drive for all units less than ½ HP): Resiliently mounted to housing, with the following features:
 - 1. Fan Shaft: Turned, ground, and polished steel; keyed to wheel hub.
 - 2. Shaft Bearings: Permanently lubricated, permanently sealed, self-aligning ball bearings.
 - 3. Pulleys: Cast-iron, adjustable-pitch motor pulley.
 - 4. Fan and motor isolated from exhaust airstream. (UL-762 Labeled)
- E. Accessories:
 - 1. Disconnect Switch: Non-fusible type, with thermal-overload protection mounted inside fan housing, factory wired through an internal aluminum conduit.
 - 2. Specify roof curbs, if they are site fabricated, in Division 07 Section "Roof Accessories."
 - 3. Minimum 1-gallon grease trap / container to prevent grease spilling on the roof.
- F. Roof Curbs: Galvanized steel; mitered and welded corners; 2-inch- thick, rigid, fiberglass insulation adhered to inside walls; and 2-inch wood nailer. Size as required to suit roof opening and fan base.

2.3 IN-LINE CENTRIFUGAL FANS

- A. Description: In-line, belt-driven centrifugal fans consisting of housing, wheel, outlet guide vanes, fan shaft, bearings, motor and disconnect switch, drive assembly, mounting brackets, and accessories.
- B. Housing: Split, spun aluminum with aluminum straightening vanes, inlet and outlet flanges, and support bracket adaptable to floor, side wall, or ceiling mounting.
- C. Direct-Driven Units: Motor mounted in airstream, factory wired to disconnect switch located on outside of fan housing[; with wheel, inlet cone, and motor on swing-out service door].
- D. Belt-Driven Units: Motor mounted on adjustable base, with adjustable sheaves, enclosure around belts within fan housing, and lubricating tubes from fan bearings extended to outside of fan housing.
- E. Fan Wheels: Aluminum, airfoil blades welded to aluminum hub.
- F. Accessories:

1. Variable-Speed Controller: Solid-state control to reduce speed from 100 to less than 50 percent. Required for all direct drive fans.
2. Volume-Control Damper: Manually operated with quadrant lock, located in fan outlet.
3. Companion Flanges: For inlet and outlet duct connections.
4. Fan Guards: 1/2- by 1-inch mesh of galvanized steel in removable frame. Provide guard for inlet or outlet for units not connected to ductwork.

2.4 DESTRATIFICATION FANS

- A. Description: Destratification fan for open ceilings. Designed to eliminate stratified air layers from ceiling to floor and help deliver energy savings.
- B. Housing: Durable and high quality .040 aluminum. Unique square design. Light weight 1.8 lbs not including the EC motor. Color is white unless otherwise noted.
- C. Motor: 92% efficient EC (Electronically Commentated) motor. Built-in thermal overload protector. Dual customizable speed levels with standard soft start feature.
- D. Fan Wheels: PA plastic fan blade material.
- E. Accessories:
 1. Cord: white UL listed 110 V cord and plug unless otherwise noted.
 2. Blade Guard: Heavy duty plated steel powder coated black.
 3. Hanging Kits: Two 5' safety cables with easy push button mechanisms by Gripple. Each cable is UL certified with a 33 lb wight restriction and a 5:1 safety factor rating.

2.5 MOTORS

- A. Comply with requirements in Division 23 Section "Common Motor Requirements for HVAC Equipment."
- B. Enclosure Type: Totally enclosed, fan cooled.

2.6 SOURCE QUALITY CONTROL

- A. Sound-Power Level Ratings: Comply with AMCA 301, "Methods for Calculating Fan Sound Ratings from Laboratory Test Data." Factory test fans according to AMCA 300, "Reverberant Room Method for Sound Testing of Fans." Label fans with the AMCA-Certified Ratings Seal.
- B. Fan Performance Ratings: Establish flow rate, pressure, power, air density, speed of rotation, and efficiency by factory tests and ratings according to AMCA 210, "Laboratory Methods of Testing Fans for Rating."

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install power ventilators level and plumb.
- B. Support units using spring isolators having a static deflection of 1 inch.

1. Secure vibration and seismic controls to concrete bases using anchor bolts cast in concrete base.
- C. Secure roof-mounting fans to roof curbs with cadmium-plated hardware
- D. Ceiling Units: Suspend units from structure; use steel wire or metal straps.
- E. Support suspended units from structure using threaded steel rods and spring hangers having a static deflection of 1 inch.
- F. Install units with clearances for service and maintenance.
- G. Label units according to requirements specified in Division 23 Section "Identification for HVAC."

3.2 CONNECTIONS

- A. Duct installation and connection requirements are specified in other Division 23 Sections. Drawings indicate general arrangement of ducts and duct accessories. Make final duct connections with flexible connectors. Flexible connectors are specified in Division 23 Section "Air Duct Accessories."
- B. Install ducts adjacent to power ventilators to allow service and maintenance.
- C. Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical Systems."
- D. Connect wiring according to Division 26 Section "Conductors and Cables for Electrical Systems."

3.3 FIELD QUALITY CONTROL

- A. Perform the following field tests and inspections and prepare test reports:
 1. Verify that shipping, blocking, and bracing are removed.
 2. Verify that unit is secure on mountings and supporting devices and that connections to ducts and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnect switches.
 3. Verify that cleaning and adjusting are complete.
 4. Disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.
 5. Adjust belt tension.
 6. Adjust damper linkages for proper damper operation.
 7. Verify lubrication for bearings and other moving parts.
 8. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in fully open position.
 9. Disable automatic temperature-control operators, energize motor and adjust fan to indicated rpm, and measure and record motor voltage and amperage.
 10. Shut unit down and reconnect automatic temperature-control operators.
 11. Remove and replace malfunctioning units and retest as specified above.
- B. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

3.4 DEMONSTRATION

- A. Train Owner's maintenance personnel on procedures and schedules related to startup and shutdown, troubleshooting, servicing, and preventive maintenance.
- B. Review data in the operation and maintenance manuals. Refer to Division 01 Section "Contract Closeout."
- C. Schedule training with Owner, through Architect, with at least 7 days' advance notice.
- D. Demonstrate operation of power ventilators. Conduct walking tour of the Project. Briefly identify location and describe function, operation, and maintenance of each power ventilator.

3.5 ADJUSTING

- A. Adjust damper linkages for proper damper operation.
- B. Adjust belt tension.
- C. Refer to Division 23 Section "Testing, Adjusting, and Balancing for HVAC" for testing, adjusting, and balancing procedures.
- D. Replace fan and motor pulleys as required to achieve design airflow.
- E. Lubricate bearings.

3.6 CLEANING

- A. After completing installation, inspect exposed finish. Remove burrs, dirt, and construction debris, and repair damaged finishes including chips, scratches, and abrasions.
- B. Clean fan interiors to remove foreign material and construction debris. Vacuum clean fan wheel and cabinet.

3.7 CONTRACTOR STARTUP AND REPORTING

- A. Final Checks before Startup: Perform the following operations and checks before startup. Startup service includes the testing, inspections and startup test reports:
 - 1. Verify that shipping, blocking, and bracing are removed.
 - 2. Verify that unit is secure on mountings and supporting devices and that connections for piping, ducts, and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnects.
 - 3. Perform cleaning and adjusting specified in this Section.
 - 4. Disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.
 - 5. Lubricate bearings, pulleys, belts, and other moving parts with factory-recommended lubricants.
 - 6. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in the fully open position.
 - 7. Disable automatic temperature-control operators.
- B. Starting procedures for fans are as follows:

1. Energize motor; verify proper operation of motor, drive system, and fan wheel. Adjust fan to indicated RPM.
 2. Measure and record motor voltage and amperage.
- C. Shut unit down and reconnect automatic temperature-control operators.
- D. Refer to Division 23 Section "Testing, Adjusting, and Balancing for HVAC" for procedures for air-handling-system testing, adjusting, and balancing.
- E. Replace fan and motor pulleys as required to achieve design conditions.
- 3.8 DEMONSTRATION AND COMMISSIONING
- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain the fans.
1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining chillers. The training will occur after the startup report has been provided to the owner and the trainer will provide two (2) Installation and Operations manuals for the use of the owner's personnel during training.
 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
 3. Schedule training with Owner, through Architect, with at least seven days' advance notice.
- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements under both local and building automation control as well as any commissioning requirements in Division 01 and 23.

END OF SECTION 233423

SECTION 233713 - DIFFUSERS, REGISTERS, AND GRILLES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes ceiling- and wall-mounted diffusers, registers, and grilles.

1.2 DEFINITIONS

- A. Diffuser: Circular, square, or rectangular air distribution outlet, generally located in the ceiling and comprised of deflecting members discharging supply air in various directions and planes and arranged to promote mixing of primary air with secondary room air.
- B. Grille: A louvered or perforated covering for an opening in an air passage, which can be located in a sidewall, ceiling, or floor.
- C. Register: A combination grille and damper assembly over an air opening.

1.3 SUBMITTALS

- A. Product Data: For each product indicated, include the following:
 - 1. Data Sheet: Indicate materials of construction, finish, and mounting details; and performance data including throw and drop, static-pressure drop, and noise ratings.
 - 2. Diffuser, Register, and Grille Schedule: Indicate Drawing designation, room location, quantity, model number, size, and accessories furnished.
- B. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 - 1. Ceiling suspension assembly members.
 - 2. Method of attaching hangers to building structure.
 - 3. Size and location of initial access modules for acoustical tile.
 - 4. Ceiling-mounted items including lighting fixtures, diffusers, grilles, speakers, sprinklers, access panels, and special moldings.
 - 5. Duct access panels.

1.4 QUALITY ASSURANCE

- A. ARI Compliance: Test and rate diffusers, registers, and grilles in accordance with ARI 650 "Standard for Diffusers, registers, and grilles".
- B. ASHRAE Compliance: Test and rate diffusers, registers, and grilles in accordance with ASHRAE 70 "Method of Testing for Rating the Air Flow Performance of Outlets and Inlets".
- C. ADC Compliance: Test and rate diffusers, registers, and grilles in certified laboratories under requirements of ADC 1062 "Certification, Rating and Test Manual".
- D. ADC Seal: Provide diffusers, registers, and grilles bearing ADC Certified Rating Seal.
- E. NFPA Compliance: Install diffusers, registers, and grilles in accordance with NFPA 90A "Standard for the Installation of Air Conditioning and Ventilating Systems".

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver diffusers, registers, and grilles wrapped in factory-fabricated fiber- board type containers. Identify on outside of container type of outlet or inlet and location to be installed. Avoid crushing or bending and prevent dirt and debris from entering and settling in devices.
- B. Store diffusers, registers, and grilles in original cartons and protect from weather and construction work traffic. Where possible, store indoors; when necessary to store outdoors, store above grade and enclose with waterproof wrapping.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provided on **LIDL National Account List**:

1. Ceiling Air Diffusers:

- a. Nailor.
- b. Titus Products Div.; Philips Industries, Inc.
- c. TROX.
- d. Price Industries.

2. Duct and Wall Registers and Grilles

- a. Nailor.
- b. Titus Products Div.; Philips Industries, Inc.
- c. TROX.
- d. Price Industries.

3. Plenum Box Diffusers

- a. Ruskin Rooftop Systems

2.2 CEILING AIR DIFFUSERS

- A. General: Except as otherwise indicated, provide manufacturer's standard ceiling air diffusers where shown; of size, shape, capacity and type indicated; constructed of materials and components as indicated, and as required for complete installation.
- B. Performance: Provide ceiling air diffusers that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Ceiling Compatibility: Provide diffusers with border styles that are compatible with adjacent ceiling systems, and that are specifically manufactured to fit into ceiling module with accurate fit and adequate support. Refer to general construction drawings and specifications for types of ceiling systems which will contain each type of ceiling air diffuser.

- D. Types: Provide ceiling diffusers of type, capacity, and with accessories and finishes as listed on diffuser schedule.

2.3 WALL REGISTERS AND GRILLES

- A. General: Except as otherwise indicated, provide manufacturer's standard wall registers and grilles where shown; of size, shape, capacity and type indicated; constructed of materials and components as indicated, and as required for complete installation.
- B. Performance: Provide wall registers and grilles that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Wall Compatibility: Provide registers and grilles with border styles that are compatible with adjacent wall systems, and that are specifically manufactured to fit into wall construction with accurate fit and adequate support. Refer to general construction drawings and specifications for types of wall construction which will contain each type of wall register and grille.
- D. Types: Provide wall registers and grilles of type, capacity, and with accessories and finishes as listed on schedule.

2.4 PLENUM BOX DIFFUSER

- A. General: Except as otherwise indicated, provide manufacturer's standard plenum box where shown; of size, shape, capacity and type indicated; construction of materials and components as indicated, and as required for complete installation.
- B. Performance: provide plenum box that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Drum Louver diffuser shall be heavy duty construction and provide minimum 30 degree vertical adjustment with full horizontal blade adjustment to deliver single point supply air for large space applications. Manufacturers published data sheet shall include CFM rating, noise, static, and throw information. Drum louver shall be in all respects equivalent to Ruskin Rooftop Systems model 578. Critical: Drum louver shall incorporate a standard 578 precision tuned internal air deflector to ensure quite even airflow to all sides with low pressure drop.
- D. Types: Provide plenum box of type, capacity, and with accessories and finishes as listed on schedule.

2.5 SOURCE QUALITY CONTROL

- A. Verification of Performance: Rate diffusers, registers, and grilles according to ASHRAE 70, "Method of Testing for Rating the Performance of Air Outlets and Inlets."

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas where diffusers, registers, and grilles are to be installed for compliance with requirements for installation tolerances and other conditions affecting performance of equipment.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install diffusers, registers, and grilles level and plumb.
- B. Ceiling-Mounted Outlets and Inlets: Drawings indicate general arrangement of ducts, fittings, and accessories. Air outlet and inlet locations have been indicated to achieve design requirements for air volume, noise criteria, airflow pattern, throw, and pressure drop. Make final locations where indicated, as much as practicable. For units installed in lay-in ceiling panels, locate units in the center of panel. Where architectural features or other items conflict with installation, notify Architect for a determination of final location.
- C. Install diffusers, registers, and grilles with airtight connections to ducts and to allow service and maintenance of dampers, air extractors, and fire dampers.

3.3 CLEANING

- A. After installation of diffusers, registers, and grilles, inspect exposed finish. Clean exposed surfaces to remove burrs, dirt, and smudges. Replace diffusers, registers, and grilles that have damaged finishes.

3.4 CONTRACTOR STARTUP AND REPORTING

- A. After installation, adjust diffusers, registers, and grilles to air patterns indicated, or as directed, before starting air balancing.

END OF SECTION 233713

SECTION 238119 - SELF-CONTAINED AIR-CONDITIONERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes packaged, outdoor, central-station air-handling units (rooftop units) with the following components and accessories:
 - 1. Direct-expansion cooling.
 - 2. Gas furnace.
 - 3. Economizer outdoor- and return-air damper section.
 - 4. Integral, space temperature controls.
 - 5. Roof curbs.

1.2 DEFINITIONS

- A. DDC: Direct-digital controls.
- B. RTU: Rooftop unit. As used in this Section, this abbreviation means packaged, outdoor, central-station air-handling units. This abbreviation is used regardless of whether the unit is mounted on the roof or on a concrete base on ground.
- C. Supply-Air Fan: The fan providing supply air to conditioned space. "Supply air" is defined as the air entering a space from air-conditioning, heating, or ventilating apparatus.

1.3 SUBMITTALS.

- A. Product Data: Include refrigerant, rated capacities, operating characteristics, furnished specialties, and accessories.
 - 1. Performance at ARI standard conditions and at conditions scheduled (provide EER).
 - 2. Performance at ARI standard unloading conditions and at conditions scheduled (provide EER).
 - 3. Refrigerant type and capacity of unit.
 - 4. Characteristics of safety relief valves.
 - 5. Minimum entering condenser-air temperature
 - 6. Sound data.
 - 7. Dimensions
 - 8. Unit construction.
 - 9. Components
 - 10. Included options
 - 11. Required service and operational clearances
 - 12. Furnished specialties and accessories.
 - 13. ASHRAE/IESNA 90.1 for energy compliance statement.
- B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 1. Wiring Diagrams: Power, signal, and control wiring.
- C. Coordination Drawings: Plans and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Structural members to which RTUs will be attached.
2. Roof openings
3. Roof curbs and flashing.

D. Field quality-control test reports.

E. Operation and Maintenance Data: For RTUs to include in emergency, operation, and maintenance manuals.

F. Submit complete warranty information as described in Article 1.7.

1.4 QUALITY ASSURANCE

A. ARI Compliance:

1. Comply with ARI 210/240 and ARI 340/360 for testing and rating energy efficiencies for RTUs.
2. Comply with ARI 270 for testing and rating sound performance for RTUs.

B. ASHRAE Compliance:

1. Comply with ASHRAE 15 for refrigeration system safety.
2. Comply with ASHRAE 33 for methods of testing cooling and heating coils.
3. Comply with ASHRAE 62.1-2004 for condensate drain pans.
4. Comply with ASHRAE/IESNA 90.1 for minimum efficiency of heating and cooling.

C. NFPA Compliance: Comply with NFPA 90A and NFPA 90B.

D. UL Compliance: Comply with UL 1995.

E. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.5 DELIVERY, STORAGE AND HANDLING

A. Follow manufacturer's instructions for unloading, rigging and storage of equipment.

B. Maintain manufacturer's recommended temperature and humidity limits during storage and installation. Protect equipment from dirt, dust and other jobsite contaminants and conditions detrimental to the equipment.

1.6 WARRANTY

A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

B. Manufacturer's standard form in which manufacturer agrees to replace components of RTUs that fail in materials or workmanship within specified warranty period. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of preliminary acceptance, whichever is longer.

1. Warranty Period for Compressors: Manufacturer's standard, but not less than five years from date of preliminary acceptance.

2. Warranty Period for Gas Furnace Heat Exchangers: Manufacturer's standard, but not less than ten years from date of preliminary acceptance.
3. Warranty Period for Solid-State Ignition Modules: Manufacturer's standard, but not less than one year from date of preliminary acceptance.
4. Warranty Period for Control Boards: Manufacturer's standard, but not less than one year from date of preliminary acceptance.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provided on **LiDL National Account List**:

1. Lennox
2. Carrier
3. TBD

2.2 CAPACITIES AND CHARACTERISTICS:

- A. Efficiency:

1. Air conditioners shall have a EER better than ASHRAE Standard 90.1 – latest edition under ARI test procedures. When air conditioners with higher efficiencies than the Standard are scheduled on the drawings, the more efficient value shall be the minimum project requirement.

2.3 CASING

- A. General: Casings shall be formed and reinforced insulated panels, fabricated to allow removal for access to internal parts and components, with joints between sections sealed.
- B. Exterior casing material shall be galvanized steel with factory-painted finish, with pitched roof panels and knockouts with grommet seals for electrical and piping connections and lifting lugs. Cabinet shall be tested 1000 hour in salt spray test in compliance with ASTM B117.
- C. Casing Insulation shall comply with NFPA 90A or NFPA 90B and ASTM C 1071, Type I. Minimum thickness shall be 1 inch, with a hydrostatic coating on all surfaces in contact with the air stream.
- D. Condensate drain pans shall be formed sections of stainless steel, with a minimum 3/4" NPT drain connection

2.4 FANS

- A. Evaporator fan shall be forward-curved, double-width, double inlet constant volume centrifugal. Fans shall be belt-driven with adjustable pulleys. Bearings shall be sealed and permanently lubricated.
1. Provide factory VFD with phase protection for variable volume fans per schedule.
- B. Condenser fans shall be propeller, mounted on shaft of permanently lubricated motor.

- C. Fan Motor: Comply with requirements in Division 23 Section "Common Motor Requirements for HVAC Equipment."
- D. Provide power exhaust fans, or standard exhaust fans per sheet schedules.
 - 1. Power exhaust fans shall be centrifugal type and be fully modulating, integrated with economizer operation.

2.5 COILS

- A. Evaporator and condenser refrigerant coils shall incorporate aluminum-plate fins mechanically bonded to seamless copper tubes. Coils shall be factory pressure tested per ARI standards and leak tested at 500 psig.
 - 1. Provide E-coating on condenser coils.

2.6 REFRIGERANT CIRCUIT COMPONENTS

- A. Compressor: Hermetic scroll, mounted on vibration isolators; with internal overcurrent and high-temperature protection. For units more than 7.5 tons, provide two compressors. Motor shall be suction gas cooled.
 - 1. For units operating with variable air volume control, provide digital scroll type compressor(s).
- B. Refrigeration Specialties:
 - 1. Refrigerant Charge: R-407 or 410A. Units shall be fully charged when delivered to the site.
 - 2. Expansion valve with replaceable thermostatic element.
 - 3. Refrigerant filter/dryer.
 - 4. Manual-reset high-pressure safety switch.
 - 5. Automatic-reset compressor motor thermal overload.
 - 6. Service ports installed in compressor suction and liquid lines.
 - 7. Crankcase heaters.
 - 8. Freezestats.

2.7 AIR FILTRATION

- A. Minimum arrestance according to ASHRAE 52.1, and a minimum efficiency reporting value (MERV) according to ASHRAE 52.2. Filters shall be glass fiber, two-inch thick (MERV 8).

2.8 GAS FURNACE

- A. Description: Factory assembled, piped, and wired; complying with ANSI Z21.47 and NFPA 54. ASHRAE 90.1 – latest edition compliant.
- B. Burners: Stainless steel with a minimum thermal efficiency of 80 percent.
 - 1. Fuel: Natural gas.
 - 2. Ignition: Electronically controlled electric spark.
- C. Heat-Exchanger: Stainless steel.

- D. Combustion system shall be induced-draft.
- E. Safeties: Provide high-temperature limit switches, flame rollout switch, flame proving controls and induced draft pressure sensor.

2.9 DAMPERS

- A. Outdoor- and Return-Air Mixing Dampers: Parallel- or opposed-blade galvanized-steel dampers mechanically fastened to cadmium plated for galvanized-steel operating rod in reinforced cabinet. Connect operating rods with common linkage and interconnect linkages so dampers operate simultaneously.
 - 1. Damper Motor: Modulating with adjustable minimum position.
 - 2. Relief-Air Damper: Gravity actuated with bird screen and hood.

2.10 ELECTRICAL POWER CONNECTION

- A. Provide for single connection of power to unit with unit-mounted disconnect switch accessible from outside unit and control-circuit transformer with built-in overcurrent protection.

2.11 CONTROLS

- A. Control equipment and sequence of operation are specified in Division 23 Section "Building Automation System (BAS)" and in the controls drawings.
- B. Provide CO2 sensor and controls where called out on drawing sheets supplied.

2.12 ACCESSORIES

- A. Duplex, 115-V, ground-fault-interrupter outlet with 15-A overcurrent protection. Include transformer if required.
- B. Filter differential pressure switch with sensor tubing on either side of filter. Set for final filter pressure loss.

2.13 ROOF CURBS

- A. Materials: Galvanized steel, insulated with corrosion-protection coating, watertight gaskets, and factory-installed wood nailer; complying with NRCA standards. Roof curb height shall be 14 inches minimum.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of RTUs.
- B. Examine roughing-in for RTUs to verify actual locations of piping and duct connections before equipment installation.
- C. Examine roofs for suitable conditions where RTUs will be installed.

- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Roof Curb: Install on roof structure, level and secure, according to ARI Guideline B. Install RTUs on curbs and coordinate roof penetrations and flashing with roof construction specified in Division 07 Section "Roof Accessories." Secure RTUs to upper curb rail, and secure curb base to roof framing or concrete base with anchor bolts.

3.3 CONNECTIONS

- A. Install condensate drain, minimum connection size, with trap and indirect connection to nearest roof drain or area drain.
- B. Install piping adjacent to RTUs to allow service and maintenance.
 - 1. Gas Piping: Comply with applicable requirements in Division 23 Section "Natural Gas Piping." Connect gas piping to burner, full size of gas train inlet, and connect with union and shutoff valve with sufficient clearance for burner removal and service.
- C. Duct installation requirements are specified in other Division 23 Sections. Drawings indicate the general arrangement of ducts. The following are specific connection requirements:
 - 1. Install ducts to termination at top of roof curb.
 - 2. Remove roof decking only as required for passage of ducts. Do not cut out decking under entire roof curb.
 - 3. Connect supply ducts to RTUs with flexible duct connectors specified in Division 23 Section "Air Duct Accessories."
 - 4. Install return-air duct continuously through roof structure.

3.4 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections. Report results in writing.
 - 1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing. Report results in writing.
- B. Tests and Inspections:
 - 1. After installing RTUs and after electrical circuitry has been energized, test units for compliance with requirements.
 - 2. Inspect for and remove shipping bolts, blocks, and tie-down straps.
 - 3. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- C. Remove and replace malfunctioning units and retest as specified above.

3.5 CLEANING AND ADJUSTING

- A. Occupancy Adjustments: When requested within 12 months of date of preliminary acceptance, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to two visits to site during other-than-normal occupancy hours for this purpose.
- B. After completing system installation and testing, adjusting, and balancing RTU and air-distribution systems, clean filter housings and install new filters.

3.6 CONTRACTOR STARTUP AND REPORTING

- A. Engage a factory-authorized service representative to perform startup service.
- B. Complete installation and startup checks according to manufacturer's written instructions and do the following:
 - 1. Inspect for visible damage to unit casing.
 - 2. Inspect for visible damage to furnace combustion chamber.
 - 3. Inspect for visible damage to compressor, coils, and fans.
 - 4. Inspect internal insulation.
 - 5. Verify that labels are clearly visible.
 - 6. Verify that clearances have been provided for servicing.
 - 7. Verify that controls are connected and operable.
 - 8. Verify that filters are installed.
 - 9. Clean condenser coil and inspect for construction debris.
 - 10. Clean furnace flue and inspect for construction debris.
 - 11. Connect and purge gas line.
 - 12. Remove packing from vibration isolators.
 - 13. Verify lubrication on fan and motor bearings.
 - 14. Inspect fan-wheel rotation for movement in correct direction without vibration and binding.
 - 15. Adjust fan belts to proper alignment and tension.
 - 16. Start unit according to manufacturer's written instructions.
 - a. Start refrigeration system.
 - b. Do not operate below recommended low-ambient temperature.
 - c. Complete startup sheets and attach copy with Contractor's startup report.
 - 17. Inspect and record performance of interlocks and protective devices; verify sequences.
 - 18. Operate unit for an initial period as recommended or required by manufacturer.
 - 19. Perform the following operations for both minimum and maximum firing. Adjust burner for peak efficiency.
 - a. Measure gas pressure on manifold.
 - b. Inspect operation of power vents.
 - c. Measure combustion-air temperature at inlet to combustion chamber.
 - d. Measure flue-gas temperature at furnace discharge.
 - e. Perform flue-gas analysis. Measure and record flue-gas carbon dioxide and oxygen concentration.
 - f. Measure supply-air temperature and volume when burner is at maximum firing rate and when burner is off. Calculate useful heat to supply air.
 - 20. Calibrate thermostats.
 - 21. Adjust and inspect high-temperature limits.
 - 22. Inspect outdoor-air dampers for proper stroke and interlock with return-air dampers.

23. Start refrigeration system and measure and record the following when ambient is a minimum of 15 deg F above return-air temperature:
 - a. Coil leaving-air, dry- and wet-bulb temperatures.
 - b. Coil entering-air, dry- and wet-bulb temperatures.
 - c. Outdoor-air, dry-bulb temperature.
 - d. Outdoor-air-coil, discharge-air, dry-bulb temperature.
24. Inspect controls for correct sequencing of heating, mixing dampers, refrigeration, and normal and emergency shutdown.
25. Measure and record the following minimum and maximum airflows. Plot fan volumes on fan curve.
 - a. Supply-air volume.
 - b. Return-air volume.
 - c. Relief-air volume.
 - d. Outdoor-air intake volume.
26. Simulate maximum cooling demand and inspect the following:
 - a. Compressor refrigerant suction and hot-gas pressures.
 - b. Short circuiting of air through condenser coil or from condenser fans to outdoor-air intake.
27. Verify operation of remote panel including pilot-light operation and failure modes. Inspect the following:
 - a. High-temperature limit on gas-fired heat exchanger.
 - b. Low-temperature safety operation.
 - c. Filter high-pressure differential alarm.
 - d. Economizer to minimum outdoor-air changeover.
 - e. Relief-air fan operation.
 - f. Smoke and firestat alarms.
28. After startup and performance testing and prior to preliminary acceptance, replace existing filters with new filters.

3.7 APPENDIX

A. Lennox Startup Instructions

1. Installing contractor must complete Contractor Acknowledgement Form and submit to Lennox National Account Services (NAS) to request the specific date for the EOC and an additional Contractor Acknowledgement Form for the specific date for the Commissioning. If either/both of the dates need to be rescheduled, the installing contractor must communicate the reschedule date to NAS at a minimum of 1 week prior in order to prevent possibility of extra charges (up to \$3000).
2. Lennox EOC and Commissioning scopes of work.
3. Letter from Lennox Technical Support documenting the proper procedure to be followed in order to operate the units during construction.
4. Directions for the installing contractor to set up the Prodigy board for units to run in temporary mode until EMS controls are installed. There is no need for temporary thermostats, as all Lennox Emergence units have standard on board sensors (outside, return, and discharge air).

5. See equipment submittals for sensors being used for Lidl projects which are ordered by and shipped to the contractor ordering the RTUs and split systems from Lennox.

B. Lennox Contacts

Equipment Quotes, Ordering, Tracking, and Missing Parts/Accessories

Lennox National Account Advocate (NAA)

Royce Hamilton

972-497-6788 800-367-6285 Opt #1

LennoxNationalAccounts@LennoxInd.com

Damaged Equipment, Installation Assistance, Equipment Alarms/Not Operating Properly

Lennox Technical Support

800-367-6285 Opt #2

NationalAccountsTechnicalSupport@LennoxInd.com

EOC and/or Commissioning Questions or Scheduling

Lennox National Account Services (NAS)

Courtney Marrero

800-333-4001

LennoxServices@LennoxNAS.com

Lidl/Lennox National Account Questions

Lennox National Account Escalation

Ron Johnston

National Account Manager

(704) 621-2606 Cell

Ronald.Johnston@LennoxInd.com



Contractor Acknowledgement Form

Requested EOC/Commissioning date _____ SAP Order# _____

Minimum 3 weeks notice is required for scheduling all services.

Contact Information

Date _____ Installing Contractor _____
 Project Name _____ Site Contact _____
 Store # _____ Phone Number _____
 Project Street Address _____ Email Address _____
 Project City/State/Zip _____

Complete checklist to indicate that ALL items will be complete and ready for Commissioning services

HVAC READY CHECKLIST	YES	NO
RTU's have permanent power connected		
RTU's have gas lines connected and gas is available		
All rooftop unit GFIs have been wired and powered		
Economizer hoods are installed and caulked according to installation instructions		
Units have clean filters installed		
All diffusers, grilles and ducts are installed and dampers are in the open position		
Mechanical contractor will be onsite during start up procedure		
MSAV & VAV Commissioning (Additional to HVAC Ready Checklist)		
All VAV boxes and zone dampers must be at and stay at 100% open during the start up.		
Static pressure transducer must be installed, wired and powered past 1 st supply air elbow from unit or at least 15 feet from unit		
Discharge air sensor shipped with unit must be installed past 1 st supply air elbow from unit or at least 15 feet from unit		
L Connection Network® Zoning Control System		
Third party control systems (Field provided and field installed)		
L Connection® Controls Commissioning (Additional to HVAC Ready Checklist)		
Communication wire pulled, labeled, and verified for validity		
Devices mounted and labeled according to site plans		

Customer Signature _____

Date _____

Fax or E Mail to Lennox NAS As Soon As Possible prior to Contractor completion of work for final Commissioning date. Failure to properly complete all items before the Lennox NAS technicians arrive on site will result in additional trip charges. Return fees are paid by the Contractor based on travel requirements, not to exceed \$3,000.00

Fax To: Lennox National Account Services 214-576-3873 Att: Services
 E-Mail lennoxservices@lennoxnas.com Phone 1-800-333-4001

Lennox® Factory Equipment Operational Check: Emergence™ (EOC)\ L Series® RTU \ Variable Air Volume (VAV)\ Constant Volume (CV)\ Constant Air Volume w/Bypass (CAVB)\ Strategos™ Rooftop Units

Performance of Factory Equipment Operations Check shall provide verification and documentation of equipment condition, integrity of installation and operational performance with regard to Lennox® specifications. The following listed items shall be performed by a Lennox® factory trained service technician. The following equipment and installation integrity checks shall be performed as part of the Lennox® factory Equipment Operational Check. Any installer defects shall be noted and any factory defects shall be repaired. A report for each unit along with a summary report for the job site will be provided to the owner, installing contractor and general contractor upon completion.

Lennox representative will perform the following services during Factory Equipment Operations Check:

Unit Installation Check:

- A. Record RTU# - Unit Model # - Unit Serial # - Unit Catalog #
- B. Check curb installation for fit, air leakage, including optional vibration isolation devices and wind, or seismic restraints – verify per manufacturers installation manuals and project specifications.
- C. Check door alignment and adjust as necessary.
- D. Check insulation is secure and clean.
- E. Check installation of condensate trap and drain lines per Lennox® installation instructions.
- F. Check unit clearances and verify unit installation per Lennox® installation instructions.
- G. Check area for cleanliness and dispose of Lennox related debris.

Electrical Systems Check:

- A. Check/Record proper incoming voltage and phase – verify per Lennox® specifications for each specific unit.
- B. Verify installation and proper sizing of electrical disconnect or circuit breaker.
- C. Check electrical connections and tighten as needed.
- D. Check/Record unit's control transformer(s) for proper secondary voltage and adjust if necessary.
- E. Check/Record installation of wiring for optional Lennox® factory installed GFCI receptacle.
- F. Verify class 2 controls wiring installation to terminal board of unit to include any project specific thermostat wiring – relative humidity sensors – indoor air quality sensors – space temperature sensors.

Prodigy™ Control System Setup and Check:

- A. Record hardware and software versions of the Controller and display panel if present.
- B. Save profile and report to USB storage device for each unit (additional charge if requested by customer).

Control System/Sensor Check:

Thermostat/Thermostat Input (DDC type controls) Check:

- A. Record thermostat or DDC controller – make and model.
- B. Verify class 2 controls wiring installation to terminal board of unit.
- C. Verify Occupancy Input for Thermostat Mode of operation (OCP input).
- D. Verify Fan, and multi stage Thermostat/DDC inputs for equipment that has multi stage operation for correct termination (G/Y1/Y2/W1/W2 – Only requires G/Y1/W1 for basic discharge air control models).
- E. Verify type of Humiditrol® control and verify correct termination has been accomplished (HUM or DI4).
- F. Verify type of economizer enable control and verify correct termination has been accomplished (GLO).

Sensor Check

Zone Temperature Sensor Check – Lennox® non/communicating style

- A. Check/Verify correct termination of sensor wires.

Relative Humidity Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.

Indoor Air Quality Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.

Supply Static Pressure Sensor Check

- A. Check/Verify correct termination of wires/installation of remote mounted supply duct static pressure sensor.

Building Static Pressure Sensor Check

- A. Check/Verify correct termination of wires/installation of building static pressure sensor.

Discharge air Temperature Sensor – Remote / Unit Mounted –

- A. Check/Verify units utilizing discharge air control mode have a remote mounted discharge air temperature sensor.
B. Check/Verify correct termination of sensor wires.

Return Air/OA Sensors

- A. Check/Verify correct termination of sensor wires

Blower System Setup and Check:

- A. Check/Record condition and type of filters.
B. Check/Record correct blower rotation.
C. Check/Record amp draw is per Lennox® specifications.
D. For Lennox® single speed belt drive blowers: check belt tension and alignment. Check pulley set screws for proper torque. Adjust as needed.
E. For Lennox® two speed belt drive motors: set two speed operation and blower over-load settings. Check pulley set screws for proper torque. Adjust as needed.
F. For Lennox® ECM motors: check ECM motor output per Lennox® specifications.
G. For Lennox® factory installed VFD belt driven motors check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
H. For Lennox® Strategos™ MSAV units: verify motor set per Lennox® specifications. Check pulley set screws for proper torque. Adjust as needed.

Outdoor Air Accessory Check:

- A. Check operation of economizer or motorized outdoor air damper by driving it full open and closed.
B. Check/Record unit economizer operation mode and sensors are present.
C. Check/Record economizer setting.
D. Check operation of barometric relief damper (if installed).
E. For Lennox® single speed belt drive power exhaust fan: check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
F. For Lennox® factory installed VFD belt driven power exhaust fan: check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
a. Check and verify power exhaust operation.

Cooling System Set Up and Check

- A. Check coil integrity and cleanliness.
B. Visual leak check all circuits. Correct as needed if operation performance suggests a possible issue.
C. Confirm proper rotation for each compressor and correct phasing only if occurs within the unit as needed (We do not correct power supply phasing to unit disconnect). Record amp draw.

- D. Adjust refrigerant charge as needed.
- E. Record temperature drop across the evaporator coil in full cooling (All compressors running).

Humiditrol® System Set Up and Check

- A. Check number of Humiditrol® stages.
- B. Verify operation of Humiditrol® demand.

Gas Heating System Set Up and Check

- A. Record fuel type.
- B. Check installation of intake and exhaust hoods. Verify per Lennox® installation instructions.
- C. Check installation of combustion air inducer intake and exhaust extensions (if applicable). Verify per Lennox® installation instructions.
- D. Check installation of gas unions for leaks.
- E. Check/Record incoming gas pressure to unit.
- F. Check/Record manifold gas pressure from the outlet of the gas valve(s) per Lennox® specifications. Adjust pressure as needed.
- G. Verify/Record temperature rise across heat exchanger in full heat. Adjust airflow if necessary.

Electric Heat System Set Up and Check:

- A. Check/Record amp draw of the individual heating elements (blower on).
- B. Check/Record temperature rise across heating elements in full heat. Adjust airflow if necessary.

Necessary requirements that must be complete prior to Factory EOC:

- A. Complete installation of rooftop unit and accessories per mechanical drawings and Lennox Installation Instructions.
- B. Rooftop unit must be started and running (24 hours before scheduled start up with no heating or cooling call).
- C. Unit's return air filters must be new and at least equivalent to factory provided filters.
- D. Static pressure transducer must be installed, wired and powered past 1st supply air elbow from unit or at least 5 feet from unit, if applicable.
- E. Discharge air sensor shipped with unit must be installed past 1st supply air elbow from unit or at least 5 feet from unit, if applicable.
- F. All VAV boxes and zone dampers must be at and stay at 100% open during the start up, if applicable.

The Factory EOC does not include the following:

- A. Any repairs or adjustment not listed above.
- B. Test and Balance.
- C. Pulling of wires or installation of components, accessories or controls.
- D. Additional jobsite visits if start-up cannot be completed due to no fault of Lennox.
- E. If EOC is scheduled and cannot be completed due to site readiness as noted above, return trip charges will apply.
- F. Any custom settings or site specific setups.

Lennox® Factory Commissioning: Emergence™ RTU/Constant Air Volume (CV) W/Humiditrol

The following listed items shall be performed by a Lennox® factory trained commissioning technician. These Integrity checks and equipment setup procedures shall be performed as part of the Lennox® factory commissioning service. Any installer related defects shall be noted and any factory defects shall be repaired/scheduled for repair. A report for each unit along with a summary report for the job site will be provided to the owner, installing contractor and general contractor upon completion. Lennox® factory commissioning is performed to ensure that the Lennox® RTU controller are configured to function per the project specifications.

Unit Check:

- A. Record RTU# - Unit Model # - Unit Serial # - Unit Catalog #.
- B. Check door alignment and adjust as necessary.
- C. Check unit's factory installed accessories for proper wiring and operation.
- D. Check and note installation of any Lennox® provided field installed accessories and verify per Lennox® and project specifications.

Electrical Systems Check:

- A. Check for and record proper incoming voltage – verify per Lennox® specifications.
- B. Check and record unit's control transformer(s) for proper secondary voltage and adjust if necessary.
- C. Verify class 2 controls wiring installation to terminal board of unit to include any project specific thermostat/communication/sensor wiring.

Prodigy™ / IMC Control System – the following options are dependent upon unit and accessories:

- A. Complete guided setup of the Prodigy™ controller.
- B. Record hardware and software versions of the controller and display panel.
- C. Check/Verify/Program Lennox® controller for proper operation in the local controller (IMC/PRODIGY™) per project specifications.
- D. Check/Verify/Program Lennox® controller for proper system mode of operation in the local controller IMC/PRODIGY™ per project specifications.
- E. Save profile and report of equipment with USB storage device for each Prodigy™ controller.
 - a. Check/Verify/Program zone temperature setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - b. Check/Verify/Program relative humidity setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - c. Check/Verify/Program IAQ setpoints in the local controller (IMC/PRODIGY™) per project specifications (if applicable).
 - d. Check/Verify/Program discharge air heating/cooling setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - e. Check/Verify/Program Fresh Air Tempering heating/cooling setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - f. Check/Verify/Program economizer control mode setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - g. Check/Verify/Program motorized damper min position setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - h. Check/Verify/Program Lennox® ERS control mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - i. Check/Verify/Program smoke shutdown mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - j. Check/Verify/Program power exhaust mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - k. Optional features. Only verified if called out in project specifications.

Control System/Sensor Check:**Thermostat/Thermostat Input (DDC type controls) Check:**

- A. Verify thermostat or DDC system make, model number.
- B. Verify occupancy input for thermostat mode of operation.
- C. Verify type of economizer enable control and verify correct termination has been accomplished (if applicable).

Sensor Check: (if applicable)**Zone Temperature Sensor Check – Lennox® non/communicating (LOCAL) style**

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify resistance value of temperature sensor if temperature reading is inaccurate.
- C. Check/Verify Lennox® sensor type matches job specification.

Relative Humidity Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify DC voltage feedback of sensor data if inaccurate reading suspected.
- C. Check/Verify Lennox® sensor type matches job specification.

Indoor Air Quality Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify DC voltage feedback of sensor data if inaccurate reading suspected.
- C. Check/Verify Lennox® sensor type matches job specification.
- D. Check/Verify IAQ Sensor is accurate.

Discharge air Temperature Sensor – Remote / Unit Mounted

- A. Check/Verify units utilizing discharge air control mode/ fresh air tempering control mode have a remote mounted discharge air temperature sensor.
- B. Check/Verify correct termination of sensor wires.
- C. Check/Verify sensor resistance values within Lennox® specification if temperature reading inaccurate.

Return Air/OA Sensors

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify sensor resistance values within Lennox® specification if temp reading inaccurate.

LON/BACnet module

- A. Check/Verify module integrity.
- B. Check/Verify control system wiring completion.

Supply Blower System Check:

- A. Check/Verify Lennox® local controller is configured for proper blower operation per project specifications.

Outdoor Air Accessory Check: (if applicable)

- A. Check operation of economizer or motorized outdoor air damper per project specifications.
- B. Check/Verify economizer sensors are present – Note deficiencies.
- C. Check/Record power exhaust type.
- D. Check damper position setting for manual OAD.

Cooling System Check:

- A. Operate stages of cooling and confirm correct staging sequence occurs based on the local controller (IMC/PRODIGY™) programmed parameters.

Heating System Check: (if applicable)

- A. Operate stages of heating and confirm correct staging sequence occurs based on the local controller (IMC/PRODIGY™) programmed parameters.

Post Operational Check:

- A. The post operational checks will be accomplished using the units primary means of control – otherwise software/local devices will be utilized to prove the units will function under normal operating conditions per project specifications.
- B. Perform cooling simulation test verify cooling stages per project specifications.
- C. Capture all report data and programming data via Lennox® USB or LConnection® software.
 - a. Perform heating simulation test verify heating stages per project specifications.
 - b. Perform ventilation operational test (if applicable) verify ventilation operation per project specifications.
 - c. Perform supply static setpoint / damper actuator control is fully functional.
 - d. Perform power exhaust operation and verify it is fully functional.
 - e. Optional features. Only verified if called out in project specifications.

Necessary requirements that must be complete prior to Factory Commissioning:

- A. Complete installation of rooftop unit and accessories per mechanical drawings and Lennox® Installation Instructions.
- B. Rooftop unit must be powered (24 hours before scheduled commissioning with no heating or cooling call).
- C. Unit's return air filters must be new and at least equivalent to factory provided filters.
 - a. Discharge air sensor shipped with unit must be installed past 1st supply air elbow from unit or at least 5 feet from unit.
 - b. Optional feature. Only verified if called out in project specifications.
- D. If gas heat units, gas supply pressure to units must be within Lennox specifications

The Factory Commissioning does not include the following and will adhere to:

- A. Any repairs or adjustment not listed above.
- B. Any sequence of operation unrelated to the Lennox® rooftop commissioning scope.
- C. Pulling of wires or installation of components/accessories/controls.
- D. Additional jobsite visits if commissioning cannot be completed due to no fault of Lennox®.
- E. If project specifications have not been presented the equipment will be configured for the Lennox® controller default parameters that will safely operate the equipment - Lennox® Commercial Controls Technical Support Team can be available to assist the contractors with technical questions throughout the project and after the commissioning by calling 1-800-308-5492 – for parts replacement and other inquiries please speak with your sales representative or call 1-800-4-Lennox.

Mechanical Contractor Equipment Startup

In regard to the startup of Lennox HVAC product, please see below excerpt from page 5 of our installations manual. This breaks down what is needed to keep from voiding the Lennox Limited Product Warranty if used during the construction phase.

Use of the unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit.

If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

- The vent hood must be installed per these installation instructions.
- A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed. (Unit may run in Prodigy Stand Alone mode)
- A pre-filter must be installed at the entry to the return air duct.
- The return air duct must be provided and sealed to the unit.
- Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- Air filters must be replaced and pre-filters must be removed upon construction completion.
- The input rate and temperature rise must be set per the unit rating plate.
- The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.
- The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.

Q4. How can I setup the unit to operate temporarily using the factory-installed return air sensor, prior to installation of the A2 zone sensor? (How do I set the unit up for temporary heating and cooling?)

You can also refer to the HowTo video at the following link for help:

<http://www.lennoxcommercial.com/landing/prodigy/how-to-videos.asp>

In order to temporarily operate the unit with the installed return air sensors, go to

SETUP > NETWORK INTERGRATION > RTU STANDALONE press the **SAVE** button. Room sensor back up mode will appear and use the adjust value buttons until you see **= RETURN AIR BACKUP** and press the **SAVE** button. Continue through and set the desired temperature set points, until the display returns to the **SETUP** menu.

Since the unit is now running off of the factory installed return air sensor we need to make the blower run continuous so it has a good sample of the air temperature. To do this go to;

SETTINGS > RTU OPTION> EDIT PARAMETER=154

Press and hold the adjust up button until a parameter **154** appears and press **SAVE**. Press the adjust up button until **=ON-CONTINUOUS 1** is displayed and press **SAVE**. Press the back button as many times as needed to return to the unit status screen. After a five minute time delay the unit will start operation based on the temperature read at the return air sensor.

(Edit parameter not required for blower operations change. It can be done through settings/blower menu. Also, blower should not have to run continuously when run in standalone)

CAUTION: The use of this unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit. If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

- ❑ The vent hood must be installed per these installation instructions.
- ❑ A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed.
- ❑ A pre-filter must be installed at the entry to the return air duct.
- ❑ The return air duct must be provided and sealed to the unit.
- ❑ Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- ❑ Air filters must be replaced and pre-filter must be removed upon construction completion.
- ❑ The input rate and temperature rise must be set per the unit rating plate.
- ❑ The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.
- ❑ The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.

Lennox® Emergence™ Prodigy 2.0: T&B Help – MSAV Units

507241-01
5/2014
Supersedes 4/2014

PRODIGY® 2.0**INSTALLATION AND SETUP GUIDE****Prodigy® 2.0 (M3 Unit Controller) Installation and Setup Guide**

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Commercial innovation at work



Introducing the Prodigy® Smartphone App

Intelligence in Control, now on the go!

Building upon our legacy of innovation, the Prodigy app represents the next generation of Lennox support tools. This unique tool is designed to help make technical support and troubleshooting easier for customers by putting the answers in the palm of their hands. Access to information while at a jobsite is the most critical and we now have an app for that!

Key Customer Benefits:

- Shorter service time with quick and easy access to information
- Enhanced troubleshooting with step by step guides
- Eliminate need for manuals at the jobsite

Feature Overview

- Watch "How to" videos on the go
- Access Prodigy related technical literature with one touch, with the ability to email links
- Alarm lookup tables with step by step troubleshooting guide
- Searchable list of configuration parameters with detailed description
- Frequently Asked Questions list

Availability

- Available **NOW** on iPhone and Android
- Free download from iTunes and Google Play
 - Keyword Search "Lennox Prodigy"



App Store



Google play

Notes for use by T&B Contractors:

- With the Prodigy 2.0 controller, MSAV is calibrated by target 100% CFM. The controller controls other CFM settings through internal calculations according to CFMs input in Setting menus.
- Any adjustments to Max. CFM airflow required by T&B must be performed by Blower sheave adjustment only and not in Prodigy controller settings or the VFD parameters.
Note: ECM/Direct Drive units will be adjusted by torque% on Prodigy controller.
- If adjustments are required to sheave, blower running amps should be re-verified.
- High and Low Minimum damper positions should be set by T&B on controller.
Note: Unit controller has factory defaults which may not be suitable in all applications.

To see current cfm settings follow these paths:

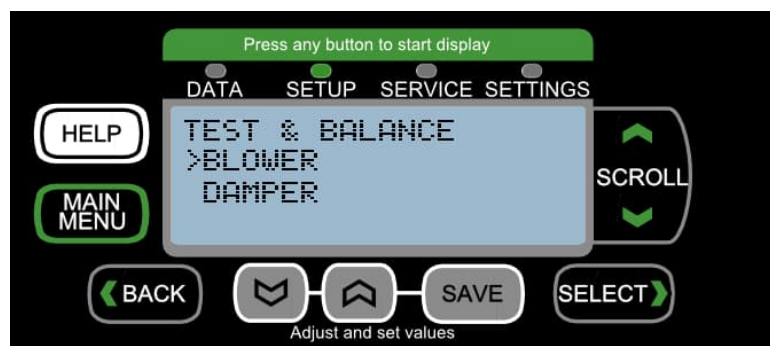
From the Main Menu, go to **SETUP** press **SELECT**.



Then **TEST & BALANCE**, press **SELECT**.



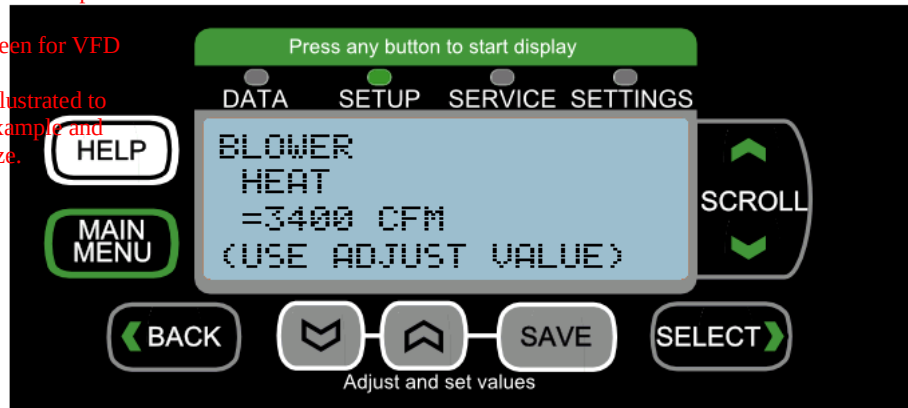
Scroll to **BLOWER** and press **SELECT**.



Heat CFM will show 1st then press SAVE.

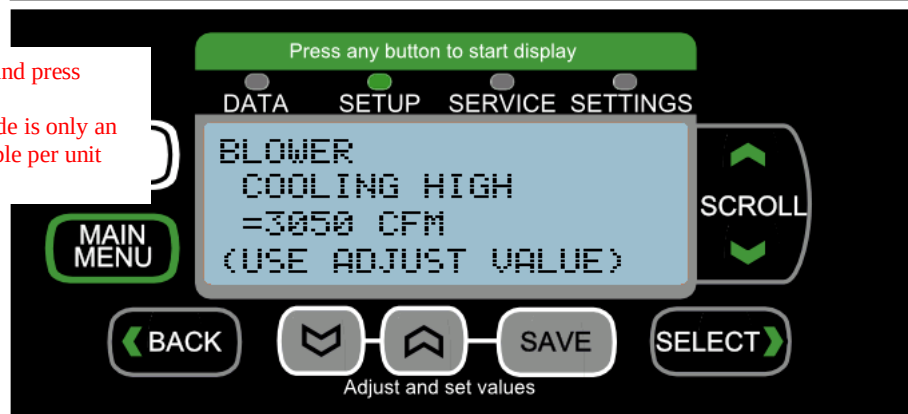
Note 1: this is the screen for VFD units.

Note 2: The setting illustrated to the right is only an example and adjustable per unit size.



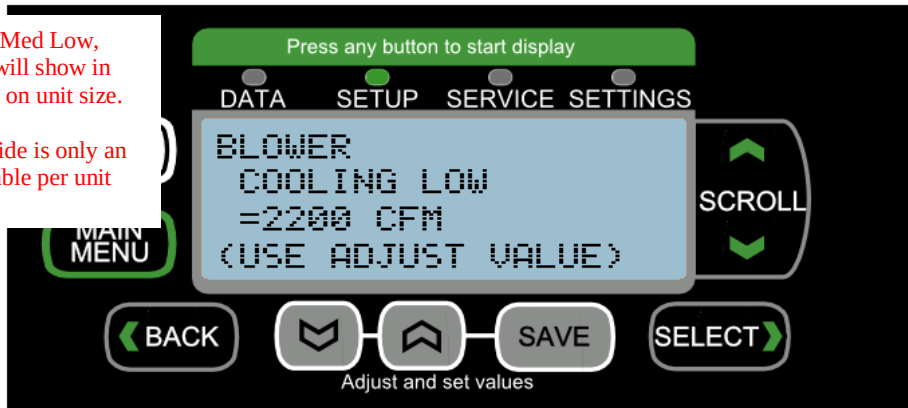
Cooling High CFM and press SAVE.

Note 1: Setting in slide is only an example and adjustable per unit size.



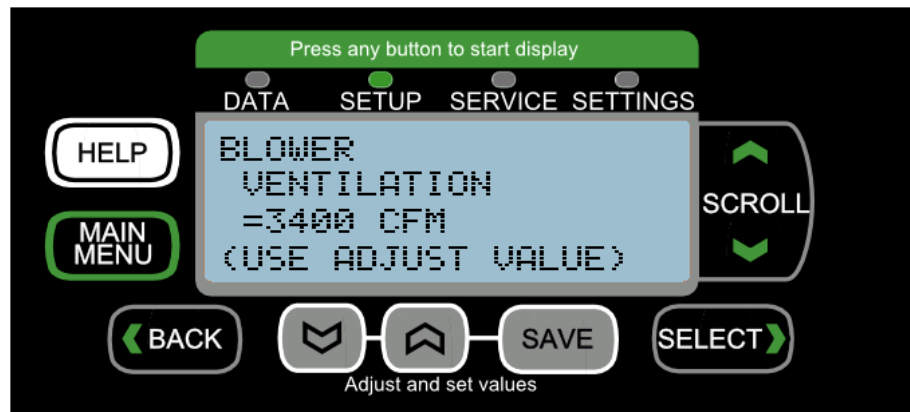
Cooling Med High, Med Low, Low CFM settings will show in sequence depending on unit size. Press SAVE again.

Note 1: Setting in slide is only an example and adjustable per unit size.



Ventilation CFM press SAVE.

Note 1: Setting in slide is only an example and adjustable per unit size.

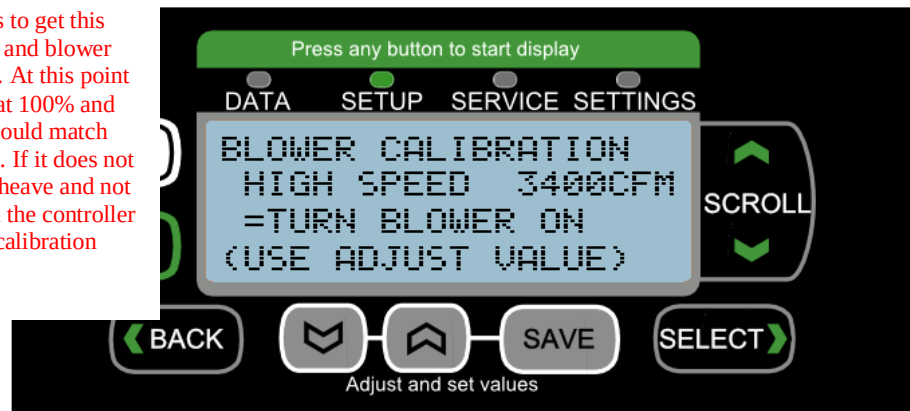


The blower calibration screen should come up next and show the highest blower CFM setting that was entered in this menu.

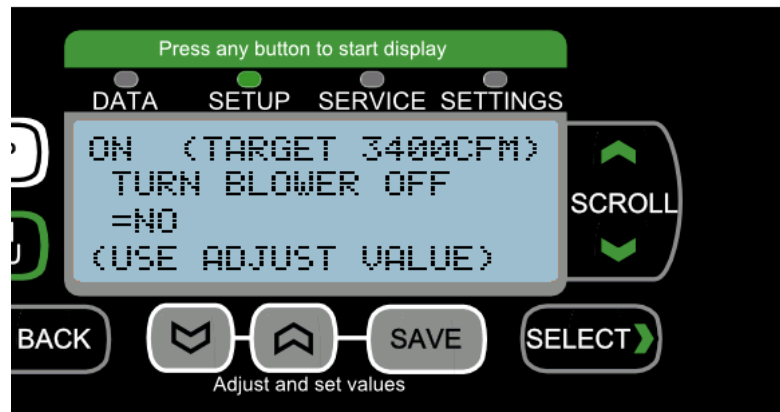
Note 1: The calibration or highest speed setting may be either High Cool or Heating CFM.



Use up/down arrows to get this screen. Press SAVE and blower will go to full speed. At this point the blower will run at 100% and measured airflow should match the calibration CFM. If it does not match, only adjust sheave and not any CFM settings in the controller until measured and calibration CFM match.



You can turn blower off after CFM measurement. If necessary, turn off unit and adjust pulley for target/calibration CFM. Retest until you have airflow set at highest CFM shown on screen. In this example, your Target would be 3400 CFM which is the highest CFM required at 100%. Once actual blower output is measured to be 3400 CFM during calibration menu, you can adjust the other CFM settings in the Prodigy directly, such as ventilation CFM.



Damper Menu Help

**All MSAV units have 2 Minimum damper positions for ventilation air.

For Damper Settings go
To SETTINGS>RTU
OPTIONS>DAMPER



*Depending on economizer type the menus will differ slightly.

Press SAVE



Press SAVE



*Depending on economizer type the menus will differ slightly.

Press SAVE.
Do not adjust.



Use UP/DOWN arrows to adjust Min. Damper For High Speed. Unit will run in high speed and damper will open to setting input, for testing, then press SAVE.



*Depending on economizer type the menus will differ slightly.

Use UP/DOWN arrows to adjust Min. Damper For Low Speed. Unit will run in low speed and damper will open to setting input, for testing, then press SAVE.

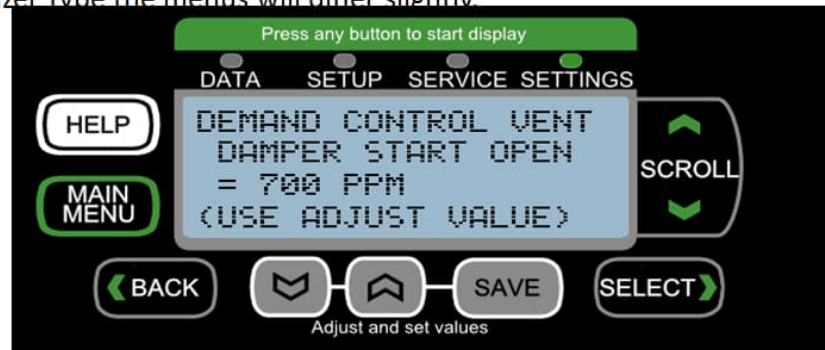


Next menus will be for DCV Options. Press SAVE.



*Depending on economizer type the menus will differ slightly.

The PPM settings should be already set, just press SAVE.



*Depending on economizer type the menus will differ slightly.

The Max Damper, FAH, and FAC options should be already set, just press SAVE until back to initial screen.



This should complete Final T&B setup required.

END OF SECTION 238119

SECTION 238126 - SPLIT-SYSTEM AIR-CONDITIONERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes split-system air-conditioning units consisting of separate evaporator-fan and compressor-condenser components.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated. Include rated capacities, operating characteristics, and furnished specialties and accessories. Include performance data in terms of capacities, outlet velocities, static pressures, sound power characteristics, motor requirements, and electrical characteristics.
 - 1. Include documentation on refrigerants, including printed statement that refrigerants are free of HCFCs.
- B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - 1. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2. Wiring Diagrams: For power, signal, and control wiring.
- C. Field quality-control reports.
- D. Operation and Maintenance Data: For split-system air-conditioning units to include in emergency, operation, and maintenance manuals.
- E. Recording(s) of training session(s).
- F. Warranty: Sample of special warranty.

1.3 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in Chicago Electric Code, by a qualified testing agency, and marked for intended location and application.
- B. ASHRAE Compliance:
 - 1. Fabricate and label refrigeration system to comply with ASHRAE 15, "Safety Standard for Refrigeration Systems."
 - 2. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1-latest edition, Section 4 - "Outdoor Air Quality," Section 5 - "Systems and Equipment," Section 6 - "Procedures," and Section 7 - "Construction and System Start-Up."
- C. ASHRAE/IESNA Compliance: Applicable requirements in ASHRAE/IESNA 90.1-most current version.

1.4 COORDINATION

- A. Ground-Mounted Remote Condensing Units: Coordinate sizes and locations of concrete bases with actual equipment provided.
- B. Roof-Mounted Remote Condensing Units: Coordinate sizes and locations of roof curbs, equipment supports, and roof penetrations with actual equipment provided.

1.5 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of split-system air-conditioning units that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period:
 - a. Compressor: Six years from date of Preliminary Acceptance or Substantial Completion.
 - b. Parts: One year from date of Preliminary Acceptance or Substantial Completion.

1.6 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Filters: One set for each rooftop unit.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - 1. Lennox HVAC
 - 2. Carrier
 - 3. TBD

2.2 INDOOR UNITS (5 TONS OR LESS)

- A. Wall-Mounted, Evaporator-Fan Components:
 - 1. Cabinet: Enameled steel with removable panels on front and ends in color selected by Architect, and discharge drain pans with drain connection.
 - 2. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and thermal-expansion valve. Comply with ARI 210/240.
 - 3. The unit must have low ambient cooling in temperatures as low as -22°F.
 - 4. Fan: Direct drive, centrifugal.
 - 5. Fan Motors:

- a. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements specified in Division 23.
 - b. Multi-tapped, multi-speed with internal thermal protection and permanent lubrication.
 - c. Enclosure Type: Totally enclosed, fan cooled.
 - d. NEMA Premium (TM) efficient motors as defined in NEMA MG 1.
 - e. Controllers, Electrical Devices, and Wiring: Comply with requirements for electrical devices and connections specified in Division 26 Sections.
 - f. Mount unit-mounted disconnect switches on exterior or interior of unit.
6. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1-latest edition.
7. Condensate Drain Pans:
- a. Fabricated with one or two percent slope in at least two planes to collect condensate from cooling coils (including coil piping connections, coil headers, and return bends) and humidifiers, and to direct water toward drain connection.
 - 1) Length: Extend drain pan downstream from leaving face to comply with ASHRAE 62.1-latest edition.
 - 2) Pan Depth: 1 inch, minimum.
 - b. Single-wall, non-corrosive non-condensing with moisture tight seal.
 - c. Drain Connection: Located at lowest point of pan and sized to prevent overflow. Terminate with threaded nipple on one end of pan.
 - 1) Minimum Connection Size: NPS 1.
8. Air Filtration Section:
- a. Comply with NFPA 90A.
 - b. Minimum Arrestance: According to ASHRAE 52.1 and MERV according to ASHRAE 52.2.
 - c. Filter-Holding Frames: Arranged for flat or angular orientation, with access doors on both sides of unit. Filters shall be removable from one side or lifted out from access plenum.
 - d. Disposable Panel Filters:
 - 1) Thickness: 1 inch.
 - 2) Frame: Galvanized steel, with metal grid on outlet side, steel rod grid on inlet side, and hinged; with pull and retaining handles.

2.3 OUTDOOR UNITS (5 TONS OR LESS)

A. Air-Cooled, Compressor-Condenser Components:

- 1. Casing: Steel, finished with baked enamel, in color selected by Architect, with removable panels for access to controls, weep holes for water drainage, and mounting holes in base. Provide brass service valves, fittings, and gage ports on exterior of casing.
- 2. Compressor: Hermetically sealed with crankcase heater and mounted on vibration isolation device. Compressor motor shall have thermal- and current-sensitive overload devices, start capacitor, relay, and contactor.
 - a. Compressor Type: Scroll.

- b. Two-speed compressor motor with manual-reset high-pressure switch and automatic-reset low-pressure switch.
 - c. Refrigerant Charge: R-407C or R-410A as scheduled.
 - d. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and liquid sub-cooler. Comply with ARI 210/240.
- 3. Fan: Aluminum-propeller type, directly connected to motor.
 - 4. Motor: Permanently lubricated, with integral thermal-overload protection.
 - 5. Low Ambient Kit: Permits operation down to -20 deg F.
 - 6. Mounting Base: Polyethylene.

2.4 ACCESSORIES

- A. Thermostat: Low voltage with sub-base to control compressor and evaporator fan.
 - 1. Compressor time delay.
 - 2. 24-hour time control of system stop and start.
 - 3. Liquid-crystal display indicating temperature, set-point temperature, time setting, operating mode, and fan speed.
 - 4. Fan-speed selection including auto setting.
- B. Automatic-reset timer to prevent rapid cycling of compressor.
- C. Refrigerant Line Kits: Soft-annealed copper suction and liquid lines factory cleaned, dried, pressurized, and sealed; factory-insulated suction line with flared fittings at both ends.
- D. Drain Hose: For condensate.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install units level and plumb.
- B. Install evaporator-fan components using manufacturer's standard mounting devices securely fastened to building structure.
- C. Install manufacturer provided condensate pump. Interlock condensate pump with operation of unit serving.
- D. Install ground-mounted, compressor-condenser components on 4-inches thick, reinforced concrete base that is 4 inches larger, on each side, than unit, with cast-in anchor-bolt inserts. Concrete, reinforcement, and formwork are specified in Division 03 Section "Cast-In-Place Concrete."
- E. Install roof-mounted, compressor-condenser components on equipment supports specified in Division 07 Section, "Roof Accessories." Anchor units to supports with removable, cadmium-plated fasteners.
- F. Install and connect pre-charged refrigerant pipes to component's quick-connect fittings. Install pipes to allow access to unit.

3.2 PIPE CONNECTIONS

- A. Pipe installation requirements are specified in other Division 22 Sections. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Pipe installations shall allow space for service and maintenance of system components.

3.3 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- B. Tests and Inspections:
 - 1. Leak Test: After installation, charge system and test for leaks. Repair leaks and repeat test until no leaks exist.
 - 2. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 3. Controls and Safeties: Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- C. Remove and replace malfunctioning units and retest as specified above.

3.4 STARTUP SERVICE

- A. Engage a factory-authorized service representative to perform startup service.
 - 1. Complete installation and startup checks according to manufacturer's written instructions.

3.5 TRAINING AND DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain unit as specified below:
 - 1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining system components. The training will occur after the startup report has been provided to the Owner and the trainer will provide four Installation and Operation manuals for the use of the Owner's personnel during training.
 - 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational troubleshooting. If the IOM does not include a written troubleshooting guide, one will be provided.
 - 3. Schedule training with Owner, through Architect, with at least seven days advance notice.
- B. Demonstrate proper operation of equipment to commissioning agent, if one, and designated Owner's personnel. The scope of the demonstration shall include functional performance requirements under local control as well as any commissioning requirements in Division 01 and 23.
- C. Video record the training session(s). The manufacturer may submit a standard training video or training CD for review as an alternate to recording of the training session. The standard video

must be reviewed and accepted by the Owner and Commissioning Authority, if one, for the alternate to be acceptable.

END OF SECTION 238126